Cambridge studies in advanced mathematics 14

# An introduction to the theory of the Riemann Zeta-Function

S.J. PATTERSON



CAMBRIDGE STUDIES IN ADVANCED MATHEMATICS 14

*Editorial Board*
D. J. H. Garling, T. tom Dieck, P. Walters

# An introduction to the theory of the Riemann zeta-function

*Already published*

1 W. M. L. Holcombe *Algebraic automata theory*
2 K. Petersen *Ergodic theory*
3 P. T. Johnstone *Stone spaces*
4 W. H. Schikhof *Ultrametric calculus*
5 J.-P. Kahane *Some random series of functions, 2nd edition*
6 H. Cohn *Introduction to the construction of class fields*
7 J. Lambek & P. J. Scott *Introduction to higher-order categorical logic*
8 H. Matsumura *Commutative ring theory*
9 C. B. Thomas *Characteristic classes and the cohomology of finite groups*
10 M. Aschbacher *Finite group theory*
11 J. L. Alperin *Local representation theory*
12 P. Koosis *The logarithmic integral I*
13 A. Pietsch *Eigenvalues and s-numbers*
14 S. J. Patterson *An introduction to the theory of the Riemann zeta-function*
15 H. J. Baues *Algebraic homotopy*
16 V. S. Varadarajan *Introduction to harmonic analysis on semisimple Lie groups*
17 W. Dicks & M. Dunwoody *Groups acting on graphs*
18 L. J. Corwin & F. P. Greenleaf *Representations of nilpotent Lie groups and their applications*
19 R. Fritsch & R. Piccinini *Cellular structures in topology*
20 H. Klingen *Introductory lectures on Siegel modular forms*
22 M. J. Collins *Representations and characters of finite groups*
24 H. Kunita *Stochastic flows and stochastic differential equations*
25 P. Wojtaszczyk *Banach spaces for analysts*
26 J. E. Gilbert & M. A. M. Murray *Clifford algebras and Dirac operators in harmonic analysis*
27 A. Frohlich & M. J. Taylor *Algebraic number theory*
28 K. Goebel & W. A. Kirk *Topics in metric fixed point theory*
29 J. F. Humphreys *Reflection groups and Coxeter groups*
30 D. J. Benson *Representations and cohomology I*
31 D. J. Benson *Representations and cohomology II*
32 C. Allday & V. Puppe *Cohomological methods in transformation groups*
33 C. Soulé et al *Lectures on Arakelov geometry*
34 A. Ambrosetti & G. Prodi *A primer of nonlinear analysis*
35 J. Palis & F. Takens *Hyperbolicity, stability and chaos at homoclinic bifurcations*
37 Y. Meyer *Wavelets and operators*
38 C. Weibel *An introduction to homological algebra*
39 W. Bruns & J. Herzog *Cohen–Macaulay rings*
40 V. P. Snaith *Explicit Brauer induction*
42 E. B. Davis *Spectral theory and differential operators*
45 R. G. Pinsky *Positive harmonic functions*

AN INTRODUCTION TO

# *the theory of the Riemann zeta-function*

S.J. PATTERSON
*Mathematisches Institut, Universität Göttingen*

CAMBRIDGE
UNIVERSITY PRESS

Published by the Press Syndicate of the University of Cambridge
The Pitt Building, Trumpington Street, Cambridge CB2 1RP
40 West 20th Street, New York, NY 10011-4211, USA
10 Stamford Road, Oakleigh, Melbourne 3166, Australia

©Cambridge University Press 1988

First published 1988
Reprinted 1989
First paperback edition 1995

Printed in Great Britain at the University Press, Cambridge

*British Library cataloguing in publication data*

Patterson, S.J.
An introduction to the theory of Riemann
zeta-function. – (Cambridge studies in
advanced mathematics; 14).

1 Calculus 2. Functions, Zeta
3. Riemann–Hilbert problems
I. Title
515.9'82 QA320

*Library of Congress cataloguing in publication data*

Patterson, S.J.
An introduction to the theory of the Riemann
zeta-function.

(Cambridge studies in advanced mathematics; 14)
Bibliography
Includes index.
1. Functions, Zeta. I. Title. II. Series.
QA246.P28 1987 515'.56 87-6652

ISBN 0 521 33535 3 hardback
ISBN 0 521 49905 4 paperback

TM

*IN MEMORIAM*

SAMUEL PATTERSON

# Contents


**Preface** xi

Notations, Cross-references, References xiii

**1 Historical introduction** 1

1.1. Definition of the zeta-function, 1.2. Euler's proof of the infinity of primes, 1.3. The values of $\zeta(2k)$, 1.4. Riemann and the zeta-function, 1.5. The Hadamard Product Formula, 1.6. The explicit formula, 1.7. The Prime Number Theorem, 1.8. The Riemann Hypothesis and the Prime Number Theorem, 1.9. The approximate functional equation, 1.10. Disclaimer, 1.11. Some number-theoretic functions.

Exercises 10

**2 The Poisson Summation Formula and the functional equation** 14

2.1. Summation formulae, 2.2. Poisson Summation Formula, 2.3. Mellin transform, 2.4. Cut-off functions, 2.5. Integral representation, 2.6. Local functional equation, 2.7. Analytic continuation and the functional equation, 2.8. Proofs, 2.9. Variant, 2.10. Further variant, 2.11. Estimates, 2.12. Growth in vertical strips, 2.13. First step towards the 'approximate functional equation', 2.14. Derivation of the Poisson Summation Formula from properties of the zeta-function, 2.15. Remarks.

Exercises 27

**3 The Hadamard Product Formula and 'explicit formulae' of prime number theory** 33

3.1. Hadamard Product Formula, 3.2. Expansion of the logarithmic derivative, 3.3. Estimates for the logarithmic derivative, 3.4. Estimate on the zeros, 3.5. Proofs, 3.6. Explicit formula, 3.7. Proof, 3.8. Explicit formula, approximate version, 3.9. Explicit formula, classical version, 3.10. Proof, 3.11. Closing remarks.

Exercises 46

**4 The zeros of the zeta-function and the Prime Number Theorem** 50

4.1. Introductory remarks, 4.2. Hadamard's zero-free region, 4.3. Prime Number Theorem, first version, 4.4. Proof of theorem 4.3, 4.5. Prime Number Theorem, second version, 4.6. The functions $N$ and $S$, 4.7. The basic estimate of $S$, 4.8. Further remarks, 4.9. Asymptotic estimate for $N$, 4.10. Proof of Theorem 4.9, 4.11. The function $N(\sigma, T)$, 4.12. The quadratic mean of the zeta-function, 4.13. Proof of Theorem 4.11, 4.14. Proof of Theorem 4.12, 4.15. General remarks on higher means.

Exercises 64

**5 The Riemann Hypothesis and the Lindelöf Hypothesis** 67

5.1. Formulation of the Riemann Hypothesis, 5.2. Formulation of the Lindelöf Hypothesis, 5.3. Equivalent forms of the Lindelöf Hypothesis, 5.4. The Lindelöf Hypothesis and the zeros, 5.5. The Riemann Hypothesis implies the Lindelöf Hypothesis, 5.6. Proof of Theorem 5.4, 5.7. A further consequence of the Lindelöf

Hypothesis, 5.8. The Riemann Hypothesis and prime numbers, 5.9. Proof of Theorem 5.8, 5.10. Reasons for believing the Riemann Hypothesis, 5.11. Zeta-functions of curves over finite fields,* 5.12. Places of a function-field,* 5.13. The zeta-function of a function-field,* 5.14. Analogue of the 'explicit formulae',* 5.15. Weil's criterion for the Riemann Hypothesis for the zeta-function of a function-field, 5.16. Weil's criterion for the Riemann Hypothesis,* 5.17. Weil's criterion and the Jacobian of the curve, *5.18. Significance of the previous section for the Riemann Hypothesis, 5.19. Concluding remarks on the Riemann Hypothesis.
Exercises 87

**6 The approximate functional equation 91**

6.1. Formulation of the approximate functional equation, 6.2. A special case, 6.3. Estimates of certain integrals, 6.4. Proof of Theorem 6.1, 6.5. Integral means, 6.6. The estimate $\zeta(1/2+it) = O(|t|^{1/6}\log|t|)$, 6.7. Van der Corput's Summation Formula, 6.8. Weyl's Lemma, 6.9. Exponential sums, 6.10. More estimates on exponential sums, 6.11. Proof of Theorem 6.6.
Exercises 108

**Appendices**

**1 Fourier theory 111**

A1.1. The Riemann–Lebesgue Lemma, A1.2. The variation of a function, A1.3. A lemma, A1.4. The Dirichlet kernel, A1.5. Representation of a periodic function by a Fourier series, A1.6. A useful estimate, A1.7. The $L^2$-theory, A1.8. Fourier integrals and the inversion formula, A1.9. Another useful estimate, A1.10. The Plancherel Theorem, A1.11. Convolution of periodic functions, A1.12. Convolution of integrable functions, A1.13. The Poisson Summation Formula.
Exercises 119

**2 The Mellin transform 122**

A2.1. Definition of the Mellin transform, A2.2. The inversion formula, A2.3. Convolution, A2.4. A variant, A2.5. Perron's Formula.
Exercises 124

**3 An estimate for certain integrals 127**

A3.1. An estimate for certain integrals, A3.2. Complements, A3.3. Further complements.
Exercises 128

**4 The gamma-function 130**

A4.1. Definition of the gamma-function, A4.2. The beta-function, A4.3. The infinite product, A4.4. The logarithmic derivative, A4.5. A further integral formula, A4.6. Stirling's Theorem, A4.7. The functional equation, A4.8. Stirling's Theorem.
Exercises 135

**5 Integral functions of finite order 138**

A5.1. Introductory remarks, A5.2. Jensen's Theorem, A5.3. Integral functions of finite order, A5.4. Integral functions of finite order without poles or zeros, A5.5. The general representation theorem, A5.6. A Maximum Principle for integral functions, A5.7. An extension of the previous section, A5.8. The Phragmén–Lindelöf Theorem, A5.9. A consequence of the Phragmén–Lindelöf Theorem, A5.10. Other convexity theorems.
Exercises 145

**6 Borel–Caratheodory Theorems** **146**

A6.1. General remarks, A6.2. The Borel–Caratheodory Theorem, A6.3. An estimate of the argument.

**7 Littlewood's Theorem** **148**

A7.1. Littlewood's Theorem, A7.2.A theorem of Backlund.

**Bibliography** **152**

**Index** **155**

# *Preface*

The theory of the Riemann zeta-function and its generalisations represent one of the most beautiful developments in mathematics. The Riemann zeta-function is a meromorphic function whose properties can on the one hand be investigated by the techniques of complex analysis, and on the other yield difficult theorems concerning the integers. It is this connection between the continuous and the discrete that is so wonderful. It is the purpose of this book to explain this connection through the example of the Riemann zeta-function. The more general zeta- and $L$-functions will not be introduced but the reader who has studied the techniques described here should have no trouble in seeing how they apply in a more general context. This book *is* intended as an introduction; there are developments in so many directions that could have been followed, but which, in the interest of conciseness, have not.

The Riemann zeta-function belongs to 'classical' mathematics, and the development of the theory here is essentially classical. Naturally this is not the first book on this subject, nor will it be the last. The aims of this book are rather different to other books, such as the classics of Landau (Landau (1)) and Titchmarsh (Titchmarsh (2)) to which the reader interested in the finer theory will turn sooner or later, or the historical treatise of Edwards (Edwards (1)), or Ivić's book (Ivić (1)) on the most delicate modern results. It is rather to bring out the central role of the Poisson Summation Formula, and, especially, of the 'explicit formulae of prime number theory' The knowledgeable reader will recognise the influence here of Weil. Consequently much has been omitted that might have been included simply because it did not fit into this scheme. From the point of view taken here the 'Riemann Hypothesis' takes on a central role in determining the direction of the investigations, and for this reason is discussed at much greater length than is usual. Indeed this book arose out of conversations, and then a lecture course, about the Riemann Hypothesis and Weil's point of view concerning it.

This book consists of a main part and a series of appendices which serve to summarise the background mathematics needed. In general the reader with a good undergraduate background in analysis and elementary number theory should be able to read the main part of the book without

having to refer to the appendices too often. The appendices contain that information which does not necessarily fit in undergraduate courses. There is one point in the discussion of the Riemann Hypothesis in Chapter 5 where rather more background is required, in this case from algebraic geometry. The sections in question are marked with asterisks, and can be omitted by a reader without this background. There are a large number of exercises, of very different levels of difficulty, so that the reader can participate more actively in the mathematics.

Finally I would like to thank J. Brüdern, M. Kneser and H. Matsumoto for having read the first manuscript and for their suggestions for improvements.

*Göttingen, 1986* *S.J. Patterson*

Note added at the second printing

The opportunity has been taken to correct a number of misprints. The author is particularly indebted to Professor O. McGuinness of Fordham University for communicating many of these to him.

## Notations

The notations which are used in this book are those which have become standard throughout mathematics in the last thirty years. In particular the usual language of set-theory is used – rather more consistently than is usual in analytic number-theory. Likewise functions are referred to by their name only and not through a representative variable; thus one writes $\zeta$ rather than $\zeta(s)$ when no particular argument is being discussed. If one wants to define a function by assigning particular values to an argument this will be done through the symbol $\mapsto$; thus if we define the function $f$ to be defined by (the assignation) $x \mapsto x^2 e^x$ then $f(3) = 9e^3$ etc.

As usual $\mathbf{Z}$ will denote the ring of rational integers, $\mathbf{Q}$ the field of rational numbers, $\mathbf{R}$ the field of real numbers and $\mathbf{C}$ the field of complex numbers. Also we shall denote by $\mathbf{N}$ the set of positive (natural) integers, not including 0, and by $\mathbf{N}_0$ the set of non-negative integers, that is, including 0. Thus $\mathbf{N}_0 = \mathbf{N} \cup \{0\}$. We shall write $F^\times$ for $F - \{0\}$, where $F$ is one of $\mathbf{Q}$, $\mathbf{R}$, or $\mathbf{C}$ and $-$ denotes the set-theoretic difference; $F^\times$ will be called the multiplicative group of $F$. Also we shall write $\mathbf{R}_+$ for $\{x \in \mathbf{R} : x \geqslant 0\}$ and $\mathbf{R}_+^\times$ for $\{x \in \mathbf{R} : x > 0\} = \mathbf{R}_+^\times \subset \mathbf{R}^\times$. If $z \in \mathbf{C}$ we shall write $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ for the real and imaginary parts of $z$ respectively.

Although it is not very important for the purposes of this book all integrations will be in the sense of Lebesgue; the Lebesgue spaces will be denoted as usual by $L^p(\cdot)$. We shall denote the closed (resp. open) interval with end-points $a$, $b$, $a < b$ by $[a, b]$ (resp. $]a, b[$) with the obvious extension to half-open intervals. We shall use the Landau $O$- and $o$-notation.

## Cross-references

Inside this book cross-references are given either to a section such as §3.2 (i.e. the second section of Chapter 3) or A3.2 (i.e. the second section of Appendix 3), or to a result such as Theorem 2.5. The latter is the only theorem in that section. To enable the reader to find such cross-references easily the running section number is given at the top of each page.

## References

References to the literature are given by the author's name or authors' names along with a reference number in brackets. These can then be found in the bibliography at the end of the book.

# 1

# *Historical introduction*

## 1.1

The Riemann zeta-function is that function defined on $\{s \in \mathbf{C}: \mathrm{Re}(s) > 1\}$ by the series

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}.$$

From Cauchy's integral test one knows that this series converges in the region described; moreover if $\mathrm{Re}(s) \geqslant a > 1$ then the series is dominated term by term by the absolutely convergent series

$$\sum_{n \geqslant 1} n^{-a}$$

so that, by Weierstrass' criterion it converges uniformly in $\{s \in \mathbf{C}: \mathrm{Re}(s) \geqslant a\}$ and therefore represents a holomorphic function in $\{s \in \mathbf{C}: \mathrm{Re}(s) > 1\}$.

There is a second representation of $\zeta$ due to Euler in 1749 which is perhaps more fundamental and which is the reason for the significance of the zeta-function. This is

$$\zeta(s) = \prod_{p} (1 - p^{-s})^{-1}$$

where the product is taken over all *prime numbers* $p$. This is called the Euler Product representation of the zeta-function and gives analytic expression to the fundamental theorem of arithmetic.

We shall prove now that the representation holds, and at the same time explicate the nature of the convergence of the product. Let $S$ be a finite set of primes and let $q(S) = \prod_{p \in S} p$. We shall prove the identity

$$\prod_{p \in S} (1 - p^{-s}) \zeta(s) = \sum_{m:(m, q(S))=1} m^{-s}.$$

To see this we argue by induction on the number of elements in $S$. By convention we shall take any product over the empty set to be 1, so that the formula is valid when $S = \varnothing$. We now have to show that if the formula is valid for $S$, and if $p'$ is a prime not in $S$ then the formula holds for $S \cup \{p'\}$. Any number $m$ can be written as $(p')^k m_1$ for some $k \geqslant 0$ where $m_1$ is coprime to $p'$. This representation is unique, and naturally we can generate all natural numbers in this way. If $m$ were coprime to $q(S)$ then so would $m_1$ be, and again all numbers coprime to $q(S)$ can be so generated The two conditions on $m_1$, that it should be coprime to $q(S)$ and to $p$ can

be paraphrased by saying that it should be coprime to $q(S)p' = q(S \cup \{p'\})$. Thus the right-hand side of the equation above can be rewritten as

$$\sum_{m_1:(m_1, q(S\cup\{p'\}))=1} \sum_{k\geqslant 0} (p')^{-ks} m_1^{-s}.$$

The sum over $k$ can be evaluated as it is a geometric series; it is $(1 - p'^{-s})^{-1}$. We then only need multiply both sides of the resulting equation by $(1 - p'^{-s})$ to obtain the identify for $S \cup \{p'\}$.

Now let $S = S(X) = \{p | p \text{ prime}, p \leqslant X\}$ where $X$ is a positive real number. The only number less than $X$ which is coprime to all the primes less than $X$ is 1 so that

$$\{m | (m, q(S(X))) = 1\} \subset \{1\} \cup \{n | n > X\}.$$

Consequently

$$\left| \prod_{p\leqslant X} (1 - p^{-s})\zeta(s) - 1 \right| \leqslant \sum_{m>X} m^{-\mathrm{Re}(s)}.$$

Since the series defining the zeta-function converges the right-hand side converges to zero as $X \to \infty$; indeed by the Cauchy integral test one can check that the right-hand side is less than $\{\mathrm{Re}(s-1)\}^{-1} X^{-\mathrm{Re}(s-1)}$. From this inequality it follows that $\zeta$ has no zeros in its domain of definition (to date) and that the family $\prod_{p\leqslant X} (1 - p^{-s})^{-1}$ converges uniformly on compact sets to $\zeta(s)$.

## 1.2

Euler made a rather simple but pregnant observation based on the Euler Product representation of the zeta-function. Suppose the set of primes were finite; then $\zeta(s)$ would remain finite as $s \to 1$ through real values greater than 1. It is, however, easy to deduce from the divergence of $\sum_{n\geqslant 1} n^{-1}$ that this is not the case. Thus there are an infinity of primes. Note that this argument is quite different from the classical one in Euclid: if $p_1, \ldots, p_r$ were the set of primes then $1 + p_1 \cdots p_r$ would be a number not divisible by any prime, a contradiction.

Euler's proof was acknowledged by Dirichlet as one of the inspirations behind his proof that in any arithmetic series of the form $am + b$, $m \geqslant 1$ and $(a, b) = 1$ there are infinitely many primes; likewise it inspired Riemann's work which we shall discuss below.

## 1.3

One of Euler's most spectacular discoveries was the formula

$$1/1^2 + 1/2^2 + 1/3^2 + \cdots = \pi^2/6$$

which appears to have motivated his interest in series of the form of the zeta-function. Very many proofs of this evaluation of $\zeta(2)$ are known; a very simple one consists in observing that the Fourier series of the function $f(x) = x$ for $x \in [0, 1]$ (which we regard as an element of $L^2(\mathbf{R}/\mathbf{Z})$) is

$$2^{-1} - \sum_{n \neq 0} (2\pi i n)^{-1} e^{2\pi i n x}.$$

The norm of this is $4^{-1} + (2\pi^2)^{-1} \sum_{n>1} n^{-2}$ by Parseval's Theorem, whereas the norm of $f$ is

$$\int_0^1 f(x)^2 \, dx = 3^{-1}.$$

Comparing these we obtain Euler's Formula.

Euler actually proved much more, namely that $\zeta(2k)$ for positive integral $k$ is of the form (rational number) $\times \pi^{2k}$. He also made very serious attempts to evaluate $\zeta(3), \zeta(5), \ldots$ which he computed to 15 places of decimals along with the corresponding quotients $\zeta(2k+1)\pi^{-(2k+1)}$. All that is known about this sequence (and the closely related 'Euler's Constant' – the definition is recalled in A4.3) is the surprisingly recent result of Apéry (1978) that $\zeta(3)$ is irrational – for a very readable account of this discovery see van der Poorten (1).

The precise form of Euler's Theorem is that

$$\zeta(2m) = 2^{2m-1} b_{2m-1} (-1)^{m-1} \pi^{2m} / (2m-1)!$$

where the sequence $b_k$ of rational numbers is defined by the Laurent expansion

$$(e^y - 1)^{-1} = y^{-1} - 2^{-1} + \sum_{n \geq 1} b_n y^n / n!$$

so that $b_{2j} = 0$ and $B_j = 2j(-1)^{j+1} b_{2j-1}$ is the $j$th Bernoulli number. One has

$$B_1 = 1/6, \quad B_2 = 1/30, \quad B_3 = 1/42, \quad B_4 = 1/30, \quad B_5 = 5/66, \ldots$$

Euler's Theorem is then equivalent to the partial fractions expansion of cot,

$$\pi \cot \pi z = z^{-1} + \sum_{n>0} 2z/(z^2 - n^2)$$

since if $|z| < 1$

$$\sum_{n>0} 2z/(z^2 - n^2) = \sum_{n>0} 2z \left( \sum_{m \geq 0} - z^{2m}/n^{2m+2} \right) = \sum_{m \geq 0} - 2z^{2m+1} \zeta(2m+2)$$

and

$$\pi \cot \pi z = \pi i \{1 + 2/(e^{2\pi i z} - 1)\} = z^{-1} + \sum_{n \geq 1} 2\pi i b_n (\pi i z)^n / n!$$

For two proofs of this formula see Exercises 1.12 and 1.13.

These evaluations of $\zeta(2k)$ will not appear again in this book but they play a fundamental role in the theory of $p$-adic zeta-functions and Iwasawa theory (which will put in a fleeting appearance in § 5.18) – the reader interested in following this further is referred to Coates(1), Iwasawa(1).

## 1.4

Riemann belonged to the generation which followed Cauchy's discovery of the basic theorems of complex analysis and he himself contributed substantially to the formalism of this theory. In modern terms he took a 'sheaf-theoretic' point of view which apparently caused his nineteenth century successors, such as Klein, a great deal of trouble but which today seems quite 'natural'. The development of complex analysis was a central preoccupation of Riemann's and so it comes as no surprise that from the beginning Riemann considered the zeta-function as an analytic function. This is one of the central ideas in the one paper that Riemann wrote on number-theory, 'Über die Anzahl der Primzahlen unter einer gegebener Größe'; this short paper – it is only ten pages long – is one of the most seminal in mathematics. In keeping with his general philosophy Riemann first showed that the zeta-function had an analytic continuation to the complex plane as a meromorphic function which has only one singularity, a simple pole of residue 1 at 1. He also showed that the zeta-function satisfied a functional equation of the form

$$\zeta(1-s) = \gamma(s)\zeta(s)$$

where

$$\gamma(s) = \pi^{1/2-s}\Gamma(s/2)/\Gamma((1-s)/2)$$

and $\Gamma$ is the Euler gamma-function (see Appendix 4).

Riemann's significant contribution here was his consideration of the zeta-function as an analytic function. In a certain sense the functional equation was already known, at least in principle. Euler had already found in 1749 by arguments involving divergent series something that is in essence the functional equation. (It is worth remarking here that complex analysis gives functions defined by, say, series a well-defined meaning outside their domain of convergence and thereby to certain 'divergent series'.) Around the time that Riemann's paper was published Malmstén, Schlömilch and Clausen found results that are very closely related to the functional equation but which refer to functions defined originally in $\{s \in \mathbf{C}: \mathrm{Re}(s) > 0\}$ so that no question of analytic continuation arises. The results of these authors do not bring out the significance of the functional equations that they prove, and only Clausen indicates any familiarity with Cauchy's work so that the results of these papers appear more as curiosities. It is very

illuminating in this connection to read these papers together with that of Riemann in order to get an impression of the intellectual background against which Riemann worked, and how much more deeply he penetrated than his contemporaries.

In his last book, Hardy (1), Hardy gives a very interesting discussion of Euler's approach to the functional equation in §§2.2 and 2.3. the reader interested in the work of Riemann's contemporaries is referred to Clausen (1), Malmstén (1) and Schlömilch (1).

## 1.5

After he gave two proofs of the functional equation Riemann deduced, without giving much in the way of details, that the zeta-function can be represented as follows

$$s(1-s)\Gamma(s/2)\zeta(s) = -e^{bs}\prod_{\rho}(1-s/\rho)e^{s/\rho}$$

where $\rho$ runs through a set $Z$ contained in $\{s\in\mathbf{C}: 0 \leqslant \mathrm{Re}(s) \leqslant 1\}$. These are the zeros of the zeta-function; let us note that from the functional equation and $\zeta(\bar{s}) = \overline{\zeta(s)}$ one has that if $\rho\in Z$ then $\bar{\rho}$, $1-\rho$, $1-\bar{\rho}\in Z$. Let now

$$N(T) = \mathrm{Card}\{\rho\in Z: 0 < \mathrm{Im}(\rho) < T\}.$$

Riemann also claimed that $N(T)$ was asymptotically equal to $(T/2\pi)\log(T/2\pi) - (T/2\pi)$. The constant $b$ in the product is given by $b = \log 2\pi - 1 - \gamma/2$ where $\gamma$ is Euler's Constant $\gamma = -\Gamma'(1) = 0.5772157\ldots$

Nowadays it is not too difficult to prove this; it follows from general principles of complex analysis that functions satisfying certain growth restrictions can be developed as infinite products. The estimate on the number of zeros is somewhat more subtle and uses properties of Dirichlet series as well as the functional equation. Note that one knows that the zeta-function is non-zero in $\{s: \mathrm{Re}(s) > 1\}$ by the Euler Product. In the course of his investigations Riemann was led to suspect that

$$Z \subset \{s: \mathrm{Re}(s) = 1/2\},$$

the Riemann Hypothesis which has been the central goal of research to the present day. Although no proof has yet appeared various weak forms of this conjecture, and in other contexts, analogues of it have been proved which have been of considerable significance.

## 1.6

Riemann's decisive idea was to combine the product representation of the previous section with the Euler Product On taking the logarithms and

then the Mellin transform Riemann found that the number of numbers less than $X$ of the form $p^k$ where $p$ is a prime and $k > 1$ (with $X$ itself being counted with weight 1/2 if it happens to be of this form) is

$$Li(X) - \sum_{\rho:\mathrm{Im}(\rho)>0} \{Li(X^{\rho}) + Li(X^{1-\rho})\} + \int_X^{\infty} \frac{1}{\xi^2 - 1} \frac{d\xi}{\log \xi} - \log 2.$$

Here $Li$ is defined by

$$Li(u) = \lim_{\varepsilon \to 0} \int_0^{1-\varepsilon} + \int_{1+\varepsilon}^{u} \frac{dv}{\log v} \qquad (u > 1).$$

If we denote the number above by $f(X)$ then since

$$f(X) = \sum_{k \geqslant 1} \pi(X^{1/k})$$

where $\pi(y)$ is the number of primes less than $y$ one deduces that

$$\pi(X) = \sum_{m>0} \mu(m) f(X^{1/m})$$

where $\mu$ is the Möbius function – see § 1.11. In conjunction with the formula above this gives a formula for the number of primes less than $X$. This remarkable formula is what is referred to in the title of Riemann's paper. Note that the summation over the zeros of the zeta-function in the expression above is conditionally but not absolutely convergent; for a discussion of this see § 3.9.

## 1.7

Riemann's formula has to be seen in conjunction with one of the outstanding problems of nineteenth century mathematics. At the end of the eighteenth century Gauss and Legendre on the basis of extensive computations conjectured that for large $X$

$$\pi(X) \sim X/\log X.$$

In view of the elementary properties of $Li$ this can also be expressed as

$$\pi(X) \sim Li(X).$$

This conjecture, now a theorem, is called the 'Prime Number Theorem'. This appeared unapproachable at the time but in the 1850s Chebychev began to obtain significant results towards this conjecture. He was able to show that for sufficiently large $X$ one had

$$0.92129\cdots X/\log X < \pi(X) < 1.0555\cdots X/\log X$$

and that if $\pi(X) \sim cX/\log X$ then $c = 1$. The methods that he used were

fairly elementary, namely that

$$\log n! = \sum_{p^k \leqslant n} [n/p^k] \log p$$

(which is quite easy to prove) in conjunction with Stirling's Formula. In the formula above $p^k$ represents a prime power and $[x]$ for $x \in \mathbf{R}$ is that integer for which $x - [x] \in [0, 1[$.

These methods, suggestive as they are, do not suffice to prove the conjecture of Gauss and Legendre, although they do make it much more credible. It appears that Riemann knew of Chebychev's work and certainly Riemann's ideas played the central role in the proof of the conjecture.

In the later years of the nineteenth century several mathematicians – such as Mertens, von Mangoldt, Hadamard and de la Vallée-Poussin – had developed Riemann's work to the point that de la Vallée-Poussin and Hadamard independently of one another in 1896 proved the Prime Number Theorem.

It is noteworthy that Riemann himself never considered the application of his explicit formula to determine the asymptotic behaviour of $\pi(X)$, although it is clear that this formula has the form of an asymptotic expansion with a leading term exactly of the form that one would expect from the conjecture of Gauss and Legendre. Indeed, to prove the Prime Number Theorem it was necessary to prove a very weak form of the Riemann Hypothesis, namely that there are no zeros of the zeta-function on the line $\{s: \mathrm{Re}(s) = 1\}$ and there is nothing in Riemann's work to indicate that he had proved anything of this kind. Nevertheless it does appear that Riemann was aware of Chebychev's work and of the relevance of his own ideas for he sent (or at least intended to send) a copy of his paper to Chebychev – see Edwards (1), p. 4.

The proof of the Prime Number Theorem represents one of the major achievements of nineteenth century mathematics and has given birth to that area of mathematics known as analytic number-theory

## 1.8

In this connection it is worth noting that the Riemann Hypothesis has a number-theoretic form. If it is true then

$$\pi(X) = Li(X) + O(X^{1/2} \log X)$$

as $X \to \infty$. As we shall see in § 5.8 the exponent 1/2 in the remainder cannot be replaced by a smaller number; thus this statement means that the approximation of $\pi(X)$ by $Li(X)$ would be optimal. On the other hand this estimate for $\pi$ implies the Riemann Hypothesis Thus one way of

regarding the Riemann Hypothesis is that it says that the error term in the Prime Number Theorem should be the smallest conceivable.

## 1.9

Since the beginning of the century a great deal has been added to the theory of the Riemann zeta-function, and wide classes of other 'zeta-functions' have been introduced and studied by similar means. In this book we shall restrict ourselves to the Riemann zeta-function (except for the discussion in §§5.11–5.18) in order to keep the subject matter within bounds.

In this century there have been considerable advances made in that part of the theory of the zeta-function associated with various estimates for different measures of the size of the zeta-function, especially in the critical strip $\{s: 0 \leqslant \mathrm{Re}(s) \leqslant 1\}$; these revolve around the 'approximate functional equation' discovered by Hardy and Littlewood in the early 1920s. This gives an approximate representation of $\zeta(s)$ in terms of two *finite* sums of the form $\sum_{n \leqslant X} n^{-s}$ and $\sum_{m \leqslant Y} m^{-(1-s)}$. There were of course previous such techniques for studying the zeta-function but the approximate functional equation is the most efficient and most useful.

The approximate functional equation is only significant in conjunction with further techniques, some of which are very general and others of which are more restricted. Particularly significant are those estimates derived from estimates for exponential sums, such as those of Weyl–Hardy–Littlewood–Vinogradov–van der Corput. In this book we shall not treat these in detail as the arguments are rather involved but in Chapter 6 a derivation of the approximate functional equation will be given along with some relatively easy applications.

Most of this chapter has dealt with the achievements of Riemann and it remains astonishing just how much Riemann's ideas have continued to guide research into the theory of the zeta-function. It is therefore of interest to note that some years after Hardy and Littlewood had found the approximate functional equation Bessel-Hagen found a manuscript in Riemann's *Nachlass*. He informed Siegel of this who analysed it in detail (Siegel (1)) and showed that the formulae noted down without explanation by Riemann indicated that Riemann had not only found the approximate functional equation but also a much more accurate 'asymptotic' form.

One can speculate a great deal concerning how much Riemann actually knew about the zeta-function and to what extent he had proved all that he had claimed. It seems unlikely that there will be much new evidence on this matter from, say, a new study of Riemann's *Nachlass* although one cannot rule this out. It is thus scarcely fruitful to let one's imagination

run too freely; from a historical point of view it is Riemann's stimulus for so many later developments that is significant.

## 1.10

In this introduction the emphasis has been on a 'developmental' history without much detail or historical apparatus. The reader interested in this is referred to Edwards' book (Edwards (1)) and will also find the remarks in Siegel's paper (Siegel (1)) very illuminating.

## 1.11

Before closing this chapter it will be useful to add a few definitions of standard number-theoretic functions which appear often in the study of the zeta-function. First let $\mu$ be the Möbius function; $\mu(m) = 0$ if $m$ is not square-free and $\mu(m) = (-1)^k$ if $m$ is square-free with $k$ prime factors.

Let

$$d_k(m) = \text{Card}\{(d_1, \ldots, d_k) \in \mathbf{N}^k : d_1 \cdots d_k = m\}$$

and for $t \in \mathbf{C}$ let

$$\sigma_t(m) = \sum_{d|m} d^t.$$

Both of these last two families of functions are multiplicative.

Let finally $\varphi$ be the Euler totient function – that is

$$\varphi(m) = \text{Card}\,((\mathbf{Z}/m\mathbf{Z})^\times).$$

The basic properties of such functions as these can be found in Hardy & Wright (1).

Finally let $\Lambda$ denote the von Mangoldt function which is defined as follows

$$\begin{aligned} \Lambda(n) &= \log p \qquad \text{if } n = p^e, \quad e \in \mathbf{N}, p \text{ prime} \\ &= 0 \qquad\quad \text{otherwise.} \end{aligned}$$

The significance of this function is that for $\text{Re}(s) > 1$ one has

$$-\zeta'(s)/\zeta(s) = \sum_{n \geqslant 1} \Lambda(n) n^{-s}.$$

To prove this we differentiate logarithmically Euler's representation

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}$$

to obtain

$$-\zeta'(s)/\zeta(s) = \sum_p p^{-s}(\log p)/(1 - p^{-s}) = \sum_p \sum_{e \geqslant 1} p^{-es} \log p = \sum_{n \geqslant 1} \Lambda(n) n^{-s}$$

## Exercises

1.1 Show that for $\mathrm{Re}(s) > 1$

$$\left| \zeta(s) - \prod_{\substack{p \leqslant X \\ p \text{ prime}}} (1 - p^{-s})^{-1} \right| \leqslant X^{1-\mathrm{Re}(s)}/\mathrm{Re}(s) - 1.$$

1.2 Let $m > 1$; show that for $\mathrm{Re}(s) \geqslant 1$

$$\begin{aligned}(1 - m^{1-s})\zeta(s) = 1^{-s} + 2^{-s} + \cdots + (m-1)^{-s} - (m-1)m^{-s} \\ + (m+1)^{-s} + \cdots + (2m-1)^{-s} - (m-1)(2m)^{-s} \\ + \cdots\end{aligned}$$

and deduce that $\zeta$ has an analytic continuation as a meromorphic function to $\{s \in \mathbf{C}: \mathrm{Re}(s) > 0\}$ with at most one simple pole situated at $s = 1$.

1.3 Show that if $\mathrm{Re}(s) > 1$

$$\zeta(s)^{-1} = \sum_{m \geqslant 1} \mu(m) m^{-s}$$

$$\zeta(s)^k = \sum_{m \geqslant 1} d_k(m) m^{-s}$$

$$\zeta(s)/\zeta(2s) = \sum_{m \geqslant 1} |\mu(m)| m^{-s}$$

$$\zeta(s)^3/\zeta(2s) = \sum_{m \geqslant 1} d_2(m^2) m^{-s}$$

$$\zeta(s)^4/\zeta(2s) = \sum_{m \geqslant 1} d_2(m)^2 m^{-s}$$

$$\zeta(s)/\zeta(s+1) = \sum_{m \geqslant 1} \varphi(m) m^{-s-1}$$

$$\zeta(s)\zeta(s-a) = \sum_{m \geqslant 1} \sigma_a(m) m^{-s} \qquad (\mathrm{Re}(s-a) > 1)$$

$$\zeta(s)\zeta(s-a)\zeta(s-b)\zeta(s-a-b)/\zeta(2s-a-b) = \sum_{m \geqslant 1} \sigma_a(m)\sigma_b(m) m^{-s}$$

$$(\mathrm{Re}(s-a) > 1, \mathrm{Re}(s-b) > 1, \mathrm{Re}(s-a-b) > 1).$$

1.4 Show that for $k_1, k_2 \in \mathbf{N}$ one has

$$d_{k_1+k_2}(n) = \sum_{d|n} d_{k_1}(d) d_{k_2}(n/d).$$

1.5 Show that if $p$ is a prime and $k \in \mathbf{N}$ then $d_2(p^k) = k + 1$ and deduce that for any $\varepsilon > 0$ there exists $n_0(\varepsilon)$ so that if $p^k \geqslant n_0(\varepsilon)$ then

$$d_2(p^k) \leqslant p^{k\varepsilon}.$$

Conclude that there exists $c(\varepsilon) > 0$ so that

$$d_2(n) < c(\varepsilon) n^{\varepsilon}.$$

1.6 By refining the argument of the previous exercise show that $d_2(n)/n^{1/3}$ takes its maximum value when $n = 2520$ and deduce that

$$d(n) \leqslant 3.527 \cdots n^{1/3}.$$

1.7 Show that for any $\varepsilon > 0$ and $k \in \mathbf{N}$ there exists $c(\varepsilon, k) > 0$ so that

$$d_k(n) < c(\varepsilon, k) n^{\varepsilon}.$$

1.8 Show that

$$\sigma_{-a}(n) = n^{-a} \sigma_a(n).$$

1.9 For $a$ with $\mathrm{Re}(a) \leqslant 0$ show that

$$|\sigma_a(n)| \leqslant d(n).$$

Deduce that for any $\varepsilon > 0$ there exists $c(\varepsilon) > 0$ so that for arbitrary $a$

$$|\sigma_a(n)| < c(\varepsilon) n^{\mathrm{Max}(0, \mathrm{Re}(a)) + \varepsilon}.$$

1.10 Show that for $k \in \mathbf{N}$ one has

$$\zeta(1 - 2k) = -b_{2k-1},$$

and

$$\zeta(-2k) = 0.$$

1.11 Show that as $k \to \infty$

$$(-1)^{k-1} b_{2k-1} (2\pi)^{2k-1} / (2k-1)! \to \pi^{-1}.$$

1.12 Show that

$$f(z) = z^{-1} + \sum_{n>0} 2z/(z^2 - n^2)$$

can be expressed as

$$f(z) = \lim_{N \to \infty} \sum_{n=-N}^{N} (z - n)^{-1}.$$

Deduce that

$$f(z+1) = f(z).$$

Show also that if $|\mathrm{Im}(z)| > 4$ then $|f(z)| < 5$. Conclude that

$$f(z) - \pi \cot \pi z$$

is a constant. Finally show that this constant is zero.

1 13 Show that with $f$ as in Exercise 1.12

$$f(z)^2 = z^{-2} + \sum_{n>0} \{4z^2(z^2 - n^2)^{-2} + 4(z^2 - n^2)^{-1}\}$$

$$+ 4z^2 \sum_{\substack{m,n>0 \\ m \neq n}} (z^2 - n^2)^{-1} (z^2 - m^2)^{-1}$$

$$= z^{-2} + \sum_{n>0} \{4z^2(z^2 - n^2)^{-2} + 4(z^2 - n^2)^{-1}\}$$

$$-8z^2 \sum_{n>0} (z^2-n^2)^{-1} \sum_{\substack{m>0 \\ m\neq n}} (m^2-n^2)^{-1}.$$

Show that

$$\sum_{\substack{m>0 \\ m\neq n}} (m^2-n^2)^{-1} = 3/4n^2$$

and deduce that

$$f'(z) = -f(z)^2 - 6\sum_{n\geqslant 1} n^{-2}.$$

Finally prove that

$$f(z) = \pi \cot \pi z$$

and

$$\sum_{n>0} n^{-2} = \pi^2/6.$$

(This argument, which avoids the techniques of complex analysis and Fourier theory, is due to Eisenstein.)

1.14 Prove that the power of the prime number $p$ dividing $n!$ is

$$\sum_{k>0} [n/p^k]$$

and deduce that

$$\log n! = \sum_{p^k\leqslant n} [n/p^k] \log p.$$

1.15 From Stirling's Formula one has

$$\log n! = (n+1/2)\log n - n + O(1).$$

Use this and Exercise 1.14 to deduce that

$$\pi(X) = O(X/\log X)$$

and then that

$$\sum_{n\leqslant X} \Lambda(n) n^{-1} \sim \log X.$$

Conclude also that there exists $c>0$ such that for $X \geqslant 2$

$$\pi(X) > cX/\log X.$$

1.16 Let $\varphi$ be the Euler totient function. Show that if $\mathrm{Re}(s+t)>1$

$$\sum_{n\geqslant 1} \varphi(n)^{-s} n^{-t} = \prod_p \{1 + p^s/(p^{s+t}-1)(p-1)^s\}$$

where the product is taken over all primes.

1.17 Show that for $\mathrm{Re}(s) > 1$, $\mathrm{Re}(s-a) > 1$ and $k \in \mathbf{N}$

$$\sum_{n \geqslant 1} \sigma_a(nk) n^{-s} = \zeta(s)\zeta(s-a) \times \prod_{p/k} (1 - p^{-(s-a)} + p^{(l_p+1)a} - p^{(l_p+1)a-s})(1-p^a)^{-1}$$

where $l_p$ is the largest integer $l$ so that $p^l | k$.

1.18 Show that

$$\sum_{d^2|n} \sigma_a(n/d^2)\sigma_{-a}(n/d^2) = \sum_{d|n} \sigma_a(d)\sigma_{-a}(n/d).$$

# 2

# *The Poisson Summation Formula and the functional equation*

## 2.1

In this chapter it will be shown how the Poisson Summation Formula leads to the functional equation for the zeta-function. This is the first case that we shall meet of a summation formula which can be summarised by simple analytic properties of a single function. One can also deduce the Poisson Summation Formula from the elementary properties of the Riemann zeta-function as we shall see in §2.14. This procedure becomes much more significant when one is dealing with broader classes of Dirichlet series about which one knows too little *a priori* to prove a corresponding summation formula. The classic example of these are the so-called Voronoi Summation Formulae in which coefficients of the form $d(n)$ appear (see Exercises 2.29–2.32). Other examples – technically a little different – are the explicit formulae of Prime Number Theory which shall be discussed in Chapter 3. This correspondence between Dirichlet series and summation formulae is a central idea in analytic number-theory.

## 2.2

The Poisson Summation Formula is usually proved as an application of Fourier theory and a proof along these lines is given in A1.13. We can reformulate it as:

**Theorem:** *Let $f \in L^1(\mathbf{R})$ be continuous and of finite total variation. Then*

$$\sum_{n \in \mathbf{Z}} f(x+n) = \lim_{N \to \infty} \sum_{\substack{m \in \mathbf{Z} \\ |m| \leqslant N}} \hat{f}(m) e^{2\pi i m x}$$

*where*

$$\hat{f}(x) = \int_{\mathbf{R}} f(y) e^{-2\pi i x y} \, dy$$

*is the Fourier transform of $f$.*

For our purposes it will be useful to reformulate this a little We shall apply it to $f_u$ instead of $f$ where $f_u(x) = f(ux)$ $(u \in \mathbf{R}^\times, x \in \mathbf{R})$; we note that

$$\hat{f_u}(x) = |u|^{-1} \hat{f}(xu^{-1}).$$

We shall also apply the formula above only in the case that $x = 0$. We shall also ensure that the right-hand side is absolutely convergent by assuming that $\hat{f} \in L^1(\mathbf{R})$ and has finite total variation; see A1.13. Thus we have the following corollary to the theorem above:

**Corollary:** *Let $f$ be a complex-valued continuous function on* $\mathbf{R}$ *such that both $f$ and $\hat{f}$ are integrable and of bounded total variation. Then one has for* $u \in \mathbf{R}^\times$

$$\sum_{n \in \mathbf{Z}} f(nu) = |u|^{-1} \sum_{m \in \mathbf{Z}} \hat{f}(mu^{-1}).$$

It is worth noting that a differentiable function is of total bounded variation if $|f'|$ is Riemann integrable and integrable on $\mathbf{R}$.

## 2.3

Let $f$ be a measurable function on $\mathbf{R}$. We shall say that $f$ *is of type* $(\alpha, \beta)$ where $\alpha, \beta \in \mathbf{R}$, $\alpha < \beta$ if for $\sigma \in ]\alpha, \beta[$ the function

$$x \mapsto f(x)|x|^{\sigma - 1}$$

is integrable.

Let $\chi$ be the characteristic function of $]-1, 1[$ and $\chi^* = 1 - \chi$. Then $f$ is of type $(\alpha, \beta)$ if $x \mapsto f(x)\chi(x)|x|^{\sigma-1}$ is integrable for all $\sigma > \alpha$ and $x \mapsto f(x)\chi^*(x)|x|^{\sigma-1}$ is integrable for all $\sigma < \beta$; the converse is also true. The set of all functions of type $(\alpha, \beta)$ has the structure of a topological vector space but we shall not discuss this here. We shall also allow $\alpha = -\infty$ or $\beta = \infty$.

Suppose now that $f$ is of type $(\alpha, \beta)$. Then for $s \in \mathbf{C}$ with $\mathrm{Re}(s) \in ]\alpha, \beta[$

$$M(f, s) = \int_{\mathbf{R}} f(x)|x|^{s-1}\,\mathrm{d}x$$

is defined and represents an analytic function on $\{s : \alpha < \mathrm{Re}(s) < \beta\}$. This is *the Mellin transform* of $f$. Under suitable circumstances which are described in Appendix 2 it has an inverse transform

$$f(x) = (1/2\pi\mathrm{i}) \int_{(\sigma)} M(f, s)|x|^{-s}\,\mathrm{d}s$$

where $(\sigma)$ describes the line $\mathbf{R} \to \mathbf{C}$; $t \mapsto \sigma + it$ and $\sigma \in ]\alpha, \beta[$.

## 2.4

We now need a lemma. Let $\theta, \theta^* : \mathbf{R} \to \mathbf{R}$ be two bounded measurable functions which satisfy the following conditions

(*a*) $\theta(x) + \theta^*(x^{-1}) = 1$,

(*b*) for all $A > 0$ $\theta(x) = O(|x|^{-A})$, $\theta^*(x) = O(|x|^{-A})$ as $|x| \to \infty$.

**Lemma:** *For $\theta$, $\theta^*$ as above one has that*

(i) *$\theta$, $\theta^*$ are of type $(0, \infty)$*

(ii) *$M(\theta, s) - 2/s$ and $M(\theta^*, s) - 2/s$ have analytic continuations to the entire complex plane, and*

(iii) *in the sense of this continuation one has*

$$M(\theta, s) + M(\theta^*, -s) = 0.$$

*Proof*: Let us define $\theta_1$ by

$$\begin{aligned} \theta_1(x) &= 1 && \text{if } |x| < 1 \\ &= 1/2 && \text{if } |x| = 1 \\ &= 0 && \text{if } |x| > 1. \end{aligned}$$

Then $\theta = \theta^* = \theta_1$ satisfies (*a*) and (*b*). One has that

$$M(\theta_1, s) = \int_{-1}^{1} |x|^{s-1} \, dx$$

which converges when $\mathrm{Re}(s) > 0$ and has the value $2/s$. Thus we see directly that in this case the lemma is valid. Now let $f = \theta - \theta_1$ and $f^* = \theta^* - \theta_1$. Conditions (*a*) and (*b*) become

$$f(x) = -f^*(x^{-1})$$

and

$$f(x) = O(|x|^{-A}), \quad f^*(x) = O(|x|^{-A}) \qquad \text{as } |x| \to \infty.$$

From these two it follows that

$$f(x) = O(|x|^{A}) \qquad \text{as } |x| \to 0.$$

As $A$ can be any positive number these conditions ensure the convergence of

$$\int f(x)|x|^{s-1} \, dx$$

for all $s \in \mathbf{C}$. This therefore represents an analytic function in the entire plane and we obtain (i), and from

$$M(\theta, s) = M(\theta_1, s) + M(f, s)$$

we obtain (ii) as well. Substituting $x$ for $x^{-1}$ in this integral we see that

$$M(f, s) = -M(f^*, -s)$$

from which (iii) follows.

## 2.5

We are now in a position to prove the analytic continuability and the functional equation of the zeta-function. We shall in fact prove a formula giving an integral representation of the zeta-function from which not only these facts follow but also several other useful consequences can be deduced. In this section we shall assume that $\theta, \theta^*$ are functions as described in the previous section.

**Theorem:** *Let $f$ be of type $(\alpha, \beta)$ where $\alpha < 1 < \beta$. Thus $f$ is integrable and $\hat{f}$ is defined. Suppose that $\hat{f}$ is of type $(\hat{\alpha}, \hat{\beta})$ where $\hat{\alpha} < 1 < \hat{\beta}$. Suppose also that $f$ and $\hat{f}$ are continuous and of bounded total variation. Then for $s \in \mathbf{C}$ satisfying $1 - \hat{\beta} < \mathrm{Re}(s) < \beta$ one has*

$$\begin{aligned} 2\zeta(s)M(f,s) = &\int_{\mathbf{R}} \left\{ \sum_{n \in \mathbf{Z} - \{0\}} f(nx) \right\} \theta(x^{-1}) |x|^{s-1} \, dx \\ &+ \int_{\mathbf{R}} \left\{ \sum_{n \in \mathbf{Z} - \{0\}} \hat{f}(nx) \right\} \theta^*(x^{-1}) |x|^{-s} \, dx \\ &- f(0) M(\theta^*, s) - \hat{f}(0) M(\theta, 1-s). \end{aligned}$$

*The sums and integrals are absolutely convergent.*

*Remark*: The left-hand side of this equation is defined only for $\mathrm{Re}(s) > 1$. As we shall see the right-hand side is convergent in the larger domain $1 - \hat{\beta} < \mathrm{Re}(s) < \beta$ and gives an analytic continuation of the function on the left-hand side to this larger domain.

*Proof*: Let $F(x) = \sum_{n \in \mathbf{Z} - \{0\}} f(nx)$. From A1.13 one knows that this is absolutely convergent for $x \neq 0$. For $s$ satisfying $1 < \mathrm{Re}(s) < \beta$ we consider

$$\int F(x) |x|^{s-1} \, dx.$$

We study this by substituting the defining series for $F$ and interchanging summation and integration, which we shall justify presently We obtain

$$\sum_{n \in \mathbf{Z} - \{0\}} \int_{\mathbf{R}} f(nx) |x|^{s-1} \, dx.$$

In the $n$th term we replace $x$ by $x/n$ which yields

$$\sum_{n \in \mathbf{Z} - \{0\}} |n|^{-s} \int_{\mathbf{R}} f(x) |x|^{s-1} \, dx$$

which is equal to

$$2\zeta(s)M(f,s).$$

To prove convergence we note that if we replace $f$ by $|f|$ (which, in view of the inequality $\|a|-|b\| \leqslant |a-b|$, is also of bounded total variation) and $s$ by $\mathrm{Re}(s)$ then the validity of the interchange follows from Lebesgue's Monotone Convergence Theorem. In view of our assumptions the final expression converges absolutely and so do then all the intermediate ones. The general case of the interchange is then justified by Lebesgue's Dominated Convergence Theorem.

Recall now that $\theta(x^{-1}) + \theta^*(x) = 1$ and thus

$$\int F(x)|x|^{s-1}\,dx = \int F(x)\theta(x^{-1})|x|^{s-1}\,dx + \int F(x)\theta^*(x)|x|^{s-1}\,dx.$$

The first term on the right-hand side is the first term on the right-hand side of the formula stated in the theorem. We thus see that it converges for $s$ satisfying $1 < \mathrm{Re}(s) < \beta$. In view of conditions on $\theta$ this remains convergent for $s$ with $\mathrm{Re}(s) < \beta$. This also shows that the second term in the formula stated in the theorem is convergent for $\mathrm{Re}(1-s) < \hat{\beta}$, i.e. for $\mathrm{Re}(s) > 1 - \hat{\beta}$ on replacing $f$ by $f\hat{}$, $\theta$ by $\theta^*$ and $s$ by $1-s$.

By the Poisson Summation Formula we know that

$$F(x) = |x|^{-1} \sum_{n \in \mathbf{Z} - \{0\}} f\hat{}(nx^{-1}) + f\hat{}(0)|x|^{-1} - f(0).$$

This we substitute into the integral $\int F(x)\theta^*(x)|x|^{s-1}\,dx$ which we know to be convergent in the region $\{s: 1 < \mathrm{Re}(s) < \beta\}$. Observe that the last two terms in the expression for $F(x)$ given above converge for $s$ with $\mathrm{Re}(s) > 0$. Thus we obtain from the substitution

$$\int \sum_{n \in \mathbf{Z} - \{0\}} f\hat{}(nx^{-1})\theta^*(x)|x|^{s-2}\,dx + f\hat{}(0)M(\theta^*, s-1) - f(0)M(\theta^*, s).$$

In the first term here we can replace $x$ by $x^{-1}$ without altering the convergence. We see that this gives the second term of the formula stated in the theorem. As in the discussion above it is also absolutely convergent when $\mathrm{Re}(s) > 1 - \hat{\beta}$ and in particular when $\mathrm{Re}(s) > 1$. Finally we note that by Lemma 2.4 one has $M(\theta^*, s-1) = -M(\theta, 1-s)$. Using this the formula of the theorem follows at once. This completes the proof.

## 2.6

The first deduction that we shall make from Theorem 2.5 is what is often called 'the local functional equation', a name which originated in 'Tate's Thesis', Tate (1). The same content is contained in investigations of the *distribution* $|\quad|^{s-1}$ and its analytic continuation (see Gelfand & Shilov (1) I.3,II.2).

**Theorem:** *There exists a function $\gamma$ meromorphic in the complex plane with the following property:*

*If $f$ satisfies the conditions of Theorem 2.5 then $s \mapsto M(f, s)$ (resp. $s \mapsto M(\hat{f}, s)$) can be analytically continued to a meromorphic function in $\{s: \mathrm{Min}(\alpha, 1-\hat{\beta}) < \mathrm{Re}(s) < \mathrm{Max}(1-\hat{\alpha}, \beta)\}$ (resp. in $\{s: \mathrm{Min}(\hat{\alpha}, 1-\beta) < \mathrm{Re}(s) < \mathrm{Max}(1-\alpha, \hat{\beta})$) and one has for $s$ in the former set*

$$M(f, s) = \gamma(s) M(\hat{f}, 1-s).$$

*In terms of the Euler gamma-function one has*

$$\gamma(s) = \pi^{1/2-s} \Gamma(s/2)/\Gamma((1-s)/2)$$

## 2.7

Before we prove Theorem 2.6 we shall state the fundamental theorem concerning the analytic continuation of the zeta-function

**Theorem:** *The function $s \mapsto \zeta(s)$ defined on $\{s: \mathrm{Re}(s) > 1\}$ can be continued analytically to a meromorphic function in the complex plane. It has but one pole which is simple and situated at $s = 1$; the residue is 1. One has also*

$$\zeta(1-s) = \gamma(s)\zeta(s).$$

## 2.8

In this section we shall prove Theorems 2.6 and 2.7. We shall first note that there exist functions $f$ of type $(0, \infty)$ such that $\hat{f}$ is also of type $(0, \infty)$. Indeed if $f$ is smooth of compact support then it is clear that it is of type $(0, \infty)$. Since by integration by parts one has

$$\hat{f}(y) = (2\pi i y)^{-K} \int (\mathrm{d}/\mathrm{d}x)^K f(x) \mathrm{e}^{-2\pi i x y} \, \mathrm{d}x$$

for any integral $K \geqslant 0$. The integral is absolutely convergent and hence

$$|\hat{f}(y)| \leqslant (2\pi|y|)^{-K} \| f^{(K)} \|_{L^1}$$

where $f^{(K)}$ represents the $K$th derivative of $f$ and $\| \quad \|_{L^1}$ the $L^1$-norm. The case $K = 0$ shows that $\hat{f}$ is bounded and the general case shows that $\hat{f}$ is of type $(0, K)$ for this $K$. Thus $\hat{f}$ is of type $(0, \infty)$. It is clear that $f$ is of bounded variation as it is of compact support and as it is smooth. One can differentiate under the integral in the integral defining $\hat{f}$ and deduce that $\hat{f}$ is differentiable. The argument above can then be applied again to show that this derivative is continuous and integrable. Hence $\hat{f}$ is also of bounded variation.

Next note that if $f$ is also zero in a neighbourhood of 0 then $f$ is of

type $(-\infty, \infty)$. We can now apply Theorem 2.5. We see first of all that any $f$ satisfying these conditions $\zeta(s)M(f,s)$ has a continuation as a meromorphic function to the complex plane. If $f$ is of type $(-\infty, \infty)$ then $M(f,\cdot)$ has an analytic continuation to the complex plane. Hence $\zeta$ has an analytic continuation to the complex plane. We define first

$$\gamma(s) = \zeta(1-s)/\zeta(s).$$

It follows next from Theorem 2.5 that $M(f,\cdot)$ can be continued as a meromorphic function to $\{s: \mathrm{Min}(\alpha, 1-\hat{\beta}) < \mathrm{Re}(s) < \beta\}$ for any $f$ satisfying the conditions of Theorem 2.5. Let us take $\theta$ such that $\theta = \theta^*$ (for example $\theta = \theta^* = \theta_1$). The right-hand side of the formula displayed in Theorem 2.5 remains unchanged when $f$ is replaced by $f\hat{}$ and $s$ by $1-s$. Thus in the common domain of definition (which by the remark above contains $\{s: 0 < \mathrm{Re}(s) < 1\}$) one has

$$\zeta(s)M(f,s) = \zeta(1-s)M(f\hat{}, 1-s).$$

This effects the analytic continuation of $M(f,\cdot)$ as a meromorphic function to $\{s: \mathrm{Min}(\alpha, 1-\hat{\beta}) < \mathrm{Re}(s) < \mathrm{Max}(1-\hat{\alpha}, \beta)\}$ and the corresponding continuation of $M(f\hat{},\cdot)$. It also proves the 'local functional equation' stated in Theorem 2.6.

This completes the proofs of the two theorems except for the evaluation of $\gamma$ and the determination of the possible singularities of $\zeta$. The most convenient approach to these is to choose a specific function $f$. We take for this purpose $f(x) = \mathrm{e}^{-\pi x^2}$. Then

$$f\hat{}(y) = \int \mathrm{e}^{-\pi x^2 - 2\pi i x y}\,\mathrm{d}x = \mathrm{e}^{-\pi y^2}\int_{\mathbf{R}} \mathrm{e}^{-\pi(x+iy)^2}\,\mathrm{d}x.$$

By Cauchy's Theorem the integral does not depend on $y$. Denote the value of the integral by $C$. By A1.8 $(f\hat{})\hat{}(x) = f(-x)$ and hence as $f\hat{} = Cf$ we have $C^2 = 1$. The integral is clearly positive when $y = 0$ and so $C = 1$. Thus $f\hat{} = f$. One has next

$$M(f,s) = \int \mathrm{e}^{-\pi x^2}|x|^{s-1}\,\mathrm{d}x.$$

In this we set $y = \pi x^2$ and we see that

$$M(f,s) = \int_0^\infty \mathrm{e}^{-y} y^{s/2-1}\,\mathrm{d}y\pi^{-s/2}.$$

By the definition of Euler's gamma-function this is $\pi^{-s/2}\Gamma(s/2)$. We can now insert this special function into the functional equation of Theorem 2.6 and obtain the quoted representation of $\gamma$.

To describe the poles of $\zeta$ we note that if $f$ is smooth and of compact support and if $f$ is zero on a neighbourhood of 0 then $f$ is of type $(-\infty, \infty)$

as we saw above. We could choose $f$ so that it is zero outside a small neighbourhood of 1, and that it is a positive function. For a given $s_0 \in \mathbf{C}$ we can choose this neighbourhood so small that for $x$ lying in it $\mathrm{Re}(|x|^{s_0}) > 0$ and hence $M(f, s_0) \neq 0$.

From Theorem 2.5 we see that for $f$ as above the only pole of $s \mapsto \zeta(s)M(f, s)$ is at $s = 1$ and that the residue here is $\hat{f}(0) = M(f, 1)$. Thus $\zeta$ has a simple pole of residue 1 at $s = 1$. By the observation above, given $s_0 \neq 1$, there exists an $f$ with $M(f, s_0) \neq 0$ and as $s \mapsto \zeta(s)M(f, s)$ is regular at $s_0$ so is $\zeta$. This completes the proofs of the two theorems.

## 2.9

The discussion above is usually given in terms of a special function $f$, often $f(x) = \mathrm{e}^{-\pi x^2}$ which gives a misleading impression as to the role of this special function. No single function of this type is fundamental. Nevertheless an appropriate choice of function $f$ can be useful, typically for estimations of $\zeta$ or for its numerical computation. There are several variant forms of the formula of Theorem 2.5 which are useful for certain applications. For example, if $f = \hat{f}$ and $0 < \mathrm{Re}(s) < 1$ one has

$$\zeta(s)M(f, s) = \int_1^\infty \left\{ \sum_{n \in \mathbf{Z} - \{0\}} f(nx) - f(0)|x|^{-1} \right\} (x^{s-1/2} + x^{1/2-s}) x^{-1/2} \, \mathrm{d}x$$

which is chiefly interesting because of its simplicity. Formulae of this type are for certain purposes quite useful as we shall see below.

## 2.10

Suppose $f$ satisfies $\hat{f} = f$ and $f(0) = 0$. Then the definition of $M(f, s)$ shows that $M(f, 1) = 0$. Thus if $f$ is of type $(0, \infty)$ we can (on noting that $\hat{f} = f$ implies that $f$ is even) represent $\zeta$ through

$$\zeta(s)M(f, s) = 2 \int_1^\infty \sum_{n > 0} f(nx)(x^{s-1/2} + x^{1/2-s}) x^{-1/2} \, \mathrm{d}x$$

with the two sides of this equation being holomorphic in the entire complex plane. If we now write $s - 1/2 = ir$ then

$$(x^{s-1/2} + x^{1/2-s})/2 = \cos(r \log x)$$

and hence

$$\zeta(s)M(f, s) = \sum_{j=0}^{\infty} a_j (-1)^j r^{2j}/(2j)!$$

where

$$a_j = \sum_{n > 0} 4 \int_1^\infty f(nx) x^{-1/2} (\log x)^{2j} \, \mathrm{d}x.$$

This is a rapidly convergent series for well-chosen functions $f$. The example taken by Riemann was

$$f(x) = (3x^2 - 2\pi x^4)e^{-\pi x^2}.$$

It has been noted by Selberg that this function can be obtained by the application of a certain differential operator to $e^{-\pi x^2}$, a remark that has important consequences in other connections. We shall verify it directly. Let $f_1 = e^{-\pi x^2}$. Then

$$\begin{aligned} M(f, s) &= 3M(f_1, s+2) - 2\pi M(f_1, s+4) \\ &= 3\pi^{-(s+2)/2}\Gamma((s+2)/2) - 2\pi\pi^{-(s+4)/2}\Gamma((s+4)/2) \\ &= (3\pi^{-1}s/2 - 2\pi\pi^{-2}(s/2)\{(s+2)/2\})\pi^{-s/2}\Gamma(s/2) \\ &= s(1-s)M(f_1, s)/(2\pi). \end{aligned}$$

As $\hat{f_1} = f_1$ and as $s(1-s)$ is invariant under $s \to 1-s$ it follows from Theorem 2.6 that $M(\hat{f}, s) = M(f, s)$. In view of the inverse Mellin transform (A2.2) we see that $\hat{f} = f$.

Note that the power series expansion of $\zeta(s)M(f, s)$ has, in this case, real coefficients which is particularly important for real $r$, i.e. for $s$ on the 'critical line'. Riemann's function has the additional feature that $f(x) < 0$ for $x > (3/2\pi)^{1/2} = 0.690988\ldots$ so that $a_j < 0$ for all $j$.

## 2.11

There is another application of the formula of §2.10. Let $\sigma = \mathrm{Re}(s)$. We obtain directly from the integral representation and the negativity of $f$ that

$$|s(1-s)\pi^{-s/2}\Gamma(s/2)\zeta(s)| \leqslant -\sigma(1-\sigma)\pi^{-\sigma/2}\Gamma(\sigma/2)\zeta(\sigma).$$

From this, Stirling's Formula (A4.8) and the obvious inequality $|\sigma - 1/2| \leqslant |s - 1/2|$ one concludes the following estimate.

**Theorem:** *There exists a constant $c > 0$ such that for all $s \in \mathbf{C}$ one has*

$$|s(1-s)\pi^{-s/2}\Gamma(s/2)\zeta(s)| < ce^{2^{-1}|s-1/2|\log^+|s-1/2|}$$

*where* $\log^+ x = \mathrm{Max}(\log x, 0)$.

## 2.12

Although the estimate of §2.11 will be useful in Chapter 3, for most purposes it is far too weak. However, we can deduce the following estimate which will later play an important role.

**Theorem:** *Let $\mu_0$ be the function defined by*

$$\begin{aligned}\mu_0(\sigma) &= 0 && (\sigma > 1)\\ &= (1-\sigma)/2 && (0 < \sigma < 1)\\ &= 1/2 - \sigma && (\sigma < 0).\end{aligned}$$

*If $\varepsilon > 0$ is given and if $a < b$ are real numbers then there exists a constant $C(a, b, \varepsilon) > 0$ such that for $s \in \mathbf{C}$, $|\mathrm{Im}(s)| > 1$ and $a < \mathrm{Re}(s) < b$ one has*

$$|\zeta(s)| < C(a, b, \varepsilon)|\mathrm{Im}(s)|^{\mu_0(\mathrm{Re}(s))+\varepsilon}.$$

*Proof*: If $a > 1$ then $|\zeta(s)| \leqslant \zeta(a)$ for $s$ with $\mathrm{Re}(s) \geqslant a$. Hence the assertion is clearly true in this case. Next Stirling's Formula (A4.8) shows that uniformly in any strip of the type described in the theorem we have for $|\mathrm{Im}(s)| > 1$

$$\gamma(s) = O(|\mathrm{Im}(s)|^{\mathrm{Re}(s-1/2)})$$

so that the statement of the theorem follows in the case $b < 0$. The general case now follows from the Phragmén–Lindelöf Theorem and Theorem 2.11 (the Phragmén–Lindelöf Theorem is given in A5.8).

## 2.13

There is another consequence of Theorem 2.5 which forms the basis of the 'approximate functional equation' which we shall discuss in Chapter 6. The statement which we shall need is:

**Theorem:** *Suppose that $f$ satisfies the conditions of Theorem 2.5 and also that* (i) *$f$ is an even function and* (ii) *$f(x) = 0$ for $|x| > 1$. Then one has for $s$ satisfying* $\mathrm{Re}(s) > \mathrm{Max}(\alpha, 1-\beta)$, $X$, $Y > 1$

$$\begin{aligned}\zeta(s) = &\sum_{1\leqslant n\leqslant X} n^{-s} + \gamma(1-s)\sum_{1\leqslant m\leqslant Y} m^{-(1-s)}\\ &-2\Bigg\{\sum_{1\leqslant n\leqslant X} n^{-s}\int_0^{n/X} f(x)x^{s-1}\,dx/M(f,s)\\ &+\gamma(1-s)\sum_{1\leqslant m\leqslant Y} m^{-(1-s)}\int_0^{mX} \hat{f}(x)x^{-s}dx/M(\hat{f},1-s)\\ &-\gamma(1-s)\sum_{m>Y} m^{-(1-s)}\int_{mX}^{\infty} \hat{f}(x)x^{-s}dx/M(\hat{f},1-s)\Bigg\}\\ &-\frac{f(0)X^{-s}}{sM(f,s)} - \frac{\hat{f}(0)X^{1-s}}{(1-s)M(\hat{f},1-s)}\cdot\gamma(1-s).\end{aligned}$$

*Remark*: This becomes significant once one can show that the right-hand side is dominated by the first two terms. This will be effected by an appropriate choice of $f$ and then a rather careful estimation of the remaining terms on the right-hand side; see Chapter 6 for details.

*Proof*: In Theorem 2.5 we take

$$\theta(x) = 1 \qquad \text{if } |x| \leqslant X$$
$$= 0 \qquad \text{if } |x| > X.$$

Since all the functions in the first term on the right-hand side of the formula of Theorem 2.5 are even we can write this as

$$4\int_{X^{-1}}^{\infty} \sum_{n>0} f(nx)x^{s-1}\,dx.$$

Since $n \geqslant 1$ and $f(nx) = 0$ if $nx > 1$ those terms with $n > X$ make no contribution. Hence we can write this term as

$$4\sum_{n\leqslant X}\int_0^{\infty} f(nx)x^{s-1}\,dx - 4\sum_{n\leqslant X}\int_0^{X^{-1}} f(nx)x^{s-1}\,dx.$$

If in both terms we replace $x$ by $x/n$ we obtain

$$2\sum_{n\leqslant X} n^{-s}M(f,s) - 4\sum_{n\leqslant X} n^{-s}\int_0^{n/X} f(x)x^{s-1}\,dx.$$

On dividing by $2M(f,s)$ we obtain the first and third terms on the right-hand side of the formula we are proving.

Next we consider the second term on the right-hand side of Theorem 2.5. We see that this is equal to

$$4\sum_{n\geqslant 0}\int_X^{\infty} \hat{f}(nx)x^{-s}\,dx.$$

We split the summation over $n$ into the terms with $n \leqslant Y$ and those with $n > Y$. The terms with the summation over $n > Y$ yield on replacing $x$ by $x/n$ the expression

$$\sum_{n>Y} n^{-(1-s)}\int_{nX}^{\infty} \hat{f}(x)x^{-s}\,dx$$

which, when divided by $M(f,s) = \gamma(1-s)^{-1}M(\hat{f}, 1-s)$ yields the fifth term of the formula we are proving.

With the same substitution the sum over $n \leqslant Y$ becomes

$$\sum_{n\leqslant Y} n^{-(1-s)}\int_{nX}^{\infty} \hat{f}(x)x^{-s}\,dx$$

which can be written as

$$\sum_{n\leqslant Y} n^{-(1-s)}\left\{M(\hat{f}, 1-s) - \int_0^{nX} \hat{f}(x)x^{-s}\,dx\right\}$$

which yields after division by $2M(f,s)$ the second and fourth terms. Finally as $M(\theta,s) = 2X^s/s$ it is clear that the last two terms on the right-hand side

of the formula of Theorem 2.5 yield directly the last two terms of the formula we are proving. This completes the proof.

## 2.14

In this chapter we have derived the general analytic properties of the Riemann zeta-function from the Poisson Summation Formula. It is also possible to work in the opposite direction and to prove the Poisson Summation Formula starting from the properties of the Riemann zeta-function, as was first pointed out by Ferrar (Ferrar(1)). This is of some interest as one only needs the Poisson Summation for one suitable function to prove the analytic continuation of the zeta-function from which one can extend its validity to a very wide class. This is much more significant when one comes to deal with other classes of Dirichlet series whose analytic properties one derives indirectly and for which the corresponding summation formula is not a direct consequence of general analytic methods. The first significant example of such a summation formula is due to Voronoi (see Landau(2), Vol. II, Part VIII, Chapter 12) and involves a summation where the divisor function $d$ weights the terms. For an example of this see Exercises 2.30 and 2.32. The technique has proven to be very significant; another example of the same kind of argument is used in Chapter 3 to prove Theorem 3.6.

The idea behind these is the following. Let $f$ satisfy the conditions of Theorem 2.5. We form

$$F(x) = \sum_{n \in \mathbf{Z} - \{0\}} f(nx)$$

and

$$F^*(x) = \sum_{n \in \mathbf{Z} - \{0\}} \hat{f}(nx).$$

Here it is advisable to define $\hat{f}$ by

$$M(\hat{f}, s) = \gamma(s) M(f, 1 - s).$$

One can argue in various ways that this is the usual Fourier transform; one possibility is to verify it for convenient functions $f$ by evaluation of the integrals concerned and then to use a density argument. Another possibility is to prove it by means of complex analysis (see e.g. Ferrar(1), Bellman(1), pp. 34–6). We shall assume that $\hat{f}$ so defined coincides with the Fourier transform. Let $f$ be an even function. Then the argument of §2.5 shows that $F$ is of type $(1, \beta)$ and has Mellin transform $2\zeta(s)M(f, s)$. Likewise $F^*$ is of type $(1, \hat{\beta})$ with Mellin transform $2\zeta(s)M(\hat{f}, s)$. From the theory of the zeta-function we know that $M(F, s)$ has an analytic

continuation to $\{s: \alpha < \mathrm{Re}(s) < \beta\}$ and $M(F^*, s)$ to $\{s: \hat{\alpha} < \mathrm{Re}(s) < \hat{\beta}\}$, both as meromorphic functions with simple poles at 1. The definition of $f\hat{}$ and the functional equation for the zeta-function show that one has on the domain where both sides are defined

$$M(F^*, s) = M(F, 1 - s).$$

This equation then defines a further analytic continuation of both sides so that this equation holds when $1 - \beta < \mathrm{Re}(s) < \hat{\beta}$. We assume now that $f$ is so chosen that the inverse Mellin transform is applicable (see A2.2); we shall not go into details here. Then

$$F(x) = (1/2\pi i) \int_{(\sigma)} M(F, s) x^{-s} \, ds/2$$

and

$$\begin{aligned} F^*(x) &= (1/2\pi i) \int_{(\sigma)} M(F^*, s) x^{-s} \, ds/2 \\ &= (1/2\pi i) \int_{(\sigma)} M(F, 1 - s) x^{-s} \, ds/2 \\ &= (1/2\pi i) \int_{(1-\sigma)} M(F, s) x^{-(1-s)} \, ds/2 \end{aligned}$$

where $(\sigma)$ denotes the vertical line with real part $\sigma$ and $1 < \sigma < \mathrm{Min}(\beta, \hat{\beta})$. Hence by Cauchy's Theorem – when we have that

$$\sup_{1-\sigma < u < \sigma} |M(F, u + it)| \to 0$$

as $t \to \infty$ – it follows that $F(x) - x^{-1} F^*(x^{-1})$ is equal to the sum of the residues of $M(F, s) x^{-s}$ at those poles between the lines $(1 - \sigma)$ and $(\sigma)$. In this case the poles are located at $s = 0$ and $s = 1$ and we obtain

$$F(x) - x^{-1} F^*(x^{-1}) = M(f, 1) x^{-1} - M(f\hat{}, 1).$$

Since $M(f, 1) = f\hat{}(0)$ and $M(f\hat{}, 1) = f(0)$ this is precisely the Poisson Summation Formula.

This method can be modified to handle functions which have several poles, a fact which can be of considerable use (see Weil (3)). We shall see an example of this in Exercises 2.30 and 2.32. See also in this connection Chandrasekharan & Narashiman (1), (2).

## 2.15

To conclude this chapter we emphasise that there are several rather special methods which are useful for conducting the analytic continuation of the zeta-function, some of which are of considerable additional significance

in that they can be used to derive more detailed information than is possible with the methods described here. Some of these are given in the exercises; for more details see Titchmarsh (2), Chapter 2.

## Exercises

2.1 Let $f(x) = e^{-A|x|}$. Show that

$$\hat{f}(y) = 2A/(A^2 + 4\pi^2 y^2)$$
$$M(f, s) = 2\Gamma(s)A^{-s} \qquad (\mathrm{Re}(s) > 0)$$
$$M(\hat{f}, s) = \pi^{-1}(2\pi/A)^{1-s}\Gamma(s/2)\Gamma(1 - s/2) \qquad (0 < \mathrm{Re}(s) < 2).$$

2.2 Use Exercise 2.1 and Theorem 2.6 to deduce that

$$\Gamma(s/2)\Gamma((1 + s)/2) = \pi^{1/2}2^{1-s}\Gamma(s).$$

2.3 Use Exercise 2.1 and the Poisson Summation Formula to show that

$$(e^A + 1)/(e^A - 1) = \sum_{n \in \mathbf{Z}} 2A/(A^2 + 4\pi^2 n^2).$$

Deduce the results of § 1.3.

2.4 Let $f_X(x) = \mathrm{Max}(0, 1 - |x|/X)$. Show that

$$\hat{f}_X(y) = X\{(\sin \pi yX)/\pi yX\}^2$$
$$M(f_X, s) = 2X^s/s(s + 1) \qquad (\mathrm{Re}(s) > 0).$$

Deduce that if $-2 < \mathrm{Re}(s) < 0$

$$\int_0^\infty (\sin^2 x)x^{s-1}\,dx = -4^{-1}\pi^s\gamma(s).$$

2.5 Let

$$f_{k,X}(x) = \mathrm{Max}(0, 1 - |x|/X)^k/k!$$

Show that

$$\hat{f}_{k,X}(y) = 2X(-1)^{(k+1)/2}(2\pi yX)^{-k-1}(\cos 2\pi yX - \{1 - (2\pi yX)^2/2!$$
$$+ (2\pi yX)^4/4! - \cdots + (-1)^{(k-1)/2}(2\pi yX)^{k-1}/(k-1)!\})$$
$$\text{if } k \equiv 1 \pmod 2$$
$$= 2X(-1)^{k/2}(2\pi yX)^{-k-1}(\sin 2\pi yX - \{(2\pi yX)/1!$$
$$- (2\pi yX)^3/3! + \cdots + (-1)^{k/2-1}(2\pi yX)^{k-1}/(k-1)!\})$$
$$\text{if } k \equiv 0 \pmod 2,$$

and that

$$M(f_{k,X}, s) = 2X^s/s(s + 1)\cdots(s + k) \qquad (\mathrm{Re}(s) > 0).$$

2.6 Let $N(x) = \sum_{n \leqslant x} n^{-1} - \log x$. Show that $N(x + 1) - N(x) = O(x^{-2})$ as $x \to \infty$ and deduce that $N(x)$ tends to a limit as $x \to \infty$. (This limit

is, by definition, Euler's Constant $\gamma$.) Show further that

$$\zeta(s) = (s-1)^{-1} + \gamma + (s-1)F(s)$$

where

$$F(s) = \int_1^\infty \{N(x) - \gamma\} x^{-s}\,dx$$

is analytic in $\{s \in \mathbf{C}: \mathrm{Re}(s) > 0\}$.

2.7 Suppose that $a_j \geqslant 0$ $(j \geqslant 1)$ is a sequence such that $\sum_{j\geqslant 1}(-1)^j a_j$ converges. For $j \geqslant 2$ let $A_j = (2a_j - a_{j-1} - a_{j+1})/4$. Show that $\sum_{j\geqslant 2}(-1)^j A_j$ is also convergent and that

$$\sum_{j\geqslant 1}(-1)^j a_j = \tfrac{3}{4}a_1 - \tfrac{1}{4}a_2 + \sum_{j\geqslant 1}(-1)^j A_j.$$

2.8 Show that if $\mathrm{Re}(s) > 0$

$$\zeta(s) - (s-1)^{-1} = 1 - s\int_1^\infty (x - [x])x^{-s-1}\,dx.$$

Deduce again that near 1 one has

$$\zeta(s) = (s-1)^{-1} + \gamma + O(|s-1|).$$

(For the definition of $[x]$ see § 1.7.)

2.9 Show that the periodic function $x \mapsto x - [x]$ can be expanded in a Fourier series

$$1/2 - \sum_{n=1}^{\infty} (n\pi)^{-1} \sin 2\pi n x$$

and for $x \notin \mathbf{Z}$ that

$$|x - [x] - 1/2 + \sum_{1\leqslant n\leqslant N} (\pi n)^{-1} \sin 2\pi n x| \leqslant (\pi N |\sin \pi x|)^{-1}.$$

2.10 From the previous two exercises deduce the functional equation of $\zeta$ in $\{s \in \mathbf{C}: 0 < \mathrm{Re}(s) < 1\}$.

2.11 Show that

$$\Gamma(s)\zeta(s) = e^{\pi i s}(2i \sin \pi s)^{-1} \int_C (e^x - 1)^{-1} x^{s-1}\,dx$$

where $x^s = e^{s \log x}$ with $0 \leqslant \mathrm{Im}(\log x) < 2\pi$ and the contour $C$ is made up of the three linear segments from $\infty - \pi i$ to $-1/2 - \pi i$, $-1/2 - \pi i$ to $-1/2 + \pi i$ and $-1/2 + \pi i$ to $\infty + \pi i$.

2.12 From Exercise 2.11 show that $\zeta$ has an analytic continuation to the complex plane as a meromorphic function. By deforming the contour to $(\sigma)$, $\sigma < 0$ and letting $\sigma \to -\infty$ prove the functional equation for $\zeta$

2.13 Use the formula of Exercise 2.11 to evaluate $\zeta(1-2k)$, $k \in \mathbf{N}$.

2.14 Show that $\zeta(s) < 0$ for $s \in ]0, 1[$ and $-2 < \zeta(1/2) < -1$. If you have a

programmable calculator show that $\zeta(1/2) = -1.46035\ldots$ (Hint: use Exercises 1.2 and 2.7.)

2.15 Show that

$$\zeta'(1/2)/\zeta(1/2) = (1/2)\log 8\pi + (1/2)\gamma + \pi/4.$$

(Use Exercise A4.8.)

2.16 Show that

$$\zeta(s) = \int_0^\infty x^{2-s}(\mathrm{d}/\mathrm{d}x)^2(\Gamma'(x)/\Gamma(x) - x^{-1})\mathrm{d}x(\sin \pi s)/2\pi s \quad (1 < \mathrm{Re}(s) < 2)$$

2.17 Let $a: \mathbf{Z} \to \mathbf{C}$ be a function of period $q$ (i.e. $a(n+q) = a(n)$). Define the finite Fourier transform $a\hat{}$ by

$$a\hat{}(k) = \sum_{0 \leqslant j < q} a(j)\mathrm{e}^{-2\pi \mathrm{i}jk/q} q^{-1/2}.$$

Let $f$ be a function satisfying the conditions of Corollary 2.2. Show that

$$\sum_{j \in \mathbf{Z}} a(j) f(j/q^{1/2}) = \sum_{j \in \mathbf{Z}} a\hat{}(j) f\hat{}(j/q^{1/2}).$$

(Remark: for functions $a$ such that $a\hat{}(j) = \tau\overline{a(j)}$ this can be used to compute $\tau$ as

$$\left\{\sum_{j \in \mathbf{Z}} a(j) f(j/q^{1/2})\right\} \Big/ \left\{\overline{\sum_{j \in \mathbf{Z}} a(j) f(j/q^{1/2})}\right\}$$

for $f$ such that $f\hat{} = \bar{f}$. For example this can be used for the computation of Gauss sums.)

2.18 Let $a$ be as in Exercise 2.17. Show that

$$\zeta(s, a) = \sum_{n \geqslant 1} a(n) n^{-s}$$

converges if $\mathrm{Re}(s) > 1$ and has an analytic continuation to a meromorphic function in the entire plane. Determine its properties. (Hint: separate cases according as to whether $a(-n) = a(n)$ for all $n$ or $a(-n) = -a(n)$ for all $n$.)

2.19 Let $a$ be as in the last two exercises. Show that $\zeta(s, a)$ has an Euler Product (i.e. $\zeta(s, a) = \prod_p P_p(s)$ where $P_p(s)$ is of the form $\sum_{e \geqslant 0} b_p(e) p^{-es}$ and $p$ runs through the set of prime numbers) only when there exists a multiplicative function $a^*$ and a divisor $q^*$ of $q$ so that

$$\begin{aligned} a(n) &= a(q^*)a^*(n/q^*) && q^* \mid n \\ &= 0 && q^* \nmid n. \end{aligned}$$

Show that if $p \nmid q$ then $P_p(s) = (1 - a^*(p)p^{-s})^{-1}$. Show that $a\hat{}$ (as in Exercise 2.17) also has the property that $\zeta(s, a\hat{})$ has an Euler Product

structure. (Hint: use Dirichlet's Theorem on primes in an arithmetic progression.)

2.20 With the notations and assumptions of Exercises 2.17 and 2.18 show that in a neighbourhood of 1 one has

$$\zeta(s,a) = a\hat{}(0)\{(s-1)^{-1} - \log q\} - \sum_{j=1}^{q} a(j)\Gamma'(j/q)/\Gamma(j/q) + O(s-1).$$

2.21 Let $0 < \theta < \pi$. Let $f$ be a function which is analytic on a neighbourhood of $C(\theta) = \{z \in \mathbf{C}: |\arg(z)| < \theta\}$. Suppose that $f$ satisfies on $C(\theta)$ the estimate $f(z) = O(|z|^{-a})$ for some $a > 1$. Let $\gamma_{\pm}$ be the contours $\mathbf{R}_+ \to \mathbf{C}; t \mapsto e^{\pm i\theta}t$. Show that for any $q > 1$

$$\sum_{n \geqslant 1} f(n) = \sum_{1 \leqslant n \leqslant q} f(n) + \int_{1/2}^{\infty} f(q+x)\,dx + \int_{\gamma_+} f(q+z)e^{2\pi iz}(1-e^{2\pi iz})^{-1}\,dz$$
$$+ \int_{\gamma_-} f(q+z)e^{-2\pi iz}(1-e^{2\pi iz})^{-1}\,dz.$$

2.22 Let $b_n(t)$ be defined by

$$z(e^{zt}-1)/(e^z-1) = \sum_{n \geqslant 1} b_n(t)z^n/n!$$

Show that $b_n$ is a polynomial of degree $n$ and that

(i) $b_n(t+1) - b_n(t) = nt^{n-1}$,

(ii) $1^k + 2^k + \cdots + n^k = b_{k+1}(n+1)/(k+1)$, for $k > 0$,

(iii) $b_n(t) = (-1)^n b_n(1-t)$,

(iv) for $0 \leqslant t \leqslant 1, n \geqslant 2$, one has $|b_n(t)| < 10n!$ (the constant 10 is not the best possible!),

(v) with $b_n$ as in §1.3

$$b_n(t) = t^n - nt^{n-1}/2 + \sum_{k<n-1} \binom{n}{k}(n-k)b_{n-k-1}t^k.$$

2.23 Show that if $f \in C^{2n}([a,b])$ with $a, b \in \mathbf{Z}$ then

$$\sum_{a \leqslant j \leqslant b} f(j) = \int_a^b f(x)\,dx + 2^{-1}\{f(a)+f(b)\}$$
$$- \sum_{1 \leqslant j \leqslant n} b_{j+1}\{f^{(2j-1)}(b) - f^{(2j-1)}(a)\}/(2j)!$$
$$-(2n)!^{-1}\int_0^1 b_{2n}(t) \sum_{a \leqslant i < b} f^{(2n)}(i+t)\,dt.$$

(This is the Euler–Maclaurin Summation Formula.)

2.24 Use Exercise 2.23 to compute $\zeta(3)$ to six decimal places.

2.25 Let $L_x$ be the contour $\mathbf{R} \to \mathbf{C}$; $t \mapsto e^{\pi i/4}t + x$. For $x \in \mathbf{R} - \mathbf{Z}$ define

$$\Phi(a,x) = \int_{L_x} e^{2\pi i(w^2/2 + aw)}/(e^{2\pi w} - 1)\,dw.$$

Show that

(i) $\Phi(a-\mathrm{i},x)-\Phi(a,x)=\mathrm{e}^{-\pi\mathrm{i}a^2}\mathrm{e}^{\pi\mathrm{i}/4}$

(ii) $\Phi(a,x+1)-\Phi(a,x)=(-1)^n\mathrm{i}\mathrm{e}^{2\pi na}\quad(n\in\mathbf{Z},x<n<x+1)$

(iii) $\Phi(a,x+1)=-\mathrm{e}^{2\pi a}\Phi(a-i,x).$

Deduce that

$$\Phi(a,x)=-\mathrm{e}^{\pi\mathrm{i}/4}\mathrm{e}^{-\pi\mathrm{i}a^2}/(1+\mathrm{e}^{-2\pi a})-(-1)^{[x]}\mathrm{i}\mathrm{e}^{2\pi[x]a}/(1+\mathrm{e}^{2\pi a}).$$

2.26 By setting $a=\mathrm{e}^{-\pi\mathrm{i}/4}u$ and multiplying the formula found in Exercise 2.25 by $u^{s-1}$ and integrating over $u$ show that for $-1<x<0$

$$\pi^{-s/2}\Gamma(s/2)\zeta(s)=\mathrm{e}^{\pi\mathrm{i}(s-1)/2}\pi^{-s/2}\Gamma(s/2)\int_{L_x}\mathrm{e}^{\pi\mathrm{i}(w^2-\mathrm{i}w)}(\mathrm{e}^{2\pi w}-1)^{-1}w^{-s}\,\mathrm{d}w$$
$$+\mathrm{e}^{\pi\mathrm{i}s/2}\pi^{-(1-s)/2}\Gamma((1-s)/2)$$
$$\times\int_{L_x}\mathrm{e}^{-\pi\mathrm{i}(w^2+\mathrm{i}w)}(\mathrm{e}^{2\pi w}-1)^{-1}w^{s-1}\,\mathrm{d}w$$

where $\bar{L}_x$ is the line $t\mapsto\mathrm{e}^{-\pi\mathrm{i}/4}t+x$.

2.27 Suppose we know the Poisson Summation Formula only for even functions. Deduce it for all functions.

2.28 Show that

$$\sum_{n\geqslant 1}n^{-1}\sigma_1(n)\mathrm{e}^{-2\pi ny}=\sum_{n\geqslant 1}n^{-1}(\mathrm{e}^{2\pi ny}-1)^{-1}.$$

Use the analytic properties of $\zeta(s)\zeta(s-1)$, including the functional equation, to show that if $f(y)$ represents the expression above then

$$f(y)-f(y^{-1})=(\pi/12)(y^{-1}-y)+(1/2)\log y$$

Deduce that if

$$\eta(z)=\mathrm{e}^{2\pi\mathrm{i}z/24}\prod_{n\geqslant 1}(1-\mathrm{e}^{2\pi\mathrm{i}nz})\qquad(\mathrm{Im}\,(z)>0)$$

then

$$(\mathrm{i}/z)^{1/2}\eta(-1/z)=\eta(z)$$

where that branch of $(\mathrm{i}/z)^{1/2}$ on $\{z\in\mathbf{C}:\mathrm{Im}\,(z)>0\}$ is taken for which $(\mathrm{i}/\mathrm{i})^{1/2}=1$.

2.29 Define

$$G(x)=(1/2\pi\mathrm{i})\int_{(\sigma)}\Gamma(s/2)^2(\pi x)^{-s}\,\mathrm{d}x\qquad(\sigma>0).$$

Prove that

(i) $G(x)=2\int_0^\infty\mathrm{e}^{-\pi(y+y^{-1})x}y^{-1}\,\mathrm{d}y,$

(ii) $G(x)=\sum_{n\geqslant 0}a_n(\pi x)^{2n}+\log x\sum_{n\geqslant 0}b_n(\pi x)^{2n}$

where

$$a_n = 2(1 + 2^{-1} + 3^{-1} + \cdots + n^{-1} - \gamma)/n!^2$$

and

$$b_n = -2/n!^2,$$

(iii) as $x \to \infty$

$$G(x) \sim 8x^{-1/2}e^{-2\pi x}.$$

2.30 With $G$ as defined in Exercise 2.29 show that

$$\sum_{n \geqslant 1} d(n)\{G(nx)x^{1/2} - G(nx^{-1})x^{-1/2}\}$$
$$= (\gamma - \log 4\pi)(x^{1/2} - x^{-1/2}) + x^{-1/2}\log x - x^{1/2}\log x^{-1}.$$

2.31 Define

$$K(x) = (1/2\pi i)\int_{(\sigma)} \{\pi^{1/2-s}\Gamma(s/2)/\Gamma((1-s)/2)\}^2 x^{-s}\,ds \qquad (\sigma > 0).$$

Show that

(i) $$K(x) = 4\sum_{k \geqslant 0} \{(2\pi)^2 x\}^{2k}(1 + 2^{-1} + \cdots + (2k)^{-1} - \gamma - \log 2\pi)/(2k)!^2$$
$$- 4\log x \sum_{k \geqslant 0} \{(2\pi)^2 x\}^{2k}/(2k)!^2,$$

(ii) $x^2K^{(iv)}(x) + 2xK^{(iii)}(x) + K''(x) - (2\pi)^2K(x) = 0.$

2.32 For $f$ of type $(\alpha, \beta)$ with $\alpha < 0$, $\beta > 1$ define $f^*$ by

$$f^*(y) = \int_0^\infty K(xy)f(x)\,dx$$

where $K$ is defined in Exercise 2.31. Show that

$$\sum_{n \geqslant 1} d(n)f(n) = \int_0^\infty (2\gamma + \log x)f(x)\,dx + \sum_{n \geqslant 1} d(n)f^*(n).$$

Explain the lack of symmetry in this formula. (This is essentially the Voronoi Summation Formula. It can also be derived from the two-dimensional Poisson Summation Formula.)

# 3

# *The Hadamard Product Formula and the 'explicit formulae' of prime number theory*

## 3.1

We have seen in Theorem 2.11 that

$$s(1-s)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

is bounded by $O(e^{|s|^a})$ for any $a > 1$ and is therefore a holomorphic integral function of order 1. From the theory of such functions, due to Hadamard, we know that it can be represented as

$$Ae^{Bs}\prod(1-s/\rho)e^{s/\rho}$$

for some $A, B \in \mathbf{C}$ and where $\rho$ runs through the zeros of the function. Moreover $\sum|\rho|^{-a}$ converges for all $a > 1$. For all this see Appendix 5.

It is known (see A4.3) that the gamma-function has no zeros in $\{s: \mathrm{Re}(s) > 0\}$. Hence the zeros of the function in question in this domain are those of the zeta-function itself. By the Euler Product Formula we see that there are no zeros of the Riemann zeta-function in $\{s: \mathrm{Re}(s) > 1\}$. Let $Z$ be the set of zeros of the Riemann zeta-function which lie in the strip $\{s: 0 \leqslant \mathrm{Re}(s) \leqslant 1\}$. Observe that by the functional equation and $\overline{\zeta(s)} = \zeta(\bar{s})$ if $\rho \in Z$ then $\bar{\rho}$, $1-\rho$, $1-\bar{\rho} \in Z$.

We shall consider $Z$ as a 'set with multiplicities'. Formally this means that there is a function $m: Z \to \mathbf{N}$ defined on $Z$, the multiplicity of the corresponding zero, and that $\sum_{\rho\in Z} a(\rho)$ or $\prod_{\rho\in Z} b(\rho)$ is to be interpreted as $\sum_{\rho\in Z} m(\rho)a(\rho)$ and $\prod_{\rho\in Z} b(\rho)^{m(\rho)}$. Such a concept is sometimes called a 'multi-set'.

It is possible to determine $A$ and $B$ in the formula above. If we let $s \to 0$ then we see that

$$A = -1.$$

There are two expressions for $B$. The first arises from the functional equation which yields, on replacing $\rho$ by $1-\rho$ in the right-hand side:

$$-e^{Bs}\prod_{\rho\in Z}(1-s/\rho)e^{s/\rho} = -e^{B(1-s)}\prod_{\rho\in Z}\{1-(1-s)/(1-\rho)\}e^{(1-s)/(1-\rho)}.$$

This transforms to

$$e^{B(2s-1)} = \prod_{\rho \in Z} \{\rho/(1-\rho)\} e^{-1/(1-\rho)} e^{-s/\rho(1-\rho)}$$

which itself yields

$$B = -\tfrac{1}{2} \sum_{\sigma \in Z} \frac{1}{\rho(1-\rho)}.$$

One can also evaluate $B$ by observing that it is the coefficient of $s$ in the power series expansion of $s(1-s)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ about 0. One can show using Exercise 2.6 that

$$B = -\gamma/2 - 1 + (1/2)\log 4\pi = -0.023\,095\,7\ldots$$

where $\gamma$ is again Euler's Constant (see A4.3). Although this formula is very satisfying it does not play a particularly important role in the theory of the zeta-function.

From the formula and the final remark of § 2.10 we have that

$$-s(1-s)\pi^{-s/2}\Gamma(s/2)\zeta(s) > 0$$

for all real $s$, and in particular there are no real elements of $Z$. We observe also that $Z = \{\rho : \mathrm{Im}(\rho) > 0\} \cup \{1-\rho : \mathrm{Im}(\rho) > 0\}$. Let $Z_+ = \{\rho \in Z : \mathrm{Im}(\rho) > 0\}$. Then we can write the product

$$e^{Bs} \prod_{\rho \in Z} (1 - s/\rho) e^{s/\rho}$$

as

$$e^{Bs} \prod_{\rho \in Z_+} (1 - s/\rho)(1 - s/(1-\rho)) e^{s/\rho(1-\rho)}$$

and because of our formula for $B$ we can write this as

$$\prod_{\rho \in Z_+} (1 - s/\rho)\{1 - s/(1-\rho)\}.$$

We can now summarise the results proved in this section:

**Theorem:** *Let $Z$ be the multi-set defined above. Then $Z \cap \mathbf{R} = \varnothing$; let $Z_+ = \{\rho \in Z : \mathrm{Im}(\rho) > 0\}$. Then $\sum_{\rho \in Z} |\rho|^{-a}$ converges for all $a > 1$ and one has*

$$-s(1-s)\pi^{-s/2}\Gamma(s/2)\zeta(s) = \prod_{\rho \in Z_+} ((1 - s/\rho)(1 - s/(1-\rho)))$$

*the product being absolutely convergent for all $s \in \mathbf{C}$.*

## 3.2

The most interesting conclusions concerning the Riemann zeta-function arise from the comparison of the Hadamard Product Formula given above

with the Euler Product Formula. For such purposes it will be convenient to take the logarithmic derivative of both sides of the formula of Theorem 3.1. This yields the following formula directly.

**Corollary:** *With the notations above one has*

$$-\zeta'(s)/\zeta(s) = s^{-1} + (s-1)^{-1} + \Gamma'(s/2)/2\Gamma(s/2) - (1/2)\log\pi - (2s-1)\sum_{\rho\in Z_+}((s-\rho)(s-(1-\rho)))^{-1}.$$

## 3.3

We now come to the first application of Corollary 3.2. It is the following theorem which involves the interplay between the Euler and Hadamard product representations of the zeta-function.

**Theorem:** *One has*

$$\zeta'(s)/\zeta(s) = O(1) \qquad \textit{in } \{s: \mathrm{Re}(s) \geqslant 2\}$$

$$\zeta'(s)/\zeta(s) - \sum_{\rho\in Z:|\mathrm{Im}(s-\rho)|<1} (s-\rho)^{-1} = O(1+\log^+|s|) \qquad \textit{in } \{s: -1 \leqslant \mathrm{Re}(s) \leqslant 2\}$$

$$\zeta'(s)/\zeta(s) = O(\log|s|) \qquad \textit{in } \{s: \mathrm{Re}(s) \leqslant -1, |s-2m| > 1 \quad \textit{for all } m\in\mathbf{N}\}$$

In this we have used $\log^+$ which is defined as

$$\log^+ x = \mathrm{Max}(0, \log x).$$

## 3.4

Before we prove Theorem 3.3 we shall state and prove a corollary which, although it will be improved upon later, can often be very useful. The proof of this corollary also gives an introduction to the techniques which shall be used in the proof of the theorem itself.

**Corollary:** *As $T \to \infty$ one has*

$$\mathrm{Card}\{\rho\in Z: T < \mathrm{Im}(\rho) < T+1\} = O(\log T).$$

Observe that this corollary implies that

$$\mathrm{Card}\{\rho\in Z: 0 < \mathrm{Im}(\rho) < T\} = O(T\log T),$$

a remark that will occasionally be useful to employ, but which will be superseded by a much sharper asymptotic estimate later.

*Proof*: In the second estimate of Theorem 3.3 we set $s = 2 + iT$. Since $-\zeta'(s)/\zeta(s) = \sum\Lambda(n)n^{-s}$ we can take absolute values and obtain (if $\mathrm{Re}(s) = 2$)

$$|\zeta'(s)/\zeta(s)| \leqslant -\zeta'(2)/\zeta(2) = O(1).$$

Thus Theorem 3.3 shows that

$$\sum (s-\rho)^{-1} = O(\log T)$$

where the sum is taken over those zeros with $|\mathrm{Im}(\rho) - T| < 1$. But one has

$$\mathrm{Re}\{(s-\rho)^{-1}\} = \{\mathrm{Re}(s) - \mathrm{Re}(\rho)\}/|s-\rho|^2.$$

Clearly one has

$$|s-\rho|^2 \leqslant 2^2 + 1^2 = 5,$$
$$\mathrm{Re}(s) - \mathrm{Re}(\rho) \geqslant 2 - 1 = 1$$

so that

$$\mathrm{Re}\{(s-\rho)^{-1}\} \geqslant 1/5$$

and as

$$\sum_{\rho \in Z : |\mathrm{Im}(\rho) - T| < 1} \mathrm{Re}\{(s-\rho)^{-1}\} = O(\log T)$$

we obtain

$$\sum_{\rho \in Z : |\mathrm{Im}(\rho) - T| < 1} 1 = O(\log T)$$

which implies directly the estimate stated in the corollary.

## 3.5

In this section we shall prove Theorem 3.3. We use the idea introduced in § 3.4, namely that by the representation

$$-\zeta'(s)/\zeta(s) = \sum \Lambda(n) n^{-s}$$

which is convergent when $\mathrm{Re}(s) > 1$ one has for $\mathrm{Re}(s) \geqslant 2$

$$|\zeta'(s)/\zeta(s)| \leqslant -\zeta'(2)/\zeta(2)$$
$$= O(1)$$

which proves the first inequality.

We shall next need the following estimate which follows from a sharper inequality to be proved in § 3.7; in the region

$$\{s \in \mathbf{C} : -1/2 \leqslant \mathrm{Re}(s) \leqslant 3/2, |s| \geqslant 1/2\}$$

one has

$$\Gamma'(s)/\Gamma(s) = O(1 + \log^+ |s|).$$

In view of this we have from Corollary 3.2 in the region

$$\{s \in \mathbf{C} : -1 \leqslant \mathrm{Re}(s) \leqslant 3, |s| \geqslant 1/2, |s-1| \geqslant 1/2\}$$

the following estimate

$$-\zeta'(s)/\zeta(s) - \sum_{\rho \in Z_+} ((s-\rho)^{-1} + (s-(1-\rho))^{-1}) = O(1 + \log^+ |s|).$$

Let $s = \sigma + it$ satisfy $-1 \leqslant \sigma \leqslant 2$ and set $s^* = 3 + it$. The estimate above

is valid both for $s$ and $s^*$; moreover $-\zeta'(s^*)/\zeta(s^*) = O(1)$. Thus if we subtract these two estimates we obtain

$$-\zeta'(s)/\zeta(s) - \sum_{\rho \in Z_+} ((s-\rho)^{-1} - (s^*-\rho)^{-1} + (s-(1-\rho))^{-1} - (s^*-(1-\rho))^{-1}) = O(1+\log^+|t|).$$

Note that

$$(s-\rho)^{-1} - (s^*-\rho)^{-1} = (3-\sigma)/\{(s-\rho)(s^*-\rho)\};$$

since by Theorem 3.1 the series $\sum|\rho|^{-2}$ converges we can rewrite this last inequality as

$$-\zeta'(s)/\zeta(s) - \sum_{\rho \in Z} \{(s-\rho)^{-1} - (s^*-\rho)^{-1}\} = O(1+\log^+|t|).$$

On the other hand if we take the logarithmic derivative of the original form of the product expansion of § 3.1 we obtain

$$\begin{aligned} &-\zeta'(s^*)/\zeta(s^*) - \sum_{\rho \in Z} \{(s^*-\rho)^{-1} + \rho^{-1}\} \\ &\quad = s^{*-1} + (s^*-1)^{-1} + \Gamma'(s^*/2)/2\Gamma(s^*/2) - B - (1/2)\log\pi \\ &\quad = O(1+\log^+|t|). \end{aligned}$$

As $\zeta'(s^*)/\zeta(s^*) = O(1)$ we deduce that

$$\sum_{\rho \in Z} \{(s^*-\rho)^{-1} + \rho^{-1}\} = O(1+\log^+|t|).$$

We observe that

$$\begin{aligned} \mathrm{Re}\{(s^*-\rho)^{-1} + \rho^{-1}\} &= \{3-\mathrm{Re}(\rho)\}/|s^*-\rho|^2 + \mathrm{Re}(\rho)/|\rho|^2 \\ &\geqslant 2\{9+(\mathrm{Im}(\rho)-t)^2\}^{-1} \end{aligned}$$

since $0 \leqslant \mathrm{Re}(\rho) \leqslant 1$. Thus we obtain

$$\sum_{\rho \in Z} (1+\{\mathrm{Im}(\rho)-t\}^2)^{-1} = O(1+\log^+|t|).$$

Consider now those $\rho$ with $|\mathrm{Im}(\rho)-t| > 1$. We will show that for any such $\rho$ and $s$ one has

$$|\{(s-\rho)^{-1} - (s^*-\rho)^{-1}\}| < 6(1+\{\mathrm{Im}(\rho)-t\}^2)^{-1}.$$

To see this we first subtract $\mathrm{Im}(\rho)\mathrm{i}$ from $s$ and $\rho$, and hence also from $s^*$. Thus we need only prove the statement when $\rho$ is real. In view of the alternative expression for $\{(s-\rho)^{-1} - (s^*-\rho)^{-1}\}$ given above we have only to verify that for real $\sigma$ with $-1 \leqslant \sigma \leqslant 2$, $t \geqslant 1$ and real $r$ with $0 \leqslant r \leqslant 1$ that

$$\frac{3-\sigma}{(\{(\sigma-r)^2+t^2\}\{(3-r)^2+t^2\})^{1/2}}(1+t^2) \leqslant 6;$$

to see this we note that

$$3-\sigma \leqslant 4,$$
$$\{(\sigma-r)^2+t^2\} \geqslant t^2$$

and

$$\{(3-r)^2+t^2\} \geqslant 1+t^2,$$

so that the left-hand side is bounded by $4(1+t^{-2})^{1/2} \leqslant 6$ as required.

From this it follows that

$$\sum_{\rho \in Z:|\mathrm{Im}(\rho)\ -t|>1} \{(s-\rho)^{-1}-(s^*-\rho)^{-1}\} = O(1+\log^+|t|)$$

Finally we note that

$$\left|\sum_{\rho \in Z:|\mathrm{Im}(\rho)-t|\leqslant 1} (s^*-\rho)^{-1}\right| \leqslant \sum_{\rho \in Z} \{1+(\mathrm{Im}(\rho)-t\}^2)^{-1}$$

since if $|\mathrm{Im}(\rho)-t| \leqslant 1$, and as $0 \leqslant \mathrm{Re}(\rho) \leqslant 1$ one has

$$|s^*-\rho|^{-1} \leqslant \{1+(\mathrm{Im}(\rho)-t)^2\}^{-1}.$$

We now have

$$-\zeta'(s)/\zeta(s)+\sum_{\rho \in Z:|\mathrm{Im}(s-\rho)|\leqslant 1} (s-\rho)^{-1}$$
$$= -\sum_{\rho \in Z:|\mathrm{Im}(s-\rho)|>1} \{(s-\rho)^{-1}-(s^*-\rho)^{-1}\} - \sum_{\rho \in Z:|\mathrm{Im}(s-\rho)|\leqslant 1} (s^*-\rho)^{-1}$$
$$+ O(1+\log^+|t|) = O(1+\log^+|t|)$$

as asserted in the theorem.

## 3.6

The fact that the Riemann zeta-function can be expressed both as an Euler Product and as a Hadamard Product shows a remarkable reciprocity between the set of primes and the set of zeros of the zeta-function. This relationship is given its full expression in the so-called 'explicit formulae' of prime number theory. We shall see two different forms of these formulae. The first form is given below and is very general; one can think of it as an identity between two distributions. The second form which is given in §3.8 is less general – at least ostensibly – but is much more precise and convenient for applications to questions of analytic number-theory. One can regard this second form as a limiting version of that given below, and correspondingly that given below as an integrated form of the second one.

**Theorem:** *Let $f$ be a function of type $(a, \beta)$ where $\alpha < -1/2$, $\beta > 1/2$ and suppose that there exist $c, \varepsilon > 0$ so that for all $s$ satisfying $\alpha \leqslant \mathrm{Re}(s) \leqslant \beta$ one*

*has*

$$|M(f,s)| < c(1+|s|)^{-1-\varepsilon}.$$

*Suppose that $f$ is of bounded total variation and that $f(x)=0$ if $x<0$. Let*

$$\Delta_\infty(f) = \lim_{N\to\infty}\left\{\int_0^\infty f(x)F_N(x)x^{-1}\,dx - f(1)\log N/\pi\right\}$$

*where*

$$\begin{aligned} F_N(x) &= x^{-1/2}(1-x^{2N})/|x-x^{-1}| && \text{if } 0<x<1\\ &= x^{+1/2}(1-x^{-2N})/|x-x^{-1}| && \text{if } 1<x\\ &= 0 && \text{if } x=1. \end{aligned}$$

*The limit in this definition exists. Then one has*

$$\begin{aligned} &\Delta_\infty(f) + \sum_p \log p \sum_{k\in\mathbf{Z}-\{0\}} p^{-|k|/2} f(p^k)\\ &\quad = M(f,1/2) + M(f,-1/2) - \sum_{\rho\in Z} M(f,\rho-1/2) \end{aligned}$$

*where the left-hand summation is carried out over the set of primes. The sums here are all absolutely convergent.*

The reader who is familiar with the theory of valuations of the rational number field (see e.g. Cassels (1)) will recognise that the left-hand side of this formula is a summation over all the classes of valuations of **Q** since these are parameterised by the set of primes (the $p$-adic valuations) and the symbol $\infty$ (the Archimedean valuation).

## 3.7

In this section we shall prove Theorem 3.6. To do this we shall evaluate in two ways

$$\lim_{T\to\infty}(1/2\pi i)\int_{C(T)}\{-\zeta'(s+1/2)/\zeta(s+1/2)\}M(f,s)ds$$

where $C(T)$ is the rectangular path with corners $-\sigma-iT, \sigma-iT, \sigma+iT, -\sigma+iT$ taken counter-clockwise, and $1/2<\sigma<\mathrm{Max}(-\alpha,\beta,1)$.

First of all we can apply the theory of residues. The zeta-function has zeros at $Z$ and a pole at $s=1$. Consider the strip $\{s: -\sigma<\mathrm{Re}(s)<\sigma\}$. We assume that $T$ is not equal to the ordinate of a zero. The function $-\zeta'(s+1/2)/\zeta(s+1/2)$ has in the strip in question a simple pole of residue 1 at 1/2 and a simple pole of residue $-$(multiplicity of $\rho$) at $\rho-1/2$ for $\rho\in Z$. Thus this integral is equal to

$$M(f,1/2) - \sum_{\rho\in Z:|\mathrm{Im}(\rho)|<T} M(f,\rho-1/2).$$

Since $\sum|\rho|^{-1-\varepsilon}$ converges so does the sum $\sum M(f,\rho-1/2)$ and hence as

$T \to \infty$ this converges to

$$M(f, 1/2) - \sum_{\rho \in Z} M(f, \rho - 1/2).$$

We shall actually let $T$ run through a special sequence. As we have seen in Corollary 3.4 there are $O(\log N)$ zeros of the zeta-function with ordinates in the interval $]N, N+1[$, $N \in \mathbf{N}$. Thus there exists a constant $c > 0$ and $T_N$ with $N < T_N < N+1$ so that $|\mathrm{Im}(\rho) - T_N| > c/\log N$ $(N \geqslant 3)$. We shall suppose that for each $N \geqslant 3$ such a $T_N$ has been chosen and that we consider only such $T$ from now on.

By Theorem 3.3 the function $\zeta'(s+1/2)/\zeta(s+1/2)$ is estimated on the line joining $-\sigma + iT_N$ to $\sigma + iT_N$ by

$$\sum_{\rho \in Z : |\mathrm{Im}(\rho) - T_N| < 1} (s-\rho)^{-1} + O(\log N).$$

By the choice of $T_N$ one has that $(s-\rho)^{-1} = O(\log N)$, and there are, by Corollary 3.4, $O(\log N)$ terms in this sum. In view of the assumption on $f$ made in the theorem the integral over this line is bounded by $(\log N)^2 N^{-1-\varepsilon}$ which tends to 0 as $N$. The same is true of the integral over the line joining $-\sigma - iT$ and $\sigma - iT$.

Next observe that on the lines $(\sigma)$ and $(-\sigma)$ one has, again by Theorem 3.3, the estimate $O(1 + \log^+ |s|)$ for $\zeta'(s+1/2)/\zeta(s+1/2)$ and hence the vertical integrals are dominated by

$$\int_0^\infty (1 + \log^+ u)(1+u)^{-1-\varepsilon} \mathrm{d}u$$

which is convergent. Thus both of these integrals are convergent.

It now follows that as $N \to \infty$ the original integral converges to

$$(1/2\pi i) \int_{(\sigma)} \{-\zeta'(s+1/2)/\zeta(s+1/2)\} M(f, s) \mathrm{d}s$$

$$-(1/2\pi i) \int_{(-\sigma)} \{-\zeta'(s+1/2)/\zeta(s+1/2)\} M(f, s) \mathrm{d}s.$$

Differentiating the functional equation logarithmically gives

$$\zeta'(s)/\zeta(s) + \zeta'(1-s)/\zeta(1-s) = -\gamma'(s)/\gamma(s).$$

In the second of the two integrals above we replace $s$ by $-s$ and then in the integrand $\zeta'(-s+1/2)/\zeta(-s+1/2)$ by $-\zeta(s+1/2)/\zeta(s+1/2) - \gamma'(s+1/2)/\gamma(s+1/2)$. We obtain now

$$(1/2\pi i) \int_{(\sigma)} \{-\zeta'(s+1/2)/\zeta(s+1/2)\} \{M(f, s) + M(f, -s)\} \mathrm{d}s$$

$$-(1/2\pi i) \int_{(\sigma)} \{\gamma'(s+1/2)/\gamma(s+1/2)\} M(f, -s) \mathrm{d}s.$$

In the first of these two integrals we represent $\zeta'(s+1/2)/\zeta(s+1/2)$ by the Dirichlet series

$$\sum_p \sum_{k>0} \frac{\log p}{p^{k(s+1/2)}}.$$

We obtain

$$\sum_p \sum_{k>0} (1/2\pi i) \int_{(\sigma)} p^{-ks} \{M(f,s) + M(f,-s)\} ds \log p / p^{k/2}.$$

The integral here is the inverse transform, and $s \mapsto M(f,-s)$ is the Mellin transform of $x \mapsto f(x^{-1})$. In view of A2.2 and the assumptions made of $f$ we therefore obtain next

$$\sum_p \sum_{k>0} \{f(p^k) + f(p^{-k})\} p^{-k/2} \log p,$$

which we simply rewrite as

$$\sum_p \sum_{k \neq 0} f(p^k) p^{-|k|/2} \log p.$$

This is precisely the second term on the left-hand side of the formula being proved.

Note that $\gamma$ has a zero of order 1 at 1 and hence an analogous application of Cauchy's Theorem to the one used above shows that

$$(1/2\pi i) \int_{(\sigma)} \{\gamma'(s+1/2)/\gamma(s+1/2)\} M(f,-s) ds$$
$$- (1/2\pi i) \int_{(0)} \{\gamma'(s+1/2)/\gamma(s+1/2)\} M(f,-s) ds = M(f,-1/2).$$

Combining all these we see that the formula of the theorem will follow if we can show that

$$\Delta_\infty(f) = -(1/2\pi i) \int_{(0)} \{\gamma'(s+1/2)/\gamma(s+1/2)\} M(f,-s) ds$$

which we shall now do. We first observe that

$$\gamma'(s+1/2)/\gamma(s+1/2) = \Gamma'((s+1/2)/2)/2\Gamma((s+1/2)/2$$
$$+ \Gamma'((-s+1/2)/2)/2\Gamma((-s+1/2)/2) - \log \pi.$$

From this it follows that we have to show that

$$\Delta_\infty(f) = f(1) \log \pi - (1/2\pi i) \int_{(0)} \Gamma'((s+1/2)/2)/2\Gamma((s+1/2)/2)$$
$$\times \{M(f,s) + M(f,-s)\} ds.$$

We represent $\Gamma(t)^{-1}$ as $\lim_{M \to \infty} t(1+t/1)(1+t/2)\cdots(1+t/M)M^{-t}$ which converges locally uniformly. Thus we can take the logarithmic

derivative and obtain

$$-\Gamma'(t)/\Gamma(t) = \lim_{M\to\infty}\left\{\sum_{0\leqslant j\leqslant M}(t+j)^{-1} - \log M\right\}.$$

We have now to estimate the error in using the $M$th member of the sequence to approximate $\Gamma'(t)/\Gamma(t)$. We have

$$-\Gamma'(t)/\Gamma(t) - \sum_{0\leqslant j\leqslant M}(t+j)^{-1} + \log M = \lim_{M'\to\infty}\sum_{M<j\leqslant M'}(t+j)^{-1} - \log(M'/M).$$

But now

$$\log(M'/M) = \sum_{M<j\leqslant M'} j^{-1} + O(M^{-1})$$

and hence the expression to be estimated is

$$\lim_{M\to\infty}\sum_{M<j\leqslant M}\{(t+j)^{-1} - j^{-1}\} + O(M^{-1}) = \sum_{j>M} t/j(t+j) + O(M^{-1}).$$

There are two possibilities (and we shall assume that $|\mathrm{Re}(t)| \leqslant M/2$). The first is when $M > |t|$. Then $t/j(t+j) = O(|t|/j^2)$ and we obtain the estimate $O(1)$. If $M \leqslant |t|$ then we divide the summation into $j > |t|$ and $M < j \leqslant |t|$. The first summation can be estimated as before and gives $O(1)$. For the summation over the second range we use $t/j(t+j) = O(j^{-1})$ and we obtain an estimate of $O(M^{-1} + \log(|t|/M))$.

Hence we have shown that subject to $|\mathrm{Re}(t)| \leqslant M/2$

$$-\Gamma'(t)/\Gamma(t) - \sum_{0\leqslant j\leqslant 2M}(t+j)^{-1} + \log M = O(1 + \log^+|t|/M).$$

From this and Lebesgue's Dominated Convergence Theorem we see that we now have to show that $\Delta_\infty(f)$ is equal to

$$f(1)\log\pi + \lim_{M\to\infty}\left((1/2\pi\mathrm{i})\int_{(0)}\sum_{0\leqslant j\leqslant M}(s+1/2+2j)^{-1}\right.$$
$$\left.\times\{M(f,s) + M(f,-s)\}\,\mathrm{d}s - f(1)\log M\right).$$

From our assumptions we have that

$$(1/2\pi\mathrm{i})\int_{(0)} M(f,s)x^{-s-a-1}\,\mathrm{d}s = x^{-a}f(x)x^{-1}.$$

Moreover, we can integrate this equation over $[1,\infty[$ and interchange the order of integration by Fubini's Theorem as the double integral is absolutely convergent for $a>0$. Thus for $a>0$ we have

$$(1/2\pi\mathrm{i})\int_{(0)} M(f,s)(s+a)^{-1}\,\mathrm{d}s = \int_1^\infty f(x)x^{-a}x^{-1}\,\mathrm{d}x;$$

our expression becomes

$$f(1)\log\pi + \lim_{M\to\infty}\left(\sum_{0\leqslant j\leqslant M}\int_1^\infty \{f(x)+f(x^{-1})\}x^{-2j-1/2}x^{-1}\,dx - f(1)\log M\right)$$
$$= f(1)\log\pi + \lim_{M\to\infty}\left(\int_1^\infty \{f(x)+f(x^{-1})\}\frac{(1-x^{-2M-2})}{(x-x^{-1})}x^{1/2}x^{-1}\,dx - f(1)\log M\right).$$

This is now almost in the form we need. We separate the part of the integrand containing $f(x)$ from that containing $f(x^{-1})$ and in the latter we replace $x$ by $x^{-1}$. In the notation introduced in the statement of the theorem we obtain next

$$\lim\left\{\int_0^\infty f(x)F_{M+1}(x)x^{-1}\,dx - f(1)\log M/\pi\right\}$$

since the part of the original integral containing $f(x)$ in the integrand yields that part of this integral over 1, and that containing $f(x^{-1})$ in the integrand yields the integral over $]0,1[$. Now the assertion of the theorem follows since

$$\log(M+1)-\log M\to 0$$

as $M\to\infty$.

## 3.8

As we indicated above there is a more special and precise form of the 'explicit formulae' which we shall now formulate. Let for $x\in\mathbf{R}$

$$\langle x\rangle = \operatorname*{Min}_{m\in\mathbf{Z}}|x-m|.$$

**Theorem:** *Let $\varepsilon_X$ be the function on* $\mathbf{N}$ *defined by*

$$\begin{aligned}\varepsilon_X(n) &= 1 && \text{if } n<X\\ &= 1/2 && \text{if } n=X\\ &= 0 && \text{if } n>X.\end{aligned}$$

*Then for $T, X>1$ one has*

$$\sum_{n\in\mathbf{N}}\varepsilon_X(n)\Lambda(n) = X - \sum_{\substack{\rho\in Z\\ |\mathrm{Im}(\rho)|<T}} X^\rho/\rho - \zeta'(0)/\zeta(0) - (1/2)\log(1-X^{-2})$$
$$+ O(X(\log X)/T + X(\log T)^2/T + \log X\,\operatorname{Min}(1, X/T\langle X\rangle))$$
$$+ O((T^{-1}\log T + X^{-T})/\log X).$$

Note that

$$\zeta'(0)/\zeta(0) = \log 2\pi;$$

see Exercise 3.1. The summation on the left-hand side of this explicit formula is, for non-integral $X$, equal to $\sum_{n<X}\Lambda(n)$.

## 3.9

On letting $T\to\infty$ in Theorem 3.8 we obtain the following very striking formula which really warrants the name of an 'explicit formula'.

**Corollary:** *With the notations of Theorem 3.8 one has for* $X>1$

$$\sum_{n\in\mathbf{N}}\varepsilon_X(n)\Lambda(n)=X-\sum_{\rho\in Z}X^\rho/\rho-\log 2\pi-(1/2)\log(1-X^{-2})$$

*where*

$$\sum_{\rho\in Z}X^\rho/\rho$$

*is defined to be*

$$\lim_{T\to\infty}\sum_{\rho\in Z:|\mathrm{Im}(\rho)|<T}X^\rho/\rho.$$

## 3.10

We shall now prove Theorem 3.8. The techniques used here are similar to those used in § 3.7. Central to the proof is Perron's Formula (A2.5); for $\sigma>0$, $T>1$

$$\begin{aligned}(1/2\pi\mathrm{i})\int_{\sigma-\mathrm{i}T}^{\sigma+\mathrm{i}T}X^s s^{-1}\,\mathrm{d}s&=\varepsilon_X(1)+O(\mathrm{Min}(X^\sigma,X^\sigma/T|\log X|)) &(X\neq 1)\\&=\varepsilon_X(1)+O(T^{-1}) &(X=1)\end{aligned}$$

We shall study the integral

$$(1/2\pi\mathrm{i})\int_{\sigma-\mathrm{i}T}^{\sigma+\mathrm{i}T}\{-\zeta'(s)/\zeta(s)\}X^s s^{-1}\,\mathrm{d}s$$

first of all using Perron's Formula and secondly by making use of the analytic properties of the zeta-function.

We assume that $\sigma>1$ and use

$$-\zeta'(s)/\zeta(s)=\sum_n\Lambda(n)n^{-s}$$

and obtain

$$\sum_n\Lambda(n)(1/2\pi\mathrm{i})\int_{\sigma-\mathrm{i}T}^{\sigma+\mathrm{i}T}(X/n)^s s^{-1}\,\mathrm{d}s$$

which, by Perron's Formula and $\varepsilon_{X/n}(1)=\varepsilon_X(n)$, is equal to

$$\sum_n\varepsilon_X(n)\Lambda(n)+O\Bigg(\sum_{\substack{n>1\\ n\neq X}}\Lambda(n)X^\sigma n^{-\sigma}\,\mathrm{Min}\{1,1/T|\log(X/n)|\}\Bigg)+O(\Lambda(X)T^{-1}).$$

This we shall now simplify. In the summation inside the first '$O$' term we consider first that part taken over $n < X/2$ and $n > 3X/2$. In these cases $|\log(X/n)| > \log(3/2) > 0$ and so the sum can be bounded by

$$O\left(\sum_{n<X/2} + \sum_{n>3X/2} \Lambda(n) X^{\sigma} n^{-\sigma}/T\right)$$

which is bounded by

$$O\left(\sum_{n>1} \Lambda(n) n^{-\sigma} X^{\sigma}/T\right).$$

We shall suppose that $\sigma < 2$ and as $-\zeta'(s)/\zeta(s)$ has a simple pole at $s = 1$ this is bounded by $O(X^{\sigma}/(\sigma^{-1})T)$.

We next consider the remaining terms in the sum. Let $X_0$ be an integer (which is unique unless $X = 1/2$) such that $|X - X_0| = \langle X \rangle$. We shall write $n = X_0 + j$; note that

$$\begin{aligned} |\log(X/n)|^{-1} &= O(X/j) \qquad (j \neq 0) \\ &= O(X/\langle X \rangle) \quad (j = 0). \end{aligned}$$

We note also that $X^{\sigma} n^{-\sigma} = O(1)$ and that $\Lambda(n) \leqslant \log n \leqslant \log(3X/2)$ so that these remaining terms give

$$O(X(\log X)^2/T) + O(\mathrm{Min}(1, X/\langle X \rangle T) \log X) + O((\log X)/T)$$

where in the last two terms the former can be omitted if $\langle X \rangle = 0$ and the latter otherwise. The last term is dominated by the first and can be omitted anyway.

We shall next apply Cauchy's Theorem. We need to observe that there exists a $T^*$ with $|T - T^*| < 1$ and $|\mathrm{Im}(\rho) - T^*| > c/\log T$ for a certain constant $c$. Altering $T$ to $T^*$ makes no difference to the estimate which we have just made. It could change the result to be proved by a sum of the form

$$\sum X^{\rho}/\rho$$

where the summation is taken over the zeros in a pair of intervals each of length at most 1. This would contribute an error of $O(X(\log T)/T)$ since $|X^{\rho}| \leqslant X$, there are $O(\log T)$ such zeros and $\rho^{-1} = O(T^{-1})$ Thus we can assume that $T$ differs from the ordinate of a zero by $O((\log T)^{-1})$ if we allow an additional error of

$$O(X(\log T)/T).$$

We next replace the path from $-\mathrm{i}T$ to $+\mathrm{i}T$ by that from $-\mathrm{i}T$ to $-M - \mathrm{i}T$, from $-M - \mathrm{i}T$ to $-M + \mathrm{i}T$ and from $-M + \mathrm{i}T$ to $+\mathrm{i}T$. Here $M$ is to be a large odd integer. We shall then add the sum of the residues of the integrand to the sum of these three integrals. We shall let $M \to \infty$; the

contribution of the integral over $-M-\mathrm{i}T$ to $-M+\mathrm{i}T$ tends to zero by Theorem 3.3 and the two other integrals converge for the same reason.

The poles of the integrand are at $s=1$ (from the pole of $\zeta$) $s=0$ (from $s^{-1}$), $s=-2j$ ($j>0$ an integer, from the 'trivial' zeros of the zeta-function) and at $\rho\in Z$ with $|\mathrm{Im}(\rho)|<T$ (from the zeros of the zeta-function). These yield

$$X-\zeta'(0)/\zeta(0)+\sum X^{-2j}/(2j)-\sum_{\rho\in Z:|\mathrm{Im}(\rho)|<T} X^{\rho}/\rho.$$

As the third term is $-(1/2)\log(1-X^{-2})$ it only remains to estimate the two integrals from $-\infty\pm\mathrm{i}T$ to $\infty\pm\mathrm{i}T$.

On these lines, by the assumption that $T$ differs from the ordinate of a zero by at least $c/\log T$ it follows from Theorem 3.3 that

$$-\zeta'(s)/\zeta(s)=O(\log|s|).$$

Consequently on these lines $-\zeta'(s)/\zeta(s)s^{-1}$ is dominated by $O((\log T)/T)$ when $-T<\mathrm{Re}(s)$ and by $O(1)$ when $\mathrm{Re}(s)\geqslant -T$. Thus the integrals are dominated by

$$\int_{-T}^{\sigma} X^{u}\mathrm{d}u(\log T)/T+\int_{-\infty}^{-T} X^{u}\mathrm{d}u\leqslant X^{\sigma}(\log T)/T\log X+X^{-T}/\log X.$$

The estimate which we now have is

$$\begin{aligned}\sum\varepsilon_X(n)\Lambda(n)=X-\zeta'(0)/\zeta(0)-\sum X^{\rho}/\rho-(1/2)\log(1-X^{-2})\\ +O(X^{\sigma}/(\sigma-1)T)\\ +X(\log T)^2/T+\mathrm{Min}(1,X/T\langle X\rangle)\log X+X(\log T)/T\\ +X^{\sigma}(\log T)/T\log X+X^{-T}(\log X).\end{aligned}$$

We let $\sigma=1+(\log X)^{-1}$ if $X\geqslant 2$ and $\sigma=2$ if $X<2$. Thus if $X\geqslant 2$ one has $X^{\sigma}=\mathrm{e}X$; if $X<2$ then $X^{\sigma}<4$. From this we have directly the estimate of the theorem; the final term of the estimate is only significant if $1<X<2$.

## 3.11

Note that the method of proof used in § 3.10 is applicable to much more general situations – see, for example, Exercise 3.17. In order to be able to use Theorem 3.8 or Corollary 3.9 it is necessary to have more information concerning the zeros of the zeta-function. This is the objective of the next chapter.

## Exercises

3.1 Show that $\zeta'(0)/\zeta(0)=\log 2\pi$.

3.2 Show that $\sum x^{\rho}/\rho$ does not converge absolutely, even when the terms $\rho$ and $\bar{\rho}$ are grouped together.

3.3 Let

$$\begin{aligned} f(x) &= x^{u}(\log x^{-1})^{v-1} && (0 < x \leqslant 1) \\ &= 0 && (1 < x < \infty) \end{aligned}$$

where $\mathrm{Re}(u) > 0$ and $\mathrm{Re}(v) > 1$. Show that

$$M(f, s) = \Gamma(v)(s+u)^{-s}$$

and

$$\Delta_{\infty}(f) = \Gamma(v)2^{-v}\sum_{l \geqslant 0}\{l + (u - 1/2)/2\}^{-v}.$$

3.4 Let

$$\begin{aligned} f(x) &= x^{2u}(1-x^2)^{v-1} && (0 < x \leqslant 1) \\ &= 0 && (1 < x < \infty). \end{aligned}$$

Show that

$$M(f, s) = 2^{-1}\Gamma(u + u/2)\Gamma(v)/\Gamma(u + v + s/2)$$

and

$$\Delta_{\infty}(f) = 2^{-1}\Gamma(u + 1/4)\Gamma(v-1)\Gamma(u + v - 3/4).$$

3.5 Show that if $\mathrm{Re}(u) > 1/2$, $\mathrm{Re}(v) > 1$

$$\begin{aligned} &\sum_{\rho \in Z}\{\rho + (u - 1/2)\}^{-v} + \Gamma(v)^{-1}\sum_{p}(\log p)^{v} Z(v, p^{-1/2-u}) \\ &= (u + 1/2)^{-v} - 2^{-v}\sum_{l \geqslant 1}\{l + (u - 1/2)/2\}^{-v} \end{aligned}$$

where

$$Z(t, x) = \sum_{k \geqslant 1} k^{t} x^{k}$$

and, as usual, $p$ runs through the set of primes.

3.6 Show that if $\mathrm{Re}(u) > 1/2$ then the function

$$v \mapsto \sum_{\rho \in Z}\{\rho + (u - 1/2)\}^{-v}$$

has an analytic continuation to the entire plane as a meromorphic function with a single pole of order 1 situated at $v = 1$ with residue $-1/2$.

3.7 Show that the function

$$v \mapsto \sum_{\rho \in Z}\{\rho + (u - 1/2)\}^{-v}$$

is not bounded in any half-plane of the form $\{s \in \mathbf{C}: \mathrm{Re}(s) > a\}$ with $a > 1$.

3.8 Prove that the function

$$v \mapsto \sum_{\rho \in Z} \rho^{-v}$$

has an analytic continuation to the entire plane as a meromorphic function with a single pole at 1 of order 1 and residue $-1/2$. Deduce that

$$s \mapsto \sum_p (\log p)^s/p$$

which is defined in $\{s \in \mathbf{C}: \mathrm{Re}(s) < 0\}$ has an analytic continuation to the entire plane as a meromorphic function with at most two simple poles located at 0 and 1.

3.9 Let

$$\begin{aligned} f(u) &= u^{-t} && (1 \leqslant u \leqslant x) \\ &= \frac{(\log y - \log u)}{(\log y - \log x)} u^{-t} && (x < u \leqslant y) \\ &= 0 && \text{otherwise.} \end{aligned}$$

Show that

$$M(f,s) = \frac{y^{s-t} - x^{s-t}}{(s-t)^2 \log(y/x)} - \frac{1}{s-t},$$

and (with a suitable modification of the definition of $\Delta_\infty$)

$$\begin{aligned} \Delta_\infty(f) = {} & \Gamma'((t+1/2)/2)/2\Gamma((t+1/2)/2) + (1/2)\log \pi \\ & + \sum_{j \geqslant 0} \frac{y^{-(1/2+t+2j)} - x^{-(1/2+t+2j)}}{(1/2+t+2j)^2 \log(y/x)}. \end{aligned}$$

3.10 Let

$$\begin{aligned} \Lambda(n; x, y) &= \Lambda(n) && (1 \leqslant n \leqslant x) \\ &= \Lambda(n) \frac{(\log y - \log n)}{(\log y - \log x)} && (x < n \leqslant y) \\ &= 0 && (n > y). \end{aligned}$$

Show that

$$\begin{aligned} \zeta'(s)/\zeta(s) = {} & -\sum_{n<y} \Lambda(n; x, y) n^{-s} + \frac{y^{1-s} - x^{1-s}}{(1-s)^2 \log(y/x)} \\ & + \{\log(y/x)\}^{-1} \sum_{j \geqslant 1} \frac{x^{-(s+2j)} - y^{-(s+j)}}{(s+2j)^2} \\ & + \{\log(y/x)\}^{-1} \sum_{\rho \in Z} \frac{x^{\rho - s} - y^{\rho - s}}{(s-\rho)^2}. \end{aligned}$$

(This formula is due to Selberg.)

3.11 Show that if $\mathrm{Re}(v) > \mathrm{Re}(u) + 1 > 3/2$ then

$$\begin{aligned} & \sum_{\rho \in Z} \frac{\Gamma(u+\rho/2)}{\Gamma(v+\rho/2)} + 2\sum_p \log p \sum_{k \geqslant 1} \frac{(p^{2k}-1)^{-u}(1-p^{-2k})^v}{(v-u)} \\ & = \frac{\Gamma(u)}{\Gamma(v)} - \frac{(u-1/2)}{(v-u)} \frac{\Gamma(u+1/2)}{\Gamma(v+1/2)}. \end{aligned}$$

Prove that

$$(u, v) \mapsto \sum_{\rho \in Z} \frac{\Gamma(u + \rho/2)}{\Gamma(v + \rho/2)}$$

has an analytic continuation to a meromorphic function on $\mathbf{C}^2$ and determine the divisors of the singularities.

3.12 Let

$$\begin{aligned} f(x) &= x^{-t}/(1-2t) \qquad (0 < x \leqslant 1) \\ &= x^{t-1}/(1-2t) \qquad (1 < x < \infty). \end{aligned}$$

Show that if $\mathrm{Re}(t) < -1/2$ then

$$M(f, s) = \{(s-t)(1-s-t)\}^{-1} \quad \text{if } \mathrm{Re}(t) < \mathrm{Re}(s) < 1 - \mathrm{Re}(t)$$

and

$$\Delta_\infty(f) = \log \pi - \Gamma'((t-1/2)/2)/\Gamma((t-1/2)/2)$$

3.13 From Exercise 3.12 and Theorem 3.6 deduce Corollary 3.2.

3.14 Derive Theorem 3.3 from Theorem 3.6.

3.15 Derive Theorem 3.6 from Theorem 3.8. And conversely?

3.16 Suppose that $(T_j)$ is a sequence $T_j \to \infty$ and $c > 0$ are such that $|\mathrm{Im}(\rho) - T_j| > c/\log T_j$ for $\rho \in Z$. Show that there exists $c' > 0$ so that for $s$ with $\mathrm{Im}(s) = T_j$ and $-1 \leqslant \mathrm{Re}(s) \leqslant 2$ one has

$$\zeta(s)^{-1} = O(\mathrm{e}^{c'(\log T_j)^2}).$$

3.17 Suppose that $\alpha < 0$, $\beta > 1$ and that $f$ is a function on $\mathbf{R}_+^\times$ such that both $f$ and $\hat{f}$ are of type $(\alpha, \beta)$, and also such that there exist constants $a, b > 0$ so that for $s$ with $\alpha < \mathrm{Re}(s) < \beta$ one has

$$M(f, s) = O(\mathrm{e}^{-a|s|^b}).$$

Show that if all the zeros of the zeta-function are simple then

$$\sum_{n \geqslant 1} \{\mu(n)/n\}\{f(n^{-1}) - \hat{f}(n^{-1})\} = \lim_{j \to \infty} \sum_{\rho \in Z : |\mathrm{Im}(\rho)| < T_j} \zeta'(\rho)^{-1} M(f, 1-\rho)$$

where $(T_j)$ is a sequence as in Exercise 3.16. (The special case of this where $f(x) = \mathrm{e}^{-ax^2}$ is a formula due to Ramanujan.)

# 4

# *The zeros of the zeta-function and the Prime Number Theorem*

## 4.1

In Chapter 3 we have seen the remarkable correspondence that relates the prime numbers with the zeros of the zeta-function. In order to derive significant conclusions concerning the asymptotic distribution of prime numbers it is necessary to describe that of the zeros. This is in its turn possible through the connection with prime numbers, i.e. through the explicit formulae. The basic example of this is given by Hadamard's zero-free region Theorem 4.2. In fact, the argument given here, apart from some minor variations, is the only one available at present. Somewhat better estimates are available but they depend on subsidiary estimates for the zeta-function (see Exercise 4.7); these in their turn are based on estimates for exponential sums of the form $\sum_{N<n\leqslant M} n^{-it}$ and the approximate functional equation. The estimation of the exponential sums involves techniques which belong to different circles of ideas than those forming the subject matter of this book. See Chapter 6 for more details.

## 4.2

**Theorem:** *There exists a constant $c>0$ so that if $\rho\in Z$ then*

$$\mathrm{Re}(\rho)<1-c\,\mathrm{Min}(1,\{\log|\mathrm{Im}(\rho)|\}^{-1}).$$

*Proof*: The proof of this theorem is based on an application of Theorem 3 3 and a remarkable but simple inequality. This is

$$3+4\cos x+\cos 2x=2(1+\cos x)^2\geqslant 0.$$

It follows from this that if $\sigma>1$ then

$$\mathrm{Re}\{-3\zeta'(\sigma)/\zeta(\sigma)-4\zeta'(\sigma+it)/\zeta(\sigma+it)-\zeta'(\sigma+2it)/\zeta(\sigma+2it)\}\geqslant 0 \quad (4.1)$$

since the left-hand side is equal to

$$\sum_{k>0,p}(\log p)p^{-k\sigma}\{3+4\cos(kt\log p)+\cos(2kt\log p)\}$$

which is a sum of positive terms.

From Theorem 3.3 we have that for $0 < \mathrm{Re}(s) < 2$, $|\mathrm{Im}(s)| > 1$

$$-\zeta'(s)/\zeta(s) = -\sum_{\rho:|\mathrm{Im}(s-\rho)|<1} (s-\rho)^{-1} + O(\log|s|).$$

If $\mathrm{Re}(s) > 1$ then $\mathrm{Re}(s-\rho) > 0$ and so $\mathrm{Re}\{(s-\rho)^{-1}\} > 0$. Thus in the region $\{s \in \mathbf{C}: 1 < \mathrm{Re}(s) < 2, |\mathrm{Im}(s)| > 1\}$ one has for a certain constant $c > 0$

$$\begin{aligned}\mathrm{Re}\{-\zeta'(s)/\zeta(s)\} &< -\mathrm{Re}\{(s-\rho_1)^{-1}\} + c(1+\log|s|) \\ &< c(1+\log|s|)\end{aligned}$$

where $\rho_1$ is a zero satisfying $|\mathrm{Im}(s-\rho_1)| < 1$. We shall now combine this with the inequality above as follows. Let $\rho$ be a fixed zero of the zeta-function and let $t = \mathrm{Im}(\rho)$. We know that $-\zeta'(\sigma)/\zeta(\sigma) < (\sigma-1)^{-1} + c'$ for $1 < \sigma$ and a certain constant $c'$. Using this and the first inequality for $\mathrm{Re}\{-\zeta'(s)/\zeta(s)\}$ for $\sigma + it$ with $\rho_1 = \rho$ and the second for $\sigma + 2it$ we obtain

$$-3(\sigma-1)^{-1} + 4\{\sigma - \mathrm{Re}(\rho)\}^{-1} + c' + c(2 + 2\log|t| + \log 2) \geqslant 0.$$

Let

$$\sigma = 1 + \{1 - \mathrm{Re}(\rho)\};$$

we then obtain

$$-\{1-\mathrm{Re}(\rho)\}^{-1} + c' + c(2 + 2\log|t| + \log 2) \geqslant 0$$

which, since $t = \mathrm{Im}(\rho)$, gives the estimate of the theorem if $|\mathrm{Im}(\rho)| > 1$.

To complete the proof it suffices to show that there are no zeros with $\mathrm{Re}(\rho) = 1$, $|\mathrm{Im}(\rho)| \leqslant 1$. Suppose there are such zeros and let $\rho$ be such that $|\mathrm{Im}(\rho)|$ is maximal. Then if $\rho = 1 + it$ in view of our assumption and what was proved above there are no zeros $\rho'$ with $\mathrm{Re}(\rho') = 1$, $|\mathrm{Im}(\rho')| > |\mathrm{Im}(\rho)|$ In particular $1 + 2it$ is not a zero. With this value of $t$ we use the inequality (4.1) again and let $\sigma \to 1$. The first term is asymptotic to $3(\sigma-1)^{-1}$, the second to $-4(\sigma-1)^{-1}$ and the third is bounded. Thus the left-hand side would tend to $-\infty$ as $\sigma \to 1$ and this contradicts the inequality. Hence there is no such zero.

## 4.3

Theorem 4.2 forms the basis of the proof of the Prime Number Theorem as we shall show. In fact the Prime Number Theorem is based on the fact that there are no zeros of the Riemann zeta-function on $1 + i\mathbf{R}$ which is much less than we have actually proved. This deduction can be made by using a Tauberian theorem (such as the Wiener–Ikehara Theorem) which, although it gives a painless proof, is not really open to improvements, such as the determination of error terms.

The first version of the Prime Number Theorem is the following:

**Theorem:** *There exists a constant $c > 0$ so that as $X \to \infty$*

$$\sum_{n<X} \Lambda(n) = X + O(X e^{-c(\log X)^{1/2}}).$$

Although this theorem does not correspond to the naive way that one might think about the Prime Number Theorem, for example, as formulated in Theorem 4.5 below, the form given here is the most useful.

Note that

$$X e^{-c(\log X)^{1/2}} = O(X/(\log X)^N)$$

as $X \to \infty$ for any $N > 0$. On the other hand we can write

$$X e^{-c(\log X)^{1/2}} = X^{1-c/(\log X)^{1/2}}$$

from which it follows that there is no $a < 1$ such that

$$X e^{-c(\log X)^{1/2}} = O(X^a).$$

## 4.4

In this section we shall prove Theorem 4.3. The idea is to use Theorem 3.8 with a suitable choice of $T$. Let us first use Theorem 4.2 to estimate

$$\sum_{\substack{\rho \in Z \\ |\mathrm{Im}(\rho)| < T}} |X^\rho|/\rho.$$

By Theorem 4.2 one has

$$X^\rho \leqslant X^{1-c(\log T)^{-1}}$$

as long as $T > 3$. Moreover by Corollary 3.4

$$\sum_{|\mathrm{Im}(\rho)| < T} |\rho|^{-1} = O\left(\sum_{0<m<T} m^{-1} \log m\right)$$
$$= O((\log T)^2)$$

by comparing the series with the corresponding integral as in Cauchy's test for convergence.

In the formula which is to be proved we may assume that $X$ is such that $X - 1/2$ is an integer since the sum varies by at most $O(\log X)$ (as $\Lambda(n) \leqslant \log n$) as $X$ varies over an interval of length 1 and on the right-hand side both $X$ and $X e^{-c(\log X)^{1/2}}$ vary by a bounded quantity.

With the assumption that $X$ is of this form, Theorem 3.8 and the estimate we have just made yield

$$\sum_{n<X} \Lambda(n) = X + O(X^{1-c/\log T}) + O(1) + O(X(\log X)/T)$$
$$+ O(X(\log T)^2/T) + O(X).$$

We now choose $T$ so that $\log T = (\log X)^{1/2}$ and, as $X = e^{\log X}$ we obtain

for the right-hand side

$$X + O(Xe^{-c(\log X)^{1/2}}) + O(1) + O(X(\log X)e^{-(\log X)^{1/2}}) + O(\log X).$$

As $\log X = e^{\varepsilon(\log X)^{1/2}}$ for any $\varepsilon > 0$ we see that for any $c_1 < c$ this is

$$X + O(Xe^{-c_1(\log X)^{1/2}}).$$

Using $c_1$ as the constant whose existence is asserted in the statement of the theorem we see that we have now proved the estimate of the theorem.

## 4.5

As a corollary of the theorem which we have just proved we shall deduce an estimate on

$$\pi(X) = \text{Card}\{p: p \text{ prime}, p \leqslant X\}.$$

The form of the Prime Number Theorem which corresponds to the original calculations and conjectures is:

**Theorem**: *There exists a constant $c > 0$ so that*

$$\pi(X) = \int_2^X (\log u)^{-1}\,du + O(Xe^{-c(\log X)^{1/2}}).$$

Before we prove this we observe that by integration by parts one has

$$\int_2^X (\log u)^{-1}\,du = X/\log X - 2/\log 2 + \int_2^X (\log u)^{-2}\,du$$

Since for $X > 4$

$$\int_2^X (\log u)^{-2}\,du = \int_2^{X^{1/2}} (\log u)^{-2}\,du + \int_{X^{1/2}}^X (\log u)^{-2}\,du < \int_2^{X^{1/2}} (\log 2)^{-2}\,du + \int_{X^{1/2}}^X (\log X^{1/2})^{-2}\,du < 3X^{1/2} + 4X/(\log X)^2.$$

Thus

$$\int_2^X (\log u)^{-2}\,du = O(X/(\log X)^2)$$

and so it follows that

$$\pi(X) = X/(\log X) + O(X/(\log X)^2)$$

and in particular

$$\pi(X) \sim X/\log X.$$

We shall now deduce this form of the Prime Number Theorem from Theorem 4.3 using the basic technique of partial summation.

First of all we note that

$$\sum_{n<X}\Lambda(n)=\sum_{p,k:p^k<X}\log p$$
$$=\sum_{p<X}\log p+\sum_{p,k\geqslant 2:p^k<X}\log p,$$

where we use the convention that $p$ represents a prime. In the second summation for any given prime $p$ there are at most $\log X/\log p$ summands, and as $k\geqslant 2$, one also has $p\leqslant X^{1/2}$. Thus the second term is bounded by

$$\sum_{2\leqslant p\leqslant X^{1/2}}\log X\leqslant X^{1/2}\log X.$$

It follows that Theorem 4.3 can be phrased as

$$\sum_{p<X}\log p=X+O(X\mathrm{e}^{-c(\log X)^{1/2}}).$$

Denote the left-hand side by $A(X)$. Then

$$\left.\begin{matrix}\{A(n+1)-A(n)\}/\log n=1\\ 0\end{matrix}\right\}\quad\begin{matrix}\text{if } n \text{ is a prime}\\ \text{if } n \text{ is not a prime.}\end{matrix}$$

Thus

$$\pi(X)=\sum_{n\leqslant X}\{A(n+1)-A(n)\}/\log n$$
$$=\{A(3)-A(2)\}/\log 2+\{A(4)-A(3)\}/\log 3+\cdots$$
$$+\{A([X]+1)-A([X])\}/\log[X].$$

Note that $A(2)=0$ and group the terms $A(3)$, $A(4)$ and so on together; one finds that

$$\pi(X)=A(3)\{(\log 2)^{-2}-(\log 3)^{-1}\}+A(4)\{(\log 3)^{-1}-(\log 4)^{-1}\}+\cdots$$
$$+A([X])\{(\log[X-1])^{-1}-(\log[X])^{-1}\}+A([X]+1)/\log[X].$$

We know that for certain constants $c,c_1>0$ and $j\leqslant X$

$$|A(j)-j|<c_1X\mathrm{e}^{-c(\log X)^{1/2}}.$$

Thus we find that

$$\pi(X)=3\{(\log 2)^{-1}-(\log 3)^{-1}\}+\cdots$$
$$+[X]\{(\log[X-1])^{-1}-(\log[X])^{-1}\}$$
$$+([X]+1)/\log[X]+E(X)$$

where

$$E(X)\leqslant c_1X\mathrm{e}^{-c(\log X)^{1/2}}(\{(\log 2)^{-1}-(\log 3)^{-1}\}+\{(\log 3)^{-1}-(\log 4)^{-1}\}$$
$$+\cdots+\{(\log[X]-1)^{-1}-(\log[X])^{-1}\}+(\log[X])^{-1})$$
$$=c_1X\mathrm{e}^{-c(\log X)^{1/2}}(\log 2)^{-1}.$$

If we now group the terms in $(\log 2)^{-1}$, $(\log 3)^{-1}$ etc. together then we have

$$\pi(X)=3(\log 2)^{-1}+\{(\log 3)^{-1}+(\log 4)^{-1}+\cdots+(\log[X])^{-1}\}+E(X).$$

By the argument used in the proof of Cauchy's criterion for the convergence of a sum in terms of the corresponding integral we have

$$(\log 2)^{-1} + (\log 3)^{-1} + \cdots + (\log [X])^{-1} < (\log 2)^{-1} + \int_2^{[X]} (\log u)^{-1} \, du$$

and

$$(\log 2)^{-1} + (\log 3)^{-1} + \cdots + (\log [X])^{-1} > \int_2^{[X]} (\log u)^{-1} \, du + (\log [X])^{-1}.$$

It follows that

$$(\log 2)^{-1} + (\log 3)^{-1} + \cdots + (\log [X])^{-1} = \int_2^{X} (\log u)^{-1} \, du + O(1).$$

The formula above for $\pi(X)$ combined with this and the estimate for $E(X)$ yields the theorem.

## 4.6

We now turn to the problem of studying the distribution of the zeros from another standpoint. Let

$$N(T) = \text{Card}\,\{\rho \in Z : 0 < \text{Im}(\rho) < T\}.$$

As was indicated in § 1.5 Riemann described without proof the asymptotic behaviour of this function and his claims were justified in 1895 by von Mangoldt. In order to describe the results it is convenient to introduce the function $S(T)$ which is defined for those $T > 0$ such that $T \notin \{\text{Im}(\rho) : \rho \in Z\}$. It is defined as the variation of $\pi^{-1} \arg \zeta(s)$ along the path consisting of a segment from 2 to $2 + iT$ and $2 + iT$ to $1/2 + iT$. The number 2 here could be replaced by any other $\sigma > 1$ without changing the value of the function.

## 4.7

The function $S$ is not very well understood. The estimate that we need is given in the following theorem.

**Theorem:** *One has as* $T \to \infty$ *that* $S(T) = O(\log T)$

*Proof*: We have that

$$\begin{aligned} |\zeta(2 + it) - 1| &= \left| \sum_{n \geqslant 2} n^{-2-it} \right| \\ &\leqslant \sum_{n \geqslant 2} n^{-2} \\ &< 1. \end{aligned}$$

Thus $\mathrm{Re}\{\zeta(2+it)\} > 0$ and hence the variation of $\arg\zeta$ along the segment joining 2 to $2+iT$ is at most $\pi/2$.

The variation along the segment joining $2+iT$ to $1/2+iT$ is the imaginary part of the integral

$$\int_{2+iT}^{1/2+iT} \zeta'(s)/\zeta(s)\,ds.$$

By Theorem 3.3 this is

$$\sum_{\substack{\rho\in Z \\ |\mathrm{Im}(\rho)-T|<1}} \mathrm{Im}\left\{\int_{2+iT}^{1/2+iT} (s-\rho)^{-1}\,ds\right\} + O(\log T).$$

In the sum here each summand is in absolute value at most $\pi$ and by Corollary 3.4 there are at most $O(\log T)$ summands. Hence the entire sum is $O(\log T)$, as asserted.

## 4.8

Davenport remarks that this is just about the only estimate that has not been improved in this century. It is known that $S$ oscillates quite violently – for example an application of Littlewood's Theorem (A7.1, see Exercise 4.9) yields

$$\int_0^T S(t)\,dt = O(\log T),$$

an estimate also due to Littlewood. If $S$ did not oscillate one would expect $O(T\log T)$ for this estimate. The numerical evidence indicates that $S(T)$ is very small; for $T < 545\,439\,823$ recent computations of te Riele, van de Lune and Winter (te Riele, van de Lune & Winter (1)) show that $S(T)$ does not exceed 2.32 and is 'usually' less than 2. At zeros it jumps by the multiplicity and hence by at least 1. Thus it cannot be estimated much more precisely than by 1. To date there are no known zeros of multiplicity greater than 1. These results might indicate that Theorem 4.7 is far from the truth. Indeed on the Lindelöf Hypothesis (§ 5.7) one can show that $S(t) = o(\log t)$ and on the (stronger) Riemann Hypothesis (§ 5.1) that $S(t) = O(\log|t|/\log\log|t|)$. In the opposite direction one can show, also under the Riemann Hypothesis, that for any $\varepsilon > 0$ there exists a sequence $t_m \to \infty$ such that

$$S(t_m) > (\log t_m)^{1/2-\varepsilon}$$

and a similar result for negative values. If the Riemann Hypothesis is not assumed then one can prove analogous results but with the exponent 1/2 replaced by 1/3. There are also similar statements for the integrated form

$\int_0^T S(t)\mathrm{d}t$. The numerical computations and these theorems are not at all at variance as $(\log t)^{1/2}$ grows very slowly – for example

$$(\log 545\,439\,823)^{1/2} = 4.485\,2\ldots$$

For more details on these questions see Titchmarsh (2) §§9.9–9.13 and 14.10–14.13 and Ghosh (1).

## 4.9

We now come to the description of the asymptotic behaviour of the function $N$.

**Theorem:** *One has, as* $T \to \infty$, $T \notin \{\mathrm{Im}(\rho): \rho \in Z\}$

$$N(T) = (T/2\pi)\log(T/2\pi) - (T/2\pi) + 7/8 + S(T) + O(T^{-1})$$

*and*

$$N(T) = (T/2\pi)\log(T/2\pi) - (T/2\pi) + O(\log T).$$

Before we prove this we note that the first estimate suffices, given the knowledge of $S(T)$, to compute $N(T)$ precisely for large enough $T$ since $N(T)$ has to be an integer. The techniques described in Chapter 2, and more often the 'approximate functional equation' suffice to compute $\zeta$ with sufficient accuracy to determine $S(T)$. On the other hand by means of the integral representation theorems such as that of §2.10, or by the 'approximate function equation', one can compute $\zeta$ on the critical line and determine the number of real zeros of $\zeta(1/2 + it)$ by real variable techniques. By comparing the number of zeros on the line $\{s: \mathrm{Re}(s) = 1/2\}$ with $N(T)$ one shows that there are no zeros which do not lie on the critical line. In this way, with various computational improvements, it has been verified that the first 1 500 000 001 zeros all lie on this line. They are simple which is what one would expect from the very small sizes of $S$ in the same range, but the difference between two consecutive zeros can be much smaller than the expected $2\pi/\log T$.

For more details on the computation of the Riemann zeta-function see Edwards (1), Titchmarsh (2); for tables of the zeros see Haselgrove (1), Haselgrove & Miller (1), Spira (1) and for the most extensive computations see te Riele *et al.* (1).

## 4.10

We shall now prove Theorem 4.9. It is clear that in view of Theorem 4.7 we need only prove the first estimate. We introduce first a notation which

will be useful. Let $f$ be any complex valued function on an open set in $\mathbf{C}$. Let $x, y$ be two points at which $f$ is defined and such that $f$ is defined, continuous and nowhere zero on the segment joining $x$ to $y$. On this segment we can define a continuous branch of $\arg f$; we let

$$\Delta_x^y \arg f = (\arg f)(y) - (\arg f)(x).$$

This does not depend on the choice of the branch taken.

Let

$$\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The zeros of this function in $\{s \in \mathbf{C}: -1 \leqslant \mathrm{Re}(s) \leqslant 2\}$ are those of $\zeta$ and it has two simple poles. Let $T$ be such that $T \notin \{\mathrm{Im}(\rho): \rho \in Z\}$. By the principle of the argument of complex analysis one has

$$\begin{aligned} &2\pi\{2N(T) - 2\} \\ &\quad = \Delta_{2-iT}^{2+iT} \arg \xi + \Delta_{2+iT}^{-1+iT} \arg \xi + \Delta_{-1+iT}^{-1-iT} \arg \xi + \Delta_{-1-iT}^{2-iT} \arg \xi. \end{aligned}$$

Recall next that $\xi(s) = \xi(1-s)$ and $\bar{\xi}(s) = \xi(\bar{s})$. It follows that

$$\Delta_2^{2+iT} \arg \xi = \Delta_{2-iT}^{2} \arg \xi = -\Delta_{-1}^{-1+iT} \arg \xi = -\Delta_{-1-iT}^{-1} \arg \xi$$

and

$$\Delta_{2+iT}^{1/2+iT} \arg \xi = \Delta_{1/2+iT}^{-1+iT} \arg \xi = -\Delta_{2-iT}^{1/2-iT} \arg \xi = -\Delta_{1/2-iT}^{-1-iT} \arg \xi.$$

Consequently we have

$$2\pi\{2N(T) - 2\} = 4(\Delta_2^{2+iT} \arg \xi + \Delta_{2+iT}^{1/2+iT} \arg \xi).$$

We next use the fact that

$$\Delta_x^y \arg f_1 f_2 = \Delta_x^y \arg f_1 + \Delta_x^y \arg f_2$$

and obtain

$$\begin{aligned} N(T) = 1 &+ \pi^{-1}(\Delta_2^{2+iT} \arg\{\pi^{-s/2}\Gamma(s/2)\} + \Delta_{2+iT}^{1/2+iT} \arg\{\pi^{-s/2}\Gamma(s/2)\}) \\ &+ \pi^{-1}(\Delta_2^{2+iT} \arg \zeta + \Delta_{2+iT}^{1/2+iT} \arg \zeta). \end{aligned}$$

The third term is by definition $S(T)$. Since $\log \Gamma$ is defined on $\{s: \mathrm{Re}(s) > 0\}$ the second term is

$$\pi^{-1} \,\mathrm{Im}\{\log \Gamma(1/4 + iT/2)\} - (T/2\pi)\log \pi.$$

By the asymptotic expansion of $\log \Gamma$ (Stirling's Formula) we have

$$\mathrm{Im}\{\log \Gamma(1/4 + iT/2)\} = (T/2)\log(T/2) - T/2 - \pi/8 + O(T^{-1}).$$

Combining these we obtain the formula stated in the theorem

## 4.11

In the following sections we shall give an estimate on the number of zeros which do not lie on the critical line $\{s \in \mathbf{C}: \mathrm{Re}(s) = 1/2\}$. We have already

seen in Theorem 4.2 that the zeros of the zeta-function cannot lie too close to the line $\{s\in\mathbf{C}: \mathrm{Re}(s)=1\}$, but this does not rule out the possibility that many of the zeros lie, say, on the line $\{s\in\mathbf{C}: \mathrm{Re}(s)=3/4\}$. We shall prove, by a method due to Littlewood, a result of Bohr and Landau to the effect that almost all of the zeros lie close to the critical line. This proof also has the feature that it demonstrates the value of certain integral estimates of the zeta-function, and in particular Theorem 4.12 below. Such estimates as this are very closely related to the 'approximate functional equation' but we shall give a proof which is a little simpler, and instructive in its own right, but which is related in its techniques with the 'approximate functional equation'.

Define for $T>0$ and $\sigma>1/2$ the quantity

$$N(\sigma, T)=\mathrm{Card}\,\{\rho\in Z: \mathrm{Re}(\rho)>\sigma, 0<\mathrm{Im}(\rho)<T\}.$$

**Theorem**: *For $\sigma>1/2$ one has (for fixed $\sigma$) as $T\to\infty$*

$$N(\sigma, T)=O(T).$$

Observe that for $\sigma>1/2$

$$N(\sigma, T)=o(N(T))$$

which expresses the fact that for any $\sigma>1/2$ almost all of the zeros lie in $\{s\in\mathbf{C}: 1-\sigma<\mathrm{Re}(s)<\sigma\}$. The estimate given above is by no means the best that is known and a great deal of work has been done in improving this estimate. For a survey of this see Heath-Brown (2).

## 4.12

For the proof of Theorem 4.11 we shall need the following estimate which is important in its own right.

**Theorem:** *For $\sigma>1/2$, $\sigma$ fixed we have as $T\to\infty$*

$$\int_0^T |\zeta(\sigma+\mathrm{i}t)|^2\,\mathrm{d}t=O(T).$$

It is possible to replace this bound by an asymptotic formula, for example the $O(T)$-term can be replaced by $(\sigma<1)$ $\zeta(2\sigma)T+O(T^{2-2\sigma}\log T+T^{1/2}(\log T)^{1/2})$ (see Titchmarsh (2), § 7.2). We shall not go into this question and its variants further here.

## 4.13

We shall first deduce Theorem 4.11 from Theorem 4.12. The crucial technical tool here is an integrated form of the principle of the argument

due to Littlewood, who first recognized its significance for such problems. This theorem is stated and proved in A7.1. Applied to our case it shows that for $\sigma_1 > 1/2$ and $T > 0$, $T \notin \mathrm{Im}(Z)$ and suitably small $\varepsilon > 0$

$$2\pi \int_{\sigma_1}^{1} N(\sigma, T)\mathrm{d}\sigma = \int_{\varepsilon}^{T} \{\log|\zeta(\sigma_1 + \mathrm{i}t)| - \log|\zeta(2 + \mathrm{i}t)|\}\mathrm{d}t$$
$$+ \int_{\sigma_1}^{2} \{\arg\zeta(v + \mathrm{i}T) - \arg\zeta(v + \mathrm{i}\varepsilon)\}\mathrm{d}v$$

where $\arg\zeta$ is determined by continuous variation along the sides of the rectangle with corners $2 + \mathrm{i}\varepsilon$, $2 + \mathrm{i}T$, $\sigma_1 + \mathrm{i}T, \sigma_1 + \mathrm{i}\varepsilon$. We have already seen that $\arg\zeta(v + \mathrm{i}T) = O(\log T)$. Clearly $\arg\zeta(v + \mathrm{i}\varepsilon) = O(1)$ and $\log|\zeta(2 + \mathrm{i}t)| = O(1)$. Thus we see that

$$2\pi \int_{\sigma_1}^{1} N(\sigma, T)\mathrm{d}\sigma = \int_{\varepsilon}^{T} \log|\zeta(\sigma_1 + \mathrm{i}t)|\mathrm{d}t + O(T).$$

By Jensen's inequality, in this case the integral form of the geometric mean -arithmetic mean inequality,

$$T^{-1} \int_{\varepsilon}^{T} \log|\zeta(\sigma_1 + \mathrm{i}t)|^2 \mathrm{d}t \leqslant \log\left\{T^{-1} \int_{\varepsilon}^{T} |\zeta(\sigma_1 + \mathrm{i}t)|^2 \mathrm{d}t\right\}.$$

By Theorem 4.12 the right-hand side is bounded which shows that

$$2\pi \int_{\sigma_1}^{1} N(\sigma, T)\mathrm{d}\sigma = O(T).$$

Choose now for a given $\sigma_2$ with $1/2 < \sigma_2 < 1$ a $\sigma_1$ with $1/2 < \sigma_1 < \sigma_2$. For $\sigma$ with $\sigma_1 < \sigma < \sigma_2$ we have

$$N(\sigma, T) > N(\sigma_2, T)$$

and hence

$$2\pi(\sigma_2 - \sigma_1)N(\sigma_2, T) \leqslant 2\pi \int_{\sigma_1}^{1} N(\sigma, T)\mathrm{d}\sigma.$$

From this it follows that

$$N(\sigma_2, T) = O(T)$$

which is what we had to prove.

## 4.14

It remains to prove Theorem 4.12. To do this we start from the inverse Mellin transform of the gamma-function

$$(1/2\pi\mathrm{i}) \int_{(\sigma)} \Gamma(s)x^{-s}\mathrm{d}s = \mathrm{e}^{-x}$$

which is valid for $\sigma > 0$. We suppose now that $\sigma > 1$ and replace $x$ by $nx$ and sum; this gives the formula

$$(1/2\pi i)\int_{(\sigma)} \Gamma(s)\zeta(s)x^{-s}ds = (e^x - 1)^{-1}$$

which is the Mellin inverse transform of one of the formulae used to prove the analytic continuation of the zeta-function. In view of Stirling's Formula the convergence is absolute. For the same reason we can move the line of integration to $(\sigma)$ with $1/2 < \sigma < 1$. The pole of the zeta-function at $s = 1$ yields a residue $x^{-1}$ and so we have

$$(1/2\pi i)\int_{(\sigma)} \Gamma(s)\zeta(s)x^{-s}ds = (e^x - 1)^{-1} - x^{-1}.$$

We note next that this formula is meaningful when $\mathrm{Re}(x) > 0$ and $x^{-s} = e^{\log x}$ with $-\pi < \mathrm{Im}(\log x) < \pi$. As both sides are analytic in this region the identity persists by the principle of analytic continuation Thus we can use this identity on $e^{i(\pi/2-\delta)}\mathbf{R}_+^\times$ where $0 < \delta < \pi/2$. Let $z = e^{i(\pi/2-\delta)}$; then we obtain

$$(1/2\pi i)\int_{(\sigma)} \Gamma(s)\zeta(s)e^{-is(\pi/2-\delta)}x^{-s}ds = (e^{zx} - 1)^{-1} - (zx)^{-1}.$$

We write this as

$$(2\pi)^{-1}\int_{-\infty}^{\infty} \Gamma(\sigma + it)\zeta(\sigma + it)e^{-i(\sigma+it)(\pi/2-\delta)}e^{-it\log x}dt$$
$$= \{(e^{zx} - 1)^{-1} - (zx)^{-1}\}x^\sigma$$

which we recognize as a Fourier transform. We can apply Parseval's Formula (A1.10) as both sides are square-integrable and obtain

$$(2\pi)^{-1}\int_{-\infty}^{\infty} |\Gamma(\sigma + it)\zeta(\sigma + it)|^2 e^{t(\pi-2\delta)}dt$$
$$= \int_0^\infty x^{2\sigma-1}|(e^{xz} - 1)^{-1} - (zx)^{-1}|^2 dx.$$

We shall now study the behaviour of both sides of this formula as $\delta \to 0$, and thus $z \to i$. The crux of the argument is to show that the right-hand side $\sim \zeta(2\sigma)(2\delta)^{-2\sigma}\Gamma(2\sigma)$ as $\delta \to 0$. It is this part of the argument that is close to those used in the proof of the 'approximate functional equation'.

To prove this we observe first that the part of the second integral taken over $]0, \pi[$ is continuous in $\delta$ in a neighbourhood of 0, and is therefore bounded. The integral over $[\pi, \infty[$ is when multiplied out a sum of four terms

$$\int_\pi^x x^{2\sigma-1}(e^{zx}-1)^{-1}(e^{z^{-1}}x-1)^{-1}\,dx - z\int_\pi^{} x^{2\sigma-1}(e^{zx}-1)^{-1}x^{-1}\,dx$$
$$-z^{-1}\int_\pi^x x^{2\sigma-1}(e^{z^{-1}}x-1)^{-1}x^{-1}\,dx+\int_\pi^x x^{2\sigma-2}x^{-2}\,dx.$$

The last term does not depend on $\delta$ and is therefore also bounded.

The first term is

$$\sum_{m,n\geqslant 1}\int_\pi^\infty e^{-(mz+nz^{-1})x}x^{2\sigma-1}\,dx.$$

Let us consider first those terms with $m=n$. As $z+z^{-1}=2\sin\delta$ these are

$$\sum_m\int_\pi^x e^{-2mx\sin\delta}x^{2\sigma-1}\,dx=\int_\pi^x (e^{2x\sin\delta}-1)^{-1}x^{2\sigma-1}\,dx.$$

After a change of variable of $x$ to $x/(2\sin\delta)$ this becomes

$$\int_{2\pi\sin\delta}^\infty (e^x-1)^{-1}x^{2\sigma-1}\,dx(2\sin\delta)^{-2\sigma}.$$

As $\delta\to 0$ the integral converges to

$$\zeta(2\sigma)\Gamma(2\sigma).$$

Thus this part is asymptotic to $\zeta(2\sigma)\Gamma(2\sigma)(2\delta)^{-2\sigma}$ which is the main term.

We have now to show that the remaining terms are of a smaller order of magnitude as $\delta\to 0$. Consider those terms in the integral with $m\neq n$. For definiteness we shall study those with $m<n$; the case of $m>n$ is similar. We have to estimate the expression

$$\sum_{m<n}\int_\pi^\infty e^{-\{(m+n)\sin\delta+(m-n)i\cos\delta\}x}x^{2\sigma-1}\,dx.$$

To the integral we apply Cauchy's Theorem and replace the integral over $]\pi,\infty[$ by the line $y\mapsto\pi+iy, y>0$. It is easy to justify this. On this line the absolute value of the integrand is

$$e^{-\pi(m+n)\sin\delta-|m-n|y\cos\delta}(\pi^2+y^2)^{\sigma-1/2}$$

so that the sum is dominated by

$$\sum_{m<n}O(|m-n|^{-1}e^{-\pi(m+n)\sin\delta})$$

since $|m-n|\cos\delta$ is bounded below and for $A>1$ one has by a change of variables the following inequality for $u>0$

$$\int_0^\infty e^{-Ax}(x^2+\pi^2)^u\,dx\leqslant A^{-1}\int_0^\infty e^{-x}(x^2+\pi^2)^u\,dx.$$

In the estimate that we obtained above let $l=n-m$ so that the variable is

restricted by $l < n$. Thus we have the estimate (on making use of $m + n > n$)

$$\sum_n \sum_{l<n} O(l^{-1} e^{-\pi n \sin\delta}).$$

The sum over $l$ is $O(\log n)$. To estimate the resulting sum

$$\sum_{n\geqslant 2} \log n e^{-\pi n \sin\delta}$$

it is most convenient to split the summation into two parts, that where $n \leqslant \delta^{-1}$ and that where $n > \delta^{-1}$. The first of these consists of $O(\delta^{-1})$ terms each of which is $O(\log \delta^{-1})$. The estimate that we thus obtain is $O(\delta^{-1} \log \delta^{-1})$. The remaining sum can be written as

$$\log \delta^{-1} \sum_{n\delta>1} e^{-\pi n \sin\delta} + \sum_{n\delta>1} \log n\delta \, e^{-\pi n \sin\delta}.$$

The first of the sums is a geometric series which can be evaluated and this yields an estimate of again $O(\delta^{-1} \log \delta^{-1})$. Finally $\log(\eta\delta)e^{-\pi n \sin\delta} = O(e^{-n\delta/2})$. Thus the second term can be estimated by $O(\delta^{-1})$. Hence the first of our four integrals is asymptotic to $\zeta(2\sigma)\Gamma(2\delta)(2\delta)^{-2\sigma}$.

It remains to examine the second and third integrals. They are complex conjugates of one another and so we need only estimate one of them. By an analogous application of Cauchy's Theorem to that used above we see that

$$\int_\pi^\infty x^{2\sigma-1}(e^{xz}-1)^{-1}dx = i\int_\pi^\infty (\pi+iy)^{2\sigma-1}(e^{z(\pi+iy)}-1)^{-1}dy.$$

For $\delta < \pi/4$ we have

$$(e^{z(\pi+iy)}-1)^{-1} = O(e^{-y/\sqrt{2}})$$

from which we see that this integral is bounded as $\delta \to 0$.

We have now completed the proof that as $\delta \to 0$

$$\int_{-\infty}^{\infty} |\Gamma(\sigma+it)\zeta(\sigma+it)|^2 e^{t(\pi-2\delta)}dt \sim \zeta(2\sigma)\Gamma(2\sigma)(2\delta)^{-2\sigma}$$

We can now deduce Theorem 4.12. From the asymptotic formula we have for $T > 1$

$$\int_{T/2}^{T} |\Gamma(\sigma+it)\zeta(\sigma+it)|^2 e^{t(\pi-2\delta)}dt = O(\delta^{-2\sigma}).$$

In this we take $\delta = T^{-1}$ and recall from Stirling's Formula that

$$|\Gamma(\sigma+it)|^2 e^{\pi t} \sim \pi |t|^{2\sigma-1}.$$

Thus we find that

$$\int_{T/2}^{T} |\zeta(\sigma+it)|^2 e^{-2t/T} t^{2\sigma-1}dt = O(T^{2\sigma}),$$

or, more simply

$$\int_{T/2}^{T} |\zeta(\sigma + it)|^2 \, dt = O(T).$$

We now replace $T$ by $T/2$, $T/4$, etc. and sum the ensuing inequalities which yields Theorem 4.12.

## 4.15

We remark that the estimate of Theorem 4.12 can be replaced by an asymptotic formula

$$\int_{0}^{T} |\zeta(\sigma + it)|^2 \, dt \sim T\zeta(2\sigma).$$

The leading term can be obtained by sharpening the argument given above through the use of a Tauberian theorem. A similar asymptotic formula is known for $\int_0^T |\zeta(\sigma + it)|^4 dt$ (see Titchmarsh (2), §7.5); beyond this the only non-trivial estimate available for higher powers, and not deducible from the results already quoted, is Heath-Brown's estimate

$$\int_{0}^{T} |\zeta(1/2 + it)|^{12} \, dt = O(T^2 (\log T)^{17}).$$

For further details of this and related questions the reader should consult Heath-Brown (1) and also Titchmarsh (2), Chapter 7.

## Exercises

4.1 For $c > 0$ let $H_c = \{s \in \mathbf{C} : |1 - c/\log|\mathrm{Im}(s)|| < \mathrm{Re}(s) < 2, |\mathrm{Im}(s)| > 2\}$. Show that for suitable $c$ one has for $s \in H_c$

$$\zeta(s) = O(\log|s|).$$

4.2 In the notation of Exercise 4.1 show that for suitable $c$ one has for $s \in H_c$

$$\zeta'(s) = O((\log|s|)^2)$$

by adapting the argument of Exercise 4.1. Deduce it also from Exercise 4.1.

4.3 Use the argument of § 4.2 to show that as $|t| \to \infty$

$$\zeta(1 + it)^{-1} = O((\log|t|)^7)$$
$$\zeta'/\zeta(1 + it) = O((\log|t|)^9).$$

4.4 Suppose that $a, b$ are increasing functions such that $a \geqslant 1$, $b > 0$, $ab = o(e^b)$ and $\zeta(s) = O(e^{b\{|\mathrm{Im}(s)|\}})$ in $\{s \in \mathbf{C} : \mathrm{Re}(s) \geqslant 1 - a(|\mathrm{Im}(s)|)^{-1}\}$. Show that there exists $c > 0$ such that if $\rho \in Z$ then

$$\mathrm{Re}(\rho) < 1 - cab(2|\mathrm{Im}(\rho)| + 1)^{-1}.$$

Hint: use Theorem A6.2; the method of §4.2, yields a much weaker result. This statement apparently represents the best one can achieve using Hadamard's inequality. Note that zero-free strips of the form $\{s \in \mathbf{C}: \mathrm{Re}(s) > \eta\}$, $\eta < 1$ are not within the range of this method.

4.5 Under the assumptions of Exercise 4.4 show that there exists $c' > 0$ so that as $|s| \to \infty$, $1 - c'ab\{2|\mathrm{Im}(s)| + 3\}^{-1} < \mathrm{Re}(s) < 2$ one has

$$\zeta'(s)/\zeta(s) = O((ab)(2|s| + 3))$$
$$\zeta(s)^{-1} = O((ab)(2|s| + 3)).$$

4.6 Deduce from Exercise 4.5 that for a suitable $c > 0$ then as $|s| \to \infty$ with $1 - c/\log\{|\mathrm{Im}(s)| + 2\} < \mathrm{Re}(s) < 2$ one has

$$\zeta'(s)/\zeta(s) = O(\log|s|)$$
$$\zeta(s)^{-1} = O(\log|s|).$$

4.7 Using new bounds for exponential sums and the approximate functional equation Vinogradov proved that there exist $c, c' > 0$ such that as $|s| \to \infty$ with

$$1 - c\{\log\log|\mathrm{Im}(s)|\}^{1/2}/\{\log|\mathrm{Im}(s)|\}^{1/2} < \mathrm{Re}(s) < 2$$

one has

$$\zeta(s) = O(\exp\{c'(\log|s|)^{1/4}(\log\log|s|)^{1/2}/(\log\log|s|)^{1/2}\}).$$

What zero-free region for $\zeta$ does this imply? What bound on $\zeta(1 + it)^{-1}$ as $|t| \to \infty$ can one deduce? What are the consequences for the error term in the Prime Number Theorem?

4.8 Show that $S(t)$ can be defined as $\pi^{-1}\Delta_{\infty + it}^{1/2 + it}\arg(\zeta)$.

4.9 Use Exercise 4.8 and the argument of §4.13 to show that

$$\int_0^T S(t)\mathrm{d}t = O(\log T) \qquad \text{as } T \to \infty.$$

4.10 Let $(a_n)$ be a sequence and $A$ a positive non-decreasing function so that $|a_n| \leqslant A(n)$. Suppose that $\sum_{n \geqslant 1} a_n n^{-s}$ converges absolutely if $\mathrm{Re}(s) > 1$ and that there exists $\alpha \in \mathbf{R}_+$ so that as $s \to 1, s > 1$

$$\sum_{n \geqslant 1} |a_n| n^{-s} = O((s-1)^{-\alpha}).$$

Let $c > 0$; show that if $\mathrm{Re}(s) > 1 - c$ and $X \in \mathbf{R}_+ - \mathbf{N}$ then

$$\sum_{n \geqslant X} a_n n^{-s} = (2\pi\mathrm{i})^{-1} \int_{c - \mathrm{i}T}^{c + \mathrm{i}T} \left( \sum_{m \geqslant 1} a_m m^{-(s+t)} \right) X^t t^{-1}\,\mathrm{d}t$$
$$+ O(X^c T^{-1}\{\mathrm{Re}(s) + c - 1\}^{-\alpha}) + O(A(2X)X^{1 - \mathrm{Re}(s)}\log X)$$
$$+ O(A(X+1)X^{1 - \mathrm{Re}(s)}T^{-1}\langle X\rangle^{-1}),$$

where $\langle X\rangle = \mathrm{Min}\{|X - m| : m \in \mathbf{Z}\}$.

4.11 Use Exercise 4.10 to prove Theorem 4.3.

4.12 Show that for any $\varepsilon > 0$

$$\sum_{n \geqslant X} \mathrm{d}(n) = X \operatorname*{Res}_{t=1} \{\zeta(t)^2 X^{t-1}/t\} + O(X^{1/2+\varepsilon})$$

$$\sum_{n \leqslant X} \mathrm{d}(n)^2 = X \operatorname*{Res}_{t=1} \{\zeta(t)^4 X^{t-1}/\zeta(2t)t\} + O(X^{3/4+\varepsilon})$$

where $d$ is the divisor function ($d = d_2$).

4.13 Show that for any $m \geqslant 1$ that there exists a polynomial $P_m$ of degree $2^{m-1}$ so that

$$\sum_{n \leqslant X} \mathrm{d}(n)^m = X P_m(\log X) + O(X^{1-2^{-m}+\varepsilon})$$

for any $\varepsilon > 0$. Deduce that for no $M > 0$ an estimate $\mathrm{d}(n) = O((\log n)^M)$ can hold.

4.14 Find a sequence $n_i \to \infty$ so that for any $\eta > 0$ there exists $i_0(n)$ so that for $i > i_0(\eta)$

$$\mathrm{d}(n_i) > \exp\{(1-\eta)(\log 2)(\log n_i)/\log\log n_i\}.$$

By sharpening the argument of Exercise 1.5 show that for any $\eta > 0$ one has

$$\mathrm{d}(n_i) = O(\exp\{(1+\eta)(\log 2)(\log n_i)/\log\log n_i\}).$$

4.15 Show that for any $\varepsilon > 0$

$$\mathrm{Card}\{n | \phi(n) \leqslant X\} = CX + O(X^{7/9+\varepsilon})$$

where

$$C = \prod_p (1 + \{p(p-1)\}^{-1}).$$

(Hint: recall Exercise 1.16.)

4.16 Estimate the 'constant', as a function of $\sigma$, in Theorem 4.11.

# 5

# *The Riemann Hypothesis and the Lindelöf Hypothesis*

## 5.1

The Riemann Hypothesis plays a central role in the theory of the Riemann zeta-function in that it indicates what one expects to be true. It is quite common first to prove some statement under the assumption of the Riemann Hypothesis and then to try to prove as good a version as one can without this assumption. In this way the Riemann Hypothesis has been a guide to the theory.

Formally the Riemann Hypothesis states that all the zeros of the zeta-function lie on the 'critical line' $\{s: \mathrm{Re}(s) = 1/2\}$, or, in our notations that

$$Z \subset \{s: \mathrm{Re}(s) = 1/2\}.$$

This can be expressed in other ways; for example that $\zeta'(s)/\zeta(s) + (s-1)^{-1}$ should be holomorphic in $\{s: \mathrm{Re}(s) > 1/2\}$, or that $\log\{(s-1)\zeta(s)\}$ should be holomorphic in the same region. There are several other ways of expressing the Riemann Hypothesis, some of which we shall see in this chapter. It is the purpose of this chapter to give some impression of the significance of the Riemann Hypothesis, of the reasons why one might expect it to be true and of the approaches which have been taken towards its proof.

## 5.2

In the theory of the Riemann zeta-function the classical approach is based on the 'approximate functional equation' which allows one to approximate the zeta-function well in the critical strip. This will be described in more detail in the next chapter. It has been clear for quite some time now that it is not possible to prove the Riemann Hypothesis purely by such methods and that further progress demands an argument improving the one of Hadamard's described in §4.2. The Lindelöf Hypothesis, which we shall formulate presently, is a further conjecture which would be a consequence of the Riemann Hypothesis but which is not sufficient to imply the Riemann

Hypothesis in the present state of knowledge. It represents the limit of what one might expect to be able to prove with classical techniques.

To formulate the Lindelöf Hypothesis we recall that by Theorem 2.12 the function $\mu$ defined by

$$\mu(\sigma) = \operatorname{Inf}\{a \in \mathbf{R}: \zeta(\sigma + it) = O(|t|^a) \quad \text{as } t \to \infty\}$$

exists and satisfies

$$\begin{aligned} \mu(\sigma) &= 0 && \text{if } \sigma > 1 \\ &= 1/2 - \sigma && \text{if } \sigma < 0. \end{aligned}$$

By the Phragmén–Lindelöf Theorem (A5.7) the function $\mu$ is convex (from which it follows also that $\mu$ is positive). The Lindelöf Hypothesis is that the function is the extreme convex function with the properties above and such that

$$\mu(\sigma) - \mu(1 - \sigma) = \sigma - 1/2$$

which is forced by the functional equation. Explicitly this means with we would have

$$\begin{aligned} \mu(\sigma) &= 0 && \text{if } \sigma \geqslant 1/2 \\ &= 1/2 - \sigma && \text{if } \sigma \leqslant 1/2. \end{aligned}$$

In other words the Lindelöf Hypothesis can be phrased as expressing that $\mu(1/2) = 0$, or that for each $\varepsilon > 0$ as $t \to \infty$

$$\zeta(1/2 + it) = O(|t|^\varepsilon).$$

In view of the discussion above it is equivalent to require that for each $\sigma > 1/2$ and $\varepsilon > 0$ as $t \to \infty$

$$\zeta(\sigma + it) = O(|t|^\varepsilon).$$

## 5.3

The formulation of the Lindelöf Hypothesis just given involves measuring the 'size' of the zeta-function by $\sup_{0<t<T}|\zeta(\sigma + it)|$. It is possible to replace the 'sup norm' here with various integral means, and this has the further advantage that it fractionates the Lindelöf Hypothesis into a family of assertions which can be investigated by special methods. The relatively trivial theorem involved here is the following:

**Theorem:** *The Lindelöf Hypothesis is true if and only if the following statement is true*:

(i) *for every* $k \geqslant 1$ *and* $\varepsilon > 0$ *one has*

$$T^{-1}\int_0^T |\zeta(1/2 + it)|^k \, dt = O(T^\varepsilon),$$

*and also if and only if*

(ii) *for every* $\sigma > 1/2$, $k \geqslant 1$ *and* $\varepsilon > 0$ *one has*

$$T^{-1}\int_0^T |\zeta(\sigma + it)|^k dt = O(T^\varepsilon) \qquad \text{as } T \to \infty$$

*holds.*

*Proof*: It is clear that if the Lindelöf Hypothesis is true then (i) and (ii) follow. Thus we have only to show that if the Lindelöf Hypothesis is false then (i) and (ii) also do not hold. We shall show this in the case of (i); the argument in the case of (ii) is analogous.

Since we are assuming that the Lindelöf Hypothesis is false we can suppose that we are given as sequence $\{t_j : j \in \mathbf{N}\}$ such that $t_{j+1} - t_j > 1$ (and so $t_j \to \infty$) and an $a > 0$ so that

$$|\zeta(1/2 + it_j)| > t_j^a.$$

Since we already know that $\mu(\sigma) \leqslant 1/2$ for $\sigma$ with $|\sigma - 1/2| \leqslant 1/4$ and hence uniformly in $\{s : |\mathrm{Re}(s) - 1/2| < 1/4\}$ we have by the Phragmén–Lindelöf Theorem in this strip

$$\zeta(s) = O(1 + |s|).$$

From this and Cauchy's inequality we have

$$\zeta'(1/2 + it) = O(|t|) \qquad \text{as } |t| \to \infty.$$

Thus we deduce that there is a constant $c > 0$ such that

$$|\zeta(1/2 + it)| > t_j^a/2 \qquad \text{if } |t - t_j| < ct_j^{-(1-a)}$$

and hence

$$t_j^{-1}\int_0^{t_j} |\zeta(1/2 + it)|^k dt > t_j^{-1}\int_{t_j - ct_j^{-(1-a)}}^{t_j} |t_j^a/2|^k dt$$
$$= 2^{-k} ct_j^{(k+1)a-2}.$$

By choosing $k$ so that $(k + 1)a > 2$ we see that (i) is false for the sequence $t_j$. This completes the proof of the theorem.

## 5.4

Although one might not expect it at first the Lindelöf Hypothesis can be expressed in terms of the distribution of the zeros of the zeta-function. The precise statement is the following where $N(\sigma, T)$ is as defined in §4.11.

**Theorem:** *The Lindelöf Hypothesis is equivalent to the following statement. For every* $\sigma > 1/2$ *one has* $N(\sigma, T + 1) - N(\sigma, T) = o(\log T)$ *as* $T \to \infty$.

## 5.5

Since the Riemann Hypothesis can be formulated as asserting that $N(\sigma, T) = 0$ for all $T$ and $\sigma > 1/2$ we have at once:

**Corollary**: *The truth of the Riemann Hypothesis implies that of the Lindelöf Hypothesis.*

This corollary is one of the reasons for the significance ascribed to the Lindelöf Hypothesis. It indicates in a certain sense that the Riemann Hypothesis represents the most extreme hypothesis that one can formulate.

## 5.6

In this section we shall prove Theorem 5.4. The proof is function-theoretic and involves ideas centered around Jensen's Theorem. It does not provide much insight into the nature of the zeta-function itself.

We shall first assume the Lindelöf Hypothesis and derive the statement formulated in the theorem. To do this we shall apply Jensen's Theorem (A5.2) to $\zeta$ on the disc $\{s \in \mathbf{C}: |s - 2 - iT| < r\}$ where $r < 3/2$ We obtain

$$\sum_{|\rho - 2 - iT| < r} \log(r/|\rho - 2 - iT|)$$

$$= (2\pi)^{-1} \int_0^{2\pi} \log|\zeta(2 + iT + re^{i\theta})| d\theta - \log|\zeta(2 + iT)|.$$

We can now use the Lindelöf Hypothesis to estimate the right-hand side; for any $\varepsilon > 0$ there exists $T(\varepsilon)$ so that for $T > T(\varepsilon)$ the right-hand side is bounded above by $\varepsilon \log T$. If we now fix $r_1 < r$ and restrict the sum on the left-hand side to those $\rho$ satisfying $|\rho - 2 - iT| < r_1$ then the sum will be decreased. It still exceeds

$$\sum_{|\rho \quad 2 \quad iT| < r_1} \log\{r/(r - r_1)\}.$$

Thus we have shown that for any $r_1 < 3/2$ one has as $T \to \infty$

$$\sum_{|\rho \quad 2 \quad iT| < r_1} 1 = o(\log T).$$

Since a rectangle with corners $\sigma + iT$, $1 + iT$, $1 + i(T + 1)$, $\sigma + i(T + 1)$ can be covered by finitely many discs of the form

$$\{s \in \mathbf{C}: |s - 2 - iT'| < r_1\}$$

for a fixed $r_1 > 2 - \sigma$ and $T' \in [T, T + 1]$ we see that

$$N(\sigma, T + 1) - N(\sigma, T) = o(\log T)$$

as required.

We shall now prove the converse. To do this we shall fix $r_1, r_2, r_3 \in \mathbf{R}$

with $1 < r_1 < r_2 < r_3 < 3/2$. We shall estimate

$$\zeta'(s)/(s) - \sum_{\rho:|\rho-2-iT|<r_3} (s-\rho)^{-1}$$

on the disc $\{s \in \mathbf{C}: |s-2-iT| < r_2\}$. In view of Theorem 3.3 and Corollary 3.4 one has on this disc

$$\zeta'(s)/\zeta(s) - \sum_{\rho:|\mathrm{Im}(\rho)-T|<4} (s-\rho)^{-1} = O(\log T);$$

we shall prove that

$$\sum_{\rho:|\mathrm{Im}(\rho)-T|<4} (s-\rho)^{-1} - \sum_{\rho:|\rho-2-iT|<r_3} (s-\rho)^{-1} = O(\log T)$$

from which it will then follow that

$$\zeta'(s)/\zeta(s) - \sum_{\rho:|\rho-2-iT|<r_3} (s-\rho)^{-1} = O(\log T). \tag{5.1}$$

To see this note that the difference is equal to

$$\sum_{\rho:|\mathrm{Im}(\rho)-T|<4, |\rho-2-iT|>r_3} (s-\rho)^{-1}$$

since the conditions $|\mathrm{Im}(\rho)-T| > 4$ and $|\rho-2-iT| < r_3$ are incompatible. In this case there are by Corollary 3.4 at most $O(\log T)$ summands and one has $|s-\rho| > r_3 - r_2$. From this the estimate above follows

We shall next use (5.1) to derive an improved version of itself. To do this we apply the Hadamard Three Circles Theorem (Titchmarsh (1), § 5.3, see also A5.7) to the function

$$\zeta'(s)/\zeta(s) - \sum_{\rho:|\rho-2-iT|<r_3} (s-\rho)^{-1}$$

on the circles $|s-2-iT| = 1/2, r_1, r_2$. On the first of the circles we have that $\zeta'(s)/\zeta(s) = O(1)$ as the circle lies in the region of absolute convergence, and the sum is $O(N(2-r_3, T-2) - N(2-r_3, T+2)) = o(\log T)$ by assumption (this is the point at which we use this assumption). Thus the function is estimated by

$$o(\log T)$$

on $\{s: |s-2-iT| = 1/2\}$. We can estimate it on the circle of radius $r_2$ by $O(\log T)$ using (5.1). If we let $k = (\log r_2 - \log r_1)/\{\log r_2 - \log(1/2)\}$ then this function is estimated by

$$o(\log T)^k O(\log T)^{1-k} = o(\log T);$$

note that this misuse of the notations verges on the inadmissible but the reader should have no trouble in convincing her- or himself that no sins have been committed here.

We have thus shown that if $s$ lies on $\{s \in \mathbf{C}: |s-2-iT| < r_1\}$ then

$$\zeta'(s)/\zeta(s) = \sum_{\rho:|s-\rho|<r_3} (s-\rho)^{-1} + o(\log T)$$

and by the Maximum Principle the same holds in the interior of the circle. Integrating we obtain for $1 > \sigma > r_1$

$$\log \zeta(\sigma + iT) - \log \zeta(2 + iT)$$
$$= \sum_{\rho : |\rho - 2 - iT| < r_3} \log(\sigma + iT - \rho) - \log(2 + iT - \rho) + o(\log T). \tag{5.2}$$

Since $\rho$ is closer to $\sigma + iT$ than to $2 + iT$ we see that

$$\mathrm{Re}\left\{ \sum_{\rho : |\rho - 2 - iT| < r_3} \log(\sigma + iT - \rho) - \log(2 + iT - \rho) \right\} < 0$$

and hence

$$\mathrm{Re}\{\log \zeta(\sigma + iT)\} < o(\log T)$$

as $\log \zeta(2 + iT) = O(1)$. This is precisely the Lindelöf Hypothesis This completes the proof of the theorem.

## 5.7

The argument of § 5.6 has a further consequence which is worth recording. If we take the imaginary part of (5.2) and note that each argument in the sum can vary by at most $\pm\pi$ and that by the Lindelöf Hypothesis there are at most $o(\log T)$ summands we can conclude the following:

**Theorem:** *On the Lindelöf Hypothesis one has as* $T \to \infty$

$$S(T) = o(\log T).$$

For a more detailed discussion of the function $S$ see § 4.8.

## 5.8

The greatest significance of the Riemann Hypothesis lies in its consequences for Prime Number Theory. This connection is expressed by the following theorem.

**Theorem:** (i) *Suppose that there exists* $\theta < 1$ *so that.*

$$Z \subset \{s \in \mathbf{C} : 1 - \theta \leqslant \mathrm{Re}(s) \leqslant \theta\}.$$

*Then as* $X \to \infty$

$$\sum_{n \leqslant X} \Lambda(n) = X + O(X^{\theta} (\log X)^2).$$

(ii) *Suppose that for* $\alpha < 1$ *one has as* $X \to \infty$

$$\sum_{n \leqslant X} \Lambda(n) = X + O(X^{\alpha}).$$

*It then follows that*

$$Z \subset \{s \in \mathbf{C}: \mathrm{Re}(s) \leqslant \alpha\}.$$

This theorem has several interesting features. Firstly since the zeta-function has infinitely many non-trivial zeros it follows in part (ii) that one must have $\alpha \geqslant 1/2$. Thus in view of part (i) it is not possible to have a substantially better error term in the Prime Number Theorem than $O(X^{1/2}(\log X)^2)$, that is, better by a power of $X$. This best possible error term is equivalent to the Riemann Hypothesis.

It is rather remarkable that it follows from this that if one were able to show that as $X \to \infty$

$$\sum_{n \leqslant X} \Lambda(n) = X + O(X^{\alpha})$$

for all $\alpha > \theta$ then one has the much more precise estimate

$$\sum_{n \leqslant X} \Lambda(n) = X + O(X^{\theta}(\log X)^2).$$

Notice that as far as the power of $X$ which appears in the error term of the Prime Number Theorem is concerned the theorem shows that this power is determined precisely by the strip in which the zeros of the zeta-function are contained. One must further remark here though that no values for $\theta < 1$ in part (i) of the theorem or for $\alpha < 1$ are known, and that any such values would represent a very substantial advance in the theory of the Riemann zeta-function.

## 5.9

In this section we shall prove Theorem 5.8. The proof is based on the 'explicit formula' Theorem 3.8 and is very similar to the argument used in §4.4. By Theorem 3.8 we have

$$\sum_{n \leqslant X} \Lambda(n) = X - \sum_{\rho \in Z: |\mathrm{Im}(\rho)| \leqslant T} X^{\rho}/\rho + O(\log X + X(\log X)/T + X(\log T)^2/T)$$

by the same arguments as were used in §4.4. The sum over the zeros can be estimated by

$$X^{\theta} \sum_{\rho \in Z: |\mathrm{Im}(\rho)| \leqslant T} |\rho|^{-1}.$$

Since there are $O(\log |t|)$ zeros with $t < |\mathrm{Im}(\rho)| < t + 1$ the sum here is dominated by $\sum_{j \leqslant T} j^{-1} \log j = O((\log T)^2)$. Thus we have shown that

$$\sum_{n \leqslant X} \Lambda(n) = X + O(X^{\theta}(\log T)^2 + \log X + X(\log X)/T + X(\log T)^2/T).$$

We now take

$$T = X^{1-\theta}$$

which gives the estimate stated in the theorem.

The proof of part (ii) is by partial summation. Let

$$A(X) = \sum_{n \leqslant X} \Lambda(n).$$

Then

$$\begin{aligned} -\zeta'(s)/\zeta(s) &= \sum_{n>1} \Lambda(n) n^{-s} \\ &= \sum_{n>1} \{A(n) - A(n-1)\} n^{-s} \\ &= \sum_{n \geqslant 1} A(n)\{n^{-s} - (n+1)^{-s}\}. \end{aligned}$$

It is straightforward using $A(n) = O(n)$ to justify this in $\{s \in \mathbf{C}: \mathrm{Re}(s) > 1\}$. We now write $A(n) = n + A_1(n)$ where $A_1(n) = O(n^{\alpha})$. We obtain

$$-\zeta'(s)/\zeta(s) = \sum_{n \geqslant 1} A_1(n)\{n^{-s} - (n+1)^{-s}\} + \sum_{n \geqslant 1} n\{n^{-s} - (n+1)^{-s}\}.$$

The second term is equal to $\zeta(s)$. This one can verify by applying the inverse transformation to the one used above.

By the Mean Value Theorem $n^{-s} - (n+1)^{-s} = O(n^{-\mathrm{Re}(s)-1})$. Thus the first term converges locally uniformly in $\{s \in \mathbf{C}: \mathrm{Re}(s) > \alpha\}$ and represents an analytic function there. The second function is analytic in the same region except for a pole at $s = 1$. These together show that $-\zeta'(s)/\zeta(s)$ is, except for a simple pole at $s = 1$, analytic in $\{s \in \mathbf{C}: \mathrm{Re}(s) > \alpha\}$ and hence $\zeta$ has no zeros in this region. This is the assertion of the theorem.

## 5.10

We now turn to the question as to why one might expect the Riemann Hypothesis to hold. The reasons are not as compelling as one might like to think. The most interesting lies in a comparison of the theory of the zeta-function with a certain class of zeta-functions 'associated with curves defined over finite fields'. This comparison will be the subject of most of the remainder of the present chapter, but before we turn to this we list the other evidence available at present.

(i) The numerical evidence is very convincing; it is now known through very extensive computer computations that the first 1 500 000 001 zeros of the Riemann zeta-function lie on the critical line. This is such a large number that it is virtually pointless to extend these calculations further. Nevertheless there have been a number of examples of assertions based on very extensive computation in analytic number-theory which have later been disproved by theoretical considerations without a counter-example being found. Thus, although it is difficult to argue with such overwhelming evidence one should still treat it with a certain degree of caution.

(ii) All the theoretical evidence points in the direction of the Riemann Hypothesis, although from a great distance Examples of this are Theorem 4.11 and the sharper versions of this that are known, and the Theorem of Hardy and Selberg that a positive proportion of the zeros are on the critical line – one knows that this proportion exceeds 34%; see Levinson (1).

(iii) Theorem 5.8 shows that the Riemann Hypothesis is equivalent to the assertion that as $X \to \infty$

$$\sum_{n \leqslant X} \Lambda(n) = X + O(X^{1/2}(\log X)^2).$$

The power $X^{1/2}$ here reminds one strongly of the Central Limit Theorem of probability theory, or, what is more or less the same, the theory of one-dimensional random walks. In a vague sense if the primes were a 'random' sequence then the error term would be 'almost certainly' of the form above. This can be made a little more precise (Lévy (1)) but naturally the primes are not 'random' in the strict sense and up to the present no positive results concerning the actual prime numbers have been achieved along these lines. Nevertheless the idea is very suggestive and this represents one of the more reassuring reasons for expecting the Riemann Hypothesis to be true.

(iv) One should probably add to this list the 'Platonic' reason that one expects the natural numbers to be the most perfect idea conceivable, and that this is only compatible with the primes being distributed in the most regular fashion possible, which as we have seen in §5.8 is equivalent to the Riemann Hypothesis. This reason is probably the most compelling for a large proportion of mathematicians.

As the reader will gather from this list the solid reasons for believing the Riemann Hypothesis to be true are meagre. In the following paragraphs we shall discuss an analogy which suggests not only that the Riemann Hypothesis might hold but also that this would be a consequence of a much deeper and extensive theory.

## 5.11

The most convincing reason that has so far been evinced for the validity of the Riemann Hypothesis is that an analogous statement is valid for the zeta-functions attached to curves over finite fields. The formal similarities are so striking that it is difficult to believe that they do not lead to even more far-reaching coincidences. The study of these zeta-functions was initiated by Artin in his thesis of 1921 who computed the zeta-functions of some specific curves (hyperelliptic curves) and conjectured the basic

analytic properties of them and, on the basis of these examples, also the analogue of the Riemann Hypothesis. This led to a great deal of activity in this area in the 1920s and 1930s. The analytic properties were established by Schmidt in 1931 but the fundamental breakthrough was due to Hasse who proved the Riemann Hypothesis for elliptic curves in 1933. This proof used new ideas taken from algebraic geometry which were not available in the classical case. Later, in 1942, Weil proved the Riemann Hypothesis in the general case for which he had to redevelop the theory of Jacobian varieties for curves in such a way that it was valid in arbitrary characteristics. This was a very substantial undertaking as the theory up to that time had been based on analytic techniques, particularly the theory of theta-functions. Consequently it was only several years later that the complete proof appeared and was based on his fundamental treatise 'Foundations of Algebraic Geometry' which played a seminal role in the development of modern algebraic geometry. More recently much simpler proofs have appeared based on an idea of Stepanov (see Bombieri (1), Voloch (1)). It appears that these new proofs, despite appearances, are logically very close to that of Weil's as has been shown in Voloch (1).

## *5.12

We shall now discuss these ideas in more detail. For a prime power $q$ denote by $\mathbf{F}_q$ the finite field with $q$ elements. We consider a finitely generated extension $k$ of $\mathbf{F}_q$ of transcendence degree 1. We may assume that $\mathbf{F}_q$ is the largest finite field contained in $k$. By the 'theorem of the Primitive Element' (van der Waerden (1), Vol. 1, §40) there are $x, y \in k$ so that $k = \mathbf{F}_q(x, y)$, $x$ is transcendental over $\mathbf{F}_q$ and there exists an irreducible polynomial $G \in \mathbf{F}_q[X, Y]$ so that $G(x, y) = 0$. Thus $k$ *is the algebraic extension of* $\mathbf{F}_q(x)$ generated by $y$ satisfying $G(x, y) = 0$ considered as a polynomial in one variable over $\mathbf{F}_q(x)$. Since $G(x, y) = 0$ defines a curve $C$ in the plane we can consider $k$ as being the field of meromorphic functions on this curve. This is more than a figure of speech since it leads to fruitful concepts, as we shall see.

We shall now introduce the notion of a place of a curve. This is a refinement of the idea of a point on a curve. We begin with the case of $\mathbf{F}_q(x)$. We shall let $\Sigma(\mathbf{F}_q(x))$ be the set consisting of the monic irreducible polynomials in $\mathbf{F}_q[x]$ and the symbol '$\infty$'. To any monic irreducible polynomial $g$ of degree $d$ we ascribe an embedding of $\mathbf{F}_q(x)$ in $\mathbf{F}_{q^d}((t))$, the field of formal power series over $\mathbf{F}_{q^d}$ as follows. We embed first the subfield $\mathbf{F}_q(g(x))$ in $\mathbf{F}_q((t))$ by sending the generator $g(x)$ to $t$. One then obtains an embedding of $\mathbf{F}_q(x)$ into $\mathbf{F}_q(x) \otimes_{\mathbf{F}_q(g(x))} \mathbf{F}_q((t))$. This latter ring corresponds

to considering the extension of $\mathbf{F}_q((t))$ given by the algebraic equation $g(x)-t$ It is not too difficult to see that there exists, by Newton's method, a power series $x(t)$ with coefficients in the splitting field $\mathbf{F}_{q^d}$ of $g$ over $\mathbf{F}_q$ giving a solution since $\mathbf{F}_q$ is a perfect field. This implies that $\mathbf{F}_q(x)\otimes_{\mathbf{F}_q(g(x))}\mathbf{F}_q((t))$ is isomorphic to $\mathbf{F}_{q^d}((t))$ with the second factor being embedded in the obvious way.

Let us define ord: $\mathbf{F}_{q'}((t))^\times \to \mathbf{Z}$ by

$$\operatorname{ord}(\sum a_n t^n) = \operatorname{Min}\{n\in\mathbf{Z}: a_n \neq 0\}$$

and

$$\| \quad \| : \mathbf{F}_q((t)) \to \mathbf{R}_+$$

by

$$\|0\| = 0$$
$$\|f\| = q'^{-\operatorname{ord}(f)} \qquad f \neq 0.$$

It is easy to verify that

$$\|x\| = 0 \qquad \text{if and only if } x = 0$$
$$\|x+y\| \leqslant \operatorname{Min}(\|x\|, \|y\|)$$
$$\|xy\| = \|x\|\,\|y\|.$$

In particular $(x, y)\mapsto \|x-y\|$ defines a metric on $\mathbf{F}_{q'}((t))$ and one can easily check that the resulting metric space is complete.

In the case discussed above the image of $\mathbf{F}_q(x)$ in $\mathbf{F}_{q^d}((t))$ is dense. To the symbol $\infty$ we ascribe the embedding $\mathbf{F}_q(x)$ in $\mathbf{F}_q((t))$ given by sending $x$ to $t^{-1}$. Every embedding of $\mathbf{F}_q(x)$ into an $\mathbf{F}_{q'}((t))$ with dense image is of the form of one of the above composed with an automorphism of $\mathbf{F}_q((t))$ – such an automorphism preserves the norm.

We can now describe the places of an arbitrary field $k$ of the type introduced above. By a place $w$ of $k$ we understand an equivalence class of embeddings $k \to \mathbf{F}_{q'}((t))$ with dense image, where two embeddings are to be considered equivalent if they differ by an automorphism of $\mathbf{F}_q((t))$ over $\mathbf{F}_q$. The embeddings are all considered as $\mathbf{F}_q$-maps. One can construct the places of $k$ as follows. We represent $k$ as above as $\mathbf{F}_q(x, y)$ and consider a place $u$ of $\mathbf{F}_q(x)$. This is represented by an embedding $\mathbf{F}_q(x)\to \mathbf{F}_{q'}((t))$ with dense image. We therefore obtain an embedding of $k$ into $k\otimes_{\mathbf{F}_q(x)}\mathbf{F}_{q'}((t))$ with dense image. One can show that this latter algebra is isomorphic to a direct sum of finitely many fields of the form $\mathbf{F}_{q''}((t''))$. The projection into any one of these yields an embedding of the type sought, and any embedding in equivalent to one of these.

We denote the set of places of $k$ by $\Sigma(k)$. Suppose that $w\in\Sigma(k)$ and that $w$ is represented by an embedding of $k$ into $\mathbf{F}_{q'}((t))$ then $q$ is of the form $q^{f(w)}$ and this defines a map $f: \Sigma(k)\to\mathbf{N}$, called the *degree* of $w$.

For our purposes it only remains to indicate the relationship of $\Sigma(k)$ with points on the curve $C$. The model of $C$ given by $G$ is, in general, not smooth but one knows from the theory of curves that there is a smooth curve in some projective space with function field $k$. It is defined over $\mathbf{F}_q$. By abuse of language we denote it also as $C$. Let $P$ be a point of $C$ defined over $\mathbf{F}_{q'}$, i.e. a map of $\mathrm{Spec}(\mathbf{F}_{q'})$ into $C$. Since $C$ is smooth this can be extended to a map of a formal neighbourhood of $P$ into $C$, i.e. of $\mathrm{Spec}(\mathbf{F}_{q'}[[t]])$ into the formal completion of $C$. But this is precisely an embedding $k \to \mathbf{F}_{q'}((t))$ after taking fractions. Two points $P$ and $P'$ give the same embedding if they are conjugate under the Galois group of $\mathbf{F}_{q'}/\mathbf{F}_q$. One can rephrase this by saying that $w \in \Sigma(k)$ corresponds to an irreducible divisor on $C$ which splits into $f(w)$ points over $\mathbf{F}_{qf(w)}$. A particular consequence of this that we shall need later is that $C(\mathbf{F}_{q^e})$ has $\sum_{f(w)|e} f(w)$ points.

As an example we have that if $k = \mathbf{F}_q(x)$ then $C$ is the projective line over $\mathbf{F}_q$ and thus $C(\mathbf{F}_{q^e})$ has $q^e + 1$ points. By the Möbius Inversion Formula this is equivalent to the fact there are

$$\sum_{f|e} \mu(f) q^{e/f}$$

monic irreducible polynomials of degree $e$ in $\mathbf{F}_q[X]$. It is a useful exercise for the reader to prove this directly. Other examples are given in the exercises for this chapter.

For the material in this section the reader is referred to Deuring (1), Swinnerton-Dyer (1) and Weil (5) for more information.

## *5.13

With the notations of the previous section we now define the zeta-function of $k$ to be

$$\zeta_k(s) = \prod_{w \in \Sigma(k)} (1 - q^{-f(w)s})^{-1}.$$

This corresponds exactly to the definition of the Riemann zeta-function by the Euler Product since the places of $\mathbf{Q}$ correspond to the embeddings $\mathbf{Q} \to \mathbf{Q}_p$ as $p$ ranges through the prime numbers and $\mathbf{Q} \to \mathbf{R}$ which gives an additional factor $\pi^{-s/2}\Gamma(s/2)$. This factor arose in Chapter 2 in the functional equation of $\zeta$.

It is not difficult to show that the infinite product defining $\zeta_k$ converges in $\{s \in \mathbf{C} : \mathrm{Re}(s) > 1\}$. It can be represented in the form

$$\zeta_k(s) = P(q^{-s})/(1 - q^{-s})(1 - q^{1-s})$$

where $P$ is a polynomial of degree $2g$ where $g$ is the genus of $C$. This

corresponds to the fact that the Riemann zeta-function has an analytic continuation into the complex plane and is an integral function of order 1. Moreover, the function $\zeta_k$ satisfies the functional equation

$$\zeta_k(1-s) = q^{(1-g)(1-2s)}\zeta_k(s).$$

These facts can be deduced from the Riemann–Roch Theorem (see e.g. the references at the end of § 5.12) or from an analogue of the Poisson Summation Formula as in Chapter 2 (see e.g. Tate (1), Weil (4)), a fact that demonstrates the very close connection between two apparently unrelated results. This connection is underlined still further by the theory of the Selberg Trace Formula which represents a non-abelian generalisation of the Poisson Summation Formula and which has the Riemann–Roch Theorem for Riemann surfaces as a special case (Selberg (4), Ihara (1)).

There is, by the above, a multiset $A$ such that

$$P(u) = \prod_{\alpha \in A}(1-\alpha u).$$

One verifies easily that the functional equation is equivalent to the fact that the mapping $\alpha \mapsto q/\alpha$ maps $A$ into itself. The analogue of the Riemann Hypothesis for $\zeta_k$ is that the zeros lie on $\{s \in \mathbf{C}: \mathrm{Re}(s) = 1/2\}$, or, what comes to the same thing, for $\alpha \in A$ that $|\alpha| = q^{1/2}$. This is true as we indicated above, and in the next few sections we shall describe some of the ideas involved in the proof. For further details see Swinnerton-Dyer (1) where a fairly comprehensive sketch of the proof is given, or Weil (5) for the full details.

## *5.14

In order to proceed further we shall derive the analogue of the 'explicit formulae' of Chapter 3. To do this we differentiate logarithmically the equation

$$\prod_{w \in \Sigma(k)}(1-u^{f(w)})^{-1} = \left\{\prod_{\alpha \in A}(1-\alpha u)\right\}(1-u)^{-1}(1-qu)^{-1}$$

and obtain first

$$\sum_{w \in \Sigma(k)} f(w)u^{f(w)}(1-u^{f(w)})^{-1} = -\sum_{\alpha \in A}\alpha u(1-\alpha u)^{-1} + u(1-u)^{-1} + qu(1-qu)^{-1}.$$

In this we compare the coefficients of $u^j$ in the power series expansion of each side at 0 and obtain for $j > 0$

$$\sum_{f(w)\mid j} f(w) = 1 + q^j - \sum_{\alpha \in A}\alpha^j. \tag{5.3}$$

Note that the left-hand side of this is the number of points of $C(\mathbf{F}_{q^j})$, a fact which will take on greater significance presently.

If we multiply both sides of (5.3) by $q^{-j/2}$ then we see that the right-hand side of the resulting equation is invariant under replacing $j$ by $-j$; this follows from the functional equation. Thus we can rewrite (5.3) as

$$q^{-|j|/2} \sum_{f(w)|j} f(w) = q^{j/2} + q^{-j/2} - \sum_{\alpha \in A} (\alpha/q^{1/2})^j \qquad (j \neq 0) \qquad (5.4)$$

Note that as in the case of the Riemann zeta-function one has, since the zeta-function has no zeros in $\{s \in \mathbf{C}: \mathrm{Re}(s) > 1\}$, that $1 \leqslant |\alpha| \leqslant q$.

Let now $\phi: \mathbf{Z} \to \mathbf{C}$ be such that $M^d(\phi, z)$ defined by

$$M^d(\phi, z) = \sum_{j \in \mathbf{Z}} \phi(j) x^j$$

converges in the annulus $\{z \in \mathbf{C}: q^{-1/2} \leqslant |z| \leqslant q^{1/2}\}$. If we multiply (5.4) by $\phi(k)$ and sum over all $k$ then we obtain the equation

$$\sum_{w \in \Sigma(k)} \sum_{l \in \mathbf{Z} - \{0\}} f(w) q^{-f(w)|l|/2} \phi(f(w)l) + (2 - 2g)\phi(0)$$
$$= M^d(\phi, q^{1/2}) + M^d(\phi, q^{-1/2}) - \sum_{\alpha \in A} M^d(\phi, \alpha/q^{1/2}). \qquad (5.5)$$

This equation is the analogue of the 'explicit formulae' of prime number theory, Theorem 3.6. The operator $M^d$ is a 'discrete' Mellin transform. Note that (5.5) also includes the functional equation of $\zeta_k$.

## *5.15

We shall now give a reformulation of the analogue of the Riemann Hypothesis for this class of zeta-function which is due to Weil. For $\phi$ as above we define $\phi^\tau$ by $\phi^\tau(k) = \overline{\phi(-k)}$. For $\phi_1, \phi_2$ as above we define the convolution $\phi_1 * \phi_2$ by

$$\phi_1 * \phi_2(k) = \sum_{j_1, j_2 : j_1 + j_2 = k} \phi_1(j_1)\phi_2(j_2).$$

Thus one has

$$M^d(\phi_1 * \phi_2, z) = M^d(\phi_1, z) M^d(\phi_2, z).$$

We now define a hermitian form $R$ on the vector space of the $\phi$ above. For $\phi_1$ and $\phi_2$ in this space $R(\phi_1, \phi_2)$ is given by one of the two equivalent expressions

$$R(\phi_1, \phi_2) = M^d(\psi, q^{1/2}) + M^d(\psi, q^{-1/2}) + (2g - 2)\psi(0)$$
$$- \sum_{w \in \Sigma(k)} \sum_{l \in \mathbf{Z} - \{0\}} f(w) q^{-f(w)|l|/2} \psi(f(w))$$
$$= \sum_{\alpha \in A} M^d(\psi, \alpha/q^{1/2})$$

where

$$\psi = \phi_1 * \phi_2^\tau.$$

**Theorem:** *The analogue of the Riemann Hypothesis holds for* $\zeta_k$ *if, and only if, the hermitian form* $R$ *is positive semidefinite.*

*Proof*: Suppose first that the Riemann Hypothesis is true. Then one would have for $\alpha \in A$ that

$$\begin{aligned} M^d(\phi_1 * \phi_2^\tau, \alpha/q^{1/2}) &= M^d(\phi_1, \alpha/q^{1/2}) M^d(\phi_2^\tau, \alpha/q^{1/2}) \\ &= M^d(\phi_1, \alpha/q^{1/2}) \overline{M^d(\phi_2, (\bar{\alpha}/q^{1/2})^{-1})} \\ &= M^d(\phi_1, \alpha/q^{1/2}) \overline{M^d(\phi_2, \alpha/q^{1/2})}. \end{aligned}$$

Thus we have

$$R(\phi_1, \phi_2) = \sum_{\alpha \in A} M^d(\phi_1, \alpha/q^{1/2}) \overline{M^d(\phi_2, \alpha/q^{1/2})}$$

which is clearly positive semidefinite. Note that the kernel of $R$ is made up of the space of such $\phi$ that $M^d(\phi, z)$ has a *simple* zero at each element of $A$.

Suppose now that $R$ is positive semidefinite but that the analogue of the Riemann Hypothesis does not hold. Then there exists $\alpha_0 \in A$ so that $\alpha_1 := q/\bar{\alpha}_0 \neq \alpha_0$. It is clear since $P$ has real coefficients that $\alpha_1 \in A$. We now choose a polynomial $F$ so that $F(\alpha_0) = \sqrt{(-1)}$, $F(\alpha_1) = -\sqrt{(-1)}$ and $F(\alpha) = 0$ for $\alpha \in A - \{\alpha_0, \alpha_1\}$. Then there exists $\phi$ with $M^d(\phi, z) = F(zq^{1/2})$. From the calculation above we have that

$$R(\phi, \phi) = \sum_{\alpha \in A} F(\alpha) \overline{F(q/\bar{\alpha})};$$

in this sum only the terms $\alpha_0$ and $\alpha_1$ give a non-zero contribution and each of these contributes $-1$. Thus $R(\phi, \phi) = -2$ which contradicts the positive semidefiniteness of $R$. This completes the proof of the theorem

## 5.16

Before we proceed with the argument here it is useful to note that an analogous statement holds for the Riemann zeta-function. We digress a little to formulate this result here; we shall not prove it as the proof uses essentially the same ideas as those in § 5.15 although a little care is required since there are infinitely many zeros.

Suppose $\phi_1$ and $\phi_2$ are functions of type $(\alpha, \beta)$ with $\alpha < -1/2$ and $\beta > 1/2$. Then we can define the convolution $\phi_1 * \phi_2$ by

$$\phi_1 * \phi_2(x) = \int_{\mathbf{R}^\times} \phi_1(y) \phi_2(xy^{-1}) |y|^{-1} \, dy$$

which is also of type $(\alpha, \beta)$, the integral being convergent almost everywhere by Fubini's Theorem. Moreover, one also has

$$M(\phi_1 * \phi_2, s) = M(\phi_1, s) M(\phi_2, s).$$

We shall also suppose that in the strip $\{s\in\mathbf{C}:|\mathrm{Re}(s)|\leqslant 1/2\}$ one has for some $c>0$ and $\varepsilon>0$ that

$$|M(\phi_j,s)|<c(1+|s|)^{-1/2-\varepsilon}.$$

We shall also define $\phi^\tau$ for such a function $\phi$ by

$$\phi^\tau(x)=\overline{\phi(x^{-1})}$$

and this satisfies similar conditions.

We can now define the hermitian form $R$ by either of the two expressions (cf. Theorem 3.6)

$$R(\phi_1,\phi_2)=M(\psi,1/2)+M(\psi,-1/2)-\Delta_\alpha(\psi)-\sum_p \log p \sum_{k\in\mathbf{Z}\setminus\{0\}} p^{-|k|/2}\psi(p^k)$$

or

$$R(\phi_1,\phi_2)=\sum_{\rho\in Z} M(\psi,\rho-1/2)$$

where

$$\psi=\phi_1*\phi_2^\tau.$$

Then one has the following criterion due to Weil (Weil (1), (2)):

**Theorem:** *The Riemann Hypothesis holds if, and only if, the hermitian form $R$ is positive semidefinite.*

This type of criterion can be extended to a very wide class of Dirichlet series definable as Euler Products which appear in number-theory. One particularly interesting feature is that when applied to the theory of Artin $L$-functions (see e.g. Martinet (1)) it implies not only the corresponding analogue of the Riemann Hypothesis but also the Artin Hypothesis i.e. that these functions are holomorphic. The analogue of this conjecture for curves over a finite field was proved by Weil at the same time as the corresponding Riemann Hypothesis.

## *5.17

We now come to the proof of the Riemann Hypothesis for the $\zeta_k$ of § 5.13. This we shall only sketch since the details of the proof depend on considerations from algebraic geometry.

To the field $k$ we introduced a curve $C$ over $\mathbf{F}_q$; let $\bar{C}$ be the curve obtained by extension of scalars to the field $\bar{k}_0=\bar{\mathbf{F}}_q$, the algebraic closure of $\mathbf{F}_q$ The field of functions of $\bar{C}$ is $k\otimes_{\mathbf{F}_q}\bar{\mathbf{F}}_q=:\bar{k}$. The map $Fr:\bar{k}\to\bar{k};\ u\mapsto u^q$ gives rise to a map $\bar{C}(\bar{k}_0)\to\bar{C}(\bar{k}_0)$ which we also denote by $Fr$ These maps are both called the 'Frobenius map'. The fixed points of $Fr^j:\bar{C}(\bar{k}_0)\to\bar{C}(\bar{k}_0)$ are precisely the points in $C(\mathbf{F}_{q^j})$ since $\mathbf{F}_{q^j}=\{x\in\bar{\mathbf{F}}_q:x^{q^j}=x\}$. Thus the

number of fixed points is, by the remark in §5.14, $\sum_{f(w)|j} f(w) = 1 + q^j - \sum_{\alpha \in A} \alpha^j$.

This one interprets as follows. The number of fixed points of a map of a variety to itself (over the complex numbers) is given by an alternating sum of the traces of the induced maps on cohomology groups. This led Weil to suggest the existence of a cohomology theory for curves over a finite field that had the same properties. This cohomology theory was developed later by Serre and, above all, Grothendieck as one example of a much broadened concept of what a cohomology theory is. For our purposes it suffices to know that for any prime $l$ not dividing $q$ one can define cohomology groups $H^*(\bar{C}, \mathbf{Q}_l)$ with coefficients in the $l$-adic numbers $\mathbf{Q}_l$. For a curve of genus $g$ one also has that $H^j(\bar{C}, \mathbf{Q}_l) = 0$ for $j > 2$, that $H^0(\bar{C}, \mathbf{Q}_l)$ and $H^2(\bar{C}, \mathbf{Q}_l)$ are one-dimensional and $H^1(\bar{C}, \mathbf{Q}_l)$ is $2g$-dimensional. Moreover as $Fr$ is of degree $q$ the map induced by it on $H^0(C, \mathbf{Q}_l)$ is the identity, and on $H^2(C, \mathbf{Q}_l)$ is multiplication by $q$. By the Lefschetz Fixed-Point Theorem we would have that the number of fixed points of $Fr^j$ on $\bar{C}(\bar{\mathbf{F}}_q)$ is

$$\mathrm{Tr}\{Fr^{*j} | H^0(\bar{C}, \mathbf{Q}_l)\} - \mathrm{Tr}\{Fr^{*j} | H^1(\bar{C}, \mathbf{Q}_l)\} + \mathrm{Tr}\{Fr^{*j} | H^2(\bar{C}, \mathbf{Q}_l)\}.$$

The first and last terms yield together $1 + q^j$ and hence we would have

$$\mathrm{Tr}\{Fr^{*j} | H^1(\bar{C}, \mathbf{Q}_l)\} = \sum_{\alpha \in A} \alpha^j,$$

so that $\alpha$ are just the eigenvalues of $Fr^* | H^1(\bar{C}, \mathbf{Q}_l)$. This is a very important interpretation of the elements of $A$ but it is not clear that it could be of much use to us. The problem is that the interpretation is essentially $l$-adic whereas we are interested in an estimate on the elements in the complex field.

Let $\phi: \mathbf{Z} \to \mathbf{Z}$ be a function taking the value 0 at all but a finite number of arguments. Then we have that

$$\phi(Fr^*) := \sum_{k \in \mathbf{Z}} \phi(k) Fr^{*k}$$

exists as an element of $\mathrm{End}\{H^1(\bar{C}, \mathbf{Q}_l)\}$. Let $\phi^\tau$ be defined as above. It follows easily from §5.15 that to prove the Riemann Hypothesis for $\zeta_k$ it suffices to show that for such one has that $\mathrm{Tr}\{\phi(q^{-1/2} Fr^*) * \phi(q^{-1/2} Fr^*)^\tau\} \geqslant 0$, once we have given this a meaning.

The remarkable fact that appears now is that although $H^1(\bar{C}, \mathbf{Q}_l)$ cannot be defined over $\mathbf{Q}$ its endomorphism ring $\mathrm{End}\{H^1(\bar{C}, \mathbf{Q}_l)\}$ can. In fact, if one considers the endomorphism ring of the Jacobian variety $J(\bar{C})$ one obtains an algebra $A$ which is defined over $\mathbf{Z}$ so that $A \otimes_{\mathbf{Z}} \mathbf{Q}_l$ is isomorphic to $\mathrm{End}\{H^1(C, \mathbf{Q}_l)\}$. There are various ways of looking at this isomorphism. One is that the $l^k$-division points of $J(\bar{C})$ determine $l^k$-fold abelian covers

of $\bar{C}$ which are unramified whereas the étale cohomology theory is based on a category of étale, or unramified, coverings of $\bar{C}$. The process of taking $H^1$ corresponds as in the Hurewicz Theorem to abelianising the fundamental group.

The Frobenius map induces an element of $\mathrm{End}\{J(\bar{C})\}$ which is invertible in $\mathrm{End}\{J(\bar{C})\}\otimes\mathbf{Q}$. On $\mathrm{End}\{J(\bar{C})\}$ there is a standard involution $a\mapsto a^{\dagger}$ called the Rosati involution. It is easy to compute the effect of this on $Fr$ and one finds $FrFr^{\dagger}=q$.

It therefore follows that in order to prove the Riemann Hypothesis for $\zeta_k$ it is necessary to prove that

$$\mathrm{Tr}\{\phi(q^{-1/2}Fr^*)*\phi(q^{-1/2}Fr^*)^{\dagger}\}\geqslant 0$$

where the arguments lie in $A\otimes\mathbf{Q}(q^{1/2})$.

This is a special case of the general inequality

$$\mathrm{Tr}(\xi\xi^{\dagger})\geqslant 0,$$

the positivity of the Rosati involution, which is proved by an application of Castelnuovo's inequality for surfaces.

For more details on this part of the argument see Swinnerton-Dyer (1). The étale cohomology theory is developed and explained in Srinivasan (1), Tamme (1) and more details on the proof sketched here are given in Weil (5).

## *5.18

One can try to see to what extent the proof sketched in the preceding paragraph can be carried over to the theory of the Riemann zeta-function. One would hope that there would be an analogue to the algebra $A$, let us say in this case a real Banach algebra with involution, and corresponding to the Frobenius endomorphism, a one-parameter group $F_0\colon\mathbf{R}\to A^{\times}$. It is better to consider this as a homomorphism $F\colon\mathbf{R}_+^{\times}\to A^{\times}$. By analogy to the case with which we have just been dealing we would hope that $A$ would be endowed with a trace map $\mathrm{Tr}\colon A_1\to\mathbf{R}$ defined on a dense subset $A_1$ of $A$ and that in $A\otimes\mathbf{C}$

$$\sum_{\rho\in Z}M(\phi,\rho-1/2)=\mathrm{Tr}\left\{\int_{\mathbf{R}_+^{\times}}\phi(t)F(t)t^{-1}\,\mathrm{d}t\right\}$$

and that for such $\phi$ as satisfy the conditions of Theorem 3.6 the trace exists. One would also hope that there existed an involution $a\mapsto a^{\dagger}$ of $A$ so that if $aa^{\dagger}\in A_1$ then $\mathrm{Tr}(aa^{\dagger})\geqslant 0$.

If we also had that $F(t)^{\tau}=F(t^{-1})$ then the Riemann Hypothesis would follow from this as it did in the analogous case of the $\zeta_k$.

Before discussing this algebra in more detail we note that under certain

assumptions (see Maurin (1), Chapter 10) the group $F: \mathbf{R}_+^\times \to A$ will have an infinitesimal generator, that is to say that the derivative $F'(1)$ should exist as a densely defined operator on $A$ commuting with right multiplication. Such operators can be regarded as giving an extension of a dense subalgebra of $A$ and in this sense as 'almost' in $A$. There have been many attempts to find an operator $F'(1)$ with the kind of properties described above as a differential operator (the space of differentiable functions on which this acts should be more or less a representation space for $A$). The eigenvalues of $F'(1)^2$ would be $(\rho - 1/2)^2$ which, according to the Riemann Hypothesis should all be negative. Thus one could expect that $F'(1)^2$ would be represented by a elliptic differential operator, presumably of the second order; to date attempts to achieve this with Laplacian or Schrödinger operators have not been successful.

One would actually expect that the algebra $A$ should be identified through the 'explicit formula'

$$\mathrm{Tr}\left\{\int_{\mathbf{R}^\times} \phi(t)F(t)t^{-1}\,\mathrm{d}t\right\} = M(\phi, 1/2) + M(\phi, -1/2) - \Delta_\infty(\phi) - \sum_p \log p \sum_{k\neq 0} p^{-|k|/2}\phi(p^k)$$

by some at present fully unknown analogue of the Lefschetz Fixed-Point Formula.

Iwasawa has proposed an analogue for $\mathbf{Q}$ and other algebraic number fields of the Jacobian of a curve. The idea is that the Jacobian classifies abelian coverings of the curve over the algebraic closure of the base field. The analogue of $\bar{k}$ of the previous section is for Iwasawa the field $C_l/Q$ which is generated by all the $l^k$ roots of unity, $k \geqslant 1$, where $l$ will be a fixed prime number. This is a cyclotomic field and is structurally very similar to $\bar{\mathbf{F}}_q$, or rather the subfield $\bigcup_{k>0}\mathbf{F}_{q^{l^k}}$. Let now $J_l/C_l$ be the maximal abelian unramified $l$-extension. The Galois group of $C_l/Q$ is isomorphic to $\mathbf{Z}_l^\times$ and this group acts on the Galois group of $J_l/C_l$. If we denote the Galois group of $K'/K$ by $Gal\,(K'/K)$ then we see that $Gal\,(J_l/C_l)$ is analogous to the set of all $l$th power division points of the Jacobian variety of a curve. Note that $J_l$ is Galois over $Q$ and hence $\mathbf{Z}_l^\times$ acts on $Gal\,(J_l/C_l)$.

This analogue of the Jacobian has very close connections with the theory of $p$-adic zeta-functions; for all this see Iwasawa (1), Coates (1), (2), Mazur & Wiles (1).

On the other hand up to the present there has been no method proposed for constructing an algebra of the type we are seeking from the various $\mathrm{End}\{Gal\,(J_l/C_l)\}$. It is clear that if such an $A$ does exist then it will be of the greatest arithmetical interest even beyond its application to proving

the Riemann Hypothesis. To find it remains a task for the future – if it exists at all.

## 5.19

Although the approach to the Riemann Hypothesis indicated in the previous paragraph would be the most interesting, and would presumably lead to the most far-reaching developments, it is far from certain if it can ever be carried out (even supposing the Riemann Hypothesis is true). There are other approaches. Several of these are based on the fact that the 'explicit formula' of Theorem 3.6 asserts that a certain distribution can be represented in two very different ways and it seems possible that this fact could suffice to imply the validity of the Riemann Hypothesis. The argument of Hadamard given in §4.2 essentially uses only the 'explicit formula' – in a very special case – and yields a zero-free region. This argument has not been improved on in this century although subsidiary estimates on the zeta-function have allowed somewhat larger zero-free regions to be determined.

An interesting approach to the Riemann Hypothesis along these lines and going beyond Weil's positivity criterion of § 5.16 was proposed in 1984 by Matsumoto who constructed a combination of spherical functions on $SL_2(\mathbf{R})$ depending *quadratically* on the distribution described by Theorem 3.6. This distribution would be 'of negative type' if, and only if, the Riemann Hypothesis holds. This approach, based on non-abelian harmonic analysis, was inspired by ideas from quantum field theory (see Matsumoto (1)). It is reminiscent of the classical technique of estimating Gauss and Jacobi sums by considering their squares (see Exercises 5.13–5.15).

An argument which is based on the 'explicit formula' should also be applicable in the function-field case. Indeed, in this case the 'explicit formulae' can sometimes be used very effectively, for example to count points on certain curves; see Drinfeld & Vladut (1), Ihara (2), Serre (1), Stark (1) and Exercises 5.17 and 5.18.

If one abstracts from the zeta-function of a curve over a finite field the data that are available in the 'explicit formulae' and assumes that this suffices to prove the corresponding 'Riemann Hypothesis' then one would have the following statement:

*Let $e_1 \leqslant e_2 \leqslant \cdots$ be a sequence of natural numbers, let $q > 1$ be real and let $A$ be a multiset of complex numbers. Suppose that one has*

(i) *the map $x \mapsto q/x$ maps $A$ into itself, and*

(ii) *for* $k \geqslant 1$ *one has*

$$\sum_{e_i | k} e_i = 1 + q^k - \sum_{\alpha \in A} \alpha^k.$$

*Then* $|\alpha| = q^{1/2}$ *for each* $\alpha \in A$.

This is false – for counter-examples see Exercises 5.20 and 5.21. This bodes ill for the possibility of an analytic proof of the Riemann Hypothesis.

Finally there are other analytic approaches to the Riemann Hypothesis that should not be entirely forgotten. There have been a number of classical function-theoretic criteria given for the validity of the Riemann Hypothesis but none of these has yet led anywhere; for references to some of these see Titchmarsh (2), § 10.23. There is also a hope that the theory of automorphic forms might lead to a proof but this is not based on any coherent philosophy and it would be very surprising if it should succeed. It should also be emphasised that the zeta-function is best understood within the context of analysis on the group of idèles (see Tate (1), Weil (4), Chapter 8) and that this should not be ignored in any attempt to prove the Rieman Hypothesis.

## Exercises

5.1 Let $d_k$ be as defined in §1.11. Show, using Exercise 4.10, that if the Lindelöf Hypothesis holds then for $\delta: 0 < \delta < 1$, $\varepsilon > 0$ and $\sigma: 1/2 < \sigma < 1$ one has as $|\mathrm{Im}(s)| \to \infty$ on $\mathrm{Re}(s) = \sigma$

$$\zeta(s)^k = \sum_{n \leqslant |\mathrm{Im}(s)|^\delta} d_k(n) n^{-s} + O((1 + |\mathrm{Im}(s)|)^{-\{1-(1-\sigma)\delta\}+\varepsilon}).$$

5.2 Show that if for each $\sigma: 1/2 < \sigma < 1$, $: 0 < \delta < 1$ there exists $b > 0$ so that one has as $|\mathrm{Im}(s)| \to \infty$ on $\mathrm{Re}(s) = \sigma$

$$\zeta(s)^k = \sum_{n \leqslant |\mathrm{Im}(s)|^\delta} d_k(n) n^{-s} + O((1 + |\mathrm{Im}(s)|)^{-b})$$

then the Lindelöf Hypothesis follows. (Here $k$ is any fixed natural number.)

5.3 Suppose that $\sigma: 1/2 < \sigma < 1$. Show that

$$\sum_{1 \leqslant m < n \leqslant X} m^{-\sigma} n^{-\sigma} |\log(m/n)| = O(X^{2-2\sigma} \log X)$$

as $X \to \infty$.

5.4 Under the Lindelöf Hypothesis show that for $\sigma: 1/2 < \sigma < 1$ as $T \to \infty$

$$T^{-1} \int_0^T |\zeta(\sigma + it)|^{2k} \mathrm{d}t \to \sum_{n \geqslant 1} d_k(n)^2 n^{-2\sigma}.$$

5.5 Use Theorem 4.12 and Hölder's inequality to show that for $\sigma: \sigma > 1/2$,

$k \geqslant 1$ there exists $C > 0$ so that

$$\int_{T/2}^{T} |\zeta(\sigma + it)|^{2k} dt > CT.$$

Show also that for $\sigma: 0 < \sigma < 1/2$ and $k > (1 - 2\sigma)^{-1}$ then

$$\int_{-\infty}^{\infty} |\zeta(\sigma + it)|^{2k} |\sigma + it|^{-2} dt$$

diverges.

5.6 Suppose that $\sigma: \sigma > 0$, $k \in \mathbf{N}$ are such that

$$\int_{-\infty}^{\infty} |\zeta(\sigma + it)|^{2k} |\sigma + it|^{-2} dt$$

converges. Show that

$$\int_{0}^{\infty} \left( \sum_{n \leqslant X} d_k(n) - \operatorname*{Res}_{s=1} \{\zeta(s)^k X^s / s\} \right)^2 X^{-2\sigma - 1} dX$$

also converges.

5.7 Deduce from Exercise 5.6 that if

$$\sum_{n \leqslant X} d_k(n) = \operatorname*{Res}_{s=1} \{\zeta(s)^k X^s / s\} + O(X^b)$$

then $b \geqslant (k-1)/2k$.

5.8 Show that if $\sum_{n \leqslant X} \mu(n) = O(X^b)$ then $\zeta$ has no zeros in $\{s \in \mathbf{C}: \operatorname{Re}(s) > b\}$ (Note that this implies that necessarily $b \geqslant 1/2$.)

5.9 Use Exercise 4.10 to show that if $b > 0$ is such that $\zeta$ has no zeros in $\{s \in \mathbf{C}: \operatorname{Re}(s) > b\}$ then for any $s$ with $\operatorname{Re}(s) > b$ one has that

$$\sum_{n \geqslant 1} \mu(n) n^{-s}$$

converges and is equal to $\zeta(s)^{-1}$.

5.10 Show that if $\zeta$ has no zeros in $\{s \in \mathbf{C}: \operatorname{Re}(s) > b\}$ then for any $a > b$ one has $\sum_{n \leqslant X} \mu(n) = O(X^a)$.

5.11 Show that if $b > 0$ is such that for all $a > b$ one has that $\pi(X) = Li(X) + O(X^a)$ then it follows that $\pi(X) = Li(X) + O(X^b \log X)$.

5.12 Show that under the Riemann Hypothesis for $\sigma > 1/2$ there exist $A, c > 0$ so that in $\{s \in \mathbf{C}: \sigma < \operatorname{Re}(s) < 2\}$ one has

$$|\zeta(s)| > c|s|^{-A}.$$

5.13 Let $q$ be a prime power and let $e$ be a non-trivial additive character of $\mathbf{F}_q$. For a divisor $n$ of $q - 1$ let $X(q, n)$ be the group of characters of $\mathbf{F}_q^{\times}$ of order dividing $n$. Show that $X(q, n)$ has $n$ elements. For any character $\chi$ of $\mathbf{F}_q^{\times}$ let

$$G(\chi, e) = \sum_{x \in \mathbf{F}_q^{\times}} \chi(x) e(x).$$

Show that

$$G(1,e) = -1$$
$$|G(\chi,e)| = q^{1/2} \qquad \text{if } \chi \neq 1.$$

Show also that

$$\sum_{x \in \mathbf{F}_q - \{0,1\}} \chi_1(x)\chi_2(1-x) = G(\chi_1,e)G(\chi_2,e)/G(\chi_1\chi_2,e) + \begin{cases} 0 & \text{if } \chi_1\chi_2 \text{ is trivial} \\ (q-1)\chi_1(-1) & \text{if } \chi_1\chi_2 \text{ is non-trivial.} \end{cases}$$

5.14 Let $C_{m,n}: x^m + y^n = 1$ be the algebraic curve defined over $\mathbf{F}_q$ where $q \equiv 1 \pmod{m}$ and $q \equiv 1 \pmod{n}$. One counts either $(m,n)$ or 0 points at infinity according to whether $-1$ is an $(m,n)$th power in $\mathbf{F}_q$ or not. Show that the number of $\mathbf{F}_q$-points on $C_{m,n}$ is

$$q + 1 + \sum_{\substack{\chi_1 \in X(q,n), \chi_2 \in X(q,n) \\ \chi_1, \chi_2, \chi_1\chi_2 \neq 1}} G(\chi_1,e)G(\chi_2,e)/G(\chi_1\chi_2,e).$$

5.15 Let $l \geqslant 1$ and let $t_l: \mathbf{F}_{q^l} \to \mathbf{F}_q$, $n_l: \mathbf{F}_{q^l} \to \mathbf{F}_q$ be the trace and norm functions. Use the Davenport–Hasse Theorem (Davenport & Hasse (1)),

$$G(\chi \circ n_l, e \circ t_l) = (-1)^{l-1} G(\chi,e)^l$$

where the left-hand side is evaluated over $\mathbf{F}_{q^l}$, to show the zeta-function of $C_{m,n}$ is

$$(1-t)^{-1}(1-qt)^{-1} \prod (1 + \{G(\chi_1,e)G(\chi_2,e)/G(\chi_1\chi_2,e)\}t)$$

where the product is taken over $\chi_1 \in X(q,m)$, $\chi_2 \in X(q,m)$ so that $\chi_1, \chi_2$ and $\chi_1\chi_2$ are all non-trivial. Verify that the genus of the curve is indeed the half of the degree of the product. Verify the analogue of the Riemann Hypothesis in this case.

5.16 Use Theorem 5.15 and Exercise 5.14 to determine the zeta-function of $C_{m,n}$ in the case $m = 2$, $n = 3$ without using the Davenport–Hasse Theorem.

5.17 Let $C$ be a curve defined over $\mathbf{F}_q$ and let $N$ be the number of points defined over $\mathbf{F}_q$. Then

$$N = q + 1 - (1/2) \sum_{\alpha \in A} (\alpha + \bar{\alpha}).$$

Use the arithmetic–geometric mean inequality applied to the set of numbers $[2q^{1/2}] + 1 + \alpha + \bar{\alpha}$ to show that

$$N \leqslant q + 1 + g[2q^{1/2}].$$

5.18 With the notation of Exercise 5.17 show that

$$N \leqslant 1 + \left(2g + \sum_{n \geqslant 1} u_n q^{n/2}\right) \Big/ \left(\sum_{n \geqslant 1} u_n q^{-n/2}\right)$$

where the sequence $(u_n)$ is such that $u_n \geqslant 0$ and

$$1 + \sum_{n \geqslant 1} u_n \cos n\theta$$

converges absolutely and is positive for all $\theta$. Use the explicit formulae. Use $(1 + \sqrt{2}\cos\theta)^2 \geqslant 0$ to deduce that if $g \leqslant q^{1/2}(q^{-1})/\sqrt{2}$ then

$$N \leqslant q^2 + 1.$$

5.19 Let $\alpha$ be a zero of the zeta-function of a curve of genus $g$ over $\mathbf{F}_q$. Let $\theta = \arg\alpha$ and suppose that $-\pi \leqslant \theta \leqslant \pi$. Show that Hadamard's method leads to the inequality

$$1 - |\alpha|/q \geqslant C(|\alpha|/q)^5 |\sin(\theta/2)|^5 \cos(\theta/2)$$

where

$$C = \{(1 - q^{-1})/2^{2g}\}^8 2^6.$$

Deduce from this that $0 \leqslant \theta \leqslant \pi$

$$|\alpha| < q\{1 - C^*\theta^5(\pi - \theta)\}$$

where

$$C^* = 2^{-1}\{(1 - q^{-1})/2^{2g}\}^8 (2/\pi)^6.$$

Show also that $C^* < 10^{-6}$.

5.20 Define the sequence $N(j)$ $(j > 0)$ by

$$\sum_{i|j} iN(i) = 1 + 5^j - \alpha^j - \alpha'^j$$

where $\alpha, \alpha$ are the roots of $X^2 - 5X + 5 = 0$. Show that $N(j)$ is a positive integer and that

$$\frac{1 - 5X + 5X^2}{(1 - X)(1 - 5X)} = \prod_{j \geqslant 1} (1 - X^j)^{-N(j)}$$

but that

$$|\alpha| \neq 5^{1/2}.$$

5.21 Define the sequence $N(j)$ $(j > 0)$ by

$$\sum_{i|j} iN(i) = 1 + 9^j - \{\sqrt{(-3)}\}^j (1 + 3^j)\{1 + (-1)^j\}.$$

Show that $N(j)$ is a positive integer and that

$$\frac{1 - 30X^2 + 81X^4}{(1 - X)(1 - 9X)} = \prod_{j \geqslant 1} (1 - X^j)^{-N(j)},$$

and that if $\alpha$ is a root of $X^4 - 30X^2 + 81$ then $|\alpha| \neq 3$.

# 6

# *The approximate functional equation*

## 6.1

The approximate functional equation has been mentioned several times in the preceding chapters. It allows the Riemann zeta-function to be well approximated by relatively 'small' finite expressions in the critical strip, whereby one can derive, in conjunction with other techniques for finding bounds for 'exponential sums', various estimates for the zeta-function itself in this region. Two examples will be given in this chapter – Theorems 6.5 and 6.6 which, in loose terms describe $L^2$ and $L^\infty$ estimates of the zeta-function.

It is also worth noting that the sharper form of the approximate functional equation which replaces the error term by an asymptotic expansion (the Riemann-Siegel Formula) can be very effectively used in numerical calculations of the zeta-function. The form of the approximate functional equation which we shall prove here is the following.

**Theorem:** *Let $X, Y \geqslant 1$ and $\sigma < 1$ be real parameters. Suppose $s \in \mathbf{C}$ satisfies $1 - \sigma < \mathrm{Re}(s) < \sigma$ and $|\mathrm{Im}(s)| = 2\pi XY$. Then*

$$\zeta(s) = \sum_{n \leqslant X} n^{-s} + \gamma(1-s) \sum_{m \leqslant Y} m^{-(1-s)} + O(X^{1/2 - \mathrm{Re}(s)}(X^{-1/2} + Y^{-1/2}) \log XY)$$

*where the implied constant depends only on $\sigma$.*

This form of the approximate functional equation can be derived from the prototypical version Theorem 2.13 by a suitable choice of the function $f$ and estimating the integrals that arise. The basic estimate will be given in § 6.3, and the application to the proof of the theorem follows in § 6.4. The proof given here is conceptual but suffers from the weakness that it yields a factor $\log XY$ in the error term which in a more sophisticated approach does not appear.

## 6.2

Before we turn to the proof we note the special case $Y = 1$ of Theorem 6.1. This can be stated as:

**Corollary:** *Suppose that $\alpha > 0$. Then one has as $|\mathrm{Im}(s)| \to \infty$ uniformly in*

$\mathrm{Re}(s) \geqslant \alpha$

$$\zeta(s) = \sum_{n \leqslant |\mathrm{Im}(s)|/2\pi} n^{-s} + O(|\mathrm{Im}(s)|^{-\mathrm{Re}(s)} \log |\mathrm{Im}(s)|).$$

This estimate suffices for many purposes but for finer estimates it is necessary to use Theorem 6.1. Note that in Theorem 6.1 we can arrange that both sums have roughly $O(|\mathrm{Im}(s)|^{1/2})$ terms whereas the sum in the corollary has a much 'larger' sum with $O(|\mathrm{Im}(s)|)$ terms. It is this difference that accounts for the much greater efficiency of Theorem 6.1 in applications.

## 6.3

In this section we shall formulate and prove an estimate for an integral which appears in many contexts related to the theory of Dirichlet series. The proof is an elementary exercise, although the same idea is important in other contexts.

**Proposition:** *Let* $s \in \mathbf{C}$ *and* $\beta > \alpha > 0$. *Then one has the inequalities*

$$\left| \int_\alpha^\beta \left\{ \frac{(\sin u)}{u} \right\}^2 u^{s-1}\, du - (\beta^{s-2} - \alpha^{s-2})/2(s-2) \right|$$
$$\leqslant |\beta - \alpha|(\alpha^{\mathrm{Re}(s)-3} + \beta^{\mathrm{Re}(s)-3})/2 \tag{1}$$

*and*

$$\left| \int_\alpha^\beta \left\{ \frac{(\sin u)}{u} \right\}^2 u^{s-1}\, du - [u^{s-2}(\sin^2 u)/(s-2) - u^{s-1}(\sin 2u)/(s-2)(s-1)]_\alpha^\beta \right|$$
$$\leqslant 2|\beta - \alpha|(\alpha^{\mathrm{Re}(s)-1} + \beta^{\mathrm{Re}(s)-1})/|(s-2)(s-1)|. \tag{2}$$

*If*

$$[\alpha, \beta] \cap \{|\mathrm{Im}(s)|/2, (|\mathrm{Re}(s)|/|\mathrm{Re}(s)-1|)|\mathrm{Im}(s)|/2\} = \varnothing$$

*then one has*

$$\left| \int_\alpha^\beta \left( \frac{\sin u}{u} \right)^2 u^{s-1}\, du - \frac{(\beta^{s-2} - \alpha^{s-2})}{2(s-2)} \right| \leqslant \frac{\alpha^{\mathrm{Re}(s)-2}}{|2\alpha - |\mathrm{Im}(s)||} + \frac{\beta^{\mathrm{Re}(s)-2}}{|2\beta - |\mathrm{Im}(s)||} \tag{3}$$

*and*

$$\left| \int_\alpha^\beta \left( \frac{\sin u}{u} \right)^2 u^{s-1}\, du - \left[ \frac{u^{s-2} \sin^2 u}{s-2} - \frac{u^{s-1} \sin 2u}{(s-2)(s-1)} \right]_\alpha^\beta \right|$$
$$\leqslant \frac{4}{|(s-2)(s-1)|} \left\{ \frac{\alpha^{\mathrm{Re}(s)}}{|2\alpha - |\mathrm{Im}(s)||} + \frac{\beta^{\mathrm{Re}(s)}}{|2\beta - |\mathrm{Im}(s)||} \right\}. \tag{4}$$

*Proof*: Since $\sin^2 u = (1 - \cos 2u)/2$ we have that

$$\int_\alpha^\beta \left\{ \frac{(\sin u)}{u} \right\}^2 u^{s-1}\, du = (\beta^{s-2} - \alpha^{s-2})/2(s-2) - \int_\alpha^\beta (\cos 2u) u^{s-3}\, du/2. \tag{6.1}$$

On the other hand integrating by parts twice yields

$$\int_\alpha^\beta (\sin^2 u)u^{s-3}\,du = [(\sin^2 u)u^{s-2}/(s-2)]_\alpha^\beta - \int_\alpha^\beta (\sin 2u)u^{s-2}du/(s-2)$$
$$= [(\sin^2 u)u^{s-2}/(s-2) - (\sin 2u)u^{s-1}/(s-1)(s-2)]_\alpha^\beta$$
$$+ 2\int_\alpha^\beta (\cos 2u)u^{s-1}\,du/(s-1)(s-2). \tag{6.2}$$

Observe that $|\cos 2u| \leqslant 1$ and that for any $t \in \mathbf{C}$ the function $u \mapsto u^{\mathrm{Re}(t)}$ is either monotonic increasing or monotonic decreasing so that

$$\left|\int_\alpha^\beta (\cos 2u)u^t\,du\right| \leqslant (\beta-\alpha)\mathrm{Sup}(u^{\mathrm{Re}(t)}) < (\beta-\alpha)(\alpha^{\mathrm{Re}(t)} + \beta^{\mathrm{Re}(t)})$$

From this inequality and (6.1) we obtain (1); from it and (6.2) we obtain (2).

We shall next prove that if

$$[a,b] \cap \left\{|\mathrm{Im}(s)|, \left|\frac{\mathrm{Re}(s)}{\mathrm{Re}(s)-1}\right||\mathrm{Im}(s)|\right\} = \varnothing$$

then

$$\left|\int_a^b (\cos u)u^{s-1}\,du\right| \leqslant \left\{\frac{a^{\mathrm{Re}(s)}}{||a|-|\mathrm{Im}(s)||} + \frac{b^{\mathrm{Re}(s)}}{||b|-|\mathrm{Im}(s)||}\right\}.$$

From this and (6.1) we obtain (3), and from it and (6.2) we obtain (4).

To prove this inequality let $\sigma = \mathrm{Re}(s)$, $t = \mathrm{Im}(s)$; we shall prove

$$\left|\int_a^b e^{iu}u^{s-1}\,d\mu\right| \leqslant (a^\sigma/|a+t| + b^\sigma/|b+t|).$$

This will suffice for if we replace $s$ by $\bar{s}$ and conjugate we obtain

$$\left|\int_a^b e^{-iu}u^{s-1}\,du\right| \leqslant (a^\sigma/|a-t| + b^\sigma/|b-t|).$$

The inequality follows on adding these two and recalling that $|a \pm t| \geqslant ||a|-|t||$ etc.

To prove the inequality we rewrite the integral under consideration as

$$\int_a^b e^{i(u+t\log u)}u^{\sigma-1}\,du$$

and apply Theorem A3.1; the inequality is an immediate consequence of this. This proves the inequality and with it the proposition.

## 6.4

In this section we shall prove Theorem 6.1 under the additional assumption that $Y \geqslant X$. This suffices, since if we can do this then by the functional

equation for the zeta-function if $X > Y$

$$\zeta(s) = \gamma(1-s)\zeta(1-s)$$
$$= \gamma(1-s)\left\{\sum_{m\leqslant Y} m^{-(1-s)} + \gamma(s)\sum_{n\leqslant X} n^{-s} + O(Y^{-\mathrm{Re}(1-s)}\log XY)\right\}$$

by the special form of the theorem which we are assuming By Stirling's Formula

$$\gamma(1-s) = O(|\mathrm{Im}(s)|^{1/2-\mathrm{Re}(s)})$$

and we know from §2.6 that

$$\gamma(s)\gamma(1-s) = 1.$$

Together these give

$$\zeta(s) = \sum_{n\leqslant X} n^{-s} + \gamma(1-s)\sum_{m\leqslant Y} m^{-(1-s)} + O(X^{1/2-\mathrm{Re}(s)}Y^{-1/2}\log XY)$$

as

$$|\mathrm{Im}(s)| = 2\pi XY.$$

This is precisely the statement of the theorem in this case.

The proof of the theorem in the special case is based here on Theorem 2.13 with the special choice

$$f(x) = \mathrm{Max}\,(1-|x|, 0).$$

Theorem 2.13 asserts that for $\mathrm{Re}(s) > 0$ and $X, Y > 1$ we have

$$\zeta(s) = \sum_{n\leqslant X} n^{-s} + \gamma(1-s)\sum_{m\leqslant Y} m^{-(1-s)} - 2R(s)$$

where

$$R(s) = X^{-s}/2sM(f,s) + X^{1-s}/2(1-s)M(f,s)$$
$$+ \sum_{n\leqslant X} n^{-s}\int_0^{n/X} f(x)x^{s-1}\,dx/M(f,s)$$
$$+ \sum_{m\leqslant Y} m^{-(1-s)}\int_0^{mX} \hat{f}(x)x^{-s}\,dx/M(f,s)$$
$$- \sum_{m>Y} m^{-(1-s)}\int_{mX}^{\infty} \hat{f}(x)x^{-s}\,dx/M(f,s),$$

where we have noted that $f(0) = 1$, $\hat{f}(0) = 1$ and we have used the 'local functional equation' $M(\hat{f}, 1-s) = \gamma(1-s)M(f,s)$. In this case, as in Exercise 2.4,

$$\hat{f}(x) = \{(\sin \pi x)/\pi x\}^2$$
$$M(f,s) = 2/s(s+1).$$

We shall first show that the sum of the first three terms is equal to

$$(X^{-s-1}/2)\{-(s/2)(X-[X])^2+(s/2)(X-[X])$$
$$-(X-[X]-1/2)X+X^2/(1-s)\}.$$

Since $X-[X]=O(1)$ the third term would then be $O(X^{-\mathrm{Re}(s)})$; as $|s|>X$ the same estimate would hold for the final term, and so this formula gives the estimate

$$(s/4)X^{-s-1}\{(X-[X])-(X-[X])^2\}+O(X^{-\mathrm{Re}(s)}).$$

It will turn out that the fourth term yields the same expression with reversed sign and with more or less the same error term.

To prove this formula we note that from the definition of $f$ one has

$$n^{-s}\int_0^{n/X} f(x)x^{s-1}\,dx = s^{-1}X^{-s}-(s+1)^{-1}X^{-(s+1)}n.$$

This leads at once to

$$\sum_{n\leqslant X} n^{-s}\int_0^{n/X} f(x)x^{s-1}\,dx = s^{-1}[X]X^{-s}-(s+1)^{-1}X^{-s-1}[X]([X]+1)/2.$$

A little manipulation now gives the asserted formula.

We consider next the fourth term. For this we shall use Proposition 6.3. From Proposition 6.3 (4) we obtain for $m$ with $mX\leqslant XY-X$ the estimate

$$m^{-(1-s)}\int_0^{mX} \hat{f}(x)x^{-s}dx/M(f,s) = -(1/2)sm^{-2}\pi^{-2}X^{-1-s}\sin^2\pi mX$$
$$-(2\pi m)^{-1}X^{-s}\sin 2\pi mX$$
$$+O(X^{1-\mathrm{Re}(s)}/\{|\mathrm{Im}(s)|-2\pi mX\}).$$

In deriving this one either uses the proposition directly, if $(\mathrm{Re}(s)/\{1-\mathrm{Re}(s)\})\pi|\mathrm{Im}(s)|$ does not lie in the interval of integration, or if this point does lie in the interval then one applies Proposition 6.3 twice, once to the interval $[0,\pi\,\mathrm{Re}(s)\{1-\mathrm{Re}(s)\}^{-1}|\mathrm{Im}(s)|]$ and once to $[\mathrm{Re}(s)\{1-\mathrm{Re}(s)\}^{-1}\pi|\mathrm{Im}(s)|,mX]$. In the latter interval we have $mX\geqslant XY-X$. But as $Y\geqslant X$ we can assume without much loss of generality that $Y\geqslant 2$ so that $2\pi mX\geqslant|\mathrm{Im}(s)|/2$ and the additional error terms which arise from the double application of Proposition 6.3 are dominated by the one given above. If $m$ satisfies $mX>XY-X$ then we use Proposition 6.3 (4) for the interval $[0,XY-X]$ and Proposition 6.3 (2). These yield for the error

$$O(X^{1-\mathrm{Re}(s)}/X+X(XY)^{-\mathrm{Re}(s)}Y^{-\{1-\mathrm{Re}(s)\}})$$

and as $Y\geqslant X$ this is $O(X^{-\mathrm{Re}(s)})$. We can now sum and find for the fourth term in $R$ the expression

$$-(1/2)s\pi^{-2}X^{-1-s}\sum_{m\leqslant Y} m^{-2}\sin^2\pi mX - \pi^{-1}X^{-s}\sum_{m\leqslant Y}(\sin 2\pi mX)/2m$$
$$+O\left(X^{-\mathrm{Re}(s)}\sum_{m\leqslant Y-1}(Y-m)^{-1}+X^{-\mathrm{Re}(s)}\right).$$

The second term here can be estimated by comparison with a harmonic series and is $O(X^{-\mathrm{Re}(s)}\log Y)$. The third term is likewise $O((1+\log Y)X^{-\mathrm{Re}(s)})$. The first term is part of a convergent series. It can be replaced by

$$-(1/2)s\pi^{-2}X^{-1-s}\sum_{m\geqslant 1} m^{-2}\sin^2\pi mX + O(|s|X^{-1-\mathrm{Re}(s)}Y^{-1}).$$

The error term here is $O(X^{-\mathrm{Re}(s)})$ as $|s| = O(XY)$. The infinite sum can be evaluated by applying the Poisson Summation Formula to $x \to f(x/X)$ We find that this sum is equal to

$$-sX^{1-s}\left\{(2X)^{-1}-1/2+\sum_{1\leqslant n\leqslant X}(1-n/X)X^{-1}\right\}.$$

This can be evaluated and is equal to

$$-sX^{-s-1}\{(X-[X])-(X-[X])^2\}/4.$$

Thus the first four terms in the expression defining $R$ can be combined to give

$$O((1+\log Y)X^{-\mathrm{Re}(s)}).$$

The first term here is now $O(X^{-\mathrm{Re}(s)-1})$ so that we have now

$$O(X^{-\mathrm{Re}(s)}(1+\log Y))$$

for the first four terms.

It now only remains to estimate the fifth term. This will be done using Proposition 6.3 (1) and (3). It is rather similar to the treatment of the fourth term. We have by Proposition 6.3 (3) that if $mX > XY + X$ then

$$m^{-(1-s)}\int_{mX}^{\infty} \hat{f}(x)x^{-s}\,dx/M(f,s)$$
$$= s(m\pi)^{-2}4^{-1}X^{-s-1} + O(|s|^2m^{-2}X^{-\mathrm{Re}(s)-1}/\{2\pi mX - |\mathrm{Im}(s)|\}.$$

If $m > Y$ is such that $XY < mX \leqslant XY + X$ then the error term has to be replaced by

$$O(|s|^2Y^{-2}X^{-\mathrm{Re}(s)-2} + |s|^2X(XY)^{-\mathrm{Re}(s)-2}Y^{-(1-\mathrm{Re}(s))}).$$

We sum over $m$ and need only use $\sum_{m>Y} m^{-2} = O(Y^{-1})$ and

$$\sum_{m\geqslant Y+1} m^{-2}(m-Y)^{-1} = \sum_{Y+1\leqslant m\leqslant 2Y} m^{-2}(m-Y)^{-1} + \sum_{m>2Y} m^{-2}(m-Y)^{-1}$$
$$= O(Y^{-2}\log Y) + O(Y^{-2})$$
$$= O((1+\log Y)Y^{-2}).$$

This gives for the fifth term

$$O(|s|Y^{-1}X^{-\mathrm{Re}(s)-1} + |s|^2X^{-\mathrm{Re}(s)-2}(1+\log Y)Y^{-2} + |s|^2Y^{-3}X^{-\mathrm{Re}(s)-1})$$

If we use $s = O(XY)$ this simplifies to

$$O(X^{-\mathrm{Re}(s)} + X^{-\mathrm{Re}(s)}(1+\log Y) + X^{-\mathrm{Re}(s)}XY^{-1}).$$

Since $Y \geqslant X$ this reduces to the error term which was claimed. This completes the proof of Theorem 6.1.

## 6.5

In §§4.12 and 4.13 we have seen that it is useful to have estimates for expressions of the form

$$T^{-1}\int_0^T |\zeta(\sigma+it)|^p\,dt$$

for the purpose of investigating the zeros of the zeta-function and in §5.3 we established a close connection between such estimates and the Lindelöf Hypothesis. The 'approximate functional equation' allows us to approach such questions, and we shall illustrate this by some results which are technically not too difficult but which make the general principles clear. The further theory of such estimates is discussed in detail in Titchmarsh (2) and for more recent results the reader is referred to Heath-Brown (2), Ivić (1).

The first theorem that we shall prove is the following:

**Theorem:** (i) *For* $\sigma \in ]1/2, 1[$ *we have as* $T \to \infty$

$$T^{-1}\int_1^T |\zeta(\sigma+it)|^2\,dt \sim \zeta(2\sigma).$$

(ii) *We have uniformly for* $\sigma$ *in* $[1/2, 3/4]$ *and* $T > 1$

$$T^{-1}\int_1^T |\zeta(\sigma+it)|^2\,dt = O(\mathrm{Min}\{\log T, T^{-1}(\sigma-1/2)^{-1}(\log T)^2\}).$$

*Proof*: (i) We apply here Corollary 6.2. It follows that

$$\zeta(\sigma+it) = \sum_{n \leqslant t/2\pi} n^{-\sigma-it} + O(t^{-\sigma}\log t).$$

It is convenient to set

$$f(t) = \sum_{n \leqslant t/2\pi} n^{-\sigma-it}.$$

We shall first show that $T^{-1}\int_0^T |f(t)|^2\,dt \sim \zeta(2\sigma)$. Indeed we have

$$\int_1^T |f(t)|^2\,dt = \int_1^T \sum_{m,n \leqslant t/2\pi} (mn)^{-\sigma}(n/m)^{it}\,dt$$

$$= \sum_{m,n \leqslant T/2\pi} \int_{2\pi\,\mathrm{Max}(m,n)}^T (n/m)^{it}\,dt(mn)^{-\sigma}.$$

In this we separate the terms with $m = n$ from the others. These terms give a contribution of

$$\sum_{n \leqslant T/2\pi} (T - 2\pi n) n^{-2\sigma}.$$

This is equal to

$$T \sum_{n \leqslant T/2\pi} n^{-2\sigma} - 2\pi \sum_{n \leqslant T/2\pi} n^{1-2\sigma} = T\zeta(2\sigma) + O(T^{2-2\sigma}) + O(T^{2-2\sigma})$$

In this case as $\sigma > 1/2$ we have $2 - 2\sigma < 1$ and this shows that these terms contribute the main term.

Now consider those terms with $m \neq n$. Again we can evaluate the integral as

$$(\log n/m)^{-1} \{(n/m)^{iT} - (n/m)^{2\pi i \operatorname{Max}(m\ n)}\} = O(|\log (n/m)|^{-1}).$$

We now have to estimate

$$\sum_{\substack{m,n \leqslant T/2\pi \\ m \neq n}} (mn)^{-\sigma} |\log (m/n)|^{-1}$$

which we shall show to be $O(T^{2-2\sigma} \log T)$. It clearly suffices to do this for the subsum taken over $m > n$. This we divide into two further sums according to whether $m > 2n$ or $n < m \leqslant 2n$. If $m > 2n$ then $\log (m/n) > \log 2$ and this subsum is bounded by

$$(2 \log 2)^{-1} \sum_{m,n \leqslant T/2\pi} (mn)^{-\sigma} = (2 \log 2)^{-1} \left( \sum_{n \leqslant T/2\pi} n^{-\sigma} \right)^2 = O((T^{1-\sigma})^2)$$

$$= O(T^{2-2\sigma} \log T).$$

To analyse the second subsum we let $j = m - n$ and take the sum over $j: 0 < j \leqslant n$ and $n \leqslant T/2\pi$. Note that $(mn)^{-\sigma} = O(n^{-2\sigma})$. Since

$$\log (m/n) = \log (1 + j/n) \geqslant j/2n$$

this subsum is dominated by

$$\sum_{n \leqslant T/2\pi} \sum_{j \leqslant n} n^{1-2\sigma} j^{-1}.$$

The sum over $j$ is $O(\log T)$ and hence the complete sum is $O(T^{2-2\sigma} \log T)$.

Thus we have now proved that

$$T^{-1} \int_0^T |f(t)|^2 \, dt = \zeta(2\sigma) + O(T^{1-2\sigma} \log T).$$

We have also, from the application of Corollary 6.2 given above, that

$$T^{-1} \int_0^T |\zeta(\sigma + it) - f(t)|^2 \, dt = T^{-1} \int_0^T O(t^{-\sigma} \log t)^2 \, dt = O(1).$$

By the triangle inequality we have that

$$T^{-1}\int_0^T |\zeta(\sigma+it)|^2\,dt = \zeta(2\sigma) + O(T^{1-2\sigma}\log T)$$

This completes the proof of (i).

(ii) To prove (ii) we need a few modifications. In the estimation of

$$T^{-1}\int_0^T |f(t)|^2\,dt$$

the terms with '$m=n$' yielded $\sum_{n\leqslant 2\pi T}(T-n/2\pi)n^{-2\sigma}$. On the one hand this is clearly less than

$$T\sum_{n\leqslant 2\pi T} n^{-1} = O(T\log T).$$

It is also less than $T\zeta(2\sigma) = O(T(\sigma-1/2)^{-1})$. Together these yield

$$T^{-1}\int_0^T |f(t)|^2\,dt = O(\mathrm{Min}\{\log T, (\sigma-1/2)^{-1}\}) + O(T^{1-2\sigma}\log T)$$

since the additional terms, those with '$m\neq n$', can be bounded as before uniformly in $\sigma$ in $[1/2, 3/4]$. The second $O$-term here is dominated by the first and can therefore be dropped.

Finally we have the estimate

$$\begin{aligned} T^{-1}\int_1^T |\zeta(\sigma+it)-f(t)|^2\,dt &= T^{-1}\int_1^T O(t^{-2\sigma}\log t)^2\,dt \\ &= T^{-1}O\left((\log T)^2\int_1^T t^{-2\sigma}\,dt\right) \\ &= T^{-1}O((\log T)^2\,\mathrm{Min}\{\log T, (\sigma-1/2)^{-1}\}). \end{aligned}$$

As before the triangle inequality and these two estimates yield the quoted estimate. This completes the proof of the theorem.

## 6.6

We turn now to a much more difficult question, that of estimating $\zeta(1/2+it)$ for large $t$. We have seen that this is for any $\varepsilon>0$ bounded by $O(|t|^{1/4+\varepsilon})$. We shall prove the following theorem which improves on this estimate.

**Theorem:** *As $t\to\infty$ one has*

$$\zeta(1/2+it) = O(|t|^{1/6}(\log|t|)).$$

Although this is not the best-known estimate of this type, any improvements require technically much more complicated proofs. At the present the best-known result obtained by classical methods (those to be

described in the following paragraphs) is that due to Kolesnik which is that for all $\varepsilon > 0$

$$\zeta(1/2 + it) = O(|t|^{35/216+\varepsilon})$$

as $|t| \to \infty$ (see Kolesnik (1)). Note that $35/216 = 1/6 - 1/216 = 0.1620\ldots$ A new method in which new estimates of cubic exponential sums has been applied by Bombieri and Iwaniec (Bombieri & Iwaniec (1)) to yield the estimate for all $\varepsilon > 0$

$$\zeta(1/2 + it) = O(|t|^{9/56+\varepsilon})$$

as $|t| \to \infty$. In this case $9/56 = 1/6 - 1/28 = 0.1607\ldots$

The proofs of these theorems are long and complicated; even the proof of the theorem announced here is not all that easy and will be given over the next five sections.

## 6.7

The first step in the proof of Theorem 6.6 is the proof of a finite, approximate form of the Poisson Summation Formula due to van der Corput. This we shall give only for a very special type of function, but one that is quite sufficient for our purposes.

**Theorem:** *Let $f$ be a real differentiable function defined on $[a, b]$, and suppose that $f'$ is monotonic decreasing. Let $\eta \in ]0, 1]$; then*

$$\sum_{a<n\leqslant b} e^{2\pi i f(n)} = \sum_{f(b)-\eta<m\leqslant f'(a)+\eta} \int_a^b e^{2\pi i\{f(x)-mx\}}\,dx + O(\log\{2 + f'(a) - f'(b)\}).$$

*Proof.* We can replace $f$ by $x \mapsto f(x) - kx$ for any integer $k$ and obtain an equivalent statement; thus we can choose $k$ so that $\eta - 1 < f'(b) - k \leqslant \eta$ and prove the theorem for $f_1(x) = f(x) - kx$. Hence we may assume without loss of generality that $\eta - 1 < f'(b) \leqslant \eta$. Now let $Q$: $Q(x) = x - [x] - 1/2$. Then as Stiltjes' integral we have

$$\int_a^b e^{2\pi i f(x)}\,dQ(x) = \int_a^b e^{2\pi i f(x)}\,dx - \sum_{a<n\leqslant b} e^{2\pi i f(n)}.$$

On the other hand by integration by parts we have

$$\int_a^b e^{2\pi i f(x)}\,dQ(x) = [e^{2\pi i f} Q]_a^b - 2\pi i \int_a^b f'(x) Q(x) e^{2\pi i f(x)}\,dx.$$

Thus we have

$$\sum_{a<n\leqslant b} e^{2\pi i f(n)} = \int_a^b e^{2\pi i f(x)}\,dx - 2\pi i \int_a^b f'(x) Q(x) e^{2\pi i f(x)}\,dx - [e^{2\pi i f} Q]_a^b.$$

Note now that $Q \in L^2(\mathbf{R}/\mathbf{Z})$ and can therefore be expanded in a Fourier series. The Fourier coefficients are easy to compute and we see that

$$Q(x) = -\pi^{-1} \sum_{l \geqslant 1} l^{-1} \sin 2\pi l x,$$

the series being convergent in $L^2(\mathbf{R}/\mathbf{Z})$, and hence also in $L^2(I)$ for any compact $I \subset \mathbf{R}$ with respect to Lebesgue measure. As $f'$ is decreasing it is measurable and so, being bounded, integrable; thus $f' e^{2\pi i f} \in L^2([a,b])$ Thus we can replace $Q$ in the integral above by its Fourier series and interchange integration and summation. We now have

$$\sum_{a<n\leqslant b} e^{2\pi i f(n)} = \int_a^b e^{2\pi i f(x)} dx + 2i \sum_{l \geqslant 1} l^{-1} \int_a^b (\sin 2\pi l x) f'(x) e^{2\pi i f(x)} dx - [e^{2\pi i f} Q]_a^b$$

$$= \int_a^b e^{2\pi i f(x)} dx + \sum_{l \neq 0} l^{-1} \int_0^b e^{2\pi i \{f(x)+lx\}} f'(x) dx - [e^{2\pi i f} Q]_a^b.$$

We shall now estimate this by the technique used in §6.3; we have the general estimate (A3.1) for two real-valued functions $F, G$ on $[a,b]$ with

(i) Inf $F' > 0$, and
(ii) $G/F'$ monotonic

then we have

$$\left| \int_a^b e^{iF(x)} G(x) dx \right| \leqslant \text{Min}\{2, F(b) - F(a)\} |G/F'(b) - G/F'(a)|$$

We shall apply this with

$$F(x) = 2\pi\{f(x) + lx\} \quad \text{or} \quad F(x) = -2\pi\{f(x) + lx\}$$

and

$$G(x) = f'(x).$$

In the first case condition (i) is satisfied if $l > -f'(b)$ and in particular if $l \geqslant \eta - f'(b)$. We have that

$$G/F'(x) = (2\pi\{1 + l/f'(x)\})^{-1}$$

which is monotonic as $f'$ is. Thus we obtain the estimate for the $l$th summand for $l \geqslant \eta - f'(b)$

$$l^{-1} \text{Min}\{f(b) - f(a) + l(b-a), 2\}(2\pi)^{-1}(\{1 + l/f'(b)\}^{-1} - \{1 + l/f'(a)\}^{-1}).$$

This is then bounded by

$$l^{-1}\{1 + l/f'(b)\}^{-1} = l^{-1} - \{l + f'(b)\}^{-1}.$$

As $l \geqslant 1$ and $-1 + \eta < f'(b) \leqslant \eta$ the Mean Value Theorem shows that this last expression is bounded by $\eta/(l - 1 + \eta)^2$ and so the sum over all $l$ with $l > \eta - f'(b)$ yields $O(1)$.

In the second case condition (i) is satisfied when $-l-f'(a)>0$; that is when $l<-f(a)$. In particular we shall apply this when $l<-f'(a)-\eta$. As before the $l$th summand is bounded by

$$\{l+f'(a)\}^{-1}-l^{-1}.$$

Let $N=[-f'(a)]$; then this is $\{l+f'(a)\}^{-1}-(l-N)^{-1}+(l-N)^{-1}-l^{-1}$. A similar argument to that used above using the Mean Value Theorem shows that

$$\sum_{l<-f'(b)-\eta}\{l+f'(b)\}^{-1}-(l-N)^{-1}=O(1).$$

Let $m=-l+N$; then the remaining terms in the sum are equal to

$$-\sum_{m>0}m^{-1}-(m+N)^{-1}+O(1).$$

This is equal to

$$-\lim_{R\to\infty}\sum_{m\leqslant R}m^{-1}-\sum_{m\leqslant R+N}m^{-1}+\sum_{m\leqslant N}m^{-1}+O(1)=\sum_{m\leqslant N}m^{-1}+O(1)$$
$$=O(\log(N+1)).$$

This is the error term quoted in the statement of the theorem. We have thereby estimated all those terms which are to be estimated, and have now proved Theorem 6.7.

## 6.8

In this section we prove an estimate for a certain type of exponential sum of a type first used by Weyl. This estimate is based on the same ideas that are used in studying quadratic Gauss sums.

**Lemma:** *Let $f$ be a real-valued function on $]a,b]\cap\mathbf{Z}$. Let $q\in\mathbf{N}$, $q\leqslant b-a$. Then one has*

$$\left|\sum_{a<n\leqslant b}e^{2\pi if(n)}\right|\leqslant 2(b-a)q^{-1/2}$$
$$+2\left\{q^{-1}(b-a)\sum_{1\leqslant r\leqslant q-1}\left|\sum_{a\leqslant m\leqslant b-r}e^{2\pi i\{f(n+r)-f(n)\}}\right|\right\}^{1/2}.$$

*Proof*: It will be convenient to extend $f$ to $\mathbf{Z}$ by defining it to be 0 on $\mathbf{Z}-]a,b]$. We shall also define $g$ on $\mathbf{Z}$ by $g(m)=1$ if $m\in]a\ b]\cap\mathbf{Z}$ and $g(m)=0$ if $m\in\mathbf{Z}-]a,b]$. For $q$ as above we have

$$\sum_{a<n\leqslant b}e^{2\pi if(n)}=q^{-1}\sum_{n\in\mathbf{Z}}\sum_{1\leqslant m\leqslant q}g(n+m)e^{2\pi if(m+n)}$$

and hence

$$\left|\sum_{a<n\leqslant b} e^{2\pi i f(n)}\right| \leqslant q^{-1}\sum_{n\in\mathbf{Z}}\left|\sum_{1\leqslant m\leqslant q} g(m+n)e^{2\pi i f(m+n)}\right|.$$

The $n$th summand on the right-hand side is non-zero only when $a-q<n\leqslant b-1$. We can restrict the summation to this interval and apply the Cauchy–Schwarz inequality. As there are at most $b-a+q$ integers in this interval we find

$$\left|\sum_{a<n\leqslant b} e^{2\pi i f(n)}\right| \leqslant q^{-1}(b-a+q)^{1/2}\left\{\sum_{n\in\mathbf{Z}}\left|\sum_{1\leqslant m\leqslant q} g(m+n)e^{2\pi i f(m+n)}\right|^2\right\}^{1/2}.$$

Since $|a|^2 = a\bar{a}$ the final term here is equal to the square root of

$$\sum_{n\in\mathbf{Z}}\sum_{1\leqslant m_1,m_2\leqslant q} g(m_1+n)g(m_2+n)e^{2\pi i\{f(m_1+n)-f(m_2+n)\}}.$$

We separate out first those terms with $m_1=m_2$; they give a total contribution of

$$q\sum_{n\in\mathbf{Z}} g(n) \leqslant q(b-a+q).$$

The remaining terms are equal to

$$\sum_{n\in\mathbf{Z}}\sum_{1\leqslant m_1<m_2\leqslant q} g(m_1+n)g(n+m_2)e^{2\pi i\{f(m_1+n)-f(m_2+n)\}}$$
$$+\sum_{n\in\mathbf{Z}}\sum_{1\leqslant m_2<m_1\leqslant q} q(m_2+n)g(m_1+n)e^{-2\pi i\{f(m_2+n)-f(m_1+n)\}}.$$

The two summands are complex conjugate to one another and it will suffice to study the first. We replace $n$ by $n-m_1$ and let $r=m_2-m_1$. With these substitutions it becomes

$$\sum_{n\in\mathbf{Z}}\sum_{1\leqslant r\leqslant q-1}(q-r)g(n+r)g(n)e^{2\pi i\{f(n+r)-f(n)\}}$$

since $r$ can be represented $q-r$ times as $m_2-m_1$. The sum over $n$ can be restricted to $n$ satisfying $a<n\leqslant b-r$ since the other summands are zero. Interchanging the order of summation shows that this is bounded by

$$\sum_{1\leqslant r\leqslant q-1}(q-r)\left|\sum_{a<n\leqslant b-r} e^{2\pi i\{f(n+r)-f(n)\}}\right|$$

which we further estimate by

$$q\sum_{1\leqslant r\leqslant q-1}\left|\sum_{a<n\leqslant b-r} e^{2\pi i\{f(n+r)-f(n)\}}\right|.$$

Using this estimate we see that we have now shown that

$$\left|\sum_{a<n\leqslant b} e^{2\pi i f(n)}\right| \leqslant q^{-1}(b-a+q)^{-1/2}\left\{q(b-a+q) + 2q\sum_{1\leqslant r\leqslant q-1}\left|\sum_{a<n\leqslant b-r} e^{2\pi i\{f(n+r)-f(n)\}}\right|\right\}^{1/2}.$$

We now use the simple inequality $(X+Y)^{1/2} \leqslant X^{1/2}+Y^{1/2}$ from which we deduce that the right-hand side is less than or equal to

$$(b-a+q)q^{-1/2} + \left\{2(b-a+q)q^{-1}\sum_{1\leqslant r\leqslant q-1}\left|\sum_{a<n\leqslant b-r} e^{2\pi i\{f(n+r)-f(n)\}}\right|\right\}^{1/2}.$$

The quoted result follows since $q \leqslant b-a$.

The importance of this lemma is that it reduces the study of exponential sums of the type under consideration from an '$f$' to a 'difference' $x \mapsto f(x+r)-f(x)$. In general this represents no improvement but for large classes of functions, such as polynomials, it allows a real saving.

## 6.9

In this section and the next we shall prove two very general estimates for exponential sums which are due to van der Corput.

**Proposition:** *Suppose* $f \in C^2([a,b], \mathbf{R})$ *where* $b-a \geqslant 1$. *Then if*

$$\operatorname{Inf} f'' > 0$$

*we have*

$$\sum_{a<n\leqslant b} e^{2\pi i f(n)} = O(\{(b-a)\operatorname{Sup} f'' + 1\}(\operatorname{Inf} f'')^{-1/2}).$$

*Proof*: If $\operatorname{Inf} f'' \geqslant 1$ then $\operatorname{Sup} f''/(\operatorname{Inf} f'')^{1/2} \geqslant 1$ and the assertion is trivially true. Thus we have to prove the estimate under the further restriction that $\operatorname{Inf} f'' < 1$.

We shall apply Theorem 6.7. This shows that the sum is equal to

$$\sum_{f'(a)-\eta<m<f'(b)+\eta}\int_a^b e^{2\pi i\{f(x)-mx\}}dx + O(\log\{2+f'(b)-f'(a)\}).$$

We shall estimate the integrals. Let

$$F(x) = f(x) - mx.$$

Then

$$F''(x) = f''(x) \geqslant \operatorname{Inf} f'' > 0.$$

Hence $F'$ is increasing and can therefore have at most one zero. Indeed

$m$ has been so chosen that $F'(a) < \eta$, $F'(b) > -\eta$. Suppose that there is a zero in the interval; call it $c$. Then we can split the integral into three parts, the integrals over $[a, c-\delta]$, $[c-\delta, c+\delta]$, and $[c+\delta, b]$ where some of these could be empty. We take the trivial estimate for the interval $[c-\delta, c+\delta]$; this is $O(\delta)$. In the other two intervals one has that

$$|F'(x)| = |F'(x) - F'(c)| > \delta \operatorname{Inf} f''$$

by the Mean Value Theorem. Thus we can apply A3.1 to these. We obtain then as an estimate for the integral

$$O(\delta) + O((\delta \operatorname{Inf} f'')^{-1});$$

we take

$$\delta = (\operatorname{Inf} f'')^{-1/2}$$

which yields the estimate

$$O((\operatorname{Inf} f'')^{-1/2}).$$

If there is no zero of $F'$ in $[a, b]$ then we have to modify the argument slightly in that we remove intervals around $a$ and $b$ of length $\delta$ and proceed as before.

We note also that

$$f'(b) - f'(a) \leqslant (b-a) \operatorname{Sup} f''.$$

Thus in the sum above there are $O(1 + (b-a) \operatorname{Sup} f'')$ terms. This leads directly to the estimate

$$O(\{1 + (b-a) \operatorname{Sup} f''\} (\operatorname{Inf} f'')^{-1/2}) + O(\log \{2 + (b-a) \operatorname{Sup} f''\}).$$

As $\operatorname{Inf} f'' < 1$ the first term dominates the second and this gives us the estimate we need.

## 6.10

In the second of these two estimates we make an alternative assumption on $f$ and shall use the previous estimate in conjunction with Lemma 6.8 to prove it. Note that this estimate does not in general supersede Proposition 6.9.

**Proposition:** *Suppose that* $f \in C^3([a, b], \mathbf{R})$ *where* $b - a \geqslant 1$. *Suppose that*

$$\operatorname{Inf} f^{(3)} > 0.$$

*Then we have*

$$\sum_{a < n \leqslant b} e^{2\pi i f(n)} = O((\operatorname{Sup} f^{(3)})^{1/2} (\operatorname{Inf} f^{(3)})^{-1/3} (b-a) + (\operatorname{Inf} f^{(3)})^{-1/6} (b-a)^{1/2}).$$

*Proof*: By Lemma 6.8 the sum on the left-hand side is bounded by

$$2(b-a)q^{-1/2}+2\left\{q^{-1}(b-a)\sum_{1\leqslant r\leqslant q-1}\left|\sum_{a<m\leqslant b-r}e^{2\pi i\Delta_r f(n)}\right|\right\}^{1/2}$$

where

$$\Delta_r f(x)=f(x+r)-f(x).$$

By the Mean Value Theorem

$$(\Delta_r f)''(x)\geqslant r\,\mathrm{Inf}\,f^{(3)}>0$$

and

$$(\Delta_r f)''(x)\leqslant r\,\mathrm{Sup}\,f^{(3)}.$$

In the expression above, $q$ is an integral parameter to be chosen later. Under our assumptions $b-a-r\geqslant b-a-(q-1)\geqslant 1$ for $q\leqslant b-a$. Again the inequality to be proved is trivial if $\mathrm{Inf}\,f^{(3)}\geqslant 1$ and so we can assume that $\mathrm{Inf}\,f^{(3)}<1$. We apply Proposition 6.9 to the inner sum and obtain the estimate

$$\begin{aligned}&O((b-a)q^{-1/2})\\&\quad+O((q^{-1}(b-a)\sum_{1\leqslant r\leqslant q\quad 1}\{1+(b-a)\,\mathrm{Sup}\,f^{(3)}r\}(r\,\mathrm{Inf}\,f^{(3)})^{-1/2})^{1/2})\end{aligned}$$

This simplifies to

$$\begin{aligned}&O((b-a)q^{-1/2}+\{q^{-1/2}(b-a)(\mathrm{Inf}\,f^{(3)})^{-1/2}\\&\qquad+q^{1/2}(b-a)^2\,\mathrm{Sup}\,f^{(3)}(\mathrm{Inf}\,f^{(3)})^{-1/2}\}^{1/2})\\&=O((b-a)q^{-1/2}+(b-a)^{1/2}(\mathrm{Inf}\,f^{(3)})^{-1/4}q^{-1/4}\\&\qquad+(b-a)(\mathrm{Sup}\,f^{(3)})^{1/2}(\mathrm{Inf}\,f^{(3)})^{-1\cdot 4}q^{1/4}).\end{aligned}$$

We now let

$$q=(\mathrm{Inf}\,f^{(3)})^{-1/3}$$

which will be admissible if

$$(\mathrm{Inf}\,f^{(3)})^{-1/3}\leqslant b-a.$$

If this condition is not satisfied then

$$\mathrm{Inf}\,f^{(3)}\leqslant(b-a)^{-3}$$

and so

$$b-a=O((\mathrm{Inf}\,f^{(3)})^{-1\ 6}(b-a)^{1\ 2})$$

and as the left-hand side here is the trivial estimate for the sum in question it follows the estimate is trivially valid in this case.

With the choice of $q$ as above we obtain the estimate

$$\begin{aligned}&O((b-a)(\mathrm{Inf}\,f^{(3)})^{1\ 6}+(b-a)^{1\cdot 2}(\mathrm{Inf}\,f^{(3)})^{-1\ 6}\\&\qquad+(\mathrm{Sup}\,f^{(3)})^{1/2}(\mathrm{Inf}f^{(3)})^{-1\cdot 3}(b-a)).\end{aligned}$$

Since the third term dominates the first we obtain the quoted estimate.

## 6.11

We can now complete the proof of Theorem 6.6. We shall first estimate sums of the form

$$\sum_{a<n\leqslant b} n^{-it}$$

using Propositions 6.9 and 6.10. Thus we shall take

$$f(x) = (2\pi)^{-1} t \log x.$$

From Proposition 6.9 if $b - a \geqslant 1$ we obtain directly the estimate

$$\sum_{a<n\leqslant b} n^{-it} = O(|t|^{1/2}(b/a)^2 + |t|^{-1/2}b).$$

We shall further assume that $b < 2a$; then we have

$$\sum_{a<n\leqslant b} n^{-it} = O(|t|^{1/2} + a|t|^{-1/2}).$$

From Proposition 6.10 we have the estimate

$$\sum_{a<n\leqslant b} n^{-it} = O(|t|^{1/6}b^2a^{-3/2} + |t|^{-1/6}b),$$

and if we further assume that $b < 2a$ then

$$\sum_{a<n\leqslant b} n^{-it} = O(|t|^{1/6}a^{1/2} + |t|^{-1/6}a).$$

Let

$$A(b) = \sum_{a<n\leqslant b} n^{-it}$$

where $a$ is held fixed and $a \leqslant b \leqslant 2a$. We then have

$$\sum_{a<n\leqslant 2a} n^{-1/2-it} = \int_a^{2a} x^{-1/2}\,dA(x)$$

$$= [X^{-1/2}A(x)]_a^{2a} + \int_a^{2a} x^{-3/2}A(x)dx/2.$$

In view of the bounds found above for $A$ we have then

$$\begin{aligned}\sum_{a<n\leqslant 2a} n^{-1/2-it} &= O(|t|^{1/2}a^{-1/2} + |t|^{-1/2}a^{1/2}) \\ &= O(|t|^{1/6} + |t|^{-1/6}a^{1/2}) \\ &= O(a^{1/2}).\end{aligned}$$

The first of these follows ultimately from Proposition 6.9, the second from Proposition 6.10 and the last is the trivial bound.

Suppose now that $|t|^{2/3} < X \leqslant |t|$. We then have

$$\left|\sum_{n\leqslant X} n^{-1/2-it}\right| \leqslant \sum_{j\leqslant \log X/\log 2} \left|\sum_{2^{-j-1}X<n\leqslant 2^{-j}X} n^{-1/2-it}\right|.$$

We split the outer sum on the right-hand side into three parts as

$$O \leqslant j < (1/3)\log X/\log 2,$$
$$(1/3)\log X/\log 2 \leqslant j < (2/3)\log X/\log 2$$

and

$$(2/3)\log X/\log 2 \leqslant j \leqslant \log X/\log 2.$$

For the first group we shall use the first estimate above, for the second the second estimate and for the third group of terms the third, trivial, estimate. Together we find since in each summation there are $O(\log X)$ terms the estimate

$$O(|t|^{1/6} + |t|^{-1/2}X^{1/2}) + O(|t|^{1/6}\log|t| + |t|^{-1/6}|t|^{1/3})$$
$$+ O(|t|^{1/6}\log|t|) = O(|t|^{1/6}\log|t|).$$

In this we take $X = |t|$ and then Theorem 6.6 follows from this estimate and Corollary 6.2.

## Exercises

6.1 By examining the integral $\int_{(w)}(\cot \pi z)z^{-s}\mathrm{d}z$ prove that if $X \geqslant 1$

$$\zeta(s) = \sum_{n \leqslant X} n^{-s} - (2\mathrm{i})^{-1}\int_{Y-\mathrm{i}\infty}^{X} (\cot \pi u - \mathrm{i})u^{-s}\mathrm{d}s$$
$$- (2\mathrm{i})^{-1}\int_{X}^{X+\mathrm{i}\infty} (\cot \pi u + \mathrm{i})u^{-s}\mathrm{d}s - X^{1-s}/(1-s)$$

and that this is valid for all $s$.

6.2 Use Exercise 6.1 to show that if $\sigma > 0$ and $C < 2\pi$ then for $X \geqslant 1$ uniformly in $\{s \in \mathbf{C}: \mathrm{Re}(s) > \sigma, |\mathrm{Im}(s)| \leqslant CX\}$ one has

$$\zeta(s) = \sum_{n \leqslant X} n^{-s} - X^{1-s}/(1-s) + O(X^{-\mathrm{Re}(s)}).$$

6.3 Suppose that $f$ is a differentiable function on $[a, b]$ and that $f'$ is increasing. Let $g$ be a positive differentiable function on $[a, b]$ so that $-g'$ is positive and decreasing. Show that if $\eta > 0$ then

$$\sum_{a < n \leqslant b} g(n)\mathrm{e}^{2\pi i f(n)} = \sum_{f'(a) - \eta \leqslant m \leqslant f'(b) + \eta} (g\mathrm{e}^{2\pi i f})\hat{}(m)$$
$$+ O(|g(a)|\log\{2 + f'(b) - f'(a)\}) + O(g'(a))$$

where the implied constants only depend on $\eta$. In this we consider $g\mathrm{e}^{2\pi i f}$ extended by 0 from $[a, b]$ to $\mathbf{R}$.

6.4 Can you deduce the Poisson Summation Formula from Exercise 6.3?

6.5 Prove the result of Exercise 6.2 using Exercise 6.3 and Exercise 2.8.

6.6 Use Theorem 6.7 to show that for $a \in \mathbf{R}$, $a > 0$

$$\sum_{0 \leqslant j < a^2} \mathrm{e}^{2\pi i j^2/a} = \int_0^{\infty} \mathrm{e}^{2\pi i x^2}\mathrm{d}x(1 + \mathrm{e}^{\pi i a/2})a^{1/2} + O(1).$$

6.7 Use the argument of Lemma 6.8 to show that if $a$ is an odd integer then

$$\left|\sum_{0\leqslant j<a} e^{2\pi i j^2/a}\right|^2 = a.$$

Deduce from this and the previous exercise that for all sufficiently large, positive, odd $a$

$$\begin{aligned}\sum_{0\leqslant j<a} e^{2\pi i j^2/a} &= a^{1/2} \quad \text{if } a \equiv 1 \pmod 4\\ &= ia^{1/2} \quad \text{if } a \equiv -1 \pmod 4.\end{aligned}$$

6.8 Let $(-)$ denote the Legendre–Jacobi symbol. Show that if

$$G(a) = \sum_{0\leqslant j<a} e^{2\pi i j^2/a}$$

and if $p, q$ are odd positive primes

$$G(pq) = \left(\frac{p}{q}\right)\left(\frac{q}{p}\right)G(p)G(q). \tag{1}$$

Deduce the law of quadratic reciprocity

$$\left(\frac{p}{q}\right)\left(\frac{p}{q}\right) = (-1)^{\{(p-1)/2\}\{(q-1)/2\}}. \tag{2}$$

Show also that (1) and (2) hold without the restriction that $p$ and $q$ should be primes (they must be assumed odd and positive), and that the result of Exercise 6.7 holds for all odd positive $a$.

6.9 Deduce the evaluation of $G(a)$ from Exercise 2.17. (Hint: Let $a(j) = e^{2\pi i j^2/a}$ and $f(x) = e^{-\pi x^2/(y - i/a)}$ and let $y \to 0$.)

6.10 Suppose that the factor $\log XY$ can be removed from the error term in Theorem 6.1. Use this theorem to prove that

$$\int_0^T |\zeta(1/2 + it)|^2 dt \sim T\log T.$$

6.11 Sharpen the result of Exercise 6.10 to

$$\int_0^T |\zeta(1/2 + it)|^2 dt = T\log T + (2\gamma - 1 - \log\pi)T + O(T^{1/2+\varepsilon})$$

for any $\varepsilon > 0$.

6.12 Under the same assumption as Exercise 6.10 prove that for $\sigma > 1/2$ as $T \to \infty$

$$\int_0^T |\zeta(\sigma + it)|^4 dt \sim \{\zeta(2\sigma)^4/\zeta(4\sigma)\}T.$$

6.13 Show that as $T \to \infty$

$$\int_0^T |\zeta(1/2 + it)|^4 dt = O(T(\log T)^4).$$

6.14 Show that if $k \in \mathbf{N}$ and $t \geqslant 2$ then

$$|\zeta(1/2 + it)|^k = O\left( (\log t)^2 \int_{-\log t}^{\log t} |\zeta(1/2 + i(t+u))|^k du \right).$$

(Start from the representation

$$\sum_{n \geqslant 1} d_k(n) n^{-s} e^{-n} = (2\pi i)^{-1} \int_{(1)} \zeta(s+u)^k \Gamma(u)\, du;$$

move the line of integration to $\{1/2 - \mathrm{Re}(s)\}$ where we demand that $\mathrm{Re}(s) = (\log t)^{-1}$ to get a bound for $|\zeta(s)|^k$ in terms of an integral over values of $\zeta$ on (1/2). Then use Cauchy's Theorem. This estimate is due to Heath-Brown.)

6.15 Use Exercises 6.11 and 6.14 to show that for any $\varepsilon > 0$

$$\zeta(1/2 + it) = O((1 + |t|)^{1/4 + \varepsilon})$$

as $t \to \infty$.

6.16 Let $m > 2$ and $M = 2^{m-1}$. Show that

$$\zeta(1 - M^{-1} + it) = O((1 + |t|)^{1/\{(m+1)M\}} \{\log(2 + |t|)\}^2)$$

as $|t| \to \infty$.

6.17 Let $f$ be a polynomial of degree $n$ with integral coefficients. Show that for large $q$

$$\left| \sum_{0 \leqslant j < q} e^{2\pi i f(j)/q} \right| \leqslant 4q^{1 - 2^{-(n-1)}}.$$

6.18 For $a \in \mathbf{N}$, $(a, q) = 1$ show that

$$\sum_{0 \leqslant j < q} e^{2\pi i a j^k / q} = O(q^{1 - k^{-1}})$$

as $q \to \infty$ for any fixed $k \in \mathbf{N}$.

# *Appendix 1*

# *Fourier theory*

## A1.1

In this appendix we shall develop sufficient Fourier theory for the applications which are made in this book The treatment which we shall give will be fairly brisk – for more background see Hardy & Rogosinski (1)

The first basic theorem which we shall need is the 'Riemann–Lebesgue Lemma', which we now give.

**Theorem:** *Let* $f \in L^1(\mathbf{R})$ *and define*

$$\hat{f}(y) = \int_{\mathbf{R}} f(x) e^{-2\pi i x y} \, dx.$$

*Then as* $y \to \infty$

$$\hat{f}(y) \to 0.$$

We shall not prove this in detail here – the simplest proof consists in noting that step functions form a dense subspace of $L^1(\mathbf{R})$ and that one can check the validity of the statement for these functions by direct calculation. It is easy to check that if the statement above holds on a dense subspace then it also holds on $L^1(\mathbf{R})$. The reader who does not want to work out the details can find a proof in Hardy & Rogosinski (1), Theorem 30; a further sketch is in Terras (1), p. 12.

## A1.2

A function $f$ defined on an interval $I \subset \mathbf{R}$ with values in $\mathbf{R}$ is said to be *of bounded variation on* $I$ if there exists $C > 0$ so that for $x_0 < x_1 < \cdots x_r$, $x_j \in I$ one has

$$\sum_{i=1}^{r} |f(x_i) - f(x_{i-1})| < C.$$

A function on $\mathbf{R}$ is said to be of *bounded total variation* if it is of bounded variation on $\mathbf{R}$ – the reason for this is that the concept of bounded variation is often construed as a local property, so that it is usual to say that a function on $\mathbf{R}$ is of bounded variation if it is of bounded variation on every compact subinterval of $\mathbf{R}$.

For an interval $I$ we define the total variation $V_a^b(f)$ for $a, b \in \mathbf{R}$ by

$$V_a^b(f) = \mathrm{Sup}\left\{ \sum_{i=1}^{r} |f(x_i) - f(x_{i-1})| \right\}$$

where the supremum is taken over all systems $x_0 < x_1 < \cdots < x_r$ $x_j \in I$ $a \leqslant x_0, x_r \leqslant b$.

It is easy to verify that $x \mapsto V_a^x(f)$ and $x \mapsto V_a^x(f) - f(x)$, where $a = \mathrm{Inf}\, I$, possibly $a = -\infty$, are increasing bounded functions and that $f$ is the difference of these. Conversely it is easy to verify that (i) functions of bounded variation on $I$ form a vector space and (ii) bounded increasing functions lie in this space.

Note also that if $f$ is continuous at a point $y$ then so is $x \mapsto V_a^x(f)$.

## A1.3

In this section we shall prove a general lemma concerning integrals of functions of bounded variation. We should recall here the simple fact (essentially first observed by Newton) that a function of bounded variation on a bounded interval is Riemann integrable. Recall also that the product of two Riemann integrable functions on a bounded interval is itself Riemann integrable.

**Lemma:** *Let $I$ be a bounded interval and let $f$ be a function of bounded variation on $I$. Let $\phi$ be a Riemann integrable function on $I$, and let $\Phi$ be any function satisfying*

$$\Phi' = \phi.$$

*Then*

$$\left| \int_I f(x)\phi(x)\mathrm{d}x - \{f(b)\Phi(b) - f(a)\Phi(a)\} \right| \leqslant \mathrm{Sup}\,|\Phi|\, V_a^b(f)$$

*where $a$ and $b$ are the end-points of $I$.*

*Proof*: It was noted above that it follows from the general theory of integration that $f\phi$ is Riemann integrable. We can therefore for $\varepsilon > 0$ find a subdivision $x_0 < x_1 < x_2 < \cdots < x_r$ of $I$ so that for any sequence $\xi_j$ with $x_{j-1} \leqslant \xi_j \leqslant x_j$ $(1 \leqslant j \leqslant r)$ we have

$$\left| \int_I f(x)\phi(x)\,\mathrm{d}x - \sum_{1 \leqslant j \leqslant r} f(\xi_j)\phi(\xi_j)(x_j - x_{j-1}) \right| < \varepsilon.$$

We choose $\xi_j$ so that $\phi(\xi_j)(x_j - x_{j-1}) = \Phi(x_j) - \Phi(x_{j-1})$ as is possible by the Mean Value Theorem. With this choice the sum is

$$\sum_{1 \leqslant j \leqslant r} f(\xi_j)\{\Phi(x_j) - \Phi(x_{j-1})\}$$

which is equal to

$$f(\xi_r)\Phi(x_r) - f(\xi_1)\Phi(x_0) + \sum_{1 \leqslant j \leqslant r} \Phi(x_j)\{f(\xi_j) - f(\xi_{j+1})\}.$$

The final sum here is bounded by

$$\operatorname{Sup}|\Phi| \sum_{1 \leqslant j \leqslant r} |f(\xi_j) - f(\xi_{j+1})|$$

and this is therefore bounded by

$$\operatorname{Sup}|\Phi| V_{\xi_1}^{\xi_r}(f).$$

From this the assertion of the lemma follows.

## A1.4

In this section we shall prove two results both in essence due to Dirichlet. In this we shall need the value of the improper integral

$$\lim_{A \to \infty} \int_{-A}^{A} \frac{\sin \pi x}{\pi x} \mathrm{d}x.$$

That this integral exists follows by Dirichlet's test (or by checking Cauchy's condition by integration by parts). It will turn out that it has the value 1; this is due to Euler. We shall derive this result in the course of our considerations. We shall denote the integral by $\gamma$. Then we have the following theorem.

**Theorem:** (i) *Let $f$ be of bounded total variation and integrable on* **R**. *Then*

$$\int f(x)\mathrm{e}^{2\pi i x y} \mathrm{d}x = O((1 + |y|)^{-1}).$$

(ii) *Let $f$ be a function on* **R** *such that $x \mapsto f(x)/(1 + |x|)$ is integrable and furthermore which is continuous at* 0 *and is of bounded variation on a neighbourhood of* 0. *Then*

$$\lim_{y \to \infty} \int f(x) \frac{\sin \pi x y}{\pi x} \mathrm{d}x \to \gamma f(0).$$

*Proof*: (i) This is a fairly direct consequence of Lemma A1.3. First of all it is trivial that

$$y \mapsto \int f(x)\mathrm{e}^{2\pi i x y} \mathrm{d}x$$

is bounded.

Next note that if $f$ is of bounded total variation then $\lim_{x \to \infty} f(x)$ and $\lim_{x \to -\infty} f(x)$ exist. If $f$ is also integrable then these limits are zero. Hence we can use the limiting case of Lemma A1.3 as $a \to -\infty$, $b \to \infty$. We take

$\Phi$: $\Phi(x) = e^{2\pi ixy}/(2\pi iy)$ and we see that

$$\left|\int f(x)e^{2\pi ixy}dx\right| \leqslant V_{-\infty}^{\infty}(f)/(2\pi|y|).$$

From this and the observation above the first assertion follows.

(ii) We shall first prove (ii) in the case that $f(0) = 0$. Let $\varepsilon > 0$. We shall show that there exists $\delta > 0$, depending on $f$, so that, for a certain absolute constant $c$, we have

$$\limsup_{y\to\infty}\left|\int_{[0,\delta]} f(x)\frac{\sin \pi xy}{\pi x}dx\right| < c\varepsilon.$$

Let

$$\Phi_0(x) = \int_0^x \frac{\sin \pi\xi}{\pi\xi}d\xi.$$

Then $\Phi_0$ is a bounded odd function. We shall apply Lemma A1.3 to $f$ and $\Phi$: $\Phi(x) = \Phi_0(xy)$ on the interval $[0, \delta]$. This yields

$$\left|\int_{[0,\delta]} f(x)\frac{\sin \pi xy}{\pi x}dx\right| \leqslant |f(\delta)\Phi_0(\delta y)| + \operatorname*{Sup}_{[0,\delta y]}|\Phi_0|\, V_0^{\delta}(f).$$

Since $f$ is continuous at 0 we can, given $\varepsilon > 0$, find $\delta > 0$ so that for $u$ with $|u| < \delta$ we have $|f(u)| < \varepsilon$. Recall next that we can represent $f$ as $f_1 - f_2$ where $f_1$ and $f_2$ are increasing. Suppose $f_1$ were not continuous at 0; then let us consider

$$f_1(0-) = \lim_{\substack{x\to 0\\ x<0}} f_1(x) \leqslant f_1(0) \leqslant f_1(0+) = \lim_{\substack{x\to 0\\ x>0}} f_1(x).$$

Define the new functions

$$\begin{aligned} f_1^*: f_1^*(x) &= f_1(x) - f_1(0-) && \text{if } x < 0\\ &= 0 && \text{if } x = 0\\ &= f_1(x) - f_1(0+) && \text{if } x > 0 \end{aligned}$$

and

$$f_2^* = f_1^* - f.$$

It is now easy to verify that $f_1^*$, $f_2^*$ are both increasing and are continuous at 0. It follows that $x \mapsto V_0^x(f)$ is continuous at 0, and thus we also require that for $x$: $|x| < \delta$ we have $V_0^x(f) < \varepsilon$.

The inequality above now yields (on taking the limit supremum as $y \to \infty$)

$$\limsup_{y\to\infty}\left|\int_{[0,\delta]} f(x)\frac{\sin \pi xy}{\pi x}dx\right| \leqslant 2\varepsilon \operatorname{Sup}|\Phi_0|.$$

By Theorem A1.1 we have

$$\lim_{y\to\infty}\int_{[\delta,\infty]} f(x)\frac{\sin \pi xy}{\pi x}\,dx = 0.$$

It follows from this and the analogous results for the negative half-line that

$$\limsup_{y\to\infty}\left|\int f(x)\frac{\sin \pi xy}{\pi x}\,dx\right| \leqslant 4\varepsilon \operatorname{Sup}|\Phi_0|.$$

Since the left-hand side is independent of $\varepsilon$ it follows that

$$\lim_{y\to\infty}\int f(x)\frac{\sin \pi xy}{\pi x}\,dx = 0.$$

Thus we have proved (ii) in the case that $f(0) = 0$. Let now $f_0$: $f(0)(x) = 1$ if $|x| \leqslant 1, f_0(x) = 0$ if $|x| > 1$. Then $f_0$ satisfies the conditions of the theorem Let us assume that we can verify the theorem for $f_0$. Then $f - f(0)f_0$ also satisfies the conditions of the theorem and $\{f - f(0)f_0\}(0) = 0$. Thus we can apply the two results we have and the linearity of the statement to derive this statement in general.

To verify the statement in this special case we note that

$$\int f_0(x)\frac{\sin \pi xy}{\pi x}\,dx = \int_{-1}^{1}\frac{\sin \pi xy}{\pi x}\,dx = \int_{-y}^{y}\frac{\sin \pi x}{\pi x}\,dx$$

and, by definition of $\gamma$ this converges to $\gamma$ as $y\to\infty$. This completes the proof of the theorem.

## A1.5

The results of the preceding paragraphs suffice to derive the main results of Fourier Theory. Let us first consider the case of Fourier series. We regard a function on $\mathbf{R}$ which satisfies $f(x+n) = f(x)$ for $x\in\mathbf{R}$, $n\in\mathbf{Z}$ as a function on the quotient group $\mathbf{R}/\mathbf{Z}$ and this is endowed with Lebesgue measure. Thus we can speak of $L^p(\mathbf{R}/\mathbf{Z})$, $C^r(\mathbf{R}/\mathbf{Z})$ (the space of $r$-times continuously differentiable functions) and so on. We can also speak of functions of bounded variation on $\mathbf{R}/\mathbf{Z}$.

Let $f\in L^1(\mathbf{R}/\mathbf{Z})$ and define

$$\hat{f}(n) = \int_{\mathbf{R}/\mathbf{Z}} f(x)e^{-2\pi inx}\,dx.$$

We can now formulate one of the central results of Fourier theory in a form due to Dirichlet.

**Theorem:** *Let $f\in L^1(\mathbf{R}/\mathbf{Z})$. Suppose that $f$ is continuous at $x$ and of bounded*

*variation in a neighbourhood of x; then*

$$\lim_{N\to\infty}\sum_{|n|\leqslant N}\hat{f}(n)e^{2\pi inx}=f(x).$$

*Proof*: We can write the right-hand side here as

$$\lim_{N\to\infty}\sum_{|n|\leqslant N}\int_{\mathbf{R}\,\mathbf{Z}}f(y)e^{2\pi in(x-y)}\,dy.$$

The finite sum and the integral can be exchanged, and the sum, which is a geometric series, evaluated. We then obtain:

$$\lim_{N\to\infty}\int_{\mathbf{R}\,\mathbf{Z}}f(y)\frac{\sin\{(2N+1)\pi(x-y)\}}{\sin\pi(x-y)}\,dy.$$

Since $[-1/2,\ 1/2[$ is a fundamental domain for $\mathbf{Z}$ on $\mathbf{R}$ we can rewrite this in the form (with $u=y-x$)

$$\lim_{N\to\infty}\int_{-1/2}^{1/2}f(x+u)\frac{\pi u}{\sin\pi u}\frac{\sin\{(2N+1)\pi u\}}{\pi u}\,du.$$

We can now apply Theorem A1.4 since $\pi u/\sin\pi u$ is increasing and we see that this is equal to $\gamma f(x)$. If $f=1$ then $\hat{f}(n)=0$ $(n\neq 0)$, $\hat{f}(0)=1$ so that we must have $\gamma=1$. This proves the theorem.

*Remark*: Note that we have also proved that $\gamma=1$.

## A1.6

The theorem of the last paragraph is clearly most useful when $\sum\hat{f}(n)e^{2\pi inx}$ is absolutely convergent. Although it is difficult to give necessary and sufficient conditions in terms of $f$ itself there is one condition which is frequently used and which is worth recording here.

**Proposition:** *Suppose* $f\in C^r(\mathbf{R}/\mathbf{Z})$. *Then*

$$\hat{f}(n)=o((1+|n|)^{-r}).$$

*Proof*: It is clear that $\hat{f}$ is bounded. By $r$-fold integration by parts we have

$$\hat{f}(n)=(2\pi in)^{-r}\int_{\mathbf{R}/\mathbf{Z}}f^{(r)}(x)e^{-2\pi inx}\,dx$$

and the assertion follows from the Riemann–Lebesgue Lemma (Theorem A1.1).

## A1.7

If $f\in L^2(\mathbf{R}/\mathbf{Z})$ then by the Cauchy–Schwarz inequality $f\in L^1(\mathbf{R}/\mathbf{Z})$ and so

$\hat{f}$ exists. It is now straightforward to check that

$$\int |f(x)|^2 \mathrm{d}x = \int_{\mathbf{R}/\mathbf{Z}} \sum_{m,n} \hat{f}(m)\overline{\hat{f}(n)}\, \mathrm{e}^{2\pi \mathrm{i}(m-1)x} \mathrm{d}x$$
$$= \sum_{n \in \mathbf{Z}} |\hat{f}(n)|^2$$

if the Fourier series of $f$ is absolutely convergent. This condition is satisfied on a dense subspace of $L^2(\mathbf{R}/\mathbf{Z})$ – one can prove using the Weierstrass Approximation Theorem that the trigonometric polynomials form a dense subspace—and it follows that $f \mapsto \hat{f}$ extends to an isometry of $L^2(\mathbf{R}/\mathbf{Z})$ onto $l^2$, where $l^2$ is the Hilbert space of functions $a: \mathbf{Z} \to \mathbf{C}$ satisfying $\sum_{j \in \mathbf{Z}} |a(j)|^2 < \infty$ and with the norm $\|a\| = \{\sum_{j \in \mathbf{Z}} |a(j)|^2\}^{1/2}$.

## A1.8

The technique of A1.5 can be carried over to the case of Fourier integrals without any difficulty. For $f \in L^1(\mathbf{R})$ we define

$$\hat{f}(y) = \int f(x) \mathrm{e}^{-2\pi \mathrm{i}xy} \mathrm{d}x.$$

**Theorem:** *Suppose that $f \in L^1(\mathbf{R})$ and that $x \in \mathbf{R}$. Suppose that $f$ is of bounded variation on a neighbourhood of $x$ and that $f$ is continuous at $x$. Then*

$$\lim_{Y \to \infty} \int_{[-Y,Y]} \hat{f}(y) \mathrm{e}^{2\pi \mathrm{i}xy} \mathrm{d}y = f(x).$$

## A1.9

Analogous to A1.6 one has:

**Proposition:** *If $f \in C^r(\mathbf{R})$ with $f^{(j)} \in L^1(\mathbf{R})$ for $1 \leqslant j \leqslant r$ then*

$$\hat{f}(y) = o((1 + |y|)^{-r}).$$

## A1.10

With a little more work one can carry over the results of A1.7 to the case of Fourier integrals. One verifies by direct computation that for $f$ of the form $\sum_{i=1}^{N} a_i f_i$, where $f_i$ is the characteristic function of an interval, one has $\|f\|^2 = \|\hat{f}\|^2$ where $\|\quad\|$ denotes the norm on $L^2(\mathbf{R})$. From this one deduces Parseval's theorem, namely that $f \mapsto \hat{f}$ extends to an isometry of $L^2(\mathbf{R})$ onto itself. We shall not go into this further here but the reader interested in constructing a proof is referred to the Exercises A1.14 and A1.15.

## A1.11

For $f_1, f_2 \in L^1(\mathbf{R}/\mathbf{Z})$ we can define the convolution $f_1 * f_2$ by

$$f_1 * f_2(x) = \int_{\mathbf{R}\,\mathbf{Z}} f_1(x-y) f_2(y)\,\mathrm{d}y.$$

It follows from Fubini's Theorem that $f_1 * f_2$ exists as an element of $L^1(\mathbf{R}/\mathbf{Z})$ and that

$$(f_1 * f_2)\hat{} = f_1\hat{}\, f_2\hat{}.$$

Note that with $*$ as multiplication $L^1(\mathbf{R}/\mathbf{Z})$ becomes a commutative Banach algebra.

## A1.12

It is rather more significant for us, especially in connection with the Mellin transform, that the analogues of the theory recalled in A1.11 are also valid for Fourier integrals. In this case we define the convolution of $f_1, f_2 \in L^1(\mathbf{R})$ to be $f_1 * f_2$ where

$$f_1 * f_2 = \int_{\mathbf{R}} f_1(x-y) f_2(y)\mathrm{d}y.$$

Again this exists as an element of $L^1(\mathbf{R})$ and one has

$$(f_1 * f_2)\hat{} = f_1\hat{}\, f_2\hat{}.$$

## A1.13

The final result that we need is the Poisson Summation Formula which we formulate below.

**Theorem:** *Suppose that* $f: \mathbf{R} \to \mathbf{R}$ *is an integrable continuous function of bounded total variation. Suppose that there exists a second function* $g: \mathbf{R} \to \mathbf{R}_+$ *so that* $g(x) = g(-x)$ *and that* $g$ *is decreasing on* $\mathbf{R}_+$ *and is integrable. Suppose that for all* $x \in \mathbf{R}$ *one has* $|f(x)| \leqslant g(x)$. *Then one has*

$$\sum_{n \in \mathbf{Z}} f(x+n) = \lim_{M \to \infty} \sum_{m:|m| \leqslant M} f\hat{}(m) \mathrm{e}^{2\pi i m x}.$$

*Proof*: By Cauchy's criterion and the Weierstrass criterion the left-hand side converges to a continuous function of $x$; also it follows from the fact that $f$ is of bounded total variation that this function is also of bounded variation Moreover this function is clearly periodic and is therefore in $L^1(\mathbf{R}/\mathbf{Z})$. We can compute its Fourier coefficients as follows: if

$$F(x) = \sum_{n \in \mathbf{Z}} f(x+n)$$

then

$$\begin{aligned}
\hat{F}(m) &= \int_{\mathbf{R}\,\mathbf{Z}} F(x)e^{-2\pi imx}\,dx \\
&= \int_{[0,1[} F(x)e^{-2\pi imx}\,dx \\
&= \int_{[0,1[} \sum_{n\in Z} f(x+n)e^{-2\pi imx}\,dx \\
&= \sum_{n\in Z} \int_{[0,1[} f(x+n)e^{-2\pi inx}\,dx \\
&= \sum_{n\in Z} \int_{[n,n-1[} f(x)e^{-2\pi imx}\,dx \\
&= \int_{\mathbf{R}} f(x)e^{-2\pi imx}\,dx \\
&= \hat{f}(m)
\end{aligned}$$

where we have made use of the Lebesgue Dominated Convergence Theorem in passing from the third to the fourth line. The statement of the theorem now follows by applying Dirichlet's Theorem (Theorem A1.5).

In general one will apply Theorem A1.13 when the right-hand side of the equation is absolutely convergent. If this is so then the formula takes on the simpler form:

$$\sum_{n\in\mathbf{Z}} f(x+n) = \sum_{m\in\mathbf{Z}} \hat{f}(m)e^{2\pi imx}.$$

In particular the choice $x = 0$ gives

$$\sum_{n\in\mathbf{Z}} f(n) = \sum_{m\in\mathbf{Z}} \hat{f}(m).$$

## Exercises

A1.1 Show that if $f$ is differentiable on $[a, b]$ with $f'$ bounded then $f$ is of bounded variation.

A1.2 Find an example of a function that is differentiable a.e. on [0, 1] with $f'$ Lebesgue integrable but $f$ not of bounded variation.

A1.3 Show that the product of two functions of bounded variation is also of bounded variation.

A1.4 Let $A \geqslant 2$ and let $f_A$ be the function periodic under $\mathbf{Z}$ so that

$$\begin{aligned}
f_A(x) &= (1 - A|x|)^2 && \text{if } |x| \leqslant A^{-1} \\
&= 0 && \text{if } A^{-1} < |x| \leqslant 1/2.
\end{aligned}$$

Show that

$$\begin{aligned} \hat{f_A} &= 2/3A && \text{if } n = 0 \\ &= (\pi n)^{-2}A - A^2(2\pi^3 n^3)^{-1}\sin(2\pi n/A) && \text{if } n \neq 0 \end{aligned}$$

A1.5 Show that

$$1^{-3} - 3^{-3} + 5^{-3} - 7^{-3} + \cdots = \pi^3/32.$$

A1.6 Show that for $n \in \mathbf{N}$

$$\sum_{j \geqslant 1} j^{-2}\{\sin(\pi j/n)\}^2 = \pi^2(n-1)/2n^2.$$

Deduce that

$$\int_0^{\infty} \left\{\frac{(\sin \pi x)}{\pi x}\right\}^2 \mathrm{d}x = 1/2.$$

A1.7 Let $f \in L^1(\mathbf{R}/\mathbf{Z})$ and let $s_n(f)$ be defined by

$$s_n(f)(x) = \sum_{j:|j| \leqslant n} \hat{f}(j)\mathrm{e}^{2\pi \mathrm{i} j x}.$$

Show that

$$N^{-1} \sum_{0 \leqslant n < N} s_n(f)(x) = \int_{\mathbf{R}\,\mathbf{Z}} K_N(x, y) f(y) \mathrm{d}y$$

where

$$K_N(x, y) = (\sin\{N\pi(x-y)\}/\sin\{\pi(x-y)\})^2/N.$$

Deduce that if $f$ is continuous at $x$ then

$$N^{-1} \sum_{0 \leqslant n < N} s_n(f)(x) \to f(x).$$

A1.8 Show that if $f \in C^1(\mathbf{R}/\mathbf{Z})$ (i.e. $f$ is continuously differentiable anc periodic) then

$$\sum_{n \in \mathbf{Z}} |\hat{f}(n)| < \infty.$$

(Hint: Use the Cauchy–Schwarz inequality.)

A1.9 Suppose that $f_1$ and $f_2$ are characteristic functions of intervals Verify that

$$\int f_1 \bar{f}_2 \, \mathrm{d}x = \int \hat{f_1}\, \bar{\hat{f}}_2 \, \mathrm{d}x.$$

A1.10 Let $f \in L^1(\mathbf{R}/\mathbf{Z})$ and let

$$R(f, x) = \hat{f}(0)x^2/2 - \sum_{n \neq 0} (2\pi n)^{-2} \hat{f}(n)\mathrm{e}^{2\pi \mathrm{i} n x}.$$

Let

$$R^*(f, x, h) = \{R(f, x+h) + R(f, x-h) - 2R(f, x)\}/h^2$$

Show that

$$R^*(f,x,h) = \hat{f}(0) + \sum_{n\neq 0} \hat{f}(n)\{(\sin \pi nh)/\pi nh\}^2 \, e^{2\pi inx}$$

and deduce that $R^*(f,x,h)\cdot h \to 0$ as $h \to 0$. Show furthermore that if for some $x$ the limit of

$$\sum_{n:|n|\leqslant N} \hat{f}(n) e^{2\pi inx}$$

exists as $N \to \infty$ then $R^*(f,x,h) \to f(x)$ as $h \to 0$.

A1.11 Suppose that $(a_n)$ and $(b_n)$ are two sequences and $S$ is a finite subset of $\mathbf{R}/\mathbf{Z}$ so that the limits $\lim_{N\to\infty} \sum_{|n|\leqslant N} a_n e^{2\pi inx}$ and $\lim_{N\to\infty} \sum_{|n|\leqslant N} b_n e^{2\pi inx}$ exist and are equal for all $x \in \mathbf{R}/\mathbf{Z} - S$. Show that $a_n = b_n$ for all $n \in \mathbf{Z}$. (Hint: use the ideas of the preceding exercise.)

A1.12 Let

$$f_n(x) = \sum_{1\leqslant j\leqslant n} j^{-1} \sin 2\pi jx.$$

Show that as $n \to \infty$

$$f_n(1/2n) \to \int_0^1 \{(\sin \pi x)/x\}\, dx.$$

The integral has the value $1.852\ldots > \pi/2$. Show that for all $x$ that $f_\infty(x) = \lim_{n\to\infty} f_n(x)$ exists, that $f_n \to f_\infty$ in $L^p(\mathbf{R}/\mathbf{Z})$ for $p < \infty$ but that there exists $C > 1$ so that for all sufficiently large $n$ one has $\|f_n\|_\infty > C\|f_\infty\|_\infty$. Compare Exercise 2.9.

A1.13 Let $f$ be a function of the form $\sum_i a_i f_i$ where the sum is finite, the $a_i$ are constants and the $f_i$ are the characteristic functions of compact intervals. Verify that

$$\|f\|_2^2 = \|\hat{f}\|_2^2.$$

A1.14 Show that the set of functions considered in Exercise A1 13 is dense in $L^2(\mathbf{R})$. Deduce that $f \mapsto \hat{f}$ can be extended to an isometry of $L^2(\mathbf{R})$. Verify also that $\tilde{\hat{f}}(x) = f(-x)$.

A1.15 Prove the assertions of A1.11.

A1.16 Determine the $f \in L^1(\mathbf{R}/\mathbf{Z})$ so that $f * f = f$.

A1.17 Show that functions of the form $f_1 * f_2$ generate a dense subset in $L^1(\mathbf{R}/\mathbf{Z})$. Use this and A1.7 to prove the Poisson Summation Formula.

# *Appendix 2*

# *The Mellin transform*

## A2.1

The Mellin transform is essentially the Fourier transform transferred to the group $\mathbf{R}_+^\times$ by means of the exponential map. However, it does contain an additional level of sophistication since one also makes essential use of the fact that if the function under consideration is reasonably well behaved then the transform is analytic. More precisely let us call a function $f: \mathbf{R}_+^\times \to \mathbf{C}$ *of type* $(\alpha, \beta)$ *for real* $\alpha < \beta$ if for $\sigma: \alpha < \sigma < \beta, x \mapsto f(x)x^{\sigma-1}$ is integrable; then

$$M(f,s) = \int_{\mathbf{R}_+^\times} f(x)x^{s-1}\,\mathrm{d}x$$

exists for $\{s: \alpha < \mathrm{Re}(s) < \beta\}$ and is analytic in this strip. This is the Mellin transform of $f$. It can also be defined for $f$ defined on $\mathbf{R}^\times$ (cf. § 2.3) but this involves only formal alterations.

If we set $x = \mathrm{e}^{2\pi u}$ and $s = \sigma + it$ then we see that

$$M(f, \sigma + \mathrm{i}t) = 2\pi \int_{\mathbf{R}} \{f(\mathrm{e}^{2\pi u})\mathrm{e}^{2\pi\sigma u}\}\mathrm{e}^{2\pi i t u}\,\mathrm{d}u$$

which exhibits the Mellin transform for fixed $\sigma$ as a Fourier transform. It is then a matter of using the theory of the Fourier transform to derive what one needs concerning the Mellin transform.

## A2.2

The first result of this kind is the inverse Mellin transform, which follows on applying Theorem A1.8 to the Mellin transform. From this we see that if $f$ is of type $(\alpha, \beta)$, continuous, and $\sigma: \alpha < \sigma < \beta$ is such that $x \mapsto f(x)x^{\sigma-1}$ is of bounded total variation (on $\mathbf{R}_+^\times$) then

$$f(x) = \lim_{Y \to \alpha} (1/2\pi\mathrm{i}) \int_{\sigma - \mathrm{i}Y}^{\sigma + \mathrm{i}Y} M(f,s)\, x^{-s}\,\mathrm{d}s.$$

The condition that $x \mapsto f(x)x^{\sigma-1}$ be of bounded variation will be met for all $\sigma$: $\alpha < \sigma < \beta$ if $f'$ is continuous and of type $(\alpha, \beta)$. If furthermore $f''$ exists, is continuous and of type $(\alpha, \beta)$ then $M(f,s) = O((1 + |s|^2)^{-1})$ uniformly for $s$: $\alpha' < \mathrm{Re}(s) < \beta'$ where $\alpha < \alpha' < \beta' < \beta$. If this is so, and it is

a condition that is typical of applications, then

$$f(x) = (1/2\pi\mathrm{i}) \int_{(\sigma)} M(f,s) x^{-s}\, \mathrm{d}s$$

where $(\sigma)$ is the line $t \mapsto \sigma + \mathrm{i}t$. Note that in this case the independence of the integral from $\sigma$ follows from Cauchy's Theorem.

## A2.3

Suppose $f_1$ and $f_2$ are of type $(\alpha, \beta)$; then the convolution

$$f_1 * f_2 : f_1 * f_2(x) = \int_{\mathbf{R}_+^\times} f_1(xy^{-1}) f_2(y) y^{-1}\, \mathrm{d}y$$

exists and is also of type $(\alpha, \beta)$. From § A1.11 one has that

$$M(f_1 * f_2, s) = M(f_1, s) M(f_2, s).$$

## A2.4

Note that from § A2.2 the function $M(f,s)$ can have a meromorphic continuation to a larger region than the original strip in which it was defined. In general one can then move the line of integration into this region and this yields asymptotic estimates for $f(x)$ as $x \to 0$ or $x \to \infty$.

Note also that one can consider the Mellin transform from the opposite point of view in which a function $\psi$ on $\{s \in \mathbf{C} : \alpha < \mathrm{Re}(s) < \beta\}$ is given which is holomorphic and satisfies certain growth conditions and one defines the transform

$$x \mapsto (1/2\pi\mathrm{i}) \int_{(\sigma)} \psi(s) x^{-s}\, \mathrm{d}s.$$

This is important in analytic number-theory.

## A2.5

There is an extreme case of the Mellin transform, known as 'Perron's Formula' which is used in analytic number-theory. This is

**Theorem:** *Let* $c > 0$, $T > 1$. *Then*

$$(1/2\pi\mathrm{i}) \int_{c-\mathrm{i}T}^{c+\mathrm{i}T} x^s s^{-1}\, \mathrm{d}s = E(x) + O(\mathrm{Min}\,(x^c/T \log x, x^c)) \quad \textit{if } x \neq 1$$

$$= E(x) + O(T^{-1}) \quad \textit{if } x = 1$$

*where*

$$\begin{aligned} E(x) &= 0 && \text{if } 0 < x < 1 \\ &= 1/2 && \text{if } x = 1 \\ &= 1 && \text{if } x > 1. \end{aligned}$$

*Proof*: Suppose first that $x = 1$. Then the integral is

$$(1/2\pi\mathrm{i})\{\log(c + \mathrm{i}T) - \log(c - \mathrm{i}T)\}$$

where we can take that branch of log on $\{s \in \mathbf{C}: \mathrm{Re}(s) > 0\}$ which is real on the real axis. The real parts of the two logarithms are equal and therefore cancel; the imaginary parts are equal in magnitude but opposite in sign and so add. Thus the integral is equal to $\arg(c + \mathrm{i}T)/\pi = (1/\pi)\arctan(T/c)$ which is equal to $(1/\pi)\{\pi/2 - \arctan(c/T)\}$. As $\arctan u = O(u)$ for $u \to 0$ the result follows.

Next let us suppose that $x > 1$. Let $b < 0$ and replace by Cauchy's Theorem the line of integration by the three linear segments from $c - \mathrm{i}T$ to $b - \mathrm{i}T$, $b - \mathrm{i}T$ to $b + \mathrm{i}T$ and from $b + \mathrm{i}T$ to $c + \mathrm{i}T$. The integrand has a pole at $s = 0$ the residue of which is $1 = E(x)$. We shall estimate the integral over each of the three segments. On the two 'horizontal' segments the integral is dominated by

$$(1/2\pi)\int_b^c 2^{1/2} x^u (|u| + T)^{-1}\,\mathrm{d}u$$

which converges as $b \to -\infty$. Moreover the integral over the vertical segment tends to 0 as $b \to -\infty$. Thus the 'error term' is

$$O\left(\left\{\int_0^\infty x^{-u}(u + T)^{-1}\,\mathrm{d}u + \int_0^c x^u (u + T)^{-1}\,\mathrm{d}u\right\}\right).$$

The second integral is clearly bounded by $\int_0^c x^u T^{-1}\,\mathrm{d}u = (x^c - 1)/T \log x$. It is also bounded by $\int_0^c x^c (1 + u)^{-1}\,\mathrm{d}u = x^c \log(1 + c)$. These together show that the second integral is bounded by the quoted function. The first integral is bounded by

$$\int_0^T x^{-u} T^{-1}\,\mathrm{d}u = T^{-1}(\log x)^{-1}$$

which is also dominated by the quoted function.

The proof in the remaining case $x$: $0 < x < 1$ is similar but now one takes $b$ large and positive. The details are very close to those given above and shall not be given here.

## Exercises

A2.1 Suppose that $f$ is of type $(\alpha, \beta)$ for some $\alpha, \beta$ with $\alpha < \beta$, $\beta > 0$ and that $f$ is, in a neighbourhood of 0, the restriction of a function which

is $(n+1)$-times differentiable. Show that $M(f,s)$ has an analytic continuation as a meromorphic function to $\{s\in\mathbf{C}:\mathrm{Re}(s)>-(n+1)\}$ and that the poles of $M(f,s)$ are simple and situated at $-k$, $k\in\mathbf{N}_0$, $k\leqslant n$. Show also that the residue of $M(f,s)$ at $-k$ is $f^{(k)}(0)/k!$

A2.2 Suppose that $f$ is of type $(\alpha,\beta)$ where $\alpha<0<\beta$ and that $f'$ is integrable on compact subintervals of $\mathbf{R}_+$. Show that

$$f(x)=\int_{(0)} M(f,s)x^{-s}\,\mathrm{d}s$$

if the interval converges absolutely.

A2.3 Suppose that $f$ is of type $(\alpha,\infty)$ and that there exists $c\in\mathbf{R}$ so that in $\{s\in\mathbf{C}:\mathrm{Re}(s)>\alpha\}$ one has

$$M(f,s)=O(\mathrm{e}^{c\,\mathrm{Re}(s)}/(1+|s|)^2).$$

Show that there exists $x_0$ so that $f(x)=0$ if $x>x_0$.

A2.4 Show that a necessary and sufficient condition that $f$ be a smooth function of compact support on $\mathbf{R}_+^\times$ is that there exist $B>0$ so that for all $N>0$ there exist $A_N>0$ so that

$$|M(f,s)|<A_N\mathrm{e}^{B|\mathrm{Re}(s)|}/(1+|s|)^N.$$

A2.5 Let $f(x)=\mathrm{e}^{-(\log x)^2/u}/2u^{1/2}$. Show that

$$M(f,s)=(\pi^{1/2}/2)\mathrm{e}^{s^2u/4}.$$

A2.6 Show that for $\sigma>0$

$$\mathrm{e}^{-x}=(1/2\pi\mathrm{i})\int_{(\sigma)}\Gamma(s)x^{-s}\,\mathrm{d}s.$$

A2.7 Show that if $\mathrm{Re}(a)>\mathrm{Re}(b)$,

$$(1/2\pi\mathrm{i})\int_{(\sigma)}\Gamma(s-a)\Gamma(b-s)x^{-s}\mathrm{d}s=\Gamma(b-a)(1+x)^{a-b}x^{-a}.$$

(Hint: Use either Exercise A2.6, Cauchy's Theorem or the Euler beta-function (§ A4.2).)

A2.8 Show that if $\sigma>\mathrm{Min}\{\mathrm{Re}(a),\mathrm{Re}(b)\}$ then

$$(1/2\pi\mathrm{i})\int_{(\sigma)}\Gamma(s-a)\Gamma(s-b)x^{-s}\mathrm{d}s=x^{-b}\int_0^\infty u^{b-a-1}\mathrm{e}^{-(u+x/u)}\mathrm{d}u.$$

A2.9 Show that if $a-b\notin\mathbf{Z}$ then

$$\begin{aligned}(1/2\pi\mathrm{i})&\int_{(\sigma)}\Gamma(s-a)\Gamma(s-b)x^{-s}\mathrm{d}s\\&=x^{-a}\sum_{k\geqslant 0}\pi x^k/\{k!\,\Gamma(k+b-a+1)\sin\pi(a-b)\}\\&\quad+x^{-b}\sum_{k\geqslant 0}\pi x^k/\{k!\,\Gamma(k+a-b+1)\sin\pi(b-a)\}.\end{aligned}$$

A2.10 Show that

$$(1/2\pi i)\int_{(\sigma)} \Gamma(s)^2 x^{-s}\,ds = 2\sum_{k\geqslant 0}(1^{-1}+2^{-1}+\cdots+k^{-1}-\gamma)x^k/(k!)^2 - (\log x)\sum_{k\geqslant 0} x^k/(k!)^2.$$

A2.11 Let

$$F(x) = (1/2\pi i)\int_{(\sigma)} \Gamma(s)^2 x^{-s}\,ds$$

where $\sigma > 0$. Show that $F$ satisfies the differential equation

$$xF''(x) + F'(x) - F(x) = 0.$$

What is the value of the differential expression on the left-hand side if $\sigma < 0$?

A2.12 Show that if $\mathrm{Max}\{\mathrm{Re}(a), \mathrm{Re}(b)\} < \sigma < \mathrm{Re}(c)$ and if $\mathrm{Re}(a) + \mathrm{Re}(b) < 0$ then

$$(1/2\pi i)\int_{(\sigma)} \Gamma(s-a)\Gamma(s-b)\Gamma(c-s)x^{-s}\,ds = x^c\Gamma(c-b)\int_0^\infty (x+u)^{b-c}u^{-a-b-1}e^{-u}\,du.$$

(Hint: Use Exercise A2.7.)

# Appendix 3

# An estimate for certain integrals

## A3.1

In this appendix we prove one theorem which is used frequently in analytic number-theory and which deserves to be singled out for attention.

**Theorem:** *Suppose $F$, $G$ are real-valued functions on $a$, $b$ and that* (i) *$F$ is differentiable and* $\mathrm{Inf}_{[a,b]} F' > 0$, *and* (ii) *that $G/F'$ is monotonic. Then*

$$\left| \int_a^b e^{iF(x)} G(x)\,dx \right| \leqslant 2|(G/F')(b) - (G/F')(a)|.$$

*Proof*: Let $F^{-1}$ be the inverse function to $F$; it is increasing. By the transformation formula and $(F^{-1})'(v) = F'\{F^{-1}(v)\}^{-1}$ one has from the substitution $x = F^{-1}(v)$ that

$$\int_a^b e^{iF(x)} G(x)\,dx = \int_{F(a)}^{F(b)} e^{iv} (G/F')\{F^{-1}(v)\}\,dv.$$

Let $H$: $H(v) = (G/F')\{F^{-1}(v)\}$; this is a monotonic function We can therefore form the corresponding Lebesgue–Stiltjes integral. By integration by parts one has with $I = [F(a), F(b)]$

$$\int_I e^{iv}\,dH(v) + \int_I H(v) i e^{iv}\,dv = e^{ib} H(F(b)) - e^{ia} H(F(a)).$$

The first integral is dominated by $|\int_I dH(v)| = |H(F(b)) - H(F(a))|$. It follows easily that

$$\left| \int_I e^{iv} H(v)\,dv \right| \leqslant 2|(G/F')(b) - (G/F')(a)|$$

as $G/F'$ is monotonic. This proves the inequality.

## A3.2

In connection with this inequality it is worth noting that often it is more efficient to use a 'trivial' inequality Thus the integral in question, $\int_I e^{iv} H(v)\,dv$, is bounded by $|\int_I H(v)\,dv|$ and this is clearly bounded by

$$\{F(b) - F(a)\}\,\mathrm{Max}\,|H|.$$

Since $H$ is monotonic one has

$$\operatorname{Max}|H| = \operatorname{Max}\{|(G/F')(b)|, |(G/F')(a)|\}.$$

## A3.3

It is possible to obtain a sequence of estimates of the type of Theorem A3.1 by making the appropriate assumptions and integrating by parts several times, rather than only once. For our purposes we do not need these refinements.

## Exercises

A3.1 Show that if $Q$ is $N$-times continuously differentiable on $[a, b]$ then

$$\int_a^b e^{iv} Q(v)\,dv = Q_N(b) - Q_N(a) + i^N \int_a^b e^{iv} Q^{(N)}(v)\,dv$$

where

$$Q_N(u) = i^{-1} e^{iu}\{Q(u) + iQ'(u) + i^2 Q''(u) + \cdots + i^{N-1} Q^{(N)}(u)\}.$$

A3.2 Show that for $1 \leqslant j \leqslant n$ there exist polynomials

$$P_j^n(x_1, \ldots, x_{n+1-j})$$

with non-negative integral coefficients so that

$$(g \circ f)^{(n)} = \sum_{1 \leqslant j \leqslant n} P_j^n(f', f'', \ldots, f^{(n+1-j)}) g^{(j)} \circ f.$$

A3.3 Show that the $P_j^n$ of Exercise A3.2 have the following properties:

(i) $P_j^n(ax_1, ax_2, \ldots, ax_{n+1-j}) = a^j P_j^n(x_1, \ldots, x_{n+1-j})$

(ii) $P_j^n(ax_1, a^2 x_2, \ldots, a^{n+1-j} x_{n+1-j}) = a^n P_j^n(x_1, \ldots, x_{n+1-j})$

(iii) $x_1 P_{j-1}^n(x_1, \ldots, x_{n+2-j}) + \sum_{1 \leqslant k \leqslant n-j} x_{k+1} \dfrac{\partial P_j^n}{\partial x_k}(x_1, \ldots, x_{n+1-j})$
$= P_j^{n+1}(x_1, \ldots, x_{n+2-j}).$

A3.4 Let $p(n,j) = P_j^n(1, 1, \ldots, 1)$. Show that

$$p(n+1, j) = p(n, j-1) + jp(n, j).$$

Deduce that

$$\sum_{1 \leqslant j \leqslant n} p(n,j) \leqslant n!$$

A3.5 Show that

$$\sum_{1 \leqslant j \leqslant n} p(n,j) l!/(l-j)! = l^n$$

and deduce that

$$p(n,j) \leqslant j^n/j!$$

and that

$$p(n,j) = \sum_{0<k\leqslant j} (-1)^{j-k} k^n/\{k!\,(j-k)!\}.$$

# *Appendix 4*

# *The gamma-function*

## A4.1

The purpose of this appendix is to summarise those properties of the gamma-function which are most significant for analytic number-theory. Since it is difficult to give a single source for all these properties, proofs will also be given here.

We introduce the gamma-function as

$$\Gamma(s) = \int_0^\infty e^{-x} x^{s-1}\, dx$$

which converges if $\mathrm{Re}(s) > 0$ and defines an analytic function. By integration by parts one has

$$s\Gamma(s) = \Gamma(s+1)$$

which yields an analytic continuation of the $\Gamma$-function to a meromorphic function in **C**. Its poles are simple and located at $0, -1, -2, -3, \ldots$ and the residue at $-m$ is $(-1)^m \cdot m!$. Note that $\Gamma(m) = (m-1)!$

## A4.2

We introduce next the beta function

$$\mathrm{B}(s,t) = \int_0^1 (1-x)^{s-1} x^{t-1}\, dx$$

(here B is a capital beta) which converges if $\mathrm{Re}(s) > 0$, $\mathrm{Re}(t) > 0$ One has

$$\mathrm{B}(s,t) = \Gamma(s)\Gamma(t)/\Gamma(s+t).$$

To see this we write $\Gamma(s)\Gamma(t)$ as a double integral

$$\Gamma(s)\Gamma(t) = \int_0^\infty \int_0^\infty e^{-(x+y)} x^{s-1} y^{t-1}\, dx dy$$

by Fubini's Theorem and make the following substitution

$$u = x + y, \quad v = x/(x+y).$$

This yields

$$\int_0^\infty \int_0^1 e^{-u}(uv)^{s-1}\{u(1-v)\}^{t-1} u\, du dv = \int_0^\infty e^{-u} u^{s+t-1}\, du \int_0^1 v^{s-1}(1-v)^{t-1} dv$$
$$= \Gamma(s+t)\mathrm{B}(s,t)$$

as required.

## A4.3

We shall next prove that $\Gamma(t)^{-1}$ is represented by the convergent infinite product

$$\Gamma(t)^{-1} = te^{\gamma t} \prod_{j=1}^{\infty} (1 + t/j)e^{-t/j}$$

where $\gamma$ is Euler's Constant. This representation is a very useful one; for the moment we note only that it shows that the gamma-function has no zeros. Recall that $\gamma = \lim_{N\to\infty} (1^{-1} + 2^{-1} + \cdots + N^{-1} - \log N) = 0.5772157\ldots$

To prove it we shall use the result of A4.2. On replacing $x$ by $x/N$ where $N$ is an integer we have

$$\Gamma(s + N)\Gamma(t)/\Gamma(s + t + N) = \int_0^N (1 - x/N)^{s+N-1}(x/N)^{t-1}\,dx/N.$$

Now define

$$\begin{aligned} f_N : f_N(x) &= (1 - x/N)^N \quad (0 < x < N) \\ &= 0 \qquad\qquad\quad (x \geqslant N). \end{aligned}$$

From the construction of the exponential function one knows that this is an increasing sequence of functions (as one can easily prove by an application of the arithmetic–geometric mean inequality) and it converges to $e^{-x}$ (as one can also verify by taking logarithms). We multiply the equation above by $N^t$ and then let $N \to \infty$. By Lebesgue's Monotone Convergence Theorem and Dominated Convergence Theorem the right-hand side converges to

$$\int_0^{\infty} e^{-x}x^{t-1}\,dx = \Gamma(t).$$

It follows now that as $N \to \infty$

$$\Gamma(s + t + N)/N^t\Gamma(s + N) \to 1.$$

We apply this to the case $s = 1$ and recall that

$$\Gamma(t + N + 1) = t(t + 1)\cdots(t + N)\Gamma(t)$$

by a repeated application of the functional equation. This yields

$$t(t + 1)\cdots(t + N)/N^t N! \to \Gamma(t)^{-1}$$

as $N \to \infty$. Thus

$$\begin{aligned} \Gamma(t)^{-1} &= \lim_{N\to\infty} t(1 + t/1)(1 + t/2)\cdots(1 + t/N)N^{-t} \\ &= \lim_{N\to\infty} t(1 + t/1)e^{-t/1}(1 + t/2)e^{-t/2}\cdots(1 + t/N)^{-t/N} \\ &\quad \times e^{t(1/1 + 1/2 + \cdots + 1/N - \log N)}. \end{aligned}$$

Now $(1+t/j)e^{-t/j} = 1 + O(j^{-2})$ and so the first product converges. Thus $1/1 + 1/2 + \cdots + 1/N - \log N$ must also converge; the constant to which it converges is, by definition, Euler's Constant $\gamma$. We have thus proved the assertion.

## A4.4

By logarithmic derivation of the formula of the previous section one has the following formula

$$-\Gamma'(t)/\Gamma(t) = \gamma + t^{-1} + \sum_{j=1}^{\infty} \{(j+t)^{-1} - j^{-1}\}.$$

From this one can already appreciate the close connection of the gamma-function with the Riemann zeta-function; see also Exercises A4 1 and 4.2.

We shall derive from this the following integral representation for $\Gamma'/\Gamma$

$$\Gamma'(t)/\Gamma(t) = \int_0^{\infty} \left( \frac{e^{-u}}{u} - \frac{e^{u(1-t)}}{e^u - 1} \right) du \qquad (\mathrm{Re}(t) > 0)$$

and for $\gamma$ the representation

$$\gamma = \int_0^{\infty} \{(e^u - 1)^{-1} - u^{-1}e^{-u}\} du.$$

To prove these one first represents $(j+t)^{-1} - j^{-1}$ as

$$\int_0^{\infty} (e^{-(j+t)u} - e^{-ju})\, du$$

and so

$$\begin{aligned}
\Gamma'(t)/\Gamma(t) &= -\gamma + \lim_{N\to\infty} \sum_{1\leqslant j\leqslant N} \int_0^{\infty} (e^{-ju} - e^{-(j+t)u}) du - \int_0^{\infty} e^{-tu}\, du \\
&= -\gamma + \lim_{N\to\infty} \int_0^{\infty} \{(1-e^{-Nu})/(e^u - 1) - (e^u - e^{-Nu})e^{-tu}/(e^u - 1)\} du \\
&= -\gamma + \int_0^{\infty} (1 - e^{u(1-t)})/(e^u - 1)\, du,
\end{aligned}$$

which is valid if $\mathrm{Re}(t) > 0$. The integral representation of $\gamma$ can be obtained in the same way. Using the two together yields the representation quoted above.

## A4.5

If we use the formula

$$\log t = \int_0^{\infty} (e^{-u} - e^{-tu})u^{-1}\, du$$

obtained by integrating $z^{-1} = \int_0^\infty e^{-zu}\,du$, then we can integrate the formula of the previous section and obtain for $\mathrm{Re}(t) > 0$

$$\log\Gamma(t) = (t - 2^{-1})\log t - t + 1 + \int_0^\infty \{2^{-1} - u^{-1} + (e^u - 1)^{-1}\}e^{-tu}u^{-1}\,du$$
$$- \int_0^\infty \{2^{-1} - u^{-1} + (e^u - 1)^{-1}\}e^{-u}u^{-1}\,du.$$

Note that both sides of this equation vanish if $t = 1$ which one uses to determine the constant of integration.

## A4.6

From the last formula it follows that

$$\log\Gamma(t) = (t - 2^{-1})\log t - t + C + o(1)$$

as $\mathrm{Re}(t) \to \infty$ where

$$C = 1 - \int_0^\infty (2^{-1} - u^{-1} + (e^u - 1)^{-1})e^{-u}u^{-1}\,du.$$

This is a weak form of Stirling's Formula. One can improve this to the following estimate valid in $\{t \in \mathbf{C}: \mathrm{Re}(t) > c\}$ for any $c > 0$

$$\log\Gamma(t) = (t - 2^{-1})\log t - t + C + O(|t|^{-1}).$$

To do this we set

$$\phi(u) = \{2^{-1} - u^{-1} + (e^u - 1)^{-1}\}u^{-1}.$$

There is no difficulty in verifying that as $u \to 0$

$$\phi(u) = 1/12 + O(u^2)$$
$$\phi'(u) = O(u),$$

and that as $u \to \infty$

$$\phi(u) = (2u)^{-1} + O(u^{-2})$$
$$\phi'(u) = O(u^{-2}).$$

Using these estimates and integration by parts in the formula of the previous section we obtain the quoted estimate.

By means of the functional equation $s\Gamma(s) = \Gamma(s + 1)$ one can extend the range of validity of the formula to a region of the form

$$\{t \in \mathbf{C}: \mathrm{Re}(t) > c\} - U$$

where $c$ is any real number and $U$ is a neighbourhood of the negative real axis. As we shall see below it can be extended to a yet larger region. The argument which we shall give also shows that

$$C = 2^{-1}\log 2\pi.$$

## A4.7

Consider now the function $F$: $F(t) = \Gamma(t)\Gamma(1-t)$. From the functional equation for $\Gamma$ it follows that

$$F(t+2) = F(t).$$

The form of Stirling's Formula which we proved above suffices to determine the asymptotic behaviour of $F(t)$ as $|\mathrm{Im}(t)| \to \infty$ Indeed we have that as $|\mathrm{Im}(t)| \to \infty$

$$\begin{aligned}\log F(t) &= (t - 2^{-1})\{\log t - \log(1-t)\} + 2C - 1 + O(|t|^{-1}) \\ &= -\pi|\mathrm{Im}(t)| + 2C + O(|t|^{-1}).\end{aligned}$$

Now $(\sin \pi t)^{-1}$ has the same poles and is also periodic with period 2. It follows that $(\sin \pi t)F(t)$ is holomorphic in the plane, and has neither poles nor zeros. Moreover, as $|\mathrm{Im}(t)| \to \infty$

$$\log\{(\sin \pi t)F(t)\} = 2C - \log 2 + O(|t|^{-1}).$$

It follows that $(\sin \pi t)F(t)$ is bounded; by Liouville's Theorem it is a constant, and the only constant which would yield the asymptotic behaviour which we have just determined is $e^{2C}/2$. Thus we have shown that

$$\Gamma(t)\Gamma(1-t) = 2^{-1}e^{2C}/\sin \pi t.$$

Evaluating the residues of the pole at $t = 1$ on both sides shows that $e^{2C} = 2\pi$. As $C$ is real we have

$$C = 2^{-1}\log 2\pi.$$

The formula which we have proved is

$$\Gamma(t)\Gamma(1-t) = \pi/\sin \pi t.$$

## A4.8

The formula which we have just proved allows us to extend the validity of Stirling's Formula into the left-hand half-plane One obtains, after a little work, that for any $\varepsilon > 0$ and

$$D_\varepsilon = \{s \in \mathbf{C}: -\pi + \varepsilon < \arg s < \pi - \varepsilon\}$$

that for $t \in D_\varepsilon$

$$\log \Gamma(t) = (t - 2^{-1})\log t - t + 2^{-1}\log 2\pi + O(|t|^{-1}).$$

The implied constant depends only on $\varepsilon$. In this we have taken that branch of log on $D_\varepsilon$ which is real on the positive real axis.

## Exercises

A4.1 Show that if $|t| < 1$

$$-\Gamma'(t)/\Gamma(t) = t^{-1} + \gamma + \sum_{k \geqslant 1} \zeta(k+1)(-t)^k.$$

A4.2 Show that in a neighbourhood of 0

$$\Gamma(t) = t^{-1} + c_0 + c_1 t + c_2 t^2 + c_3 t^3 + \cdots$$

where

$$c_0 = -\gamma$$
$$c_1 = \{\gamma^2 - \zeta(2)\}/2$$
$$c_2 = -\{\gamma^3 - 3\gamma\zeta(2) + \zeta(3)\}/6$$
$$c_3 = \{\gamma^4 - 12\gamma^2\zeta(2) + 8\gamma\zeta(3) + 3\zeta(2)^2 - 6\zeta(4)\}/24$$
$$c_4 = -\{\gamma^5 - 10\gamma^3\zeta(2) + 20\gamma^2\zeta(3) + 15\gamma\zeta(2)^2 - 30\gamma\zeta(4) + 24\zeta(5)\}/120$$

Give a general expression for $c_n$.

A4.3 Show that in a neighbourhood of 0

$$\Gamma(-n+t) = (-1)^n (n!)^{-1}(t^{-1} + (1^{-1} + 2^{-1} + \cdots + n^{-1} - \gamma) + 2^{-1}\{(1^{-2} + 2^{-2} + \cdots + n^{-2} - \pi^2/6) + (1^{-1} + 2^{-1} + \cdots + n^{-1} - \gamma)^2\}t + \cdots).$$

A4.4 Deduce from Exercise A4.1 that

$$-\Gamma'(t)/\Gamma(t) = (2t)^{-1} - \log t - (2\pi i)^{-1} \int_{(\sigma)} \pi\zeta(s+1)(\sin \pi s)^{-1} t^s \, ds$$

if $-2 < \sigma < -1$ and $t \in \mathbf{C} - \{x \in \mathbf{R} : x < 0\}$.

A4.5 Deduce from Exercise A4.4 that given $\varepsilon > 0$, $N > 0$ then

$$\log \Gamma(t) = (t - 2^{-1}) \log t - t + C - \sum_{1 \leqslant i \leqslant N} \zeta(1-2i) t^{1-2i}(2i-1) + O(|t|^{-2N})$$

uniformly in $\{z \in \mathbf{C} : |\arg z| < \pi - \varepsilon\}$ as $|t| \to \infty$ where $C$ is some constant. Deduce from A4.7 that $C = 2^{-1} \log 2\pi$.

A4.6 Show that

$$\Gamma(s)\Gamma(s+1/n)\,\Gamma(s+2/n)\cdots\Gamma(s+(n-1)/n) = n^{1/2-ns}(2\pi)^{(n-1)/2}\Gamma(ns).$$

A4.7 Show that

$$\Gamma'(1/2)/\Gamma(1/2) = -\gamma - 2\log 2.$$

A4.8 Show that

$$\Gamma'(1/4)/\Gamma(1/4) = -3\log 2 - \gamma - \pi/2$$

and that

$$\Gamma'(1/3)/\Gamma(1/3) = -(3/2)\log 3 - \gamma - \pi/2\sqrt{3}.$$

A4.9 Show that

$$\Gamma(s+1)^{-1}+\Gamma(s+2)^{-1}+\cdots$$
$$=\mathrm{e}(s^{-1}-1!^{-1}(s+1)^{-1}+2!^{-1}(s+2)^{-1}-\cdots)/\Gamma(s).$$

A4.10 Let $a_1, a_2, \ldots, a_r > 0$ and let $f$ be a positive measurable function on $\mathbf{R}_+$. Show that

$$\int_Q f(x_1+x_2+\cdots+x_r)x_1^{a_1-1}x_2^{a_2-1}\cdots x_r^{a_r-1}\,\mathrm{d}x_1\cdots\mathrm{d}x_r$$
$$=\Gamma(a_1)\Gamma(a_2)\cdots\Gamma(a_r)\Gamma(a_1+\cdots+a_r)^{-1}\int_{\mathbf{R}_+} f(x)x^{\Sigma a_j-1}\,\mathrm{d}x$$

where $Q=(\mathbf{R}_+)^r$.

A4.11 Prove that the volume of $\{(x_1,\ldots,x_r)\in\mathbf{R}^r \mid |x_1|^{k_1}+|x_2|^{k_2}+\cdots+|x_r|^{k_r}\leqslant X\}$ is equal to

$$2^r\{\Gamma(1+k_1^{-1})\Gamma(1+k_2^{-1})\cdots\Gamma(1+k_r^{-1})/\Gamma(1+k_1^{-1}+\cdots+k_r^{-1})\}$$
$$\times X^{k_1^{-1}+k_2^{-1}+\cdots+k_r^{-1}}$$

if the exponents $k_i$ are all positive.

(This integral plays an important role in the treatment of Waring's problem by the Hardy–Littlewood method.)

A4.12 Show that the volume of the unit sphere in $\mathbf{R}^r$ is

$$\pi^{r/2}/\Gamma(1+r/2).$$

A4.13 Show that

$$\Gamma(s)\Gamma(s+a)=\Gamma(a+1)^{-1}\sum_{i\geqslant 0}(-1)^i\{\Gamma(a-i)i!(s+i)\}^{-1}.$$

A4.14 Show that if $\mathrm{Re}(t)>1$

$$\sum_{j\geqslant 0}\mathrm{B}(s+j,t)=\mathrm{B}(s,t-1).$$

A4.15 Define

$$G(s)=(2\pi)^{s/2}\mathrm{e}^{-s(s+1)/2-\gamma\cdot s^2/2}\prod_{j\geqslant 1}\{(1+s/j)^j\mathrm{e}^{-s+s^2/2j}\}.$$

Verify that the product on the right converges and that

$$G(s)=\Gamma(s)G(s-1)$$
$$G(1)=1$$
$$G'(s)/G(s)=2^{-1}\log 2\pi+2^{-1}-s+s\Gamma'(s)/\Gamma(s)$$

Deduce that

$$\log G(-s)/G(s)=\int_0^s t\cot\pi t\,\mathrm{d}t-s\log 2\pi.$$

(This function appears in the functional equation for the Selberg zeta-function which bears the same relation to the Selberg trace formula as the standard Riemann zeta-function does to the explicit formulae of prime number theory.)

# *Appendix 5*

# *Integral functions of finite order*

## A5.1

In analytic number-theory almost all the analytic functions which are used satisfy certain growth conditions at infinity. These can be subsumed in the general condition that the functions are 'integral of finite order'. In particular this class of functions can be represented by infinite products (as we used in § 3.1) and the norms describing the growth satisfy various convexity properties (as we used in § 2.12, for more details see Appendix 6). In this appendix we shall summarise those aspects of the theory of integral functions of finite order which are useful for analytic number-theory. For the sake of completeness we shall also sketch proofs.

## A5.2

The basic theorem in all investigations of this type is Jensen's Theorem, which is stated below. It is in the same cadre as Cauchy's Theorem but is a 'real variable' theorem and it is this fact that accounts for its usefulness.

**Theorem:** *Let $r > 0$ and let $f$ be a function analytic in a neighbourhood of $\{z \in \mathbf{C}: |z| \leqslant r\}$. Suppose that $f(0) \neq 0$ and that $f$ has no zeros on $\{z \in \mathbf{C}: |z| = r\}$. Then*

$$(2\pi)^{-1} \int_0^{2\pi} \log |f(re^{it})| dt = \log f(0) + \sum_{\zeta: f(\zeta)=0} \operatorname{ord}_\zeta(f) \log (r/|\zeta|).$$

Here $\operatorname{ord}_\zeta(f)$ denotes the order of the zero of $f$ at $\zeta$. One can also give a version of this theorem valid when $f(0) = 0$, but this is not necessary for our purposes.

*Proof*: Suppose first that $f$ has no zeros. Then $\log|f|$ is a harmonic function and the assertion of the theorem is simply the Mean Value Property, namely that the integral over the boundary of a circle, divided by the length of the circle, is the value of the harmonic function at the centre of the circle. This proves the assertion in this case, or at least reduces the assertion to one that will not be proved here.

Next let

$$Q(z, \zeta) = (z - \zeta)\,(r^2 - z\bar{\zeta}),$$

where we shall usually assume that $|z| \leqslant r$, $|\zeta| < r$. For $|z| = r$ one has

$$\begin{aligned} |Q(z,\zeta)| &= |z-\zeta|/|r^2 - z\bar{\zeta}| \\ &= |z-\zeta|/|z\bar{z} - z\bar{\zeta}| \\ &= |z-\zeta|/|z|\,|\bar{z}-\bar{\zeta}| \\ &= |z|^{-1}. \end{aligned}$$

Thus

$$(2\pi)^{-1}\int_0^{2\pi} \log|Q(re^{it},\zeta)|\,dt = -\log r.$$

By the first observation

$$(2\pi)^{-1}\int_0^{2\pi} \log|r^2 - re^{it}\bar{\zeta}|\,dt = 2\log r,$$

and hence we deduce that

$$(2\pi)^{-1}\int_0^{2\pi} \log|re^{it}-\zeta|\,dt = \log r.$$

Let now

$$F(z) = f(z)/\prod_{\zeta: f(\zeta)=0} (z-\zeta)^{\mathrm{ord}_\zeta(f)}.$$

Then $F$ has no zeros in the closed disc and our first case yields

$$\begin{aligned} (2\pi)^{-1}\int_0^{2\pi} \log|F(re^{it})|\,dt &= \log|F(0)| \\ &= \log|f(0)| - \sum_{f(\zeta)=0} \mathrm{ord}_\zeta(f)\log|\zeta|. \end{aligned}$$

But we also have

$$(2\pi)^{-1}\int_0^{2\pi} \log|F(re^{it})|\,dt = (2\pi)^{-1}\int_0^{2\pi} \log|f(re^{it})|\,dt - \sum_{f(\zeta)=0} \mathrm{ord}_\zeta(f)\log r$$

by the definition of $F$ and the result above. Combining these gives the assertion of the theorem.

## A5.3

We now define a function $f$ on $\mathbf{C}$ to be 'integral of order $\alpha$' if it is meromorphic with finitely many poles and if for each $\beta$: $\beta > \alpha$ we have as $|z| \to \infty$

$$f(z) = O(e^{|z|^\beta}).$$

This is not the usual definition in that poles are allowed but it is the most convenient form for the analytic theory of numbers and differs from the

usual theory only in inessential details. We have first of all the following theorem which will follow from Jensen's Theorem.

**Theorem:** (i) *Let $f$ be integral of order $\alpha$; then for any $\beta > \alpha$ we have*

$$\sum_{\zeta: f(\zeta)=0, \infty |\zeta| \leqslant r} \operatorname{ord}_\zeta(f) = O(r^\beta).$$

(ii) *For any $\beta > \alpha$ the series*

$$\sum_{\zeta: f(\zeta)=0} \operatorname{ord}_\zeta(f) |\zeta|^{-\beta}$$

*is convergent.*

*Proof*: By changing the origin if necessary we can assume, without loss of generality, that $f(0) \neq 0$. There is a polynomial $p$ such that $pf$ is entire; it is also integral of order $\alpha$ and the assertions of the theorem are unaltered if $f$ is replaced by $pf$. Thus, again without loss of generality, we can assume that $f$ is entire.

We can now apply Theorem A5.2. We obtain at once from the growth condition on $f$ and the fact that the left-hand side of the following inequality is positive, the estimate

$$\sum_{\zeta: f(\zeta)=0, \infty |\zeta| \leqslant r} \operatorname{ord}_\zeta(f) \log(r/|\zeta|) = O(r^\beta)$$

for any $\beta > \alpha$. Apply this with $r$ replaced by $2r$ and then restrict the sum to that over $\zeta$ with $|\zeta| \leqslant r$. This yields the first estimate of the theorem For the second we note that from the estimate that we have just proved it follows that

$$\sum_{\substack{\zeta: f(\zeta)=0 \\ 2^k < |\zeta| \leqslant 2^{k+1}}} \operatorname{ord}_\zeta(f) = O(2^{k\beta}).$$

Thus the series in question is dominated by $\sum_k 2^{(\beta' - \beta)k}$ for any $\beta' > \alpha$. By choosing $\beta'$ so that $\alpha < \beta' < \beta$ we obtain the convergence of the series in question.

## A5.4

We now come to the other central ingredient of the theory of integral functions of finite order. This is the following result, which will effectively reduce the study of such functions to the study of their zeros.

**Theorem:** *Suppose $f$ is an integral function of order $\alpha$ which has neither poles or zeros. Then there exists a polynomial $g$ of degree not exceeding $\alpha$ so that $f = e^g$.*

*Proof*: Let $g = \log f$. This is an analytic function defined in $\mathbf{C}$ satisfying $\operatorname{Max}\{0, \operatorname{Re} g(z)\} = O(|z|^{\beta})$ as $|z| \to \infty$ and $\beta > \alpha$. We have to show that $g$ is a polynomial. Let us write

$$g(z) = \sum_{m \geqslant 0} (a_m + \mathrm{i}b_m) z^m$$

and so

$$\operatorname{Re}\{g(re^{it})\} = \sum_{m \geqslant 0} (a_m \cos mt + b_m \sin mt) r^m.$$

Multiply this first by $1 + \cos mt$ and integrate over $(0, 2\pi)$. The left-hand side is bounded above by $O(r^{\beta})$ and the right-hand side gives $\pi(2a_0 + a_m r^m)$. Thus $a_m \leqslant cr^{\beta - m}$ and so if $m > \beta$ we deduce that $a_m \leqslant 0$. Multiply the equation above now by $1 - \cos mt$ and proceed as before. We obtain $-a_m \leqslant 0$ and together these yield $a_m = 0$ for $m > \beta$. An analogous argument shows that $b_m = 0$ if $m > \alpha$. This completes the proof of the theorem.

## A5.5

We come now to the fundamental representation theorem for integral functions of finite order. This shows that such functions behave rather like polynomials. This theorem is due to Hadamard and is closely related to a similar theorem for a more general class of functions due to Weierstrass.

**Theorem:** *Let $f$ be an integral function of order $\alpha$. Let $m = [\alpha]$ and let $g_\zeta(z) = (z/\zeta) + 2^{-1}(z/\zeta)^2 + \cdots + m^{-1}(z/\zeta)^m$ There exists a polynomial $G$ of degree not exceeding $m$ such that*

$$f(z) = z^{\operatorname{ord}_0(f)} e^{G(z)} \prod_{\substack{\zeta : f(\zeta) = 0, \infty \\ \zeta \neq 0}} \{(1 - z/\zeta) e^{g_\zeta(z)}\}^{\operatorname{ord}_\zeta(f)}$$

*and the infinite product is absolutely convergent.*

*Proof*: We shall merely sketch this proof as the details are not particularly important here. Let

$$F(z) = z^{\operatorname{ord}_0(f)} \prod_{\substack{\zeta : f(\zeta) = 0, \infty \\ \zeta \neq 0}} \{(1 - z/\zeta) e^{g_\zeta(z)}\}^{\operatorname{ord}_\zeta(f)}$$

By Theorem A5.3 there is a sequence $(r_k)$, $2^k < r_k \leqslant 2^{k+1}$ $(k \geqslant 1)$ and a constant $c > 0$ such that

$$|r_k - |\zeta|| \geqslant c r_k^{(1 - \beta)} \qquad \text{if } f(\zeta) = 0, \infty$$

where $\beta$ is as in Theorem A5.3. A direct but somewhat tedious estimate

shows that there exist constants $c', c'' > 0$ such that if $|z| = r_k$ then $|F(z)| > c' e^{-c'' r_k^\beta}$. This estimate uses only the properties of the sequence of zeros described in Theorem A5.3. From this it follows that $f(z)/F(z)$ has no zeros or poles and that there exist constants $c^*, c^{**}$ such that if $|z| = r_k$ then

$$|f(z)/F(z)| \leqslant c^* e^{c^{**} r_k}.$$

By the Maximum Principle and the definition of the sequence $r_k$, which satisfies $4^{-1} \leqslant r_{k-1}/r_k \leqslant 1$, one has

$$f(z)/F(z) = O(e^{c^{**} 4|z|^\beta})$$

in the entire plane. We can now apply Theorem A5.4 and we see that there exists a polynomial $G$ of degree at most $\beta$ such that $f(z)/F(z) = e^{G(z)}$. From this the assertion follows directly.

## A5.6

We shall need in the next appendix a certain extension of the Maximum Principle in non-compact regions. For $\sigma \in \mathbf{R}$ we let, as before, $(\sigma) = \{z \in \mathbf{C}: \mathrm{Re}(z) = \sigma\}$.

**Theorem**: *Let $f$ be an integral function of finite order. Let $\sigma_1 < \sigma_2$ be real, and suppose that $f$ is bounded on $(\sigma_1)$ and $(\sigma_2)$. Then $f$ is bounded on* $\{z \in \mathbf{C}: \sigma_1 \leqslant \mathrm{Re}(z) \leqslant \sigma_2\}$.

*Proof*: We suppose that $f$ is bounded on $(\sigma_1)$ and $(\sigma_2)$ by 1. By a harmless change of coordinates we can assume that $\sigma_1 = -\pi/4$ and $\sigma_2 = \pi/4$. Also by multiplying $f$ by a suitable rational function of degree 0 we may assume that $f$ is entire in the strip. Then for $\varepsilon > 0$ we consider the function

$$g_\varepsilon(z) = f(z) e^{-\varepsilon \cos z}.$$

Suppose that $c_1 > 0$ and $\beta$ are such that

$$|f(z)| \leqslant c_1 e^{|z|^\beta}.$$

Then we have

$$|g_\varepsilon(z)| \leqslant c_1 e^{|z|^\beta - \varepsilon \cos \mathrm{Re}(z) \cosh \mathrm{Im}(z)}.$$

Thus there exists $c^* > 0$ such that if $-\pi/4 \leqslant \mathrm{Re}(z) \leqslant \pi/4$, $|\mathrm{Im}(z)| > c^*$

$$|g_\varepsilon(z)| \leqslant e^{-|\mathrm{Im}(z)|^\beta / 2}.$$

On $(-\pi/4)$ and $(\pi/4)$ we have $|g_\varepsilon(z)| \leqslant 1$. Thus on the boundary of $\{z \in \mathbf{C}: |\mathrm{Re}(z)| \leqslant \pi/4, |\mathrm{Im}(z)| \leqslant Y\}$ the function $g_\varepsilon$ is bounded by 1 when $Y$ is large enough. By the Maximum Principle it follows that $g_\varepsilon$ is bounded by 1 in any such rectangle, and hence also in the strip $\{z \in \mathbf{C}: -\pi/4 \leqslant$

$\mathrm{Re}(z) \leqslant \pi/4\}$. This does not depend on the choice of $\varepsilon$ and so we obtain that $f$ is bounded by 1 in this strip.

## A5.7

It is worth noting that we have proved a little more than was stated in formulation of Theorem A5.6. We have shown that if $|f| \leqslant 1$ on the two lines $(\sigma_1)$ and $(\sigma_2)$ then it is bounded by 1 in the whole strip. We can modify this a little in that we let $M_1 = \mathrm{Sup}_{(\sigma_1)}|f|, M_2 = \mathrm{Sup}_{(\sigma_2)}|f|$ and consider the function

$$f^*: f^*(z) = f(z)\mathrm{e}^{tz}$$

where

$$t = -\log(M_1/M_2)/(\sigma_1 - \sigma_2).$$

Then the supremum of $f^*$ on $(\sigma_1)$ is $M_1^{-\sigma_2/(\sigma_1-\sigma_2)}M_2^{\sigma_1/(\sigma_1-\sigma_2)}$, and this is also the supremum on $(\sigma_2)$. Thus, under the assumption that $f$ is entire in the strip we see that $f^*$ is bounded by this quantity in the strip. This gives the convexity theorem.

**Theorem**: *Suppose that $f$ is an integral function of finite order entire in the strip $\{z \in \mathbf{C}: \sigma_1 \leqslant \mathrm{Re}(z) \leqslant \sigma_2\}$. Suppose that $M_j = \mathrm{Sup}_{(\sigma_j)}|f|$ exists for $j = 1, 2$. Then for $z$ satisfying $\sigma_1 \leqslant \mathrm{Re}(z) \leqslant \sigma_2$ we have*

$$|f(z)| \leqslant M_1^{\{\mathrm{Re}(z)-\sigma_2\}/(\sigma_1-\sigma_2)}M_2^{\{\sigma_2-\mathrm{Re}(z)\}/(\sigma_1-\sigma_2)}.$$

This theorem can be reformulated saying that the function $\sigma \mapsto \log \mathrm{Sup}_{(\sigma)} f$ is convex. One can also regard this as an 'extreme' case of Hadamard's 'Three Circles Theorem' which states that

$$r \mapsto \log \mathrm{Sup}_{C_r}|f|,$$

with $C_r = \{z \in \mathbf{C}: |z| = r\}$, is convex.

## A5.8

The idea used in the proof of Theorem A5.7 can be modified to give other estimates. The most useful from our point of view is the Phragmén–Lindelöf Theorem. The standard proof of this uses $\Gamma(az + b)^{-1}$ in place of $\mathrm{e}^{tz}$ where $a$ and $b$ are suitable constants. In carrying out the proof one has to use Stirling's Theorem and here the constants involved are not under control. This is often a great disadvantage for numerical work, and it is possible to get around this by constructing a function with similar properties to $\Gamma(az + b)^{-1}$ but where all the estimates are under control. This construction uses ideas deriving from the proof of the Riemann

Mapping Theorem. Thus we shall quote without proof this sharper version of the Phragmén–Lindelöf Theorem and refer the reader to Rademacher (1), § 33, p. 66 for a proof.

**Theorem:** *Let $f$ be integral of finite order and suppose that $Q \leqslant \sigma_1 < \sigma_2$ are parameters. Suppose that $a_1 > a_2$ and that*

$$M_j = \sup_{\mathrm{Re}(z)=\sigma_j} |f(z)||Q+z|^{-a_j} \qquad (j=1,2)$$

*exist. Then if $\sigma_1 \leqslant \mathrm{Re}(z) \leqslant \sigma_2$ we have*

$$|f(z)| \leqslant M_1^{\{\sigma_2-\mathrm{Re}(z)\}/(\sigma_2-\sigma_1)} M_2^{\{\mathrm{Re}(z)-\sigma_1\}/(\sigma_2-\sigma_1)} |Q+z|^{(a_1\{\sigma_2-\mathrm{Re}(z)\}+a_2\{\mathrm{Re}(z)-\sigma_1\})/(\sigma_2-\sigma_1)}.$$

The conditions given here on $f$ are actually stronger than are strictly necessary (*loc. cit.*) but from the point of view of analytic number-theory this seems to be the most convenient form.

An interesting consequence of this theorem is that one can also apply it to the gamma-function itself and one obtains effective constants in the estimates for this function which one does not obtain from Stirling's Theorem. See Exercises A5.5–5.7.

## A5.9

One almost immediate consequence of Theorem A5.8 is the following convexity theorem.

**Theorem**: *Suppose $f$ is an integral function of finite order and define*

$$\mu(\sigma) = \limsup_{\substack{|\mathrm{Im}(z)|\to\infty \\ \mathrm{Re}(z)=\sigma}} \log|f(z)|/\log|z|.$$

*Suppose that $\mu(\sigma_1)$ and $\mu(\sigma_2)$ are finite. Then $\mu(\sigma)$ is finite for $\sigma_1 \leqslant \sigma \leqslant \sigma_2$ and on this interval $\mu$ is a convex function.*

## A5.10

There are many other convexity theorems for functions of finite order which are of importance for analytic number-theory. These concern the asymptotic behaviour of such expressions as

$$\left\{\int_0^T |f(\sigma+it)|^p dt\right\}^{1/p}$$

so that the results described above can be considered as the case $p=\infty$ of these estimates. We shall not go into this here but refer the reader to the following works for more details: Prachar (1), Titchmarsh (2).

## Exercises

A5.1 Show that $\sum_{n\geqslant 0} z^n/(n!)^2$ represents an integral function of order 1/2.

A5.2 Let $a>0$. Construct an integral function of order $a$.

A5.3 Construct a non-polynomial integral function of order 0.

A5.4 Let $f$ be an integral function of finite order such that there exist $\sigma_1<\sigma<\sigma_2$, $a_1<a_2$ so that

$$f(\sigma_i+\mathrm{i}t)=O((1+|t|^{a_i})) \qquad \text{as } |t|\to\infty.$$

Show that for $N>2a_2$ and suitable $m,n$

$$f(z)\{\Gamma(mz+n)\cos(\pi z/2)\}^N$$

is bounded on $\{s\in\mathbf{C}|\sigma_1\leqslant \mathrm{Re}(s)\leqslant\sigma_2\}$ (without using Theorem A5.8) and, using Stirling's Formula, deduce that for $\sigma_1<\sigma<\sigma_2$

$$f(\sigma+\mathrm{i}t)=O((1+|t|^{\{a_1(\sigma_2-\sigma)+a_1\}/(\sigma_2-\sigma_1)}) \qquad \text{as } |t|\to\infty.$$

A5.5 Show that

$$|\Gamma(1/2+\mathrm{i}t)|\leqslant 2\pi \mathrm{e}^{-\pi|t|}$$

and

$$|\Gamma(1+\mathrm{i}t)|\leqslant 2\pi(1+|t|^2)^{1/2}\mathrm{e}^{-\pi|t|}.$$

A5.6 Show that if $0<c<1$, $\mathrm{Re}(s)\geqslant(1-c)/2$

$$|\Gamma(s+c)/\Gamma(s)|\leqslant|s|^c.$$

A5.7 Show that if $-1/2\leqslant\mathrm{Re}(s)\leqslant 1/2$

$$|\Gamma((1-s)/2)/\Gamma(a/2)|\leqslant|(1+s)/2|^{1/2-\mathrm{Re}(s)}.$$

# *Appendix 6*

# *Borel–Caratheodory Theorems*

## A6.1

In the main text of this book the use of such theorems as we shall briefly describe here has been avoided. For more sophisticated applications this is not possible and so for the sake of completeness two relatively simple results of this type will be given which are often useful. This type of theorem is closely related to the Maximum Principle but characteristic of the class of Borel-Caratheodory Theorems is that one assumes only a bound on the real part and deduces one on the imaginary part from this.

## A6.2

The first result of this type is the following:

**Theorem:** *Let $f$ be analytic on $\{z\in\mathbf{C}:|z-z_0|\leqslant r\}$ where $z_0\in\mathbf{C}$ and $r>0$ are given. Then one has*

$$|f(z)-f(z_0)|\leqslant 2\Big(\sup_{|w-z_0|=r}\operatorname{Re}\{f(w)-f(z_0)\}\Big)|z-z_0|/(r-|z-z_0|)$$

*Proof*: We can assume that $z_0=0$ and $f(z_0)=0$. Suppose that $f(z)=\sum_{n\geqslant 1}a_nz^n$. If we write $a_n=|a_n|e^{i\varphi_n}$ then on $|z|=r$

$$\operatorname{Re}\{f(re^{i\theta})\}=\sum_{n\geqslant 1}|a_n|\cos(n\theta+\varphi_n)r^n.$$

Taking the $n$th Fourier coefficient yields

$$|a_n|r^n=\pi^{-1}\int_0^{2\pi}\operatorname{Re}\{f(re^{i\theta})\}\cos(n\theta+\varphi_n)\,d\theta.$$

By the Mean Value Theorem for harmonic functions

$$0=\pi^{-1}\int_0^{2\pi}\operatorname{Re}\{f(re^{i\theta})\}1\,d\theta$$

and thus

$$\begin{aligned}|a_n|r^n&=\pi^{-1}\int_0^{2\pi}\operatorname{Re}\{f(re^{i\theta})\}\{1+\cos(n\theta+\varphi_n)\}\,d\theta\\&\leqslant 2\sup_{|z|=r}\operatorname{Re}\{f(z)\}.\end{aligned}$$

From this the result follows immediately as

$$|f(z)| \leqslant \sum_{n \geqslant 1} |a_n| r^n (|z|^n / r^n)$$
$$\leqslant 2 \operatorname*{Sup}_{|w|=r} \operatorname{Re}\{f(w)\} |z|/(r - |z|)$$

as asserted.

## A6.3

The second theorem is a logarithmic version.

**Theorem:** *Suppose $f$ is analytic in $\{z \in \mathbf{C}: |z - z_0| \leqslant r\}$. Let $a, b \in ]0, r[$ and suppose that $f$ has no zeros in $[z_0 - a, z_0 + b]$. Then one has*

$$|\arg f(z_0 + b) - \arg f(z_0 - a)|$$
$$\leqslant \pi + \pi \frac{\log(\operatorname{Sup}_{|w|=r} |f(w)|) - \log|\operatorname{Re}\{f(z_0)\}|)}{\log r - \log \operatorname{Max}(a, b)}.$$

*Proof*: We may suppose that $z_0 = 0$. Let $f^*(z) = \overline{f(\bar{z})}$. Let $q$ be the number of zeros of $\operatorname{Re}(f)$ on $[-a, b]$. In each of the segments between the zeros the argument varies by at most $\pi$ and we therefore have that

$$|\arg f(-a) - \arg f(b)| \leqslant (q + 1)\pi.$$

The zeros of $\operatorname{Re}(f)$ are the zeros of the analytic function $g = f + f^*$, at least on the real axis. By Jensen's Theorem

$$(2\pi)^{-1} \int_0^{2\pi} \log|g(re^{i\theta})| d\theta = \operatorname{ord}_0(g) \log r + \log|g(w)/w^{\operatorname{ord}_0(g)}\Big|_{w=0}|$$
$$+ \sum \operatorname{ord}_\zeta(g) \log r/|\zeta|;$$

see A5.2. In particular we have

$$\log(r/\rho) \sum_{\substack{\zeta: g(\zeta)=0 \\ |\zeta| \leqslant \rho}} 1 \leqslant (2\pi)^{-1} \int_0^{2\pi} \log|g(re^{i\theta})| d\theta - \log|g(w)/w^{\operatorname{ord}_0(g)}\Big|_{w=0}|$$

by ignoring the terms with $|\zeta| > \rho$ and using the fact that $\operatorname{ord}_\zeta(g) \geqslant 1$. In this case $g(0) \neq 0$ and so the second term here is $-\log(2 \operatorname{Re}\{f(0)\})$. The first term, the integral, is clearly bounded by $\log(2 \operatorname{Sup}_{|w|=r} |f(w)|)$. As the left-hand side exceeds $\log(r/\rho)q$ the quoted result follows on taking $\rho = \operatorname{Max}(a, b)$.

# *Appendix 7*

# *Littlewood's Theorem*

## A7.1

Littlewood's Theorem is an average version of the principle of the argument.

**Theorem:** *Let $a < b$, $T' < T$ and let $f$ be a function analytic in a neighbourhood of $R(a, b, T', T) = \{z \in \mathbf{C}: a \leqslant \mathrm{Re}(z) \leqslant b, T' \leqslant \mathrm{Im}(z) \leqslant T\}$. Let $N(u)$ be the number of zeros of $f$ in $R(a, b, T', T) \cap \{z \in \mathbf{C}: \mathrm{Re}(z) > u\}$. Suppose that $f$ has no zeros on the boundary of $R(a, b, T', T)$. Then*

$$2\pi \int_a^b N(u)\mathrm{d}u = \int_{T'}^{T} \{\log|f(a+\mathrm{i}t)| - \log|f(b+\mathrm{i}t)|\}\mathrm{d}t + \int_a^b \{\arg f(v+\mathrm{i}T) - \arg f(v+\mathrm{i}T')\}\mathrm{d}v$$

*where* 'arg' *is determined by continuous variation along the boundary from $b+\mathrm{i}T$ and $b+\mathrm{i}T'$ respectively.*

*Proof*: We have that

$$N(u) = (2\pi\mathrm{i})^{-1} \int_{C(u)} \{f'(z)/f(z)\}\mathrm{d}z$$

where $C(u)$ is the rectangle with the corners $b+\mathrm{i}T', b+\mathrm{i}T, u+\mathrm{i}T, u+\mathrm{i}T'$. This is the following sum

$$-(2\pi\mathrm{i})^{-1} \int_{T'}^{T} f'(u+\mathrm{i}t)/f(u+\mathrm{i}t)\mathrm{i}\mathrm{d}t + (2\pi\mathrm{i})^{-1} \int_{T'}^{T} \{f'(b+\mathrm{i}t)/f(b+\mathrm{i}t)\}\mathrm{i}\mathrm{d}t$$

$$+(2\pi\mathrm{i})^{-1} \int_u^b \{f'(v+\mathrm{i}T')/f(v+\mathrm{i}T')\}\mathrm{d}v - (2\pi\mathrm{i})^{-1} \int_u^b \{f'(v+\mathrm{i}T)/f(v+\mathrm{i}T)\}\mathrm{d}v.$$

We need only consider the real part of this. The real parts of the last two integrals together are

$$(2\pi)^{-1}(\{\arg f(b+\mathrm{i}T') - \arg f(u+\mathrm{i}T')\} - \{\arg f(b+\mathrm{i}T) - \arg f(u+\mathrm{i}T)\}).$$

Integrating this over $a, b$ gives

$$(2\pi)^{-1}(b-a)\{\arg f(b+\mathrm{i}T) - \arg f(b+\mathrm{i}T)\} + (2\pi)^{-1} \int_a^b \{\arg f(u+\mathrm{i}T) - \arg f(u+\mathrm{i}T')\}\mathrm{d}u.$$

Now we consider the first two integrals. We integrate these with respect to d$u$ over $[a,b]$ and interchange the order of integration. We obtain

$$-(2\pi)^{-1}\int_{T'}^{T}\{\log|f(b+it)|-\log|f(a+it)|\}\,dt$$

$$+(2\pi)^{-1}(b-a)\int_{T'}^{T}\{f'/f(b+it)\}\,dt$$

where the logarithms have been continued along the horizontal lines – this is defined for almost all $t$. The real part of the first integral here is

$$-(2\pi)^{-1}\int_{T'}^{T}\{\log|f(b+it)|-\log|f(a+it)|\}\,dt.$$

The last integral is equal to

$$\begin{aligned}&\mathrm{Re}((2\pi i)^{-1}(b-a)\{\log f(b+iT)-\log f(b+iT')\})\\&=(2\pi)^{-1}(b-a)\{\arg f(b+iT)-\arg f(b+iT')\}\end{aligned}$$

which cancels the analogous term above. These evaluations now give the theorem.

## A7.2

There is a development of the ideas of the previous section that is often quite useful, and is due to Backlund. We let $\log^+ = \mathrm{Max}(\log, 0)$.

**Theorem:** *Suppose that $a<b$ and $r=b-a$ and that $f$ is a function analytic on $\{z\in\mathbf{C}: a-r\leqslant \mathrm{Re}(z)\leqslant b+r\}$. Suppose also that $f(\bar{s})=\overline{f(s)}$, and that $\mathrm{Re}\{f(b+it)\}\geqslant c>0$ for all $t\in\mathbf{R}$. Let $N_T(u)$ be the number of zeros of $f$ in $\{z\in\mathbf{C}: u\leqslant \mathrm{Re}(z)\leqslant b, |\mathrm{Im}(z)|\leqslant T\}$. Then if $T>r$*

$$\int_a^b N_T(u)\,du < r+4^{-1}+(4T+1)(\log c^{-1})/2\pi$$

$$+\operatorname*{Sup}_{b':|b-b'|\leqslant r}\int_{-3T}^{3T}\log^+|f(b'+ix)|\,dx/4\pi$$

$$+\int_{-2T}^{2T}\log^+|f(a+it)|\,dt/2\pi.$$

*Proof*: In Littlewood's Theorem we shall estimate the terms involving $\arg f(v+iT)-\arg f(v-iT)$ and we shall apply the estimate to $R(a,b,-T,T)$. Then $\arg f(v+iT)-\arg f(v-iT)$ vanishes when $\mathrm{Re}\{f(v+iT)+f(v-iT)\}=0$, and is thus bounded by

$$2\pi+\pi\times\text{Number of real zeros of } g \text{ in } [v,b]$$

where

$$g(z) = f(z + iT) + f(z - iT).$$

Let us denote by $n(\rho)$ the number of zeros of $g$ in $\{z \in \mathbf{C}: |z - b| \leqslant \rho\}$. Thus we have

$$|\arg f(v + iT) - \arg f(v - iT)| \leqslant 2\pi + \pi n(b - v).$$

We shall next use Jensen's Theorem to estimate $n$. We have

$$\int_0^R n(\rho)^{-1} \mathrm{d}\rho = (2\pi)^{-1} \int_0^{2\pi} \log|g(b + Re^{i\theta})| \mathrm{d}\theta - \log|g(b)|.$$

We note that

$$\int_a^b n(b - v)\mathrm{d}v \leqslant r \int_0^r n(\rho)\rho^{-1} \mathrm{d}\rho$$

from which follows

$$\int_a^b |\arg f(v + iT) - \arg f(v - iT)| \mathrm{d}v$$
$$\leqslant 2\pi r + 2^{-1} \int_0^{2\pi} \log|g(b + re^{i\theta})| \mathrm{d}\theta - \log|g(b)|.$$

Recall that $-\log|g(b)| \leqslant -\log 2c$. We shall also use the inequality

$$\log^+(A + B) \leqslant \log 2 + \log^+ A + \log^+ B$$

from which follows

$$\log|g(z)| \leqslant \log^+|g(z)|$$
$$\leqslant \log 2 + \log^+|f(z + iT)| + \log^+|f(z - iT)|.$$

Thus from Littlewood's Theorem we obtain

$$\int_a^b N_T(u)\mathrm{d}u \leqslant r + 2^{-1}\log 2 - 2^{-1}\log 2c$$

$$+ (4\pi)^{-1}\left[\int_0^{2\pi} \log^+|f(b + iT + re^{i\theta})|\mathrm{d}\theta + \int_0^{2\pi} \log^+|f(b - iT + re^{i\theta})|\mathrm{d}\theta\right]$$

$$+ (2\pi)^{-1} \int_{-T}^{T} \{\log|f(a + it)| - \log|f(b + it)|\}\mathrm{d}t.$$

In order to bring this into a more convenient form we note that as a function of $T \mapsto N_T(u)$ is increasing

$$N_T(u) \leqslant T^{-1} \int_T^{2T} N_x(u)\mathrm{d}x.$$

Substituting this in the inequality above yields

$$\int_a^b N_T(u)\mathrm{d}u \leqslant \{r + 2^{-1}\log 2 - 2^{-1}\log 2c\}$$
$$+ (4\pi)^{-1}\int_0^{2\pi}\int_T^{2T} \log^+ |f(b + \mathrm{i}x + re^{\mathrm{i}\theta})|\mathrm{d}x\,\mathrm{d}\theta$$
$$+ (4\pi)^{-1}\int_0^{2\pi}\int_T^{2T} \log^+ |f(b - \mathrm{i}x + re^{\mathrm{i}\theta})|\mathrm{d}x\,\mathrm{d}\theta$$
$$+ (2\pi)^{-1}\int_T^{2T}\int_{-x}^{x} \log|f(a + \mathrm{i}t)|\mathrm{d}t\,\mathrm{d}x/T$$
$$- (2\pi)^{-1}\int_T^{2T}\int_{-x}^{x} \log|f(b + \mathrm{i}t)|\mathrm{d}t\,\mathrm{d}x/T.$$

The second and third terms here can be more conveniently estimated from above by

$$(4\pi)^{-1}\operatorname*{Sup}_{b'}\int_{T-r}^{2T+r} \log^+ |f(b' \pm \mathrm{i}x)|\mathrm{d}x$$

where the supremum is taken over $b'$ satisfying $|b' - b| \leqslant r$. Then as $T > r$ we can estimate the sum of these two integrals by

$$(4\pi)^{-1}\operatorname{Sup}\int_{-3T}^{3T} \log^+ |f(b' + \mathrm{i}x)|\mathrm{d}x.$$

The final two terms can be estimated by

$$(2\pi)^{-1}\int_{-2T}^{2T} \log^+ |f(a + \mathrm{i}t)|\mathrm{d}t + (2\pi)^{-1}\int_{-2T}^{2T} \log^+ |f(b + \mathrm{i}t)|^{-1}\mathrm{d}t.$$

The second is bounded by

$$(2/\pi)\,T\log c^{-1}.$$

These estimates give the assertion of the theorem since

$$2^{-1}\log 2 - (2\pi)^{-1}\log 2 < 4^{-1}.$$

# *Bibliography*


R. Bellman: (1) *A brief introduction to theta functions.* Holt, Rinehart & Winston, New York, 1961.

E. Bombieri: (1) Counting points on curves over finite fields (*d'après* S.A. Stepanov). Sém. Bourbaki, No. 430. In: *Springer Lecture Notes, 383.* Springer, Berlin, 1974.

E. Bombieri & H. Iwaniec: (1) On the order of $\zeta(\frac{1}{2}+it)$. *Ann Sc. Norm. Sup. Pisa* **XIII** (1986) 449–72.

J.W.S. Cassels: (1) Global fields. In: *Algebraic Number Theory.* Eds J.W.S. Cassels, A. Fröhlich. Academic Press, London, 1967.

K. Chandrasekharan & R. Narashiman: (1) Functional equations with multiple gamma factors and the average order of arithmetical functions. *Ann. Math.* **86** (1962) 93–136.

K. Chandrasekharan & R. Narashiman: (2) On the mean value of the error term for a class of arithmetical functions. *Acta Math.* **112** (1964) 41–67.

T. Clausen: (1) Beweis des von Herrn Schlömilch aufgestellten Lehrsatzes. *Arkiv für Math. u. Phys.* **XXX** (1858) 166–70.

J.H. Coates: (1) $p$-adic $L$-functions and Iwasawa theory. In: *Algebraic Number Fields,* Ed. A. Fröhlich. Academic Press, London, 1977.

J.H. Coates: (2) The work of Mazur and Wiles on cyclotomic fields. Sém. Bourbaki, No. 575. In: *Springer Lecture Notes, 901.* Springer, Berlin, 1981.

J.B. Conway: (1) *Functions of one complex variable.* Springer, Berlin, 1973.

H. Davenport: (1) *Multiplicative Number Theory.* 1st edn, Markham, Chicago, 1967. 2nd edn, Ed. H. Montgomery. Springer, Berlin, 1980.

H. Davenport & H. Hasse: (1) Die Nullstellen der Kongruenzzetafunktionen in gewissen zyklischen Fällen. *J. f. d. r. u. ang. Math.* **172** (1934) 151- 82.

M. Deuring: (1) Zeta functions of algebraic varieties. *J. Indian Math. Soc.* **20** (1956) 89 - 101.

V.G. Drinfeld & S.G. Vladut: (1) On the number of points of an algebraic curve. (Russian) *Funct. Anal. and Appl.* **17** (1983) 68–9. (English translation) *Funct. Anal. and Appl.* **17** (1983) 53–4.

H.M. Edwards: (1) *Riemann's Zeta Function.* Academic Press, New York, 1974.

T. Estermann: (1) *Introduction to modern prime number theory.* Cambridge University Press, Cambridge 1952.

W.L. Ferrar: (1) Summation formulae and their relation to Dirichlet series. *Compositio Math.* **1** (1935) 344 60.

I.M. Gelfand & G.E. Shilov: (1) *Generalised Functions, I.* Academic Press, New York, 1964.

A. Ghosh: (1) On Riemann's zeta function – sign changes of $S(T)$. In: *Recent Advances in Analytic Number Theory,* Vol. 1. Eds. H. Halberstam, C. Hooley. Academic Press, London, 1981.

G.H. Hardy: (1) *Divergent Series.* Oxford University Press, Oxford, 1949.

G.H. Hardy & W.W. Rogosinski: (1) *Fourier Series.* 3rd edn, Cambridge University Press, Cambridge, 1956.

G.H. Hardy & E.M. Wright: (1) *An Introduction to the Theory of Numbers.* 4th edn, Oxford University Press, Oxford, 1960.

C.B. Haselgrove: (1) The first 900 zeros of the Riemann zeta function. Royal Society Depository for Unpublished Mathematical Tables, Table No. 82, 1961.

C.B. Haselgrove & J.C.P. Miller: (1) Tables of the Riemann zeta function. *Royal Society Mathematical Tables,* Vol. 6. Cambridge University Press, Cambridge, 1960.

H. Hasse: (1) Über die Riemannsche Vermutung in Funktionenkörpern. In: *Comptes rendus d. Cong. Int. des Math., Oslo, 1936*, Tome 1. Brøggers, Oslo, 1937, 189–206.

D.R. Heath-Brown: (1) The twelfth-power moment of the Riemann zeta-function. *Quarterly J. Math.* (2) **29** (1978) 443–62.

D.R. Heath-Brown: (2) The Riemann zeta-function. In: *Journées Arithmétiques, 1980.* Ed. J.V. Armitage. Cambridge University Press, Cambridge, 1981.

Y. Ihara: (1) Non-abelian class-fields over function fields in special cases. In: *Actes d. Cong. Int. des Math., Nice,* 1970, Vol. 1. Gauthier–Villars, Paris, 1971.

Y. Ihara: (2) Some remarks of the number of rational points of algebraic curves over finite fields. *J. Fac. Sci. Tokyo* **28** (1981) 721–24.

A. Ivić: (1) *The Riemann zeta function.* J. Wiley, New York, 1985.

K. Iwasawa: (1) On $p$-adic $L$-functions. *Ann. Math.* **89** (1969) 198–205.

E. Janke & F. Emde: (1) *Tables of functions.* Dover Publications, New York, 1945.

G. Kolesnik: (1) On the order of $\zeta(\frac{1}{2}+it)$ and $\Delta(R)$. *Pacific J. Math.* **98** (1982) 107–22.

E. Landau: (1) *Handbuch der Lehre von der Verteilung der Primzahlen* (2 Vols). Teubner, Leipzig, 1909. (Reprint: Chelsea Publishing Company, New York, 1953.)

E. Landau: (2) *Vorlesungen über Zahlentheorie* (3 Vols). Hirzel, Leipzig, 1927.

N. Levinson: (1) Deduction of a semioptimal mollifier for obtaining a lower bound for $N_0(T)$ for Riemann's zeta-function. *Proc. Nat. Acad. Sci. U.S.A.* **72** (1975) 294–97.

P. Lévy: (1) Arithmétique et calcule des probabilités. *Oeuvres*, Vol. VI, No. 249. Gauthier–Villers, Paris. 1980. (= *Cong. Int. Phil. Soc. (1949)* 125–33.)

C.J. Malmstén: (1) De integralibus quibusdam definitis, seribusque infinitis. *J. f. d. r. u. ang. Math.* **38** (1849) 1–39.

J. Martinet: (1) Character theory and Artin $L$-functions. In: *Algebraic Number Fields.* Ed. A. Fröhlich. Academic Press, London, 1977, 1–87.

H. Matsumoto: (1) Représentations projectives d'un groups associées à certains accouplements invariants. *C. R. Acad. Sci. Paris* **298** (1984) 55–8.

K. Maurin: (1) *Methods of Hilbert spaces.* PWN, Warszawa, 1972.

B. Mazur & A. Wiles: (1) Class fields of abelian extensions of **Q** *Inventiones Math* **76** (1984) 179–330.

G. Polya: (1) Über trignometrische Integrale mit nur reellen Nullstellen. *J. f. d. r. u. ang. Math.* **158** (1927) 6–18.

A. van der Poorten: (1) A proof that Euler missed... Apéry's proof of the irrationality of $\zeta(3)$. An informal report. *The Mathematical Intelligencer* **1** (1979) 195–203.

K. Prachar: (1) *Primzahlverteilung.* Springer, Berlin, 1957.

H. Rademacher: (1) *Topics in Analytic Number Theory.* Springer, Berlin, 1973.

H.J.J.te Riele, J. van der Lune & D.T. Winter: (1) On the zeros of the Riemann zeta function in the critical strip, IV. *Math. of Computation* **46** (1986) 667–81.

B. Riemann: (1) Über die Anzahl der Primzahlen unter einer gegebenen Größe. *Monatsb. der Berliner Akad.*, 1858/60, 671–80. (= *Gesammelte Mathematische Werke.* 2nd edn, Teubner, Leipzig, 1982, No VII, 145–55.)

O. Schlömilch: (1) Lehrsatz. *Arkiv für Math. u. Phys.* **XII** (1849) 415.

A. Selberg: (1) Contributions to the theory of the Riemann zeta function. *Arch. for Math. ogNaturv. B.* **48** (1946) No. 5.

A. Selberg: (2) The zeta-function and the Riemann hypothesis. In: *10 Skand. Math. Kongress* (1946). Gjellerup, København, 1947, 187–200.

A. Selberg: (3) The general sieve method and its place in prime number theory. In: *Proc. Int. Cong. Math. 1950, Cambridge, USA*, Vol. 1. AMS, Providence, RI 1952.

A. Selberg: (4) Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces with applications to Dirichlet series. *J. Indian Math. Soc.* **20** (1956) 47–87.

J.P. Serre: (1) Sur le nombre de points rationels d'une courbe algèbrique sur un corps fini *C. R. Acad. Sci. Paris* **296** (1983) 397–402.

J.P. Serre: (2) Résumé de Cours. Annuaire du Collège de France, 1983–1984 *I, Sciences math., phys., et natur.* 81–4.

C.L. Siegel: (1) Über Riemanns *Nachlass* zur analytischen Zahlentheorie, *Quellen and Studien zur Geschichte der Mathematische, Astronomie und Physik*, **2** (1932) 45–80. (= C.L. Siegel, *Gesammelte Abhandlungen*, Vol. 1. Springer, Berlin, 1986, No. 18.)

R. Spira: (1) Tables of the Riemann zeta function. (Description of tables.) *Math. of Computation* **18** (1964) 519–21.

B. Srinivasan: (1) Representations of finite Chevalley groups. In: *Springer Lecture Notes, 764*, Springer, Berlin, 1979.

H.M. Stark: (1) On the Riemann Hypothesis in hyperelliptic function fields. In: *Proc. Symp. Pure Math.* XXIV. AMS, Providence, RI, 1973.

H.P.I. Swinnerton-Dyer: (1) Applications of algebraic geometry to number theory. In: *1969 Institute on Number Theory, Proc. Symp. Pure Math.* XX. AMS, Providence, RI, 1971.

G. Tamme: (1) Einführung in die étale Kohomologie. *Regensburger Trichter*, No. 17, Mathematics Institute, Regensburg, 1979.

J.T. Tate: (1) Fourier analysis in number fields and Hecke's zeta function. In: *Algebraic Number Theory*. Eds. J.W.S. Cassels, A. Fröhlich. Academic Press, London, 1967.

A. Terras: (1) *Harmonic Analysis on Symmetric Spaces and Applications*, Vol. 1, Springer, New York, 1985.

E.C. Titchmarsh: (1) *The Theory of Functions*. 2nd edn, Oxford University Press, Oxford, 1950.

E.C. Titchmarsh: (2) *The Theory of the Riemann Zeta Function*. Oxford University Press, Oxford, 1951.

J.F. Voloch: (1) Curves over finite fields. Ph.D. Thesis, Cambridge, 1984.

B.L. van der Waerden: (1) *Modern Algebra* (2 Vols). Ungar, New York, 1950.

A. Weil: (1) Sur les "formules explicites" de la théorie des nombres premiers. *Comm. Lund (Vol. dédié à Marcel Riesz)* 252-65. (= *Oeuvres sci.* [1952b], Vol. II. Springer, New York, 1979.)

A. Weil: (2) Sur les formules explicites de la théorie des nombres. *Izv. Akad. Nauk. (Ser. Math.)* **36** (1972) 3–18. (= *Oeuvres sci.* [1972], Vol. III. Springer, New York, 1979.)

A. Weil: (3) Remarks on Hecke's lemma and its use. In: *Algebraic Number Theory, International Symposium, Kyoto, 1976*. Ed. S. Iyanaga, Japan Society for the Promotion of Science, 1977. (= *Oeuvres sci.* [1977a], Vol. III. Springer, New York, 1979.)

A. Weil: (4) *Basic Number Theory*. 3rd edn, Springer, Berlin, 1974.

A. Weil: (5) *Courbes algébriques et variétés abéliennes*. Hermann, Paris, 1971.

E.T. Whittaker, G.N. Watson: (1) *Modern Analysis*. 4th edn, Cambridge University Press, Cambridge, 1927.

D. Zagier: (1) The first 50 million prime numbers. *The Mathematical Intelligencer* **0** (1977) 7–19.

# *Index*


algebraic geometry 76, 82–4
Apéry, R. 3
approximate functional equation 8, 23, 50, 67, 91ff
Artin, E. 75
Artin *L*-functions 82

Bernoulli numbers 3
beta function (B) 130

Cauchy, A.L. 4
Chebychev, P. 6, 7
cohomology theory 83
computations 56–7, 74
convolution 118, 123
v.d. Corput, J.G. 100
critical line 22, 67
curve
  over a finite field 75ff
  Jacobian of 83–5
  place of 76–8
  point of 78

Davenport, H. 56
Dirichlet, P.G.L. 2, 30, 113, 115
Dirichlet integral 113
distributions 18
divisor functions ($d_k, \sigma_t$) 9–10, 31–2

Eisenstein, G. 12
Euclid 2
Euler, L. 1–4
Euler constant ($\gamma$) 3, 34, 131–2
Euler function ($\varphi$) 9
Euler product 1–2, 33, 35, 38–9, 86–7
explicit formulae of prime number theory 6, 38–9, 43, 44, 79–80, 85

Ferrar, W. 25
finite field 76
Fourier integral 117
Fourier series 115, 116
Fourier transform 116, 117
Frobenius map 82, 83
function field 76

gamma function ($\Gamma$) 4, 130ff
gamma function, product representation 131
Gauss, C.F. 6, 7
Grothendieck, A. 83

Hadamard, J. 7, 31, 50, 90, 141
Hadamard product 31–5
Hardy, G.H. 8, 75
Hasse, H. 76
Heath-Brown, D.R. 64, 110

integral function 138–39
integral function, order of 139
integral means of the zeta-function 59–64, 68–9, 97
Iwasawa theory 4, 58

Legendre, A.M. 7
Legendre–Jacobi symbol 109
Lindelöf Hypothesis 67–72
Littlewood, J.E. 8, 59–60
local functional equation 18
logarithmic integral (*Li*) 6

v. Mangoldt, H. 7, 55
v. Mangoldt function (A) 9
Matsumoto, H. 86
maximum principle, extended 142
Mellin transform 6, 15, 122ff
Möbius function ($\mu$) 9, 49
multi-set 33

numbers, *l*-adic 83

partial summation 53
Perron's Formula 44, 123
prime numbers 1, 2, 5–9, 38, 44, 51–3, 72–5

Ramanujan, S. 49
Riemann, B. 4–9, 22, 55, 91
Riemann Hypothesis 5, 7, 67, 70, 72–6, 80–2, 86–7
Riemann Hypothesis, Weil's criterion 80–82
Riemann–Lebesgue lemma 111
Riemann–Siegel Formula 91

Schmidt, F.K. 76
Selberg, A. 22, 48, 79
Selberg Trace Formula 79, 137

set with multiplications 33
Siegel, C.L. 8, 9, 91
Stepanov, S.A. 76
Stirling's Formula 22, 133, 134
Summation Formula
  Euler–Maclaurin 30
  Poisson 14, 25 6, 108, 118–19
  Voronoi 14, 25, 32

Tate's thesis 18
Theorems
  Backlund 149–51
  Bohr–Landau 59
  Davenport–Hasse 89
  Hadamard's 'Three Circles' 71, 143
  Hardy–Selberg 75
  Jensen 70, 138
  I ittlewood 60, 148
  Phragmén–Lindelöf 23, 68, 143, 144
  Plancherel 117
  Prime Number 6, 7, 51–3
  Riemann Mapping 144
  Riemann–Roch 79
  Wiener–Ikehara 51
type of a function 15, 122

variation 111–13
  function of bounded 111
  function of total bounded 111
Vinogradov, I.M. 8, 65

Weierstrass, K. 141
Weil, A. 76, 78, 80, 82, 84
Weyl, H. 102

zero set of Riemann zeta-functic
    5, 33, 50
zeta-function
  functional equation 4, 19
  integral representation 17, 23,
  of curve over a finite field 78
  *p*-adic 4, 85
  Riemann (ζ) 1ff.